

Locality at the Interface

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Note on this version

This thesis is presented substantially as it was defended in December 2025. Since its completion, my thinking has continued to develop, and I have begun to reconsider some of the assumptions pursued here - particularly those concerning the proposed Interface system and its implications for broader questions about the architecture of the language faculty. I therefore see the thesis not as an endpoint, but as one stage in a continuing investigation of questions that remain central to my research.

If you notice any errors, or have comments, questions, or requests for clarification, please feel free to contact me at giacomo.presotto@startmail.com.

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Abstract

There are many approaches to locality, which differ in how they answer a central question: where do locality constraints originate? All available theories successfully explain specific subsets of the data, yet none captures the full range of locality phenomena. This suggests a “division of labour” among locality accounts, whereby variation in locality effects (within and across languages) is partitioned into distinct explanatory domains. This observation raises two possibilities: either “locality effects” comprise ontologically distinct phenomena stemming from heterogeneous sources, or there is a common denominator linking phenomena such as strong and weak islands, superiority effects, and subject–object asymmetries in A’-dependencies. If the latter is correct, a fundamental question arises: why has no unified account emerged? If these effects share an underlying basis, one might expect a unified theoretical framework to capture them. The persistence of this gap remains largely unaddressed in the literature.

This thesis addresses this issue by developing a framework that seeks to explain why locality effects are multifaceted and resistant to a single overarching explanation. The core proposal is that locality effects arise from the interaction between purely linguistic factors and cognitive dynamics that extend beyond the language faculty. Most (if not all) locality-related phenomena are argued to result from the interplay between cognitive mechanisms responsible for language processing and the grammatical representations underlying them. This interaction is located at the Interface, understood as the point of interaction between grammar (conceived as a set of mental representations and their combinatory rules) and the domain-general principles and mechanisms through which language emerges and is put to use (Chomsky et al. 2023).

In pursuing these questions, the dissertation offers novel insights into specific cases, including argument–adjunct asymmetries in extraction from English weak and strong islands and intervention effects in Spanish object relative clauses, both investigated experimentally. In the final chapter, the long-standing issue posed by the (subject part of the) CED is addressed theoretically.

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List of Abbreviations

ACC	Accusative
AJT	Acceptability Judgement Task
BC	Blocking Category
CED	Condition on Extraction Domains
CF	Computational Form
CI	Conceptual-Intentional system
CLLD	Clitic Left Dislocation
CNPC	Complex Noun Phrase Constraint
CP	Complementiser Phrase
DAT	Dative
DO	Direct Object
DOM	Differential Object Marking
DP	Determiner Phrase
EPP	Extended Projection Principle
ERP	Event-Related brain Potential
FL	Faculty of Language
fRM	Featural Relativised Minimality
GB	Government and Binding
GEN	Genitive
GG	Generative Grammar
GP	Generative Procedure
GT	Government Theory
IM	Internal Merge
IO	Indirect Object
IP / TP	Inflectional Phrase / Tense Phrase
LA	Labelling Algorithm
LCA	Linear Correspondence Axiom
LF	Logical Form
MLC	Minimal Link Condition

NEG	Negation
NP	Noun Phrase
NPAH	Noun Phrase Accessibility Hierarchy
OBL	Oblique
ORC	Object Relative Clause
OVS	Object-Verb-Subject
PF	Phonological Form
PIC	Phase Impenetrability Condition
RC	Relative Clause
RM	Relativised Minimality
RT	Reaction Time
SI	Strong Island
SM	Sensory-Motor system
SMT	Strong Minimalist Thesis
SRC	Subject Relative Clause
SRT	Sentence Repetition Task
SVO	Subject-Verb-Object
UG	Universal Grammar
VP / vP	Verb Phrase
WI	Weak Island
WO	Word Order
WM	Working Memory
WS	(derivational) Workspace

1

Introduction

Reduced to its essentials, natural language can be described as a system that generates an infinite number of structures from a finite set of elements, a property known as *discrete infinity* (Huybregts 2019; Chomsky et al. 2023). Generative linguists call these elements *features*. Features constitute the building blocks of linguistic structures and combine to form bigger units. These units are organised hierarchically into full-fledged sentences. At this level, syntactic mechanisms form bonds that establish stronger connections between specific (sets of) elements. When two or more such elements are linked, they are said to form a syntactic dependency. This dissertation is concerned with a particular subset of syntactic dependencies: long-distance dependencies, also known as filler–gap dependencies. Take English *wh*-questions, for example. In (1a), a long-distance dependency occurs between the *wh*-element at the beginning of the sentence (the filler) and its gap-position, where it is semantically selected by a θ -assigning element (in this case, the verb).¹ In principle, this type of dependency can be extended indefinitely, as the distance between a gap and its antecedent does not have a fixed upper limit. However, in some cases long-distance dependencies are not quite as unrestricted. In (1b) for instance, it is not possible to establish a dependency between the fronted *wh*-element and its embedded gap.

- (1) a. What does Britt think that Rowan said ___?
b. * What did Britt meet the mechanic who fixed ___?

Indeed, human language exhibits two contrasting traits in this respect. While there is no aprioristic boundary to the length and depth of its structures, individual syntactic dependencies are sometimes restricted to specific domains (see, a.o., Stroik 2009; Rizzi 2013a; Aboh et al. 2014 for discussion). The effort to describe, understand, and predict

¹Throughout this thesis, I will use the term *gap-position* or simply *gap* to refer to the endpoint of long-distance dependencies, maintaining a relatively theory-neutral approach. In the examples provided, gaps will be denoted by an underscore.

the make-up of these domains forms the basis of the theory of locality, which, as the title suggests, is the focus of this dissertation.

Over the years, locality has been studied in diverse manifestations, leading to numerous descriptions, classifications, formalisations, and theories. This thesis, however, does *not* aim to review the myriad instances of locality effects or the ever expanding array of interpretations they have inspired. Instead, it seeks to approach locality through a simpler, basic set of foundational questions: Why do locality phenomena exist? Why are they so unstable and changeable across and within different languages? Do they change over time? And most importantly, what is the *source* of locality constraints?

Many prominent linguists have explored some of these questions before me. Above all, the problem of *where* locality effects actually originate stands out as particularly central, and has captured the attention of researchers from various backgrounds (sparking, at times, fairly impassioned debates). Although individual proposals are marked by inevitable and undeniable peculiarities, there is broad consensus in categorising answers to the question of locality's source into three distinct approaches (Sprouse and Hornstein 2013). First, *resource-based approaches* attribute locality to domain-general cognitive principles hard-wired into human cognition. From this perspective, locality emerges naturally as the human parser avoids overly demanding computations due to limited processing capacity. At the other extreme, *grammar-based approaches* argue that locality effects arise from deeper, language-specific constraints mentally encoded in a speaker's grammar. In this view, locality constraints are formalised as grammatical rules. Finally, a third, undeservedly less-known class of accounts consists of *grounded approaches*, which combine elements of the first two: locality constraints are represented in the grammar but are shaped over time by pressures from non-language-specific cognitive mechanisms.

I will discuss these approaches more extensively in Chapter 2. For now, I simply note that determining why locality exists and where it is encoded has been a subject of considerable debate among language scientists, and advocating for or against any single approach is not an easy task, as each framework tends to excel at explaining certain phenomena while struggling with others. This raises an important question. There are numerous convincing proposals on the market, but while these admittedly account for many phenomena, it remains unclear why they can typically address only fairly specific subsets of data. Somehow, there appears to be a 'division of labour' among locality accounts, dividing the vast variation in locality effects (both within and across languages) into distinct zones of competence. However, although this compartmentalisation is evident, little has been said about why such a multitude of seemingly conflicting accounts is, in fact, necessary for a comprehensive understanding of locality.

This touches on a crucial point, as we really need to look deeper into why so many,

and such different, theories can account for similar empirical facts. After all, if we assume that phenomena such as strong and weak islands, superiority effects, and subject–object asymmetries in A'-dependencies all stem from some form of locality constraint, why has no single, unified account emerged? If these effects originate from the same underlying principles, should the theoretical framework used to interpret them not also be unified? Yet this is not the current state of affairs, and the question of why a common denominator remains elusive is largely unaddressed.

This thesis seeks to fill this gap. Specifically, it aims to explore a framework that explains why locality effects are so multi-faceted and resistant to a single, overarching explanation, while also yielding novel insights into specific cases, such as extraction from English weak and strong islands, and intervention effects in Spanish object relative clauses. At the same time, it represents an attempt to determine whether, to what extent, and which locality effects can be ascribed to syntactic constraints proper, and which should instead be traced to alternative sources. Clearly, this issue is crucial not only for locality theory, but for syntactic theory more broadly. Understanding the source of locality is fundamental when considering that countless syntactic analyses depend upon locality-effects as diagnostics for movement or variable binding. Determining when and whether such effects are genuinely syntactic is therefore essential for developing well-founded hypotheses about other aspects of structure and derivation in natural language.

Before outlining some key aspects of the proposal, let me first clarify the scope of this work. I do not aim to provide a definitive solution to the vast problem of locality. Attempting to reduce the complexity and heterogeneity of locality effects to a single, one-size-fits-all explanation would seem overly ambitious to me, at least at this stage of our understanding. That said, this does not mean we should shy away from addressing the fundamental questions surrounding locality. Much remains to be explored, and this dissertation focuses on what is arguably one of the most pressing of these questions: the source of locality.

To pursue this enquiry, I propose something of a reversal of the standard approach: rather than starting from empirical data and working towards theoretical unification, I begin by critically examining the range of existing locality accounts. From this perspective, as I will argue, it becomes clearer why seemingly incompatible theories have each achieved a degree of success, despite their limited scope. The reason, I contend, is that the source of locality is inherently unstable and multifaceted. Rather than proposing an entirely new account of locality, then, this thesis advances an alternative perspective, one that seeks to integrate the most compelling insights from existing theories.

I should add that this approach is not entirely new. There have already been efforts to rethink the problem of locality by integrating elements from different theories and frame-

works. Notably, Boeckx’s (2012) book exemplifies such an approach, to which I will frequently refer throughout this thesis, as I share its implicit assumption that the complex problems posed by locality call for flexible solutions. I will address in more detail how this flexibility can be achieved when I outline the main hypothesis of this dissertation in Chapter 3. The core idea, however, is as follows: locality effects arise from the interaction between purely linguistic factors and cognitive dynamics that extend beyond the language faculty. Put differently, I propose that most (if not all) locality-related phenomena result from the interplay between cognitive mechanisms responsible for language processing and the grammatical representations underlying them. This interaction is not uniform, as the factors involved carry varying weight depending on the specific case. For instance, in some scenarios, grammatical rules are more rigid and operate almost independently of constraints on sentence processing. In others, the influence of grammar appears more flexible or indirect, with domain-general cognitive dynamics playing a more decisive role. We should think of this interaction as dynamic and versatile, where the demands of language-specific and non-language-specific factors create a spectrum of variability, as visualised in Figure 1.1.



Figure 1.1: A conceptual illustration of the Interface-spectrum and its interactions between language-specific and non-language-specific factors.

The various outcomes classified as locality effects likely span the entire spectrum, with some being more ‘grammar-driven’ and others more ‘processing-driven.’ In this thesis, I will refer to this spectrum as the *Interface*. By this, I mean the interface between grammar (understood as a set of mental representations and their combinatory rules) and the domain-general principles and mechanisms through which language emerges and is put to use. I will spell out the details in Section 3.2, showing that this definition of interface aligns naturally with standard minimalist assumptions, and seamlessly comprises the ‘traditional’ PF/LF interfaces.²

Finally, while this space of interaction may support the interpretation of a variety of linguistic phenomena, I will, for now, restrict my focus to locality-related ones, proposing that this is precisely where they originate. In essence, I will argue for a theory of *locality at the Interface*.

²Throughout this thesis, I use the capitalised term *Interface* to refer to my own conceptualisation, as defined in Chapter 3. When referring to the standard acceptance of interface(s) in the minimalist literature, I use the lowercase *interface*.

1.1 The structure of the thesis

Chapter 2 traces the development of locality theory, outlining the empirical phenomena at stake, the ways in which different approaches account for them, and the challenges each approach faces. Chapter 3 introduces the Interface Hypothesis, developing the proposal that locality effects arise from the interaction between grammatical representations and domain-general pressures, and outlines its main predictions. Chapters 4 and 5 form the empirical core of the thesis: a series of experiments test the predictions of the Interface Hypothesis and inform its refinement by focusing on amelioration patterns across intervention and impenetrability locality, respectively. Specifically, Chapter 4 examines intervention locality through four experiments on Spanish relative clauses across adult and child populations, while Chapter 5 reports two experiments on the acceptability of extraction from weak and strong islands in English. Chapter 6 integrates these results into a broader theoretical discussion, showing how they support and refine the Interface Hypothesis. In this context, I propose a rethinking of the notions of intervention and impenetrability, and, in addressing impenetrability, revisit the long-standing issue of the CED: after reviewing prominent accounts and their strengths and weaknesses, I put forward my own proposal. Finally, Chapter 7 concludes.

Some Background: What, Why, and Where of Locality

Hofmeister et al. (2013) open their chapter with a particularly apt metaphor to introduce the problem of the source of locality effects. They consider the case of computer chess programs, and how different factors might affect the selection of a particular move during a game. In particular, they wonder whether and to what extent a specific move is determined by the move-selection algorithm of the software or by the processing power of the computer's RAM. This analogy becomes even more compelling when picturing a scenario in which two different computers play two different moves. For instance, given the same board configuration, a computer from the 1990s might select a move that does not match the choice of an older machine from the 1970s. Now, assuming that the newer computer's move is objectively better, the question arises as to what led its predecessor to select a less optimal one. There are essentially two possibilities: either the 1970s computer lacked the processing power to compute the better move, or its software employed a search strategy and decision algorithm that caused it to overlook it. As Hofmeister et al. (2013) note, this is not too hard a nut to crack: one need only inspect the code and track the move-selection procedures. In the case of the human mind, however, it is skulls rather than nuts that would need cracking open. As of now, however, that does not seem to be an available option, bringing us to the theoretical challenge underlying the linguistic facts attributed to locality effects. What is responsible for, say, an island violation? Is it a matter of 'RAM' or 'software'? In less metaphorical terms, is locality driven by grammar or by the inherent limitations of the language processor—or, perhaps, a combination of the two?

This chapter examines this question by reviewing the main theoretical approaches and their proposed solutions to this conundrum. In doing so, I will attempt to trace, as

faithfully as possible, the key developments in the theory of locality: when they arose, why they were proposed, and how they evolved.

2.1 How locality came to be: grammar-based approaches

The history of locality theory is long and convoluted, with countless intertwining Principles, Conditions, and assorted formalisations that, over time, have created a labyrinth of notions increasingly difficult to untangle. As is often the case in scientific research, however, beneath the steaming brains and the complex theories they generate lie very basic observations (Feynman 1999). In the case of locality, it all began with the simple recognition that certain English sentences are just plainly bad. Boeckx (2012: 4) notes that in his PhD thesis, Noam Chomsky considered the following example:¹

- (1) * Whom did [your interest in ____] seem to me rather strange?

Native speakers of English universally agree that this sentence sounds odd and unnatural, yet unless they hold a degree in linguistics, they are unlikely to explain *why*. In fact, most people never reflect on why they are so adept at distinguishing well-formed from ill-formed sentences in their native language. Chomsky, for his part, thought about this extensively and developed the generative enterprise around the conviction that a set of innate rules, represented in each speaker's mind, provides them with a firm grasp of what is grammatical and what is not.² In the specific case of (1), Chomsky argued that there must be a particular constraint preventing this transformation (the then term for movement). This allowed for the possibility that different types of transformations could be subject to different constraints. Already in his famous (1964) *Current Issues in Linguistic Theory*, however, Chomsky sought to formulate a more general rule, defining what is broadly considered the first constraint on long-distance dependencies: the A-over-A condition. This principle states that in a context where an element of category A is dominated by an element of the same category, movement must target that bigger, dominating element:

(2) **A-over-A condition**

If a rule ambiguously refers to A in a structure of the form of (i), the rule must apply to the higher, dominating node A.

- (i) ... [A ... [A ...

¹A historical note. Chomsky's PhD dissertation, *Transformational Analysis* (Chomsky 1955), was completed in 1955 at the University of Pennsylvania. A significant portion of its content was later incorporated into *The Logical Structure of Linguistic Theory*, which was written between 1955 and 1956 but remained unpublished until 1975. These ideas were further refined and presented in *Syntactic Structures* (Chomsky 1957), which became the foundational text of transformational-generative grammar.

²For an overview of the principles and development of generative theory, see, e.g., Freidin (1992, 2012) and Chomsky (2004b).

Shortly after its formulation, however, the A-over-A condition was called into question by John Robert Ross, who is widely regarded as something of ‘the father of locality theory.’ In his ground-breaking (1967) PhD thesis,³ Ross observed that the A-over-A condition was somewhat inadequate, as it was both too weak and too strong. Too weak, for it cannot rule out sentences like (3b), where the moving element and the domain containing it are not of the same category:

- (3) a. How nice a car do you have?
 b. * How nice do you have [___ a car]?

(Boeckx 2012: 5)

And too strong, since it rules out perfectly grammatical sentences like (4), in which the extraction involves the movement of a nominal category out of a larger NP.

- (4) Which tennis player did you watch [a documentary about ___]?

Thus, Ross had to redefine the *Constraints on Variables in Syntax* and, for the first time, described a number of linguistic facts that laid the groundwork for all subsequent discussions on locality. It was Ross who, with a particularly inspired metaphor, introduced a first list of the locality constraints that have since become known as *island constraints*. Much like their geological counterparts, syntactic islands are domains that are difficult to escape. More specifically, islands can be described as environments that inhibit, to varying degrees, the extraction of syntactic material. Ross’s original list of islands included *complex noun phrases* (5a), *coordinate structures* (5b), *sentential subjects* (5c), and *left branches* (5d).

- (5) a. * What did you make [NP the claim that Britt said ___]?
 b. * What did you buy [ConjP apples and ___]?
 c. * What did [CP that Alan sold ___] bother you?
 d. * Whose did you ride [NP ___ bike]?

The inventory of syntactic islands has since expanded significantly, and I will discuss it more extensively in Section 5.1. I will, for the time being, limit myself to noting that defining a set of specific constraints was only the first step towards developing a comprehensive theory of locality. The next natural stage was to derive a universal rule that could account for individual instances of locality effects. Borrowing Aboh et al.’s (2014) words, “[I]t was hoped that an *intensional* characterisation of the notion of island could be arrived at, in place of an *extensional* list” (Aboh et al. 2014: 6, italics mine). The

³A revised version of Ross’ thesis was published as a book in Ross (1986).

first effort in this direction was undertaken by Chomsky in his *Conditions on Transformations* (Chomsky 1973), where he ambitiously attempted to derive all (then known) island constraints under a unique structural principle: the Subjacency condition.

(6) **Subjacency condition**

No rule can involve X , Y in the structure:

... X ...[α ... Z ... – WYV ...]...

where

- (a) Z is the specified subject of WYV ,
- or (b) Y is in COMP and X is not in COMP,
- or (c) Y is not in COMP and α is a tensed S.

There are two basic ideas behind this formalisation. First, there exist specific syntactic domains that preclude the formation of syntactic dependencies out of their boundaries, namely tensed clauses and clauses containing an overt lexical subject.⁴ Second, such restrictions can be circumvented, which, according to (6), happens when the dependency targets a COMP position, and if the dependency originates in a domain where there is an unoccupied COMP position. This way, it was possible to explain why cases like (7) are ungrammatical, while at the same time ruling-in sentences like (8). Specifically, in (7), syntactic material is extracted out of, respectively, a tensed clause (7a) and a clause containing a lexical subject (7b):

- (7) a. * John seems [that __ is smart]
- b. * John seems [that Mary likes __]

(Boeckx 2012: 9)

In (8), extraction of the *wh*-element is made possible by the fact that its movement proceeds in two steps. It first targets the free, unoccupied COMP position in the embedded clause, and then proceeds to the higher COMP in the matrix clause:

- (8) Who do you believe [Mary likes __]?

(Boeckx 2012: 10)

⁴Originally, this was stated via the *Tensed-S Condition* (i) and the *Specified Subject Condition* (ii):

- (i) No rule can involve X , Y in the structure
... X ...[α ... Y ...]
where α is a tensed sentence.
- (ii) No rule can involve x , Y in the structure
... X ...[α ... Z ... – WYV ...]
where Z is the specified subject of WYV in α .

Note that according to the definition in (6), there is no inherent limit to the number of steps involved in a syntactic dependency, so long as all the nodes that are involved are adjacent, or using Chomsky's terminology, *subjacent*.⁵ Consider, for instance, the sentence in (9), adapted from Haegeman (1994):

- (9) [CP₁ How did you say [CP₂ that Jeeves thinks [CP₃ that Lord Emsworth will solve this problem ____]]]? (Haegeman 1994: 403)

Here, the *wh*-element *how* does not target the highest CP (using more up-to-date terminology) in one fell swoop. Rather, it first targets all the intermediate CPs, completing three steps. These 'steps' are what have come to be known as *cycles*, and long-distance movement was since maintained to proceed *successive-cyclically*.⁶ In Chomsky (1973), the relevant cycles for the syntactic derivation were taken to be NP and S (the current TP), which were defined *bounding nodes*. With this in mind, a simplified definition of subjacency can be given as follows:

(10) **Subjacency (simplified)**

No rule can involve *X*, *Y* in the structure ...*X*...[β ...[β ...*Y*...] where:

- (a) *X* is separated from *Y* by more than one bounding node β
- (b) a bounding node is a node of category S or N.

At this point, the next crucial step was to derive an intensional definition of a bounding node. Before addressing how this was attempted, let me first highlight another pivotal development in the theory of locality, introduced in Chomsky's (1977) seminal paper *On wh-movement*. This work revolutionised the field in two major ways.⁷

First, he emphasised a point already hinted at by Ross (1967): island constraints, though driven by universal grammatical principles, are not necessarily constant across languages. Crucially, what counts as a bounding node can vary cross-linguistically. For instance, notions like Specified-subject or Tensed-S might be characterised differently depending on the language. This insight opened the door to the parametrisation of locality constraints, further explored by Rizzi (1978, 1982).

Second, Chomsky divided transformational rules into two subtypes: *Move NP* and *Move wh-phrase*, later redefined as *A-* and *A-bar movement*, respectively. He also ar-

⁵This clarifies the origin of the term, which refers to the configurational relation between two nodes, defined by Chomsky (1986: 30) as follows: If (α_i, α_{i+1}) is a link in a chain, then α_{i+1} is subjacent to α_i .

⁶For discussion of the empirical evidence for successive-cyclicity see Freidin (1978, 1999), Kayne and Pollock (1978), Torrego (1984), Adger (2003: 377ff.), and Lasnik (2006).

⁷For a detailed analysis of the paper and its impact on the field, see Cheng and Corver (2010) and in particular Chapter 1 therein.

gued that island constraints specifically restrict transformations of the second type, those involving wh-movement.⁸

Returning to the question of an intensional definition of a bounding node, this was addressed in *Barriers* (Chomsky 1986), where an attempt was made to characterise bounding nodes for subjacency in terms of government relations, and thereby connect two of the main concepts of Government and Binding theory. In short, the gist of Chomsky's (1986) proposal is as follows: bounding nodes are redefined in terms of *barriers*, which are defined through the notion of *blocking category* (BC), itself dependent on the definition of *L-marking*, which is a specific type of (a specific type of) government. Here are the relevant definitions:

(11) **Barrier**

X is a barrier for Y iff one of the following holds:

- (a) X immediately dominates Z , where Z is a BC for Y .
- (b) X is a BC for Y and $X \neq IP$.⁹

(12) **Blocking Category**

X is a BC for Y iff X is not L-marked and X dominates Y

(13) **L-marking**

X L-marks Y iff X is a lexical category that θ -governs Y

(14) **θ -government**

X θ -governs Y iff X governs Y and X theta-marks Y ¹⁰

The list could go on to include the definitions of (regular) government, m-command, c-command (see below), dominance, θ -marking, and more. That said, aside from some updated terminology, the underlying principle remains consistent with the original subjacency condition: movement across two barriers is not permitted.¹¹ As a matter of fact, some might argue that much of the subsequent theory on locality restrictions has substantially retained Chomsky's (1973) initial intuitions.

To some extent, this appears true even for the otherwise innovative ideas of the Minimalist Program (Chomsky 1995), in which the necessity to 'keep movement short' remained

⁸As Sprouse and Hornstein (2013: fn. 3) note, the original formulation of island constraints by Ross referred to "chopping" rules. The restriction to movement, and specifically wh-movement, was developed later by Chomsky (1973, 1977, 1981)

⁹Here, IP corresponds to the former S and the current TP.

¹⁰Essentially, θ -government defines the relation between a lexical head and its sister node.

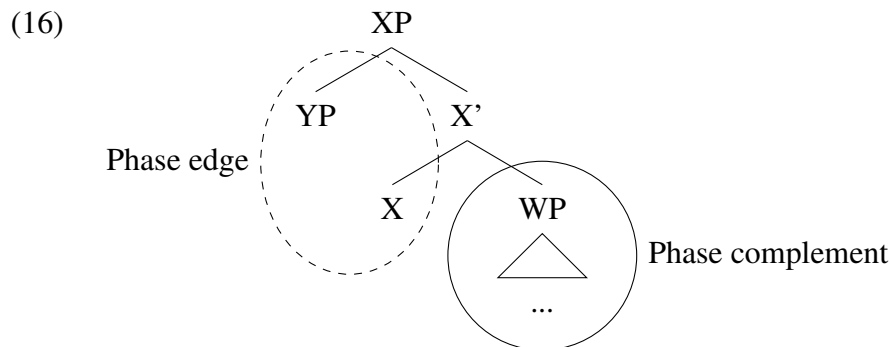
¹¹Note, however, that the new system based on barrierhood was an improvement over the framework of Chomsky (1973), in that it could also capture extraction asymmetries between arguments and non-arguments, as observed by Huang (1982). I will return to this point in later chapters.

central. Whereas Subjacency limited movement length by ensuring it did not cross more than one cyclic node, minimalist theorising captured the same constraints through the notion of *phases* (CP and vP, see below) and the *Phase Impenetrability Condition* (PIC), stated in Chomsky (2000: 108) as follows:

(15) **Phase Impenetrability Condition**

In phase α with head H, the domain of H is not accessible to operations outside α ; only H and its edge are accessible to such operations.

Essentially, the PIC states that once a phase is completed (Spelled-Out), its internal structure becomes inaccessible for further syntactic operations. However, material located at the edge of the phase (the specifier of the phase head or the head itself) remains accessible, functioning as an “escape hatch” for movement (Chomsky 2000, 2001, 2004a, 2008). So, assuming the XP in (16) is a phase, the material contained in its complement cannot be extracted once the phase is completed.¹²



This is, of course, just a simplified and condensed illustration of phase theory, whose scope is not limited to the (re-)interpretation of locality constraints. Rather, it is framed within a broader ambition: to constrain syntactic operations in accordance with principles of derivational economy.¹³ Though innovative and compelling in many respects, the notion of phase has also prompted a number of serious questions. Perhaps most crucially, it remains unclear how to arrive at an operationally precise definition. That is, what projections actually count as phases? As hinted above, vP and CP are traditionally assumed to be phases, but the theory still lacks a clear intensional definition, making it difficult to

¹²It should be pointed out, though, that Chomsky (2001) introduced a partial revision of the PIC, according to which the complement of a phase becomes inaccessible only once the next higher phase head has been merged:

PIC (Chomsky 2001: 14)

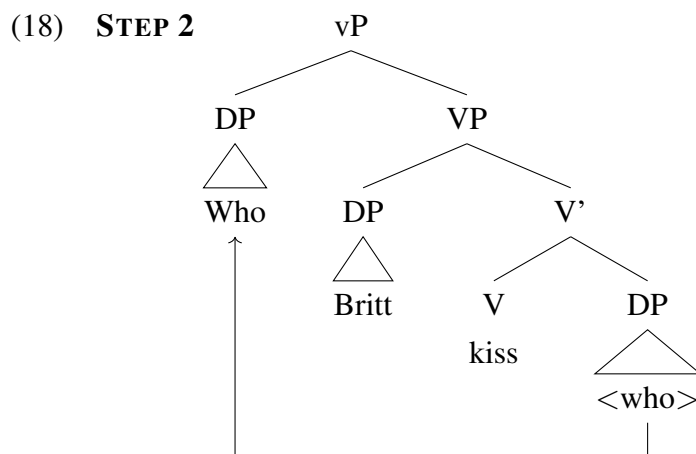
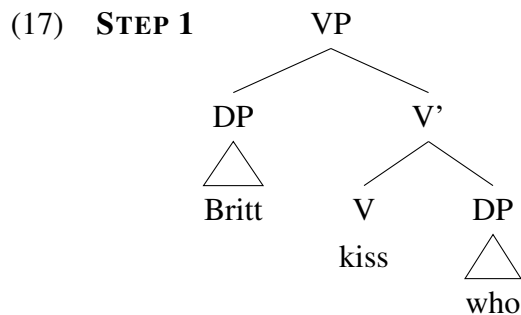
Given the structure [_{XP} Z [_{HP} α [H YP]]], with H and Z the heads of phases:

The domain of H is not accessible to operations at ZP; only H and its edge are accessible to such operations.

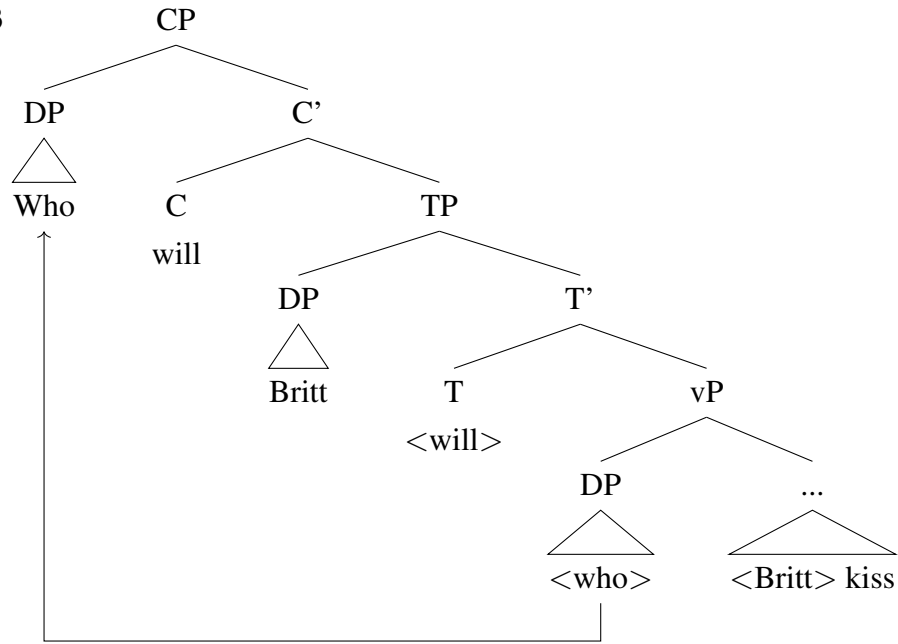
¹³See Hornstein et al. (2005: 330–365) and Boeckx (2006a: 44–61) for accessible yet precise overviews. For a more comprehensive treatment of phases, see Gallego (2010).

determine what constitutes a phase in principled terms (see Uriagereka 1999a, 2000; Epstein and Seely 2002; Boeckx 2003; Grohmann 2003; Nissenbaum 2000; Bruening 2001; Abels 2003; Svenonius 2004; Legate 2003; Boeckx and Grohmann 2004, 2007; Richards 2005; Müller 2004b, 2010; Gallego 2011; Gallego and Uriagereka 2007 for discussion). I will not delve into this issue here. Instead, I will focus on a related problem that bears directly on the theory of locality and movement constraints.

The PIC, as originally formulated, is clearly too rigid: even the simplest cases of wh-movement (as in, e.g., [_{CP} *Who* [_{TP} *will Britt* [_{VP} *kiss*]]?) would be ruled out. To avoid this unwanted result, Chomsky proposed that movement from the phase complement to the phase edge is permitted *prior* to the application of the PIC. This is achieved by moving the relevant material to one of the specifiers projected by the phase head (which appear to be positable unconstrainedly, see, e.g., Ura 1996, Richards 2001). Such movement is triggered by general movement-driving features, variously labelled *P(eripheral)-features* (Chomsky 2000), *(generalised) EPP-features* (Chomsky 2001, 2004a), *edge features* (Chomsky 2005, 2008), [$\bullet$ X $\bullet$] (Müller 2010), and so on. A very simplified illustration of the derivational steps required to generate the sentence just mentioned is shown in the following trees:



(19) STEP 3



This mechanism was introduced to solve the problem of undergeneration: under the PIC, the complement of a phase head becomes inaccessible once Spelled-Out, making certain grammatical movements impossible unless intermediate movement to the phase edge is permitted. However, the theory now faces the opposite problem: overgeneration. In its current form, there is nothing to prevent any element from moving to the escape hatch (i.e., the phase edge), thereby neutralising the need for independently deriving locality constraints on extraction. As noted by Boeckx (2012), this is potentially an intractable flaw: the system lacks a principled mechanism to block overgeneration while preserving empirical coverage. One proposed solution is to restrict the availability of edge/EPP features based on structural conditions (see, e.g., Ceplová 2001; Müller 2010). Yet, as Boeckx (2012: 62–73) argues in detail, such restrictions are problematic on both conceptual and technical grounds. Long story short, even the newer formalisation based on phases cannot escape a certain degree of stipulative arrangement in order to account for locality phenomena, a shortcoming to which I will return in Section 2.1.3.

Phase theory is, of course, far removed in many respects from the early-days constraints on locality.¹⁴ Yet, as noted by Sprouse and Hornstein (2013), it appears that all ‘Subjacency-like’ accounts of island effects share four essential properties:

- (i) a way to determine the size of a movement step
- (ii) a constraint on the permissible step size
- (iii) an indication of which structural positions allow extraction (“escape hatches”)

¹⁴Even though Boeckx and Grohmann (2007) put forth a series of arguments suggesting that phases are essentially just traditional bounding nodes/barriers in disguise.

- (iv) a specification of the number of available positions within an escape hatch

Starting from these common traits, we observe that much work on locality has focused on two main points: a) the parametrisation of one or more of these properties to account for cross-linguistic variation in islandhood, and b) the theoretical legitimisation of these four properties. Once again, while this two-fold effort has driven the continuous evolution of different accounts, they can still be grouped under a relatively cohesive class.

Nonetheless, a separate strand of research has given rise to yet another set of theories. These fall under the framework of Relativised Minimality, which addresses locality phenomena that only partially overlap with those underlying Subjacency-like constraints. Before introducing Relativised Minimality, it is useful to first distinguish between two types of locality: impenetrability locality and intervention locality.

2.1.1 From impenetrability to intervention: the advent of Relativised Minimality

Thus far, I have considered cases of particular syntactic domains—*islands*—that are essentially impervious to movement, in that they resist extraction (almost) unconditionally. At this point, I should mention that in fact syntactic islands come in two types: *strong islands* and *weak islands* (see Szabolcsi 2006; Szabolcsi and Lohndal 2017 for an extensive overview). Descriptively, the difference between the two is rather straightforward. Strong islands block movement of any type, while weak islands allow it depending on the nature of the element that is extracted. For this reason, some scholars prefer the labels *absolute* and *selective* islands, suggesting that the latter only preclude a subtype of extractions. The standard diagnostic to distinguish absolute/strong islands from selective/weak islands is based on the extraction asymmetries between arguments and adjuncts (Huang 1982; Chomsky 1986; Cinque 1990). If extracting an argument/NP or an adjunct/PP produces an equally firm disruption, we are likely dealing with a strong/absolute island. For instance, this is the case with Complex DPs (in this case, a relative clause):

- (20) a. * Who/Which kid must you call [the teacher who punished ___]?
 b. * Where/In which room must you call [the teacher who put the book ___]?

(Adapted from Szabolcsi and Lohndal 2017: 5)

If, instead, the island-effect only emerges (or gets more severe) when an adjunct is extracted, the island belongs to the weak/selective class, as is the case for so-called *whether islands*.¹⁵

¹⁵In Chapter 5, I will discuss results that call this type of asymmetry into question. Here, I simply report the standardly accepted observations for expository reasons.

- (21) a. ? Who/Which customer do you wonder [whether Ellie hugged ___]?
 b. * How/In what way do you wonder [whether Ellie hugged Adam ___]?

Interestingly, this distinction is not merely a matter of organising the variational space of island phenomena into descriptively different categories. Rather, strong and weak islands have traditionally been linked to two distinct concepts of locality, leading to partially parallel theoretical trajectories. On the one hand, strong islands exemplify impenetrability locality. On the other, weak islands are thought to arise from intervention locality.

Impenetrability locality is the type of locality that has driven the definition of all the Subjacency-like accounts revised in the previous section (along many others), and can be expressed as follows (see Rizzi 2013a: 172):

(22) **Impenetrability locality**

X and Y cannot undergo a syntactic dependency in the configuration:

... X ...[α ... Y ...]

Where α is a subjacency-sensitive domain (e.g. a sentential subject, a complex DP, an adverbial, ...)

Intervention locality, by contrast, developed somewhat independently, with its roots tracing back to the *Minimal Distance Principle*, introduced by Rosenbaum (1967) in the context of control theory. The Minimal Distance Principle states that PRO (the null pronominal subject of infinitival clauses) must be controlled by the closest nominal in the main clause, as illustrated in (23).¹⁶

- (23) Britt told Ellie_{*i*} [PRO_{*i*} to leave]

The first comprehensive characterisation of intervention within a formal theory of movement, however, was developed only years later by Rizzi in his groundbreaking monograph, *Relativized Minimality* (Rizzi 1990).

Relativised Minimality is a syntactic principle that requires movement to occur within a *Minimal Configuration*. Informally, this means that movement must take place in the smallest possible environment. Formally, it was originally expressed as follows (see Rizzi 1990: 27):

(24) **Relativised Minimality (original formulation)**

X α -governs Y only if there is no such Z that:

- (i) Z is a base-generated position¹⁷

¹⁶For an overview of the theory of control, see Landau (2000, 2013, 2015, 2024) and Boeckx et al. (2010).

¹⁷This clause was provided “to limit the blocking effect to heads and specifiers, as adjoined positions do not seem to have this capacity” (Rizzi 1990: 27).

- (ii) Z is a α -GT compatible with Y ¹⁸
- (iii) Z c-commands Y and does not c-command X

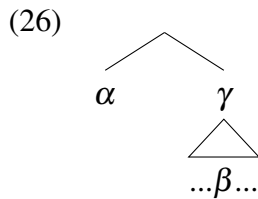
Essentially, the idea is that a trace created by movement can only be effectively governed by its antecedent if no other potentially compatible governor intervenes. Still, the very reference to government relations already marks a certain distance from how we formally describe syntactic relations today, which is likely why Rizzi (2013a: 172) saw the need to simplify his original definition as follows:

(25) **Relativised Minimality (simplified)**

In the configuration: $\dots X \dots Z \dots Y$, X and Y cannot be connected by movement (or other local relations) if Z intervenes between them, and Z is of the same structural type as X

We can now more clearly identify the two main components that define Relativised Minimality. The central insight is that a syntactic dependency between two elements is blocked when a third element satisfies two conditions: the *intervention condition* and the *similarity condition*.

The intervention condition is defined in terms of c-command. First introduced in Tanya Reinhart's dissertation (Reinhart 1976), c-command has since become a fundamental tool for describing syntactic relations. At its core, c-command captures the hierarchical relationship between two elements in syntactic structure. In (26), α c-commands β because β is contained within γ , the sister node of α :



I should note that the definition of c-command is far from univocal and has undergone various refinements over the years (see Barker and Pullum 1990; Kayne 1994; Barker 2012 for discussion). This may explain why Aboh et al. (2014: 21) sought to generalise the notion of intervention as follows:

- (27) α is an intervener for a relation $R(\beta, \gamma)$ iff:

$$R(\beta, \alpha) \quad \wedge \quad R(\alpha, \gamma) \quad \wedge \quad \neg R(\alpha, \beta)$$

¹⁸Two elements are α -Government Theory compatible (α -GT compatible) when they are in a configuration corresponding to one of the four possible government types (Head government under m-command, A-government, A-bar government, Head-government under c-command). See Rizzi (1990: 26-27) for further details.

There is, however, an irreducible characteristic of traditional definitions of c-command that remains central for Relativised Minimality. Regardless of how c-command is defined, the relation relevant for determining intervention must be computed hierarchically rather than linearly. Consider the following pair:

- (28) **When** did [_{DP} the uncertainty [_{PP} about **who** won]] dissolve __?
 (Rizzi 2013a: 173)

- (29) *? **When** did Rowan wonder [_{CP} **who** entered the store __] ?

In both cases, *who* linearly intervenes between *when* and its gap. In (28), however, *who* is embedded within the subject DP and, as such, does not intervene in hierarchical terms. By contrast, in (29), *when* c-commands *who*, and *who* c-commands the gap created by the movement of *when*. Crucially, only the hierarchically intervening *who* in (29) leads to ungrammaticality. Based on these considerations, it appears clear that we must express the first of the two conditions defining Relativised Minimality—the intervention condition—in terms of a (version of) c-command relation.

The second relevant condition expressed in (25) is based on the concept of similarity. The core idea is that not just any intervening element between two linked positions qualifies as a proper intervener, but only one that shares some degree of similarity with the target of the operation in question. While intervention has consistently been defined in terms of c-command, the criteria for computing syntactic similarity have been subject to continuous revision over the years. The original intuition presented in Rizzi (1990) was that a violation is triggered when an intervening element occupies a structural position that matches the typology of the target of the relation. At the time, the taxonomy of syntactic positions was broadly divided into three categories: heads, A-specifiers, and A-bar specifiers. This categorisation aligned with the typology of movement operations, which were defined based on the type of position they target. Thus, head movement targets head positions, while phrasal movement is split into A-movement and A-bar movement, which target A-specifiers (typically subject positions) and A-bar specifiers (e.g., the landing sites of wh-movement in the left periphery), respectively. According to this categorisation, two elements are “similar” if they belong to the same structural class. It follows that, for instance, heads can block head movement, as in (30).¹⁹

- (30) a. The guests could have left
 b. Could the guests __ have left?
 c. * Have the guests could __ left?

(Rizzi 2013a: 176)

¹⁹Note, however, that treating head displacement as analogous to phrasal movement is not uncontroversial. See Dékány (2018) for an overview.

The same applies to A-movement, which can be obstructed by an intervening phrase occupying an A-position. This is illustrated in (31), where the subject can be extracted from a predicative small clause (i.e., an instance of A-movement) only if no intervening A-specifier, such as *John* in (31c), is present.

- (31) a. Everyone considers [[John] [a good doctor]]
 b. John is considered [___ [a good doctor]]
 c. * A good doctor is considered [John ___]

(Rizzi 2013a: 176)

The situation with A-bar movement is somewhat more problematic, however. I have already specified that certain islands are ‘selective,’ in that they tolerate extraction with particular types of extractees. Now, if the distinction between heads, A-, and A-bar specifiers laid out in Rizzi (1990) were sufficient, we would expect such asymmetries to be derivable solely by reference to these three categories. Yet it soon became clear that this was not possible without introducing additional rules and conditions (see below). Moreover, the system originally defined by Rizzi blocks all instances of A-bar movement when this crosses over another A-bar position. If we look again at the sentence in (21a), though, we note it already represents a counterexample, as movement of a wh-phrase (*Who/Which customer*) over another wh-phrase (*whether*) does not result in ungrammaticality. Things get even more complicated considering that such asymmetries are not limited to the distinction between argumental and adverbial material. In French, for instance, the wh-element *combien* ‘how’ can move on its own (32b) or pied-piping its lexical restriction (32a). As famously observed by Obenauer (1984, 1994), however, movement of *combien* alone stops being an option when another wh-element intervenes (here *comment*), and pied-piping is the only option (33a).

- (32) a. Combien de problèmes sais-tu résoudre ___?
 how of problems know-you solve
 ‘How many problems can you solve?’
 b. Combien sais-tu résoudre [___ de problèmes] ?
 how know-you solve of problems
 ‘How many problems can you solve?’
- (33) a. ? Combien de problèmes sais-tu comment résoudre ___?
 how of problems know-you how solve
 ‘How many problems do you know how to solve?’
 b. * Combien sais-tu comment résoudre [___ de problèmes] ?
 how know-you how solve of problems
 ‘How many problems do you know how to solve?’

In a similar vein, Baltin (1992) notes that in English, the predicate of a small clause can be normally extracted (34), but not out of a *whether*-island (35):

- (34) a. How many people do you consider __ intelligent?
 b. How intelligent do you consider John __?
- (35) a. ?? How many people do you wonder whether I consider __ intelligent?
 b. * How intelligent do you wonder whether I consider John __?

So, once again, extraction asymmetries in the A-bar domain define an intricate frame that cannot be immediately captured by the original notion of “structural similarity,” nor could all these exceptions be accounted for based on the sheer distinction between arguments and non-arguments. Indeed, even within the argumental class there seem to be different conditions depending on some referential properties of the moved element. Cinque (1990) suggested that arguments can only be moved out of a weak island when they are *D-linked*, meaning that their interpretation is pre-established by the discourse (Pesetsky 1987; Comorovski 1989). This would account for the contrast between the ungrammaticality of (36a) and the grammaticality of (36b), in which a specific interpretation of the extracted *wh*-phrase is enforced through the partitive construction:

- (36) a. * Quanti soldi non sai come guadagnare __?
 how money not know.2SG how earn
 ‘How much money don’t you know how to make?’
- b. ? Quanti dei soldi che ti servono non sai come guadagnare __?
 how of money that DAT need not know.2SG how earn
 ‘How much of the money that you need don’t you know how to make?’

(Adapted from Cinque 1990)

A number of analyses have been put forth to explain these facts (a.o., Cinque 1990, Rizzi 2001, 2004, 2018), but exploring them in detail would exceed the scope of this chapter. Suffice it to say that, while often quite elaborate, various approaches existed to accommodate these asymmetries within the Relativised Minimality framework. The real challenge, so to speak, arose from a different set of observations. Let us take a closer look at the case of *combien*-movement in French. The extraction of bare *combien* is not banned just in the presence of another *wh*-element (33b), but also with an intervening negation (37b), or a quantificational adverb (38b):

- (37) a. ? Combien de problèmes ne sais-tu pas résoudre __?
 how of problems neg know-you NEG solve
 ‘How many problems can’t you solve?’

- b. * Combien ne sais-tu pas résoudre [__ de problèmes] ?
 how NEG know-you NEG solve of problems
 ‘How many problems can’t you solve?’
- (38) a. ? Combien de livres a-t-il beaucoup consultés __?
 how of books has-he much consulted
 ‘How many books has he consulted a lot?’
- b. * Combien a-t-il beaucoup consultés [__ de livres] ?
 how has-he much consulted of books
 ‘How many books has he consulted a lot?’

This partially follows from the original definition of Relativised Minimality: assuming the negation in (37b) and the quantificational adverb in (38b) occupy an A-bar specifier (Cinque 1999; Rizzi 2004: 232), A-bar movement of a bare *wh*-element across them is expected to result in ungrammaticality.²⁰ Crucially, however, this does not hold with all adverbs. In (39), for instance, both *Combien* and *Combien de livres* can freely move across *attentivement* (Obenauer 1994; Laenzlinger 1998).

- (39) a. Combien de livres a-t-il attentivement consultés __?
 how of books has-he carefully consulted
 ‘How many books has he carefully consulted?’
- b. Combien a-t-il attentivement consultés [__ de livres] ?
 how has-he much consulted of books
 ‘How many books has he carefully consulted?’

In addition, extraction of an argument (40a) or an adjunct (40b) over a left dislocated phrase (which is standardly assumed to occupy an A-bar position, Rizzi 1997) is not ruled out in Italian:²¹

- (40) a. Non so a chi pensi che, queste cose, le dovremmo
 NEG know.1SG to whom think.2SG that these things them should.1PL
 dire __
 say
 ‘I don’t know to whom we should say these things’
- b. Non so come pensi che, a Gianni, gli dovremmo parlare __
 NEG know.1SG how think.2SG that to Gianni DAT should.1PL talk
 ‘I don’t know how you think that we should talk to Gianni’

(Adapted from Rizzi 2004: 232)

²⁰What is perhaps less clear is why, according to the original theory, movement of the same *wh*-element together with its pied-piped material results in a weaker violation.

²¹Note, however, that the case of CLLDs is likely more complex. These structures have been argued to exhibit a mixed nature, potentially involving both movement and base-generation components (see Cinque 1990; Grohmann 2000; Ott 2014; den Dikken and Surányi 2017 for discussion).

These examples make it clear that the A/A-bar distinction is too rigid, as it incorrectly excludes sentences like, for instance, (40a, b), where A-bar movement is possible despite the presence of intervening A-bar specifiers. This meant that Relativised Minimality needed some rethinking.

A first attempt in this direction can be seen in the *Minimal Link Condition* (MLC) proposed by Chomsky (1995), stated as follows:

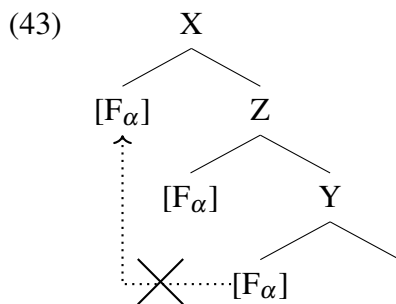
(41) **Minimal Link Condition**

α can raise to target K only if there is no legitimate operation Move β targeting K, where β is closer to K.

As Chomsky (1995: 273) put it, the definition of the MLC gets “more natural” if reinterpreted in terms of *Attract F*, i.e., the syntactic operation responsible for feature checking, later reformulated as *Agree* in Chomsky (2000):

(42) K *attracts* F if F is the closest feature that can enter into a checking relation with a sublabel of K.

This mechanism differs from the original formulation of Relativised Minimality in two key ways: i) it operates during the derivation, and ii) it assumes that intervention is based on *featural identity*. The discussion of the differences between a representational model (like Relativised Minimality) and a derivational model (like the MLC) is problematic to say the least. Readers interested in this debate can refer to Rizzi (2001, 2013a), Brody (2002), Epstein and Seely (2002) and Boeckx (2012).²² In this context, I focus instead on the second key innovation of the MLC, namely the condition on featural identity. This condition states that movement is blocked if an intervener has the same featural specification as the movement target, as illustrated in the following tree:



This revision successfully accommodates some of the problematic cases discussed above; however, it continues to face challenges with others. Centred around feature identity, the MLC is highly selective, to the point that it does not apply to the derivations of

²²I will, however, address this aspect to some extent in Chapter 6, leaning towards a representational approach to locality constraints.

(39) and (40), where the intervener and the target do *not* share exactly the same features. As a result, it correctly predicts their grammaticality. At the same time, however, the MLC is even too selective, as it fails to block cases like (37b) and (38b), where wh-movement is prevented by a negative element and a quantificational adverb, respectively. Since wh-elements are not featurally identical to either negations or quantificational adverbs, the MLC incorrectly predicts that the derivations of (37b) and (38b) should be possible.

At this stage, it became evident that further refinement was needed. Ideally, this would result in a theory that strikes a balance between the overly rigid Relativised Minimality and the overly permissive MLC. The wait was not long. In his (2001) PhD dissertation, *Move dissolves into Merge*, Michal Starke proposed reinterpreting Relativised Minimality in terms of syntactic features, demonstrating that an element with a richer featural specification can be extracted across one with a more impoverished specification (Starke 2001; Rizzi 2004). Starke's approach served as the foundation for a paradigmatic revision of Relativised Minimality, which soon came to be rebranded as *featural* Relativised Minimality. This renewed version of the theory was first programmatically presented in Rizzi (2004), where the similarity condition was reformulated to incorporate Starke's (2001) insights. The distinction between heads, A-, and A-bar positions remained central, with the crucial innovation that the class of A-bar specifiers was now organised into distinct subclasses:

- (44) **Argumental:** person, number, gender, case
 Quantificational: Wh, Neg, measure, focus, ...
 Modifier: evaluative, epistemic, Neg, frequentative, celerative, measure, manner, ...
 Topic
- (Rizzi 2004: 243)

The rationale behind this subdivision lies in the featural specification of A-bar positions. If two (types of) phrases share certain features, they are grouped into the same class. Given this revised taxonomy of the A-bar system, it was then natural to reformulate syntactic similarity as follows:

- (45) Two elements are syntactically similar if:
- (i) They are both heads
 - (ii) They are both A-specifiers
 - (iii) They are both A-bar specifiers *and* belong to the same A-bar category (e.g., Quantificational, Modifier, ...)

This adjustment introduced a new level of flexibility to the theory, allowing it to successfully distinguish between grammatical sentences like (39b) and ungrammatical ones like (37b, 38b). Indeed, an adverb such as *attentivement* in (39b) belongs to the Modifier class and therefore does not interfere with a wh-chain. By contrast, both a wh-element (38b) and a negation (37b) do, since they belong to the same featural class as the element undergoing movement, namely a wh-phrase. The paper discusses several additional examples that further support the newly proposed featural typology and explain, for instance, why the Topic class stands as a distinct category. Due to space constraints, I will not review all such examples here. Instead, I turn to yet another development of the theory, one that in fact dispenses with the notion of featural classes in favour of a more categorical application of the concept of featural similarity.

This latest version of Relativised Minimality evolved through a series of influential papers that examined intriguing asymmetries between subject and object A-bar dependencies from both acquisitional and psycholinguistic perspectives (Grillo 2008; Belletti et al. 2012; Guasti et al. 2012a; Adani et al. 2017; Villata 2017 a.o.). I will return to the specifics of this system throughout my thesis, particularly when discussing the results of my own experiments on relative clauses in Chapter 4. For now, I focus on the key innovations that distinguish the newer version(s) of Relativised Minimality from its first major revision in Rizzi (2004).

As always, the intervention condition remains firmly anchored in the notion of c-command. Also the definition of syntactic similarity remains consistent with previous statements. As a matter of fact, the new system not only preserves but reinforces the centrality of syntactic features, as similarity is now rigorously computed based solely on the featural specification of the intervener and the target of a syntactic dependency. The key innovation is that the set of features considered in this computation is restricted to those that are “syntactically active.” That is, only features that function as triggers for syntactic movement are relevant. Or, in yet more recent formulations, only those features that “define syntactic positions by licensing applications of external and internal merge” (Rizzi 2018: 347).

But this is not the only novelty. The newer versions of featural Relativised Minimality introduce a third, crucial specification: similarity is not binary. That is, two syntactic objects are not simply similar or non-similar; rather, similarity exists on a scale, determined by a set-theoretical computation. Specifically, the syntax evaluates the set of syntactically active features that define the bundles of both the target and the intervener positions and determines which set-theoretic relation within the Distinctness Hierarchy applies. At its minimum, the Distinctness Hierarchy includes identity, inclusion, and disjunction, as in

(46).²³

(46) **Distinctness Hierarchy**

In a given configuration ... X ... Z ... Y ... the degree of similarity between X and Z is determined by the set-theoretic relation between X and Z , where similarity increases along the following scale:

$$\text{Disjunction} > \text{Inclusion} > \text{Identity}$$

This more fine-grained computation of similarity allows the system to account for a wide range of variation in the perceived penalty associated with structures involving intervention locality configurations. Specifically, intervention effects are predicted to be stronger when the features of the target and the intervener define a relation of identity and weaker in the case of inclusion. In other words, this new version predicts that the more similar the intervener is to the target, the higher the penalty (Rizzi 2018: 346).

This was a particularly welcome innovation, given that multiple observations indicate a wide spectrum of graded acceptability in locality-affected structures (Villata et al. 2016; Villata 2017). The reference to the hierarchy in (46), albeit probably still too coarse, represents a valuable effort to formally predict such variation. For instance, Rizzi (2018) illustrates that the increasing penalty in the sentences in (37) can be correlated with their respective degrees of similarity in set-theoretical terms. That is, (37a) is more deviant than (37b) because the former represents a case of identity, whereas the latter represents a case of inclusion, while (37c) is unproblematic as it constitutes a case of disjunction.²⁴

- | | | |
|------|---|------------------|
| (47) | a. * What do you wonder who could buy ___ ? | Identity |
| | <div style="display: flex; justify-content: space-around; margin: 0;"> [+Q] [+Q] </div> | |
| | b. ?? What book do you wonder who could buy ___ ? | Inclusion |
| | <div style="display: flex; justify-content: space-around; margin: 0;"> [+Q; +N] [+Q] </div> | |

²³A number of other relevant relations have been proposed: e.g., intersection (Belletti et al. 2012), complex identity, inverse inclusion (Villata et al. 2016).

²⁴Let me briefly address a terminological (and conceptual) issue here. Rizzi likely uses *disjunction* to refer to *disjointness*, which is a standard set-theoretic notion, whereas disjunction is traditionally a logical connective ($\vee$), meaning ‘or’, rather than a set-theoretic relation. Moreover, if Rizzi is right in assigning no active feature to the trivial intervener in (37c), then describing the set relation in question as disjunction is problematic not just from a terminological perspective. In set theory, two sets are disjoint if they share no elements, formally: $A \cap B = \emptyset$. At the same time, the empty set is *vacuously* a subset of any set, formally: $\forall A (\emptyset \subset A)$. Thus, given A and $\emptyset$, the relation between the two is both disjointness *and* inclusion, formally:

$$\forall A (\emptyset \subset A) \quad \wedge \quad (A \cap \emptyset = \emptyset)$$

While this does not necessarily entail a mathematical contradiction, it does present a challenge in a system where these two relations are expected to yield contrasting predictions. For a formal overview of these set-theoretic aspects, see e.g. Halmos (1974); for a more detailed discussion of this formal inconsistency in RM, see Presotto and Mattiuzzi (in prep.).

c. What do you think he could buy ___ ?

Disjunction

[+Q] []

(Adapted from Rizzi 2018: 348)

In this light, I can finally present the formal and, to the best of my knowledge, most recent definition of Relativised Minimality in (48), which will serve as the reference point for the remainder of this thesis.

(48) **Featural Relativised Minimality**

In a given configuration ...*X*...*Z*...*Y*... a local syntactic relation between *X* and *Y* is disrupted iff:

- (i) *Z* c-commands *Y* and *Z* does not c-command *X* (intervention condition).
- (ii) *Z* matches *X* in terms of active syntactic features (similarity condition).
- (iii) The degree of disruption is a function of the featural distinctness of *X* with respect to *Z*, in accordance with the Distinctness Hierarchy.

2.1.2 A shared assumption

In the preceding two subsections, I have outlined a concise map of the main stages in the historical development of locality theory. In particular, I have shown that these theories have been shaped by examples of ill-formed sentences that reveal the illicitness of certain movement operations in specific syntactic contexts. I have also sought to do justice to the diverse theoretical landscape that has emerged from these observations over the past six decades. While I could only consider a small subset of the published work, I hope to have at least manifested that numerous accounts address these issues, each with its own rules and theoretical nuances. Yet, despite their divergence, all these theories share a fundamental common denominator: they attribute the source of locality effects to grammar, and nothing else. The central hypothesis is that grammar alone imposes constraints requiring certain operations to be completed within configurationally defined limits. In this sense, locality is viewed as an inherent primitive of human grammars.

Bringing back the chess metaphor introduced at the beginning of this chapter, we could say that grammar-based accounts assign sole responsibility for locality constraints to the ‘software’, rather than the ‘RAM:’ islands are rules, a ban imposed by the algorithm.

I will now turn to proposals that take the opposite stance on locality, arguing that the true culprit behind locality effects is, in fact, the RAM, or more precisely, the limited computational resources available to the sentence parser. This perspective is represented by resource-based approaches, which I will discuss in Section 2.2. Before doing so, however, I will first outline some of the challenges faced by grammar-based theories.

I believe that clarifying these issues can help us comprehend why some scholars have sought alternative explanations for locality phenomena.

2.1.3 Some issues with grammatical approaches

Grammar-based accounts are, for the most part, enthusiastically embraced by generative linguists, who remain firmly convinced that grammatical rules must be central to the debate on locality. Such a steadfast theoretical stance has not been immune to critique, however. In fact, almost as soon as island effects were first described in the 1960s, alternative theories emerged, challenging the assumption that these effects must be explained syntactically. Over the years, contributions from various fields have increasingly highlighted a number of issues common to grammar-based approaches. First and foremost, maintaining the innateness of locality constraints comes with the drawback of lacking a principled explanation. That is, the question of why island constraints exist and why they look as they do is substantially left unanswered (Boeckx 2012: 27).²⁵ What is more, there is an acknowledged, though partial, redundancy in the accounts provided by distinct syntactic principles. For instance, although intended to account for, respectively, strong- and weak-islands, subjacency-like principles and Relativised Minimality happen to overlap in predicting the ill-formedness of some sentences (Boeckx 2012: 23). So much so that proposals have been made to unify locality constraints under a unique theory based on Relativised Minimality alone (Starke 2001; Abels 2003; Rizzi 2009; Zeijlstra 2023; Keine and Zeijlstra 2025).

At a closer look, even when we limit our focus to strong islands, certain inconsistencies emerge. For instance, Boeckx (2012), Boeckx and Grohmann (2004, 2007) and Boeckx and Gallego (2011) point out that the advent of phases and the PIC sits somewhat uneasily with some of the most basic insights from earlier stages of locality theory—that is, non-complements are opaque to movement. Let me unpack this. At least since Ross' Sentential Subject Constraint (i.e., no element can be extracted from a CP if that CP is a subject), and later the Adjunct Condition (i.e., the ban on extraction out of adjuncts), it has been clear that such syntactic domains behave differently from complements. This insight, which essentially associates subjects and adjuncts while distinguishing them from complements, was later formalised and generalised by Huang's (1982) *Condition on Extraction Domain* (CED), building on an original intuition by Cattell (1976). The CED has been revised extensively over the years (see, e.g., Kayne 1984; Chomsky 1986; Cinque 1990; Manzini 1992), but its core can be summarised as follows: movement must not cross a barrier, where a barrier is an XP that is not a complement (see Müller 2010: 36).

²⁵That said, we should not forget that phase-based analyses are ultimately grounded in more general principles of computational efficiency, and not just in strictly syntactic considerations.

This is straightforwardly exemplified in the contrast considered by Boeckx (2012: 55):

- (49) a. Who did you see [pictures of ___]?
 b. *Who did [pictures of ___] cause Bill to leave the room?
 c. *Who did Bill arrive [after seeing (pictures of) ___]?

In other words, extraction out of complements is allowed, or at least less rigidly constrained (49a), whereas extraction out of non-complements is prohibited (49b, c). Phase theory, however, revolves around a different logic: that the complement of a phase head becomes inaccessible (i.e., opaque to further syntactic operations), while the phase edge, including specifiers, remains accessible (see above). On this view, specifiers are exactly the domains from which extraction should be licit. But this prediction seems directly at odds with the contrasts in (49): the PIC appears to predict that complements should be more opaque than specifiers, effectively inverting the generalisation captured by the CED. I could not put it more directly than Boeckx (2012: 62), who writes: “[t]he PIC simply gets the CED backward.”

Several attempts have been made to resolve this apparent contradiction by refining the formulation of the PIC, many of which I have cited above. Perhaps the most formally precise of these is Müller’s (2010) proposal. Albeit generally well-argued and technically rigorous, such systems have not always met with unanimous acceptance (to put it mildly). For instance, Boeckx (2012: 62–72) offers a rather uncompromising critique of Müller’s approach. Without delving into the technical details, I align myself with the general sentiment that, even if one finds these solutions reasonable, they inevitably require a substantial degree of stipulation in order to derive traditional CED effects from the PIC. This is, in fact, a broader and frequently voiced criticism of grammar-based theories of locality (see below). Moreover, it seems to me that in recent years a growing number of scholars have acknowledged the intrinsic limitations of phase-based approaches to locality constraints. Indeed, even Chomsky et al. (2023: 65) concede that “[...] mechanisms like the PIC are inadequate on their own to model islandhood.”

At a more empirical level, moreover, many have pointed out that the datasets upon which the original generalisations on locality were built (Chomsky 1964, 1977, 1986; Ross 1967; Rizzi 1978; Huang 1982) were rather limited and, above all, have been weakened by an extensive number of counterexamples over the years. These counterexamples, in the words of Hofmeister and Sag (2010), “have been labeled as ‘special’, ‘peripheral’, or ‘unique’, in the absence of independent motivation” and have led towards refinements of the theory that have the inevitable flavour of ad-hoc stipulations (see also Chung 1994; Hofmeister 2011; Hofmeister et al. 2013). More generally, the very existence of so-called amelioration effects (discussed extensively in the following chapters)

and cross-linguistic variation in the acceptability of island-violating structures should prompt us to reconsider the uncritical treatment of all locality effects as purely syntactic phenomena. Within a formal, syntax-based account, it is extraordinarily easy to handle such (apparent) exceptions by introducing, for instance, language-specific parameters. In this sense, syntactic accounts might even appear “dangerously powerful” (Phillips 2013a: 106).

Typical cases of cross-linguistic variation in island effects include the well-known Swedish (and more generally, Scandinavian) data. For instance, it has often been observed that Swedish appears to allow extraction from relative clauses, as in (50).²⁶

- (50) Den teorin känner jag ingen som tror på.
 that theory know I nobody that believes in
 ‘That theory, I know nobody who believes in.’

(Adapted from Phillips 2013a: 75)

Another oft-cited instance comes from Rizzi (1982), who shows that Italian allows certain *wh*-extractions (52) that are ungrammatical in English (51).

- (51) *What_i do [_{IP} you wonder [_{CP} who_j [_{IP} ____j likes ____i]]]?

(Phillips 2013a: 76)

- (52) Tuo fratello, [_{CP} a cui_i mi domando [_{CP} che storie_j abbiamo
 your brother to whom 1SG.REFL wonder which stories have.3PL
 raccontato ____i ____j], era molto preoccupato]
 told was very worried
 ‘Your brother, to whom I wonder what stories they told, was very worried.’

(Adapted from Phillips 2013a: 76)

Not only does such variability in the emergence of island effects pose a challenge to the internal consistency of the theory, but it also raises the question of how these exceptions can be acquired by language learners. As Phillips (2013a: 107) observes, “[...] to date there are almost no accounts of how island constraints could be learned from the limited data available to children.” A problem, he continues, that “[t]hose who are skeptical of formal accounts of islands accurately see [...] as a serious flaw”.

In short, formal syntactic accounts face the following issues:

- They lack a principled explanation.
- They involve a substantial degree of stipulation.

²⁶See, however, Engdahl (1980, 1982), Nordgaard (1985) and Kush et al. (2018) for a fuller presentation of the data, as well as the references in fn. 37 for alternative analyses questioning whether these are genuine island violations.

- Amelioration effects and extraction asymmetries, in particular, are difficult to account for without resorting to ad hoc mechanisms.
- Exceptions and cross-linguistic variation pose a challenge for learnability. While universal locality constraints do not need to be learned, language-specific ones do, along with exceptions and ameliorations within a given language. How do children acquire these patterns given the very limited input available to them?

2.2 The alternative: reductionist approaches

What if locality constraints are not determined by innate, domain-specific principles of human grammar? What if what we call *syntactic islands*, *intervention effects*, or *blocking categories* are merely the epiphenomenal outcome of language-independent factors? Questions like these form the foundation of reductionist theories. *Reductionist* because, as Sprouse et al. (2012) and Phillips (2013a) aptly put it when coining this (by now standard) label, all such theories share the goal of *reducing* locality effects to extra-grammatical factors. Within this perspective, then, locality is understood as the product of mechanisms external to the syntactic component and, in essence, independent of it.

It is crucial to distinguish between two subgroups within the broader family of reductionist approaches and not treat them as equivalent (a point that Phillips himself underlines, Phillips 2013a: 81). The first subgroup encompasses semantic and pragmatic approaches to locality. While these approaches reject a syntactic explanation, they remain firmly anchored within the realm of formal theories. They still argue that island effects arise from grammatical rules, but they maintain that these rules are governed by principles rooted not in syntax, but in semantics or pragmatics. Semantic and pragmatic accounts are thus *reductionist* in a very different sense from resource-based accounts, which form the second subgroup within reductionism. Whereas the former maintain that grammatical principles govern locality effects (albeit through semantics or pragmatics rather than syntax), the latter claim that grammar plays little to no role in determining locality effects. The various theories of locality mentioned so far can therefore be classified into three groups: at one end are syntax-based accounts (e.g., subjacency-like principles and Relativised Minimality); at the other are resource-based accounts, which attribute locality effects to domain-general cognitive mechanisms (see below). Positioned between them are semantic and pragmatic accounts, which, again, exclude syntax as the determining factor but still recognise grammar as the source of locality (Fig.2.1).

I will first briefly review the main ideas behind semantic and pragmatic accounts. Then, I will turn to resource-based approaches, which by far constitute the largest group of alternatives to traditional locality accounts.

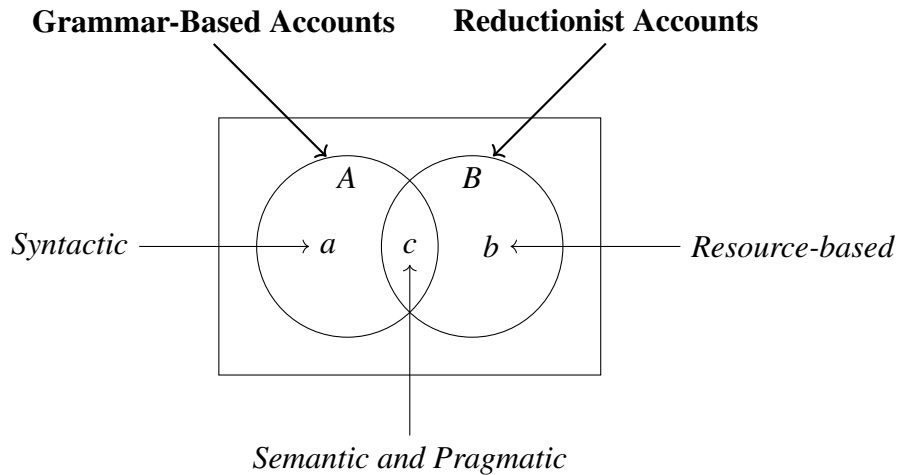


Figure 2.1: Classification of locality theories based on current terminology.

2.2.1 Semantic and pragmatic accounts

The general, intuitive idea shared by this reductionist subgroup is that some locality effects do not arise because of illicit syntactic operations, but rather from semantic and pragmatic inconsistencies. These proposals typically focus on weak islands, arguing that their oddness stems from their meaning. Put simply, proponents of these theories maintain that weak islands are syntactically well-formed but fail to make sense.

This perspective was first introduced by Szabolcsi and Zwarts (1990), who proposed that downward-entailing operators (such as negation) trigger intervention effects. In later work (Szabolcsi and Zwarts 1993, 1997), they refined their theory by adopting a Boolean algebra approach to scope-taking expressions (e.g., negation and quantifiers), offering a precise formalisation of certain intervention effects. The core idea is that each scopal operator behaves like a Boolean function over a set of individuals. For example, negation corresponds to taking the Boolean complement of a set ($\neg$), existential quantification corresponds to join ($\vee$), and universal quantification corresponds to meet ($\wedge$). This allows for a formal characterisation of scope-taking but also predicts when wh-extraction is blocked. Specifically, Szabolcsi and Zwarts argue that weak islands emerge when a wh-phrase must take scope over an operator whose associated Boolean operation is undefined for the relevant semantic domain. This explains why negation and universal quantification systematically block wh-movement: when a wh-phrase ranges over a partially ordered domain, the required Boolean operation (e.g., complement, meet) may be ill-defined, rendering extraction impossible (see the cited sources for details, and Szabolcsi 2006 for an overview). However, While Szabolcsi and Zwarts' Boolean approach provides a compelling mathematical formalisation of intervention, it does not fully capture contextual and pragmatic factors that have been shown to influence weak island acceptability, as highlighted

in work by Kroch (1989), Fox and Hackl (2007), Abrusán (2007, 2011a,b, 2014).

In particular, Abrusán (2014) presents an account whose central claim is that weak islands are ill-formed because they inevitably entail a contradiction. This contradiction arises in two possible ways. First, when the question, which is assumed to inherit the presuppositions of all its potential answers, requires a set of presuppositions that are mutually incompatible. Second, when the *Maximal Informativity Principle* (Dayal 1996), which requires that questions have a maximally informative answer, is not met, making it impossible to provide a coherent response without contradiction.

The first scenario applies to cases of *presuppositional* or *factive* islands, where factive verbs such as *regret* make extraction out of their complement CP unacceptable, as seen in the following pair:

- (53) a. How does John think that he behaved at the party?
 b. * How does John regret that he behaved at the party?

Abrusán argues that this effect arises because factive verbs introduce *universal presupposition projection*, meaning that all possible answers must share the same presupposition (Abrusán 2014: 60ff.). When extracting a manner or degree wh-word, these presuppositions become mutually contradictory, leading to ungrammaticality. Crucially, this effect does not extend to extraction of individuals, as their presuppositions do not introduce conflicting alternatives (Abrusán 2014: 62). This explains why sentences like the following are acceptable:

- (54) Who does John regret that he invited?

The second scenario arises when no maximally informative answer can be provided, violating Dayal's Maximal Informativity Principle. As mentioned, this principle requires that a wh-question have a single most informative answer. In the case of negative island contexts, however, negation prevents this from being established (Abrusán 2014: Ch. 3). Much like in factive islands, the distinction between questions about manner and questions about individuals is crucial, as only the former lead to ungrammaticality:

- (55) a. Who didn't John invite to the party?
 b. * How didn't John behave at the party?

There are further technical aspects of Abrusán's account that I leave aside for present purposes. Independently of these formal details, however, there are aspects of her account that appear problematic.

First, if questions about individual entities should make extraction acceptable, why do sentences like (56a, b) remain marginal?

- (56) a. ? In which way does Rowan regret that he behaved __?
 b. ??? In which way didn't Ellie behave __?

In (56a), the use of the lexically restricted manner-wh should ensure that the question pertains to a single, specific behaviour that Rowan regrets adopting, making it comparable to cases like (54). Yet (56a) appears somewhat degraded in comparison to (54). This contrast is even more pronounced in (56b), which is markedly worse than (54).

Second, I believe Abrusán's account cannot explain the perceived difference in the following pair:

- (57) a. ? With whom do you regret that John came __?
 b. Whom do you regret that John came with __?

The logical form of (57a) and (57b) is the same, yet (57b) is more acceptable than (57a). If acceptability in intervention-affected sentences were solely a matter of semantic contradiction computed at LF, these two sentences should not exhibit any difference. I will show that a more minimal interpretation can account for this and similar cases. Indeed, a closer examination reveals that (57a) is ambiguous, whereas (57b) is not. This is because in (57a), the extracted adjunct *with whom* can function as either a modifier of the medial verb *regret* or a modifier of the embedded predicate *come*. This ambiguity does not arise in (57b), as the stranded preposition in the embedded clause creates a gap that must necessarily be filled. In other words, given an appropriate context, a legitimate answer to (57a) could be one that specifies the person with whom the addressee regrets that John came (e.g., *With Britt, Britt and I regret that John came*). In this case, the question would have the following representation:

- (58) With whom do you regret __ that John came?

I will explore this proposal in greater detail in Chapter 5, where I will also present experimental evidence in support of it. Here is a spoiler, though: this explanation naturally extends to most argument–adjunct asymmetries. Whereas adjuncts can reconstruct more freely, arguments cannot, as they typically have only one possible gap to fill. Thus, the contrast in (55) follows straightforwardly. In fact, under this view, the distinctions based on the idiosyncratic differences between individuals and manners in Abrusán's account would potentially no longer be necessary.

I still believe that semantic and pragmatic factors play a crucial role in many locality effects, and I agree, for instance, with Abrusán's suggestion that negative islands constitute one such case (Abrusán 2014: 87ff). In fact, it seems likely that semantic arguments could be extended to other instances of intervention effects involving negation. Abrusán focuses on cases of wh-extractions in negative questions, but I suspect a semantic analysis

should also cover cases like those examined in Rizzi (2004) concerning adverb fronting. The relevant minimal pair is displayed in (59).

- (59) a. Rapidamente, i tecnici hanno risolto il problema
 rapidly the technicians have solved the problem
 ‘Rapidly, the technicians solved the problem.’
 b. * Rapidamente, i tecnici non hanno risolto il problema
 rapidly the technicians not have solved the problem
 ‘Rapidly, the technicians didn’t solve the problem.’

(Adapted from Rizzi 2004)

Rizzi explains this contrast by arguing that the manner adverb *rapidamente* moves from its base-generated position to a Mod(ifier)P in the left periphery. In doing so, it crosses over negation. Since Negation-features and Modifier-features belong to the same A-bar class (see above), intervention effects arise. I do not believe this explanation is correct for at least two reasons. First, moving a manner adverb to ModP over another manner adverb should trigger the same effects, or even stronger ones. Yet, this prediction is not borne out, as shown in (60):

- (60) a. Rapidamente, il velocista ha fatto ripetutamente il giro della pista.
 rapidly the sprinter has done repeatedly the lapse of.the track
 ‘Rapidly, the sprinter repeatedly ran around the track.’
 b. Ripetutamente, il velocista ha fatto rapidamente il giro della pista.
 repeatedly the sprinter has done rapidly the lapse of.the track
 ‘Repeatedly, the sprinter rapidly ran around the track.’

Note that debating the relative height of the two adverbs in Cinque’s (1999) hierarchy is futile. If *ripetutamente* is higher than *rapidamente*, then (60a) should exhibit intervention effects. Conversely, if *rapidamente* is higher than *ripetutamente*, then (60b) should exhibit intervention effects. While it is true that the two sentences convey slightly different meanings,²⁷ neither sentence is unacceptable, contrary to RM’s predictions.

Second, the contrast is more intuitively understood by considering the semantics involved. Fronting *rapidamente* extends its scope over the entire sentence. Now, much like in English, *rapidamente* ‘rapidly’ is arguably a factive adverb, meaning that it presupposes the occurrence of the event it modifies. If that same event is then negated, a contradiction arises. Unsurprisingly, fronting another manner adverb that is not factive results in clear amelioration:

²⁷In (60a), the sprinter could have performed some laps poorly, but their overall performance was good. In (60b), the sprinter must have made every single lap very fast.

- (61) Coraggiosamente, i tecnici non hanno risolto il problema.
bravely the technicians not have solved the problem
'Bravely, the technicians didn't solve the problem.'

I agree that this sentence sounds somewhat quirky, but I assume that a specific context could make it acceptable (unlike the example in 59b). For instance, it could mean that the technicians were bold enough not to fix a particular problem as a form of protest against their employer, who, say, had cut their preventive measures. Another, perhaps more natural, example illustrating this can be found in (62):

- (62) Coraggiosamente, mio nonno non si piegò al regime.
bravely my grandpa not himself bow to.the regime
'Bravely, my grandpa did not bow to the regime'

I include these examples to emphasise the actual necessity of incorporating semantics (in a broad sense) into the set of factors crucial for explaining locality effects.²⁸ In fact, I will demonstrate in Chapter 5 that contextual and discourse aspects play a pivotal role in ameliorating weak islands.

Before turning to resource-based accounts it is worth noting that, in addition to semantic explanations, some proposals attribute locality phenomena to more bona fide pragmatic factors. Certain accounts, for instance, focus on discourse cohesiveness and information-structural constraints, arguing that island effects arise from disruptions in the natural flow of the conversation rather than from strictly syntactic or semantic constraints (see, e.g., Erteschik-Shir 1973; Goldberg 2006, 2013; Abeillé et al. 2020; Cuneo and Goldberg 2023). Goldberg (2006), for example, argues that islands resist extraction because they introduce backgrounded information, which clashes with the foregrounding requirements of *wh*-questions. Since *wh*-questions seek salient, new information, extracting a *wh*-element from a backgrounded environment creates a pragmatic inconsistency (Goldberg 2006: ch.7). Moreover, based on experimental evidence from Kluender and Kutas (1993), Goldberg argues that backgrounded material is harder to process, making extraction from these contexts more cognitively costly (Goldberg 2006: 151ff).

It would be foolish to dismiss the role of discourse entirely in island phenomena. The very fact that D-linking has been identified as a key factor in extraction from weak islands (see above) underscores the importance of considering contextual factors when evaluating locality effects. However, purely pragmatic accounts fall short of explaining certain effects that appear to be irreducibly syntactic in nature. For instance, Boeckx (2012: 29) points out that cases like **What did John eat ham and ___?* remain ungrammatical regardless of context manipulation. He also cites a note by Benjamin Bruening, written in a

²⁸Due to space constraints, I must leave a fuller elaboration of these observations for future work.

2010 blog entry (Bruening 2010), which presents a series of compelling counterexamples to accounts like Goldberg's. In essence, Bruening casts doubt on the key assumption common to pragmatic accounts that some (allegedly) non-syntactic phenomena exhibit island sensitivity, which these accounts take as evidence that islands are not syntactic constraints. For instance, Goldberg (2006: 132–133) claims that direct replies to questions are island-sensitive. In (63), for example, the reading *She is dating someone new* is only available when the phrase is not contained within an island.

- (63) Q: Why was Laura happy?
- a. The woman who lives next door thought she was dating someone new.
 - b. #The woman who though she was dating someone new lives next door.
 - c. It's likely that she's dating someone new.
 - d. #That she's dating someone new is likely.
 - e. John said that she was dating someone new.
 - f. #John shouted that she was dating someone new.

Yet, Goldberg herself acknowledges exceptions, and Bruening provides additional ones, demonstrating that direct replies *can* occur inside islands like, e.g., complex NPs (64).

- (64) Q: Why was Laura so happy?
- A: I talked to someone who said she was dating someone new.
(cf. *Who did you talk to [someone who said she was dating ___]?)

Moreover, Bruening shows that a pragmatic account cannot explain the following contrasts:

- (65) A: Yesterday I met a man who claimed that John stole something.
B1: What?
B2: *What did you meet a man who claimed tha John stole ___?
- (66) A: John was furious because his wife had taken something of his.
B1: What?
B2: *What was John furious because his wife had taken ___?
- (67) A: We've been assigned to read a book and write a certain kind of paper.
B1: What kind of paper?
B2: *What kind of paper have we been assigned to read a book and write ___?

This presents a major challenge for any pragmatic account that rejects the role of syntax in determining island constraints. He summarises his argument as follows:

In each of these cases, A makes a statement containing an indefinite. B seeks to determine the referent of that indefinite. B can do that with sluicing in all of the (B1) cases. B cannot do that by repeating the entire sentence, with extraction of a wh-phrase corresponding to the indefinite (the B2 cases). Yet the discourse contexts for B1 and B2 are identical. It is simply impossible to explain the contrast between B1 and B2 as being due to discourse. [...] I conclude from all of the above that discourse approaches to islands simply do not work, and should be abandoned.

Thus, while pragmatic factors are likely to influence the emergence and modulation of locality effects, there are serious challenges to the claim that they are the primary, let alone sole, cause of (weak) islands' unacceptability.

2.2.2 Resource-based accounts

I began this chapter by comparing human language to a chess game on a computer, borrowing Hofmeister et al.'s (2013) clever analogy. As a matter of fact, though, the distinction between the linguistic counterparts of RAM and software was popularised in the scientific community long before these concepts lent themselves to an intuitive metaphor. As early as (1963), Miller and Chomsky discussed the challenges posed by centre-embedded clauses such as *The student (that) everyone (that) I talked to trusts arrived late*. Although no grammatical rule prohibits the formation of such structures, they often feel awkward, if not outright unacceptable (cf. *The man who the boy who the student who the teacher admired knew saw arrived*). In the absence of an evident (and rational) grammatical constraint, Miller and Chomsky concluded that their general unacceptability must stem from processing limitations. That is, formal grammars allow indefinitely deep recursive structures, but human cognitive capacities impose strict constraints on how much recursion can actually be processed. Sentences with multiple centre-embedded clauses, they argued, overwhelm memory because they require holding several incomplete dependencies simultaneously. In terms of our present-day analogy, then, Miller and Chomsky's observations drew an early distinction between RAM and software in the language system, highlighting the contrast between what a grammar allows and what humans can realistically process. This distinction became a cornerstone of Chomsky's later theories, which famously established the divide between *competence* and *performance*. Most formal linguists have since focused on competence, aiming to rigorously model the formal properties of Universal Grammar, often at the expense of considering processing effects in shaping linguistic theory. This somewhat narrow focus has led to some dissatisfaction within a part of the scientific community. Some scholars, skeptical of the increasingly intricate rules and stipulations invoked to explain linguistic phenomena, sought to reassert

the role of processing in linguistic explanation. Locality effects were no exception. Inspired by Miller and Chomsky's (1963) original hypothesis, a strand of research emerged advocating for processing-based explanations to locality. These accounts constitute the second major group of reductionist approaches and form the focus of this subsection.

One of the earliest resource-based interpretations of island phenomena dates back to (1979), when Givón argued that the unacceptability of sentences like those in (68) were due to processing factors.

- (68) a. *The man who I saw [the dog that bit ____] fell down.
 b. *Who did you see [the dog that bit ____]?

Essentially, he maintained that (68a) and (68b) are not ruled-out by grammatical constraints but are instead simply difficult for the parser to integrate. While Givón's arguments were not grounded in particularly developed cognitive notions, subsequent research has drawn on more refined psychological categories. For instance, Deane (1991) identified *attention* as the crucial resource in sentence parsing. He argued that extraction phenomena create a situation in which the language user must simultaneously attend to two parts of the syntactic structure, placing strain on the limited working memory available for automatic syntactic processing, and that attention should be regarded as the "critical variable for a cognitive theory of extraction" (Deane 1991: 4). This is, in fact, the logic at the core of all resource-based accounts, which build on the idea that sentences' well-formedness is a function of their computational cost. In this regard, Boeckx (2012: 32) writes:

The general intuition behind [resource-based] reductionist arguments is that island constraints reflect cognitive limits, e.g., on attention: island domains are defined by elements that divert the limited resource of attention away from filler and gaps. Put differently, the idea is that there is a processing cost associated with the operations necessary to build the syntactic structures that we have come to call islands, but, according to the reductionists, islands do not reflect structural constraints. Rather, the perception of unacceptability arises as a by-product of the processing requirements of the relevant sentences.

This line of reasoning laid the foundation for a prolific body of research aimed at identifying (or, more precisely, demonstrating) the higher processing cost associated with locality-affected structures (Kluender 1991, 1992, 1998; Kluender and Kutas 1993; Hofmeister and Sag 2010 a.o.). The reason why particular structures prove more demanding than others have been connected to disparate factors, like, for instance, the higher cost of integrating a referential expression (Warren and Gibson 2002, 2005; Hofmeister and Sag 2010),

the difficulties associated with keeping similar information in the working memory (Gordon et al. 2001, 2004), or the misanalysis caused by temporal ambiguity (Pritchett 1991, and Siloni 2014 for an overview).

Yet another strand of research has specifically investigated how and to what extent islands affect on-line processing. In the vast majority of cases, these studies have concluded that islands have a significant impact on the parser. For instance, Stowe (1986) demonstrated the *Filled Gap Effect* is sensitive to the boundaries of island domains. She found that the reading times at the boldfaced pronouns in the sentences in (69) were slower when a wh-dependency is involved, like in (69a). This suggests that readers initially attempted to interpret the wh-element (*who*) as the direct object of *bring* and were forced to reanalyse the structure upon encountering the pronoun. These findings align with the hypothesis that fillers are “active,” meaning they follow a mechanism that seeks to complete a filler-gap dependency as soon as possible (Crain and Fodor 1985; Frazier and Flores D’Arcais 1989; Aoshima et al. 2002).

- (69) a. My brother wanted to know who Ruth will bring **us** home to ___ at Christmas.
 b. My brother wanted to know if Ruth will bring **us** home to Mom at Christmas.

Interestingly, the *Active Filler Strategy* does not apply when a potential gap is located within an island boundary. Stowe observed that readers exhibited no slowdown upon encountering *Greg* in (70), suggesting that they did not attempt to reconstruct the wh-filler as the argument of the preposition *about*, which appears within a subject island.

- (70) a. The teacher asked what the silly story about **Greg’s** older brother was supposed to mean ____.
 b. The teacher asked if the silly story about **Greg’s** older brother was supposed to mean anything.

Similar results have been reported across various experimental paradigms, including eye-tracking studies (Traxler and Pickering 1996) and event-related brain potential (ERP) measures (Garnsey et al. 1989).

Yet another class of experiments has explored the impact of islands on on-line processing by examining the disruptive effects that arise at island boundaries. For instance, McKinnon and Osterhout (1996) measured ERPs at the word *when* in cases where it marked the boundary of an adjunct island (71a) and in cases where it simply introduced an adverbial (71b). Crucially, only in the island condition did *when* elicit a P600 effect, a neural response typically associated with syntactic anomalies or processing difficulty.

- (71) a. I wonder which of his staff members the candidate was annoyed **when** his son was questioned by ____.

- b. I wonder whether the candidate was annoyed **when** his son was questioned by his staff member.

These findings have since been replicated in further studies, which are thoroughly reviewed and discussed in Sprouse et al. (2013b: 83ff). At this point, however, I believe it is important to emphasise that while quantitative confirmation of island effects through processing measures is both valuable and informative, it does not resolve the debate on the source of locality constraints. Sprouse et al. (2013b: 85) make this point explicitly:

It is certainly interesting to find that so many of the island effects that we can observe in off-line judgments also impact the online search for gap sites. But this correlation between off-line and on-line island effects does not help to decide between competing accounts of the cause of island effects. Formal grammatical accounts can straightforwardly capture the correlation by assuming that island constraints are deployed quickly and effectively by the parser. Meanwhile, resource-based reductionist accounts can equally well explain the correlation, by claiming that the same resource limitations that are responsible for island effects in acceptability judgements constrain on-line dependency formation.

In sum, the question of the source of locality remains open and cannot be resolved merely by measuring idiosyncratic on-line processing strategies in locality-affected sentences. Though promising and, I should add, necessary, this type of evidence can only take us so far in exploring questions on the origin of locality. As Phillips notes, indeed, most competing accounts can generate predictions compatible with the findings of on-line processing studies.

There are other critical aspects related to processing accounts, some of which are often vigorously emphasised by opponents of reductionism. I will discuss these in the next subsection. Before turning to them, however, I will briefly examine two representative accounts that illustrate how resource-based approaches attempt to explain *intervention* locality. Up to this point, I have reviewed theories and experiments concerned with strong islands (i.e., cases of impenetrability locality, see above). Yet some processing-based proposals have also sought to interpret intervention effects as arising from non-syntactic factors. In particular, many studies have focused on object relative clauses and, more broadly, object A-bar dependencies, which are a classic test-bed for intervention effects.

It is notorious that moving an object across an intervening subject often creates some sort of degradation. I will dedicate a whole chapter to these effects, focusing on object relative clauses in Spanish (Chapter 4). For now, suffice it to say that there are many different accounts to interpret and predict these phenomena. In syntactic theory, the standard analysis of intervention in A-bar dependencies relies on Relativised Minimality. Yet, as said, there have been alternative proposals that attempt to explain the asymmetry between

subject and object A-bar dependencies in terms of independent cognitive factors. To introduce them, let us look at some relevant examples. Object relative clauses like (72b) have been repeatedly found to be more difficult to process than their subject counterparts (72a).

- (72) a. The barber that ___ admired the lawyer climbed the mountain.
b. The barber that the lawyer admired ___ climbed the mountain.

(Gordon et al. 2001: 1414)

The same holds for cleft sentences, where object extraction (73b) is harder than subject extraction (73a).

- (73) a. It was the barber that ___ saw the lawyer in the parking lot.
b. It was the barber that the lawyer saw ___ in the parking lot.

(Gordon et al. 2001: 1418)

There are two main processing-based accounts to these facts, the *Increased referential processing approach* (Warren and Gibson 2002, 2005), and the *Similarity-based approach* (Gordon et al. 2001, 2004).

The Increased referential processing approach predicts the asymmetries in (72) and (73) based on the hypothesis that a syntactic dependency becomes hard to integrate when its computation is obstructed by elements requiring substantial referential processing. In other words, Warren and Gibson argue that if an element in the intervening subject position imposes a high processing cost for the assignment of its discourse referents, it will hinder the integration of the object dependency. Clearly, this approach can generate precise predictions only when combined with a hierarchy of nominal elements ranked according to their (independent) processing cost. The elements occupying the highest positions in this hierarchy are elements that, when appearing in the critical site, cause the greatest parsing difficulty. For instance, Warren and Gibson maintain that definite descriptions (i.e., definite DPs headed by a common noun) are harder to process than proper names and pronouns. This explains why replacing *the lawyer* in a sentence like (73b) with a proper name (74a) or a pronoun (74b) results in faster reading times and improved accuracy in comprehension.

- (74) a. It was the lawyer that Dan saw ___ at the party.
b. It was the lawyer that we saw ___ at the party.

(Warren and Gibson 2005: 757)

However, the Increased referential processing approach cannot explain why sentences like (75a) and (75b) also exhibit faster reading times:

- (75) a. It was Dan that the businessman saw ___ at the party.
 b. It was you that the businessman saw ___ at the party.

(Warren and Gibson 2005: 757)

Because these sentences display a definite description in the subject positions, they should yield the same processing difficulties of sentences like (73). Yet, again, this prediction is not borne out.

The *Similarity-based approach* proposed by Gordon et al. (2001, 2004) fares better in this respect. In this account, the focus is on the relation holding between the moved object and the intervening subject, rather than the subject alone. Gordon et al. argue that processing difficulties arise whenever these two elements are similar. Consequently, sentences like (75a) and (75b) are predicted to be processed more quickly because of the dissimilarity between the object, a common noun, and the subject, which in these cases is a proper name or a pronoun. At the same time, when the subject and the object do match, a higher processing cost is expected, successfully accounting for the slower reading times of (72b) and (73b).

One might notice that the *Similarity-based approach* closely resembles Relativised Minimality, as both rely on the notion of similarity to explain and predict intervention effects. However, two crucial differences set them apart. First, Relativised Minimality pairs its similarity condition with an intervention condition, defined hierarchically in terms of c-command (see above). Second, in Relativised Minimality, similarity is determined through a formal syntactic computation that involves only specific features. Because of this, general similarity between two elements is insufficient to trigger an intervention effect; they must be similar in terms of relevant syntactic features, and the degree of similarity is structured along a hierarchy of set-theoretic relations. Note that this difference is not merely conceptual, the two accounts generate contrasting predictions in cases like (76).

- (76) It was you that we avoided ___ at the party.

(Warren and Gibson 2005: 757)

In this case, both the moved object and the intervening subject are pronouns. As such, the *Similarity-based approach* predicts a disruptive effect similar to (73b). In contrast, Relativised Minimality predicts no intervention effects, as the two syntactic objects are endowed with different person features. Experimental results from Warren and Gibson (2005) show that (76) yields significantly faster reading times than (73b), confirming the predictions of Relativised Minimality (see Belletti and Rizzi 2013 for further discussion).

In sum, neither the Increased referential processing approach nor the *Similarity-based approach* fully accounts for the complex pattern of emergent intervention effects in ob-

ject A-bar dependencies. While the Increased referential processing approach is too permissive, since it predicts a disruption only when the intervening subject is a full DP, the Similarity-based approach is too rigid, as it incorrectly predicts intervention effects when both the subject and object are pronominal.

2.2.3 Some issues with reductionist approaches

I have already discussed some limitations of semantic and pragmatic accounts in my review of them. To reiterate, I have emphasised (as, I am sure, most formal linguists would) that considering semantic and pragmatic factors is indispensable for a thorough understanding of locality effects. Any credible theory of locality must incorporate both at some level. However, I have also shown that relying solely on semantico-pragmatic properties is insufficient to explain the intricate pattern of locality effects and that some of the assumptions underlying these theories face serious challenges. Moreover, these accounts explicitly aim to explain only a limited subset of locality effects—namely, weak islands. Even though I agree that recognising distinctions among the wide range of phenomena grouped under the broad label of locality effects is important, this should not lead us to abandon the pursuit of a comprehensive formal theory of locality. Any account that deliberately restricts its scope to a narrow subset of locality effects is, in my view, limited in its very premise. Unless we categorically distinguish weak and strong islands as entirely unrelated phenomena, I believe it is always preferable to seek a theoretical framework that, while acknowledging their differences, also unifies them as related phenomena.

The other major class of reductionist theories—resource-based accounts—is also not entirely immune to criticism. In fact, I have already highlighted some of the difficulties these accounts face in generating predictions that align with empirical evidence. But processing-based approaches encounter several additional challenges that, to date, keep eluding a conclusive resolution (see Sprouse et al. 2013b, Phillips 2013a,b, and Boeckx 2012: 37–46 for a complete list and extended discussion).

To begin with, as with grammar-based theories, cross-linguistic variation in the emergence (and intensity) of locality effects poses a serious problem for reductionist approaches. Still, whereas theoreticians can appeal to parametric rules to account for such differences, it is far less clear how purely processing-based accounts can address this issue. Under the assumption that human cognition is one and the same for speakers of different languages, the reason why cross-linguistic variation in island taxonomy should emerge is not obvious (Phillips 2013a: 97–100).

This also relates to the problem of learnability, which I previously mentioned when reviewing the challenges of syntactic approaches (see Section 2.1.3). That is, how can a child acquire all (language-specific) locality constraints based on so little (positive) input?

One might assume that reductionism circumvents this issue altogether. After all, if locality effects emerge as surface phenomena caused by limited computational resources, children would have no syntactic constraints to acquire. Yet Phillips (2013a: 103) reasons that this assumption may very well introduce even greater problems. If locality constraints are universal, then there is nothing for children to learn, and the absence of (locality-violating) sentences in their input would not be surprising, as it would align with their innate knowledge. However, if no such innate knowledge exists, children might infer grammatical patterns from these gaps in the target grammar, potentially leading to misguided or incorrect generalisations. The issue becomes even more complex with language-specific constraints. If these are encoded as syntactic rules, they surely pose a challenge, yet so do many other language-specific rules that are not universal. If, however, resource-based theories are correct, then cross-linguistic variation in locality effects should be linked to cross-linguistic differences in resource deployment, which brings us back to the problem outlined above: cognitive resources are not expected to vary depending on the language one speaks. And even if we concede that speakers of different languages manage processing resources differently, “[i]t is unclear how children could learn about cross-language differences in resource deployment” (Phillips 2013a: 104).

Another empirical puzzle often raised in evaluations of resource-based approaches concerns the phenomenon of satiation, which refers to the effect whereby repeated exposure to an unacceptable sentence leads to its amelioration. If this held true for islands, it would pose a challenge for syntactic accounts: why would repeated exposure improve a sentence if it cannot be generated by the grammar in the first place?²⁹ Snyder (2000) tested this effect using standard island violations and found mixed evidence, with only some islands exhibiting satiation effects. Interpreting his results, Snyder suggested that those islands showing satiation effects might indeed call for an extra-grammatical explanation. However, Sprouse (2009) attempted to replicate Snyder’s (2001) findings but observed no satiation effects, and concluded that Snyder’s results were likely compromised by design flaws (see Sprouse 2009 and Boeckx 2012: 41 for further discussion and implications). Even though the absence of satiation effects may not constitute a full-fledged objection to processing accounts, it certainly weakens the empirical case against grammatical accounts.

But what perhaps constitutes the most compelling counterargument to reductionism has to do with the very premises on which resource-based approaches explain locality effects. These theories claim that the unacceptability of island-violating sentences arises because *wh*-dependencies and island structures are independently costly, and their co-occurrence in the same structure causes significant disruption. A first problem with this

²⁹I am here abstracting away from the distinction between derivational and representational theories.

explanation is that, whereas numerous studies have confirmed the independent processing cost of wh-dependencies (Gibson 1998; Phillips et al. 2005; Sprouse et al. 2012; Wagers 2013), evidence for the independent cost associated with island structures remains sparse. In fact, Sprouse et al. (2012) found that only one out of the four island types they examined significantly impacted acceptability ratings, suggesting that, in the absence of wh-dependencies, island structures do not necessarily impose a particular burden on the parser. At a more conceptual level, moreover, it is not immediately clear which factors make island structures so problematic, and why. Consider, for example, the idea that many island structures, such as complex NPs or relative clauses, place a burden on the parser because they contain “informationally rich” nominals, with complex referential and lexical semantic properties (Kluender 1991; Warren and Gibson 2002, 2005; Hofmeister and Sag 2010). If complex referentiality is indeed taxing for the parser, why is it that extracting D-linked, information-rich elements from islands is easier? In other words, “[i]f it’s easier to maintain D-linked elements in working memory, why is accessing their mental referents so disruptive?” (Boeckx 2012: 37). In addition, the idea that severe degradation arises when a wh-dependency must be processed simultaneously with another costly operation is neither sufficient nor entirely accurate. For instance, this cannot explain why unacceptability arises only when the gap is *inside* an island (77a), and not when it is merely linearly separated from its filler by the island (77b).³⁰

(77) a. * Who did John read [the report that Mary kissed ___]?

b. Who did John give [the report that Mary died] to ___?

(Boeckx 2012: 37)

Given that in both cases a long-distance wh-dependency cannot be completed before encountering an island structure, the wh-filler must be kept in working memory while another (allegedly) burdensome operation is processed. Yet (77b) is much better than (77a). What is more, the very notion of “simultaneity” remains hard to formalise in capacity-based theories (Sprouse et al. 2012: 89–90).

Now, let us consider once more why locality effects are expected to emerge under a processing account. As said, resource-intensive operations, such as those associated with wh-movement and island structures, are assumed to carry independent cognitive costs. When both are present within the same sentence, they are said to combine in a way that overwhelms the parser, resulting in unacceptability. According to accounts such as Kluender’s (1991; 1992; 1998; 2004) and Hofmeister and Sag’s (2010), it is the conjunction of these independent costs that overloads the processing system and gives rise

³⁰Boeckx (2012: 37) does not mark (77a) with an asterisk. Given the context in which it is presented, however, I assume it is simply a typo.

to island effects. As emphasised by Sprouse (2007), however, this effect is not merely additive, but rather *superadditive*: the perceived unacceptability of island-violating sentences is greater than the sum of the individual effects associated with wh-dependencies and island structures in isolation. To better illustrate this, let me briefly summarise how this conclusion was supported.

In a highly influential paper, Sprouse et al. (2012) designed a 2x2 factorial experiment manipulating two independent variables with two levels each: DEPENDENCY LENGTH (short vs. long), and TYPE OF STRUCTURE (island vs. non-island). The four resulting conditions are exemplified in (78).

- | | | |
|------|--|-------------------|
| (78) | a. Who ___ thinks that John bought a car? | short; non-island |
| | b. What do you think that John bought ___? | long; non-island |
| | c. Who ___ wonders whether John bought a car? | short; island |
| | d. What do you wonder whether John bought ___? | long; island |
- (Sprouse et al. 2012: 86)

The use of a factorial design like this makes it possible to isolate the effects of the individual factors using subtraction.³¹ In this case, the difference between (78a) and (78b) isolates the effect of the length of the wh-dependency, and the difference between (78a) and (78c) isolates the effect of the island structure. These effects can be quantified by comparing the mean acceptability ratings for each condition (i.e., the dependent variable considered in the study). Crucially, Sprouse et al. found that the effect size observed in condition (78d) exceeded the linear sum of the isolated effects of dependency length and island structure. Statistically, this indicates an interaction between the two factors—that is, a *superadditive* effect. In mathematical terms, this can be expressed as: $(78a - 78d) = (78a - 78b) + (78a - 78c) + X$, where X represents the superadditive effect. This X is also referred to as the *difference-in-differences* or *DD score* (Maxwell and Delaney 2004), and it will play a central role in the discussion that follows.

Now, a basic resource-based theory might not predict such an interaction at all, assuming instead that island effects are simply the outcome of ‘piling up’ multiple costly operations. Such a prediction would make sense only if the parser had unlimited capacity. Under a more realistic assumption of limited processing resources, however, superadditive effects are (according to reductionists) expected. This is precisely what resource-based accounts *à la* Kluender propose: the superadditive effects reflect a psychological interaction between the two sets of processes represented by the two experimental factors, stemming from their reliance on a shared pool of limited resources (Sprouse et al. 2012:

³¹For a detailed description of this methodology, see Sprouse and Villata (2021), on which the present discussion draws.

87). Grammar-based approaches, on the other hand, interpret the emergent interaction as the effect of a grammatical constraint that penalises only sentences that contain wh-dependencies that enter island structures. Both frameworks can thus accommodate the superadditive effect. Crucially, though, they generate contrasting predictions with respect to a central (yet often under-represented) dimension of locality effects: individual variation. Specifically, resource-based approaches predict that the magnitude of island effects should co-vary with processing capacity. That is, individuals with lower working memory or attentional resources should exhibit stronger island effects, while individuals with greater cognitive capacity should show weaker effects. Under a strictly grammatical theory, instead, the parser's resource limitations are not expected to modulate the effect. Any individual variation in acceptability judgements would therefore not be attributable to processing capacity *per se*. Back to the DD-score, it was exactly this measure that Sprouse et al. used to test these contrasting predictions. Specifically, they examined whether participants' DD-scores correlated with two independent measures of working memory capacity: serial recall and the *n*-back task.³² Crucially, they found no significant correlation, which led them to conclude that their results align more closely with grammar-based theories of locality effects.

To summarise, there appear to be several challenges to the explanatory power of reductionist accounts of locality effects. In this section, I have only reviewed the objections that I personally find most compelling, but a number of other interesting counterarguments have been raised in recent years (see the references cited throughout this subsection, and in particular Sprouse and Hornstein's 2013 volume). A condensed list of issues is outlined below:

- Semantico-pragmatic accounts have a very limited scope, typically addressing only weak islands, and some have been shown to generate incorrect predictions.
- Processing-based accounts of intervention effects are either too permissive or too restrictive, and fail to capture all and only those cases of degradation observed in object A-bar dependencies.
- Resource-based reductionism poses difficulties for theories of the learnability of locality effects.

³²A possible criticism is that these specific measures tap into cognitive components that are not directly involved in the processing of island structures. The authors acknowledge this possibility but consider it unlikely, given that most working memory measures are known to be highly correlated. As Sprouse et al. (2013b: 94) note, moreover, substantiating this concern would require the existence of a specific measure that (i) correlates with island effects, but (ii) does not correlate with either serial recall or the *n*-back task—something for which there is currently no evidence.

- Cross-linguistic variation in locality effects is difficult to explain within current reductionist frameworks.
- Satiation effects do not provide convincing support for extra-grammatical explanations of island effects.
- There is an apparent inconsistency regarding referentiality: information-rich structures are assumed to burden the parser, yet D-linking appears to facilitate referent maintenance in working memory.
- No clear correlation has been found between working memory capacity and the acceptability of island-violating structures.

2.3 And the compromise: grounded approaches

There is a third class of approaches to locality, which Sprouse et al. (2012) term *grounded theories*. In a certain sense, grounded theories reconcile the hypotheses of both reductionist and grammar-based accounts. Indeed, grounded approaches tend to reference constraints on language processing while maintaining that island constraints are grammar-based. That is, they are similar to resource-based accounts in advocating arguments based on computational limitations of the parser, but crucially differ from these in ascribing island phenomena to core grammatical rules. More concretely, grounded theories characterise locality constraints as the grammar's adaptive response to cognitive pressures: sentences that are too difficult to process have been progressively ruled out by the syntax (Fodor 1978, 1983; Berwick and Weinberg 1984; Hawkins 1999). It is thus worth stressing that grounded accounts, rather than representing a true alternative to grammar-based theories, can be seen as a complementary attempt to explain why grammars feature locality constraints and how these have evolved over time. Synchronically, these two approaches are in fact largely equivalent and generate comparable predictions.

To better illustrate the logic of grounded theories, let me consider Hawkins's (1999) proposal, which offers a representative formulation of this framework. The central hypothesis defended by Hawkins is that grammars evolve to favour structures that are easier to process. A claim later formalised as the *Performance–Grammar Correspondence Hypothesis* (PGCH):

(79) **Performance–Grammar Correspondence Hypothesis**

Grammars have conventionalised syntactic structures in proportion to their degree of preference in performance, as evidenced by patterns of selection in corpora and by ease of processing in psycholinguistic experiments.

(Hawkins 2004: 3)

As seen in (79), Hawkins grounds his hypothesis in two types of evidence: typological patterns and psycholinguistic data, which I will discuss below. His motivation to explore an alternative explanation for locality arises from two central concerns with traditional grammar-based theories. First, subadjacency-like principles often seem either too strong or too weak, and frequently fail to capture the cross- and intra-linguistic variation observed in locality effects. Second, traditional grammatical constraints exhibit a substantial degree of ad-hocness and generally fail to unify all the relevant syntactic, semantic, and lexical factors. In this regard, he writes (Hawkins 1999: 245):

There is [...] a large amount of stipulation, and an absence of motivation or explanation for constraints of generality. Why should complex NPs be less hospitable environments for gaps than other structures? Why are there so-called wh-islands? Why are resumptive pronouns found in place of gaps in some syntactic positions and not others? We are simply told that this is the way things are. It is also apparent that there are numerous factors that contribute to well-formedness in this area—syntactic, semantic, and lexical—yet no unifying principle has been proposed for why all of them should be relevant and for why they combine together in the ways they do to give us the variation we observe.

This excerpt nicely illustrates what is perhaps the central goal of grounded theories: the search for a *motivation* for locality constraints. This motivation, according to Hawkins, lies in the tendency of grammar to adapt itself to the cognitive constraints under which it is processed.³³

In developing his theory, Hawkins starts from four basic assumptions.

First, the Active Filler Hypothesis (already discussed in Section 2.2.2) holds:

(80) **Active Filler Hypothesis**

When a filler of category XP has been identified in a non-argument position, such as COMP, rank the option of assigning its corresponding gap to the sentence over the option of identifying a lexical phrase of category XP.

(Clifton and Frazier 1989: 292)

In essence, this means that the filler must be held in working memory until the gap is encountered. Moreover, the filler is said to be “active” in the sense that the parser tries to associate it with the first possible gap site to resolve the dependency as quickly as possible.

Second, the complexity of a filler-gap dependency can be computed based on structural size: the more intervening nodes between the filler and the gap, the more complex

³³I should note that this is in fact a fairly standard assumption, even within formal frameworks, at least since the advent of Minimalism, which rests on the idea that grammar is a “perfect solution” to the needs of the interface systems. I will return to these considerations in more detail in Chapter 3.

the dependency. In sum, the complexity of a *Filler-Gap Domain* (FGD) is a function of its structural depth:

(81) **Filler-Gap Domain (FGD)**

FGD consists of the smallest set of terminal and non-terminal nodes dominated by the mother of a filler and on a connected path that must be accessed for gap identification and processing; for subcategorised gaps the path connects the filler to the gap's subcategoriser and includes, or is extended to include, the gap's dependent and disambiguating arguments (if any); for non-subcategorised gaps the path connects the filler to the gap site; all constituency relations and co-occurrence requirements holding between these nodes belong to the description of the FGD.

(Hawkins 1999: 248)

Third, the sentence processor favours smaller FGDs, as formalised by the *Minimise Domain* principle:

(82) **Minimise FGDs**

The human processor prefers FGDs to be as small as possible.

(Hawkins 1999: 251)

Fourth and finally, these processing pressures are assumed to shape grammars over time. In particular, if a complex FGD is allowed by the grammar of a given language, then all simpler FGDs should also be allowed. In other words, grammatical constraints exhibit implicational patterns, whereby increased complexity implies the availability of all less complex configurations:

(83) **Filler-Gap Complexity Hypothesis**

If an FGD of complexity n on a complexity hierarchy H is grammatical, then FGDs for all less complex variables on $H(n - 1)$ will also be grammatical.

(Hawkins 1999: 252)

What is particularly interesting about this hypothesis is that it makes rather robust predictions. Let us assume that a costly syntactic strategy is available to a language in a specific context, and that this context occupies a particular point on an independently defined scale of complexity. Then, the strategy in question is predicted to be applicable to all less complex contexts on that scale. In sum, when some degree of complexity is permitted, "it appears to be hierarchically organized, with grammaticality cutting off in different languages at fixed and implicationally arranged points" (Hawkins 1999: 252).

Hawkins supports this hypothesis by considering well-known typological patterns in relativisation, first thoroughly described by Keenan and Comrie (1977) in their *Noun*

Phrase Accessibility Hierarchy (NPAH) (SU = subject; DO = direct object; IO = indirect object; OBL = oblique; GEN = genitive/possessor):

(84) **Noun Phrase Accessibility Hierarchy**

SU $\succ$ DO $\succ$ IO $\succ$ OBL $\succ$ GEN

(Keenan and Comrie 1977)

The NPAH, assumed to be universal, indicates that the higher a function ranks, the more likely it is to be relativisable—that is, *accessible* to relativisation. For instance, if a language can relativise direct objects, it must also allow for subject relativisation. In many languages (included English) relativisation is formed via a filler-gap dependency, in which the head of the relative clause is the filler and its interpretive site within the relative clause is the gap.³⁴ Building on this, Hawkins motivates the NPAH by showing that higher-ranked positions correspond to smaller FGDs. He argues that “there is a clear general correlation between the number of dominating and co-occurring nodes required for relativizable positions and their [NP]AH ranking” (Hawkins 1999: 253). For example, the structural domain between a relative clause head and its gap is smaller in subject relative clauses than in object relative clauses, since subjects are merged in a higher position (Spec,vP). In contrast, object relative clauses require a larger FGD, one that includes at least the subject nodes as well (see Hawkins 1999: 253ff. for details). Thus, if a grammar tolerates the complexity of object relativisation via filler-gap dependencies, it follows that it will also permit subject relativisation via the same strategy.

Note that these predictions apply to relativisation *strategies*, not merely to the general possibility of relativising a given function in the NPAH (see Salzmann 2017: 22ff. for discussion). For example, if the filler-gap strategy is licensed, it must be available for subject relativisation, then for direct objects, and so on, as formalised below:

(85) **Relative Clause Gap Hierarchy**

If a relative clause gap is grammatical in position *P* on a complexity hierarchy *H*, then gaps will be grammatical in all higher positions on *H*.

(Hawkins 1999: 256)

This prediction is strongly supported by Keenan and Comrie’s (1977) data, which show that all sampled languages allowing for gap-relativisation limit gap positions to exactly those permitted under the implicational hierarchy in (85).³⁵ See the list in (86), adapted from Hawkins (1999: 256–257):

³⁴I will discuss the theory and derivations of relative clauses in detail in Chapter 4.

³⁵This applies to what Hawkins calls [–Case] gap relativisation. That is, cases where a filler-gap dependency is formed and the relative is introduced by a relativiser, not a relative pronoun bearing case morphology. A different implicational scale applies in those cases (see Hawkins 1999: 260). I will return to the distinction between relativisers and relative pronouns in Section 4.1.3.

- (86) Languages that allow [-Case] relative clause gaps in Keenan and Comrie's data:

SU only: Aoban, Arabic (Classical), German, Gilbertese, Iban, Javanese, Kera, Malagasy, Maori, Minang-Kabau, Tagalog, Toba Batak

SU and DO only: Chinese (Pekinese), Dutch, English, Finnish, Fulani, Genoese, Greek, Hebrew, Italian, Malay, Persian, Swedish, Tongan, Welsh, Yoruba, Zurich German

SU and DO and IO/OBL only: Basque, Catalan, Korean, North Frisian, Hausa, Roviana, Shona, Sinhala, Spanish, Tamil, Turkish

SU and DO and IO/OBL and GEN: Japanese

Other “simpler strategies” may be employed by grammars to facilitate the parser when relativising lower-ranked positions. For instance, when relativisation involves a resumptive (or “copy”, in Hawkins’ terms) pronoun, processing becomes easier, “since an empty category no longer needs to be inferred from its environment but is given formal expression” (Hawkins 1999: 257). As a result, gap strategies are expected to target higher positions in the NPAH, while lower positions are more likely to involve strategies such as resumption. This idea is formalised in the following principles:

(87) **Gap Avoidance Hypothesis**

If a (potential) FGD of complexity n on a hierarchy H is avoided by employing a semantically equivalent alternative structure, then gaps in all more complex environments on $H(n+1)$ will be avoided as well.

(Hawkins 1999: 258)

(88) **Relative Clause Copy Pronoun Hierarchy**

If a copy pronoun in a relative clause is grammatical in position P on a complexity hierarchy H , then copy pronouns will be grammatical in all lower positions that can be relativised at all.

(Hawkins 1999: 258)

One might wonder why gap relativisation exists at all. If resumption is easier, why is it not employed across the board? Hawkins suggests that gaps may have other advantages. While resumptive pronouns offer clear benefits in terms of explicit dependency marking, they introduce additional morphological material. Gaps, by contrast, provide “economy of expression” (Haiman 1983) as they involve “less form processing” (Hawkins 1999: 259). In essence, there seems to be a competition between reduced morphological (form) processing and explicit dependency marking. These complexity differences are thus expected to map onto contexts of varying complexity: gaps to less complex ones (smaller FGDs), and resumption to more complex ones (larger FGDs). We could say, somewhat simplifying, that there appears to be a trade-off between morphological simplicity and

syntactic complexity, with grammars evolving to balance these demands. I will return to this idea of grammatical trade-offs in Chapter 4, where I show that Spanish-acquiring children benefit from explicit cues in the comprehension of object relative clauses, but are hindered when too much disambiguating information is provided, essentially reflecting Hawkins' intuitions on typological patterns.

Interestingly, Hawkins further expands his hypothesis to include classic examples of islands, such as Ross' Complex NP Constraint (see above). As a matter of fact, he generalises his predictions to all wh-gap and pronoun-gap dependencies, laying down a complexity hierarchy similar to the NPAH, the *Clause embedding hierarchy*:

(89) **Clause embedding hierarchy**

$$s[\dots VP[\dots O_i \dots] \dots] \succ s[\dots S'[\dots O_i \dots] \dots] \succ s[\dots NP[Det N S'[\dots O_i \dots] \dots]]$$

(Hawkins 1999: 263)

The hierarchy is to be interpreted as follows: infinitival phrases are more hospitable to gaps than finite embedded clauses and complex NP environments, with the latter being the most resistant to extraction. The rationale behind this distribution is the same as before: the positions of the hierarchy are defined in terms of FGD size, and the prediction is that if a gap is allowed in a position *n* of the hierarchy, it will be allowed also in all higher ones.

A language like Russian, for instance, allows for the formation of gaps in infinitival subordinates (90b) but not in finite embedded clauses (90a).³⁶

- (90) a. *Vot [ogurcy_i s[kotoeoye_i ja obeščal s' [čto prinesu O_i]].

'Here are the cucumbers which I promised that I'd bring.'

- b. Vot [ogurcy_i s[kotoeoye_i ja obeščal VP [prinesti O_i]].

'Here are the cucumbers which I promised to bring.'

(Hawkins 1999: 263, adapted from Comrie 1973: 297)

French (91, 92) or English (93, 94), by contrast, tolerate gaps also in finite subordinate clauses.

- (91) a. la femme_i s[que_i j'espere s' [que vous allez épouser O_i]]

'the woman who I hope that you are going to marry'

- b. la femme_i s[que_i j'espere VP [épouser O_i]]

'the woman who I hope to marry'

(Hawkins 1999: 263)

³⁶In (90), (91), (93), and (94) I report the notation used by Hawkins, where 'O' signals a gap.

- (92) a. Qui avez-vous essayé de voir ___?
 who have-you tried to see
 ‘Who did you try to see?’
 b. Qui espériez-vous qu’ elle voie ___?
 who hoped.2SG-you that she see.SUBJ.3SG
 ‘Who did you hope that she would see?’
- (93) a. Who_i S[did you try VP[to see O_i]]?
 b. Who_i S[did you hope S[that you would see O_i]]?
 (Hawkins 1999: 264)
- (94) a. The person_i S[that you tried VP[to see O_i]] is Harry.
 b. The person_i S[that you hoped S[that you would see O_i]] is Harry.
 (Hawkins 1999: 264)

But neither language allows gaps in complex NPs:

- (95) *Who do you know NP[the professor that taught ___]?
 (Adapted from Hawkins 1999: 264)
- (96) *Qui est-ce que tu connais NP[le professeur qui a embrassé ___]?
 who is-it that you know the professor who has kissed

Only a minority of languages allow gaps in the lowest rank of the hierarchy in (89). Swedish seems to be one such language, as shown in (97), where a gap is licensed within a relative clause.³⁷

- (97) ett ben_i som_j jag ser [en hund [som gnager på ___i]]
 a bone which I see a dog which is-gnawing on
 (Adapted from Hawkins 1999: 262)

Again, then, Hawkins supports his hypothesis by examining typological patterns and showing that the distribution of gap availability in relativisation (98) and wh-dependencies (99) conforms to the implicational predictions of the hierarchy in (89).

- (98) Permitted FGDs for relative clause heads:

- a. S[... VP’ ...]: Swedish, Japanese, English, French, Russian, German
 b. S[... S’ ...]: Swedish, Japanese, English, French

³⁷Note, however, that some have argued that the Swedish examples (and possibly others) are only apparent violations of the CNPC. Kush and Lindahl (2011), for instance, show that these are in fact extractions out of small clauses rather than true complex NPs, from which extraction is consistently banned (see also Phillips 2013a and Boeckx 2012: 39–40 for discussion).

- c. $s[... NP[Det N S'] ...]$: Swedish, Japanese

(Hawkins 1999: 263)

(99) Permitted FGDs for wh-gaps:

- a. $s[... VP' ...]$: Swedish, English, French, Russian, German
 b. $s[... S' ...]$: Swedish, English, French, Russian, German
 c. $s[... NP[Det N S'] ...]$: Swedish

(Hawkins 1999: 264)

Note that Hawkins does not restrict his predictions to FGD size alone. He further develops his theory by arguing that, even when two FGDs are equally complex, the human sentence processor still shows a preference for minimising syntactic processing effort:

(100) **Reduce Additional Syntactic Processing in FGDs**

The human processor prefers to minimise the syntactic rules and processing operations that apply in FGDs.

(Hawkins 1999: 269)

According to Hawkins, this preference is evidenced in classic weak island effects. While in (101a) there is only one filler-gap dependency to resolve, (101b) introduces an additional dependency, making integration harder and resulting in ungrammaticality.

- (101) a. What did you hope that they would bake __?
 b. *What_i did you wonder how_j they would bake ___i ___j?

(Hawkins 1999: 269)

From examples like these, Hawkins derives yet another hierarchy:

(102) **Subordinate Gap/No Gap Hierarchy**

If a matrix filler can be matched with a gap in a subordinate clause of complexity n containing another gap, then it can be matched with a gap in a subordinate clause of complexity n containing no other gap

(Hawkins 1999: 269)

Put simply, this means that if an environment allows for two gaps, then it must also permit all single-gap dependencies within that same environment. As expected, the likelihood of tolerating multiple gaps increases in structurally less complex environments. For instance, if we revise the examples in (101) to involve extraction from an infinitival

subordinate clause (which involves a smaller FGD than finite clauses, as shown in 89) we observe that multiple gaps are indeed permitted:³⁸

- (103) a. What did you hope to bake ___?
 b. What_i did you wonder how_j to bake ____i ____j?
 (Hawkins 1999: 269)

Similar considerations apply to semantic processing, as expressed by (104):

- (104) **Reduce FGDs Semantically**
 The human processor prefers to minimise the amount of semantic information that needs to be processed in an FGD.
 (Hawkins 1999: 271)

This principle should help explain why languages often favour semantically ‘lighter’ bridging verbs when a filler–gap dependency is involved. Kluender (1992) discusses examples originally reported by Culicover and Wilkins (1984), which show a gradient of degradation as the bridging verb intervening between *how* and its gap becomes semantically richer. Hawkins’ adaptations of the relevant sentences are reported in (105), where ‘>’ denotes ‘better than’ and ‘=’ denotes ‘equal to’:

- (105) a. How angry did Mary *say* that John was? >
 b. How angry did Mary *say softly* that John was? =
 c. How angry did Mary *whisper* that John was?
 (Hawkins 1999: 272)

These contrasts are taken by Hawkins to support the following hierarchy:

- (106) **Bridging Verb Hierarchy**
 If a bridging verb or verb complex *V* of semantic specificity *n* is grammatical in an FGD, then all verbs or verb complexes *V'* with less semantic specificity than *V* will also be grammatical.
 (Hawkins 1999: 272)

³⁸As Hawkins himself acknowledges, however, there are cases of island effects involving infinitival complements, such as *What_i did you ask [why_j to bake ____i ____j]?. Yet he rather nonchalantly leaves this observation uncommented. I would be more cautious in brushing aside the relevance of such examples, as they pose a serious challenge to the predictions of (102). Along similar lines, it is unclear to me how Hawkins’ theory would exclude a sentence like (i), which involves a single wh-gap inside an infinitival subordinate clause:

(i) ??How do you wonder [whether to bake ___]

This definition, of course, raises the problem of how to determine and quantify semantic specificity. Hawkins does not provide a full-fledged formalisation, somewhat glossing over how such quantification (and the resulting hierarchy) should be achieved. He briefly alludes to the possibility of referring to general “semantic components or features,” suggesting that, for example, *whisper* involves elements of both *say* and *softly*. Another possible criterion, he suggests, could be entailment: since *whisper* entails *say*, but not vice versa, the former should occupy a lower position on the hierarchy. These remarks remain impressionistic, however, and as such cannot support precise, testable predictions.

This anticipates some of the limitations that Hawkins’, and more generally grounded, approaches occasionally face, as discussed in Section 2.3.2: despite their theoretical appeal, they often lack the precision and formal rigour found in more established frameworks, chiefly grammatical ones. Moreover, leaving predictions at such a preliminary stage makes it difficult to distinguish competing hypotheses. For example, Hawkins interprets the contrasts in (107) as “a clear example of the preference” stated in (104): since definite NPs introduce more semantic and pragmatic specificity than indefinites, and possessive modifiers add further referential relations, the increasing degradation from (107a) to (107c) is explained in terms of growing semantic processing demands.

- (107) a. Who did you see *a* picture of? >
 b. Who did you see *the* picture of? >
 c. Who did you see *John’s* picture of?

(Hawkins 1999: 272)

Yet, once more, only a formal semantic theory of “specificity” and “complexity” could yield robust, falsifiable predictions. Furthermore, the contrasts in (107) can plausibly be attributed to syntactic complexity instead. After all, semantic distinctions are often mirrored in syntactic structure, and degrees of specificity are no exception: arguably, there is a syntactic inclusion relation between definites and indefinites (see Pinzin 2024 and references therein). Likewise, possessives such as *John’s* are structurally higher than simple definites. From this perspective, the pattern in (107) may reflect increasing *syntactic* processing demands, rather than *semantic* ones. This illustrates how Hawkins’ theory, in the absence of further development, may come across as redundant, as the same data could be accounted for by alternative mechanisms: the contrasts in (107) could be explained via (104), via (100), or simply by appealing to FGD size, thus simply referring to (81).

The list of examples in Hawkins’ paper continues with several other cases of degradation, each analysed through the same logic we have seen so far: observable contrasts are interpreted as reflections of increasing processing demands, and from these contrasts

Hawkins derives implicational hierarchies. For reasons of space, I will not review all of these cases here. My aim has simply been to illustrate the core rationale behind grounded theories: the attempt to motivate grammatical constraints by appealing to processing limitations. This brings me to one final, crucial consideration, which I explore in the next section.

2.3.1 A crucial distinction: grounded vs resource-based approaches

One might be tempted to think that Hawkins' approach is not fundamentally different from a resource-based one, perhaps representing something of a 'covert' version of the latter, distinguished primarily by terminology. After all, several (though certainly not all) processing-based theories are in fact closely tied to grammatical principles.³⁹ This is well illustrated by the frameworks presented in Berwick's (1992) edited volume, aptly titled *Principle-Based Parsing: Computation and Psycholinguistics*, which features contributions emphasising that the sentence processor is not independent from grammatical constraints but, rather, operates in accordance with grammatical constraints. Processing, in these accounts, is substantially grounded in grammatical structure. For example, in one chapter, Pritchett (1991) argues that island violations arise "from the combination of (a) local parsing decisions that ultimately prove not to be consonant with a global grammatical representation, and (b) the parser's inability to perform the structural reanalysis necessary to obtain a grammatical representation" (Pritchett 1991: 304). While his account clearly centres on parsing dynamics, it still relies crucially on a well-defined grammatical architecture. As Pritchett himself puts it, this is "a theory that itself necessarily makes crucial use of principles of grammar" (Pritchett 1991: 305).

Similar conclusions have also been drawn in more recent contributions, which maintain that grammatical structures are closely tied to parsing mechanisms (Lidz 2022), and that there is likely a single sentence analyser mediating between grammar and on-line processing (Lewis and Phillips 2015; Parker 2019).⁴⁰ In sum, even some resource-based accounts appear to fully acknowledge the fundamental role of grammar, arguing for a processing approach that explicitly incorporates grammatical considerations. So what, then, sets these apart from grounded theories, which are grammatical approaches that explicitly incorporate processing considerations? This is what Pritchett (1991: 302) has to say on the matter (*italics mine*):

³⁹For instance, Hofmeister and Sag (2010: 406) concede that "it is impossible to prove, given the current state of our knowledge of both grammar and processing, that grammar has no part in creating the observed processing differences [in island violations]."

⁴⁰See Section 3.2 for further discussion.

[T]he characterization of island constraint violations in terms of processing will have implications for grammatical theory since principles such as subadjacency will have no grammatical status if the data they attempt to characterize represent instead epiphenomena of performance. It is important to distinguish this claim from one that holds that Subadjacency is a grammatical constraint whose functional motivation is to be found in the structure of the parser. [...] Analyses of this latter sort *continue to maintain that Subadjacency violations are ungrammatical but that the syntax proper must incorporate such a constraint given the parser's intrinsic (possibly deterministic) structure* (see Fodor, 1985 for critical discussion). *This contrasts with the hypothesis defended here that the global representations associated with island violations are themselves grammatical, but locally lead to processing breakdown.* This approach is potentially stronger in that it must account for the empirical data involving island violations rather than simply motivate a grammatical constraint which in turn accounts for the empirical facts.

So while both resource-based and grounded theories may acknowledge the relevance of both on-line processing and grammatical principles, only the latter maintain that locality effects are grammatically encoded. In contrast, pure reductionist accounts treat locality as a by-product of processing overload. This is not merely a terminological or speculative distinction, it leads to concrete and testable differences in empirical predictions. This thesis is dedicated to exploring precisely these divergences, offering an attempt to identify cases where competing theories of locality generate distinct predictions. The identification and discussion of such (potentially) informative cases constitute the core of the coming chapters, but I can already offer a preview of how the aforementioned frameworks diverge in interpreting and predicting linguistic phenomena. To do so, I draw on what I consider to be a particularly insightful observation from Hawkins (1999), who discusses the following example:

(108) Who do you think that Fred believes that Bill suspects...that Mary saw ___?

The ellipsis indicates that the sentence could be recursively extended to include *n* other instances of finite subordinate clauses. Indeed, once a grammar conventionalises the possibility to include a wh-gap in a certain environment, “the possibility of necessity extends to several such clauses, because grammars cannot count” (Hawkins 1999: 267). So while the increasing depth of a gap will lead to increasing “processing complexity in performance”, its grammaticality remains intact.⁴¹ By contrast, if a non-licensed environment (here, a complex NP) occurs at any point along the path from filler to gap, the sentence will be ungrammatical:

⁴¹A similar observation was already included in Ross (1967), where it is emphasised that the distance between a gap and its antecedent in a long-distance dependency does not have a fixed upper limit:

- (109) *Who do you think that Fred believes that Bill heard the rumour that Mary saw ___?

Hawkins uses this contrast to emphasise that the availability of a gap is determined grammatically, via conventionalised rules. It is not entirely clear, however, whether he fully recognised just how crucial this contrast is for distinguishing his approach from, say, Pritchett's. If he did, he certainly stated it rather subtly. Let me spell it out more explicitly. The two sentences show that recursive embedding is permitted indefinitely, provided the recursive structure remains *grammatically licit*, but becomes ungrammatical the moment a grammatical constraint is violated. This is entirely expected if sentence generation is governed by grammatically encoded constraints. It is less straightforward, however, under a view in which unacceptability stems solely from processing difficulty. After all, in both examples, a wh-filler must be maintained in working memory while substantial information is processed. So why should only (109) be degraded? A purely resource-based approach would likely struggle to account for this contrast.⁴²

In sum, while bona fide resource-based accounts seek to derive locality (and possibly other) phenomena purely in terms of domain-general processing principles, grounded approaches propose that such principles are the *reason* why grammatical constraints exist and take the form they do. In this sense, “[g]rammatical conventions are ‘frozen’ processing preferences” (Hawkins 1999: 279), but they are not *just* processing preferences.

To return to my beloved computer metaphor: grounded approaches argue that RAM limitations force the software to adapt. But suppose we run the same software on a far more powerful computer. Under a grounded approach, the same constraints would still apply, unchanged (consistent, for that matter, with the lack of correlation between working memory differences and island acceptability discussed earlier). A purely resource-based theory, on the other hand, would predict a difference: having more resources would allow the system to compute more and generate more complex chess moves, or in our case, more complex sentences. Note, however, that even within a straightforward grounded approach, we cannot (and arguably should not) exclude the possibility that, over time, the software might evolve to permit less strict rules in response to increased computational capacity.

(i) What did Bill buy ___?

(ii) What did you force Bill to buy ___?

(iii) What did Harry say you had forced Bill to buy ___?

(iv) What was it obvious that Harry said you had forced Bill to buy ___? (Ross 1986: 5)

⁴²Especially if we consider that even a much shorter version of (109) like *Who did John hear the rumour that Mary saw? is clearly worse than (108).

2.3.2 Some issues with grounded approaches

As I did for grammatical and reductionist approaches, I conclude this section on grounded theories by briefly noting some of the challenges and open questions they face. In my review of Hawkins' proposal, I have already pointed out that several of his generalisations are only partially supported by empirical data. Although the core idea (namely, that syntactic complexity modulates cross- and intra-linguistic patterns) is well backed by typological evidence, many of the hierarchies derived from general principles such as reducing syntactic or semantic processing effort are in fact based on relatively limited datasets, often restricted to English. In addition, I noted that some of the factors posited to explain increases in processing cost are not sufficiently formalised, making the theory difficult to falsify and, at times, prone to internal redundancy or contradiction.

Also the problem of learnability of locality constraints, of course, still remains. It is hard to overlook the fact that language-specific constraints on movement are only sparsely represented in the input, making it difficult for the learner to draw reliable inferences based solely on negative evidence. Hawkins does offer a different perspective on this issue, however. In a brief note towards the conclusion of his paper (Hawkins 1999: 280–281) he writes:

My approach suggests a different solution to the negative evidence problem (which is a genuine problem regardless of one's theoretical position on innateness, cf. Bowerman 1988). Instead of regarding the constraints on filler–gap dependencies as innate, we can hypothesise that learners will comprehend their input and will postulate grammars in accordance with ease of processing. [...] In this way the simplest environment on each hierarchy will be postulated first on account of processing ease, and any extension beyond that to lower hierarchy positions will be forced on the learner by positive evidence from the relevant language. Hierarchies of complexity like the [NP]AH have already been shown to predict the order of language acquisition (see Hawkins 1987). I propose now that the processing complexity that motivates these hierarchies may also explain the negative evidence problem by providing a principled limit on initial hypotheses and by requiring extensions beyond this to be justified by the data of experience. [...] It is also possible that learning is triggered by encountering grammatical instances of a low hierarchy position gap with a certain filler, whereupon the grammaticality of all simpler instances of that construction can be inferred.

In short, Hawkins suggests that it is not the constraints themselves that are innate, but rather the complexity hierarchies, which are naturally derived from an innate drive towards processing ease. Again, though, this hypothesis needs to be tested against empirical data that validate its predictions, ideally for all the hierarchies proposed by the

author. Moreover, albeit intriguing, Hawkins' take on the learnability problem remains largely speculative, and he himself concludes his note acknowledging that "these questions require further research."

One final issue I would like to address concerns, in fact, all the approaches to locality reviewed so far, though it arguably affects grammatical accounts to a lesser extent. Throughout this chapter, I have tried to illustrate just how complex and multi-faceted the landscape of locality effects really is. These effects vary widely in their severity and invite interpretations across syntactic, semantic, pragmatic, and processing domains. At the same time, I hope to have shown that this variation is not entirely random: there remains a somewhat clear divide between more severe and less severe constraints, with differing degrees of susceptibility to circumvention.

For instance, strong islands are not only 'strong' in the sense of being rigid or 'absolute' in blocking extraction, they also appear to resist rescuing strategies more systematically, or altogether. In contrast, weak islands allow for more flexibility: even extractions that are in principle illicit can sometimes be salvaged through appropriate manipulation (see Chapter 5). To reiterate a point made in Section 2.2.3, I believe that a robust theory of locality must be able to account for these differences, and grounded approaches such as Hawkins' fall short in this regard.

As I will illustrate in the next chapter, this thesis seeks to make two central contributions. The first is to reaffirm the importance of grammatical representations in accounting for locality constraints. The second (closely tied to the issue just described) is to do justice to the differences between strong and weak islands. An issue that, to varying degrees, remains problematic for all the theories reviewed so far.

3

The Hypothesis: Locality as an Interface Phenomenon

Now that the main theoretical approaches to locality have been surveyed, I am in a better position to articulate what is perhaps the overarching research question of this thesis: which of these accounts offers a more adequate explanation? And since each essentially locates the source of locality in a different domain, the question can also be framed as: where do locality effects originate? Independently of how it is formulated, however, this question presupposes several others. How can we evaluate which (class of) account(s) proves more compelling? Is there a way to pinpoint the origin of specific locality effects? That is, a reliable ‘diagnostic’ for determining *where* a given effect arises? Moreover, the rich array of theories reviewed in Chapter 2 cannot be forced into just three main categories: reductionist, grammar-based, and grounded approaches. Within each class, there are multiple proposals, each with significantly different implications for the theoretical status of locality. For instance, assuming locality is driven by syntactic principles, do these operate on derivations or representations? And within resource-based approaches, do processing constraints interact directly with grammatical structure, or do they apply later, at the surface string level? Finally, as noted in the Introduction, there seems to be a kind of ‘division of labour’ among existing theories: each explains a particular subset of data, but none accounts for all the facts. How is it possible that so many (often conflicting) proposals are, at least partially, successful? Why are locality effects so heterogeneous and fragmented?

All these questions could easily inform a full-length treatise. Yet I believe there is a way to approach them within a unified perspective. As outlined in the Introduction, I will tackle the main question of locality’s source by examining when and how amelioration effects arise, and what kinds of factors give rise to them. Indeed, I will show that

amelioration effects offer a promising yet underexplored domain for advancing our understanding of locality. In what follows, I lay out in more detail how I will pursue this approach through the investigation of two main empirical phenomena. Then, after briefly reviewing some core theoretical assumptions of the Minimalist framework, I will present my hypothesis, showing how it follows naturally from these assumptions. I conclude the chapter with a summary and some remaining open questions.

3.1 Amelioration effects as a window into locality

As I have tried to show throughout the last chapter, locality effects are diverse and multifaceted. Their empirical manifestations span a wide range of phenomena, with varying degrees of severity and interpretability across syntactic, semantic, pragmatic, and processing domains. So much so, that it is extremely difficult to produce clear hypotheses (and predictions) embracing all its possible expressions. And in this landscape, many approaches appear at least partially creditable and ‘right.’ This, I believe, raises two problems. First, the ultimate theory of locality may not exist. Or if it does, it will need to be flexible enough to integrate multiple explanatory dimensions. Indeed, a credible account cannot be ‘just syntactic’ or ‘just resource-based’ (see, e.g., Hofmeister and Sag 2010; Boeckx 2012; Dillon and Hornstein 2013). Second, the empirical and theoretical scope of locality makes the pursuit of a comprehensive theory too ambitious a goal for a single thesis. Nonetheless, there remain important and tractable questions. Of these, one in particular stands out: how are locality effects ameliorated?

It is well known that island constraints are not uniformly rigid. Under certain conditions, even ‘illicit’ extractions can be repaired or circumvented.¹ However, the relationship between these amelioration effects and the broader questions of why locality constraints exist and where they originate remains rather underexplored. Some scholars have suggested that island obviation may shed light on the nature of islandhood itself (Merchant 2001, 2008; Lasnik 2001; Boeckx 2003, 2008a, 2012), arguing that “island circumvention data can provide a rich source of insights into the nature of islandhood” (Boeckx 2012: 89). Yet, to my knowledge, little work has taken this suggestion further in a systematic way.² Moreover, most such discussions have focused on repair strategies for (strong) islands, leaving aside other well-documented amelioration effects, such as those seen in other object A-bar dependencies like, e.g., relative clauses.

This thesis builds and elaborates on that insight. Therefore, instead of focusing on degradation *due to* locality constraints, I will focus on *amelioration of* locality constraints. The logic is as follows: if the factors that regulate amelioration reflect, at least in part, the factors responsible for degradation, then the study of amelioration effects may offer a productive (and relatively underexplored) avenue for understanding the source of locality itself. Another way to frame this connection emerges when applying a single circumven-

¹I use the terms *amelioration*, *circumvention*, *obviation*, and *repair* rather interchangeably here. However, the choice of a specific term is neither trivial nor innocuous, as each might bespeak distinct theoretical assumptions. For instance, if something is ‘repaired,’ it must have been generated first, which strongly suggests a derivational approach to locality (see Boeckx 2012: 79ff. for discussion). I will return to these considerations in Chapter 6.

²A notable exception is represented, however, by recent (and to my knowledge still unpublished) work by Jon Sprouse and Sandra Villata (see fn. 21).

tion strategy to different structures: if the same strategy yields different outcomes across, for example, weak- and strong-islands, then the constraints at play must differ in kind. I will pursue this line of inquiry looking at the two prototypical types of locality—namely, intervention and impenetrability.

Let us begin with intervention, the focus of Chapter 4. To reiterate, the central question is whether intervention effects are best explained by pure processing mechanisms or grammar-based theories, particularly in light of amelioration patterns. More specifically, I ask: is amelioration of intervention effects guided by grammatical principles, or by general processing constraints? I approach this question through the best-known and most developed syntactic account of intervention: Relativised Minimality (reviewed in Chapter 2). This framework makes relatively precise predictions, which I assess against both prior literature and, chiefly, new experimental evidence. My empirical contribution focuses on Spanish subject and object relative clauses, with particular attention to the roles of word order and Differential Object Marking (DOM) in facilitating their comprehension and processing. Beyond reinforcing the well-attested asymmetry between subject and object relatives (see, e.g., Lau and Tanaka 2021), I will argue that the data are, in fact, incompatible with the predictions of Relativised Minimality. Instead, they point towards an *ambiguity-resolution* account, suggesting that amelioration arises when the parser can quickly rule out misleading alternatives and posit the relative head gap in the correct position. As I will discuss, this explanation resembles garden-path resolution accounts of the kind mentioned in Section 2.3.1 (e.g., Pritchett 1991). More importantly, however, this kind of explanation aligns more naturally with processing-based, reductionist theories than with strictly syntactic ones. A conclusion that, I will reason, accords well with certain foundational principles of the Minimalist Program (see Sections 3.1.1 and 3.2).

This, however, is only a first step. As discussed above, processing-based accounts are not monolithic. So a question arises as to which, among them, better accounts the data. Whereas some approaches posit a close connection between processing and grammatical representations, others are fully grammar-independent, attributing locality effects solely to domain-general computational pressures. The former rest on the assumption that sentence processing is intimately tied to syntactic or, more broadly, grammatical knowledge. By contrast, the latter deny any strictly necessary role for grammatical competence and appeal instead to general cognitive mechanisms that are not language-specific.³ Crucially, these two perspectives yield different empirical predictions, which I will test by further

³I will elaborate on these distinctions in Chapter 4. For now, note that grammar-independent accounts typically invoke strategies that do not presuppose access to deep grammatical structures, such as probabilistic biases or shallow processing heuristics. For instance, they might associate the ameliorative effect of a cue with the mere statistical co-occurrence between its phonetic form and a preferred reading, without involving any structured grammatical representation.

examining *how* the cues mentioned above (i.e., word order and DOM) can support ambiguity resolution in Spanish object relative clauses. The key question is this: how do these cues generate an amelioration effect? Does the parser rely on their internal grammatical representation, or does it simply exploit a surface-level association between their presence and, say, a statistically preferred interpretation? Put differently, is knowledge of the grammar underlying these cues a necessary condition for their ameliorative effect?

To address this, we would ideally examine a population exposed to Spanish and capable of parsing linguistic input, but not (fully) sensitive to the grammatical function of, say, DOM.⁴ As it turns out, children offer an ideal test case. In Chapter 4, I show that children who have not yet acquired the grammar of DOM do not benefit from its presence, while those who have show comprehension patterns more closely aligned with adult speakers. This suggests that while locality amelioration effects may be compatible with processing-based explanations, they cannot be fully reduced to grammar-independent heuristics: access to underlying representations remains crucial. In other words, these findings rule out an entire class of resource-based theories: those that rely exclusively on general surface cues. In contrast, they support *principled* processing approaches such as those proposed by Pritchett (1991), Kluender (1991, 1992, 2004) and Hofmeister and Sag (2010) (see Chapter 2). In essence, I argue for an approach in which locality effects may arise because of processing demands, but can be circumvented when appropriate and, crucially, *acquired* grammatical cues are available. An important point I will further emphasise when discussing these findings is that they offer direct support to the view that grammatical knowledge and its abstract representations play a central role in shaping both the emergence and the mitigation of locality effects. An idea frequently assumed, but, to my knowledge, rarely if ever demonstrated empirically.

This opens up a further question: can locality *always* be circumvented? Despite claims that truly ‘absolute’ islands may not exist (see, e.g., Ross 1967; Postal 1999; Boeckx 2003, 2012), it is difficult to deny that certain configurations (typically those classified as strong islands) are significantly more resistant to extraction.⁵ In Chapter 5, I examine both weak and strong islands and test whether extraction from them can be ameliorated. Once again, I show that supporting the positioning of the gap (in this case, a wh-gap) plays a central role: weak islands can be mitigated by manipulating the context such that the syntactic representation of the wh-question licenses the interpretation of the wh-

⁴In particular, the role of the morpheme *a* in marking direct objects (see Section 4.1.3 for details)

⁵The very fact that the degradation associated with intervention effects is expected to vary as a function of the (featural) similarity between the target and the intervener of a chain suggests that weak islands are more amenable to amelioration—thus generating a somewhat *milder* penalty overall. By contrast, strong islands tend to block extraction more categorically. Note, however, that even within the latter class, there is notable variation in the severity of degradation, particularly in cases involving adjuncts (Dal Farra 2020; Porta 2024).

gap within the island. Interestingly, however, the same strategy proves ineffective in the case of strong islands, which remain severely degraded even under comparable contextual manipulations. An asymmetry, this, that points to qualitatively different constraints, ultimately suggesting that there is a fundamental divide between strong and weak islands.

This is just one aspect of the results. They also offer a novel perspective on the traditional claim that weak islands are sensitive to an argument–adjunct asymmetry (see, e.g., Cinque 1990; Rizzi 1990; Szabolcsi and Lohndal 2017 for discussion). Indeed, they provide evidence that adjunct extraction is just as viable as argument extraction (again, provided the context supports the interpretation of the adjunct’s gap within the island environment). On this view, the long-standing observation that *whether*-islands (i.e., the weak-island type considered in my experiments) ban adjunct extraction may be an artefact. More precisely, it may simply reflect the parser’s default tendency, in the absence of a licensing context, to interpret the adjunct as modifying the matrix rather than the embedded verb, thereby favouring a shorter dependency over a longer one, in accordance with general processing principles such as the Active Filler Hypothesis (see Section 2.3). Arguments, by contrast, are more constrained in distribution, as they must satisfy θ -theoretic requirements (see Chapter 5 for details). I will argue that the apparent asymmetry between arguments and adjuncts is thus emergent rather than structural, and does not reflect any deep constraint tied to their categorial status.

Acknowledging the empirical and, presumably, ontological differences between strong and weak islands should not deter us from the broader ambition of identifying a unified account of locality phenomena, however. As emphasised in Section 2.2.3, I believe it is important to keep alive the prospect of a unified theory. To be clear, we must certainly recognise and account for the differences between weak and strong islands. At the same time, however, it remains desirable to develop a framework that can interpret these differences in a principled and coherent way.

So, returning to the central research question posed at the beginning of this chapter: which of the available accounts best captures locality phenomena and, crucially, both the differences and the similarities between impenetrability and intervention? The answer is, in a way, both all of them and none of them. That is, each approach offers valuable insights when considered on its own, but none suffices as a complete explanation of all relevant phenomena. As argued above, a more flexible and integrative perspective is needed. And while I do not claim to fully develop a new formal theory of locality in this thesis, I will outline a hypothesis that aims to incorporate the most compelling elements of existing theories—importantly, in a principled and explanatory way. The core idea was already sketched in the Introduction: locality effects originate at the interface between, loosely speaking, grammar and the performance system. This implies, essentially, that locality

is determined neither by grammar alone nor by the incidental outcome of independent constraints on sentence processing. In a certain sense, locality effects surface as these two components naturally interact at their point of connection.

Before offering a more detailed formulation of this hypothesis, I will show that it follows quite naturally if we adopt the core assumptions of the Minimalist framework. One might even say it *comes for free*. To make this point, I will briefly outline the specific aspects of the Minimalist Program that, in my view, provide the theoretical backdrop for this proposal. The following subsection is thus dedicated to a selective overview of relevant notions. I simply aim to summarise the foundational ideas most pertinent to the discussion that follows. These are standard and widely accepted within the field, and the informed reader may wish to skip ahead to Section 3.2.

3.1.1 The Minimalist Program and the Strong Minimalist Thesis

In *The Minimalist Program* (1995), Chomsky articulates a research agenda aimed at radically simplifying the theory of Universal Grammar by deriving linguistic structures from a small set of optimal, conceptually necessary principles. The central methodological shift is from the descriptive richness of GB (Chomsky 1981, 1986; see Haegeman 1994 for a complete overview) to explanatory minimalism: rather than positing numerous language-specific rules, the model seeks to uncover the most economical and computationally efficient mechanisms underlying the human faculty of language. As Hornstein (2018) notes, this development is consistent with the trajectory of scientific inquiry more broadly: to move beyond impressionistic speculation, one must first build a formal model of the system in question; only then does the opportunity arise to generalise and explain its properties in a principled way. In this light, GB can be seen as the necessary first stage of inquiry into human language: a highly modular theory that identifies with precision the properties specific to human grammars. The next step is to ask *why* these (and not other) rules exist, and *why* they exhibit their particular characteristics. In Hornstein's words (Hornstein 2018: 51–52):

That humans have Gs [grammars] and the capacity to acquire them is trivial. But what the actual properties of these Gs and FL [the faculty of language] are is decidedly not trivial and requires scientific inquiry of the standard variety. Thus, early research in GG [generative grammar] focused on describing the kinds of rules and operations [i.e., generative procedures (GPs)] found in particular Gs and, just as importantly, not found in such Gs. The resulting plausible inventory of GPs feeds inquiry into the structure of FL, with the goal of finding those features of FL that

would explain why the available GPs are in fact available and why the nonexistant GPs are absent.

As the name suggests, moreover, linguistic minimalism is not only about asking ‘why questions’; it is also about *minimising* the set of postulated grammatical mechanisms. In practical terms, “[...] one should start with GB and simplify, simplify, simplify” (Hornstein 2018: 54). But why is simplicity so important? As it turns out, it is not (just) a matter of decluttering the intricate network of GB’s definitions. Unlike architectural minimalism, linguistic minimalism is not merely a quest to reduce overabundant ornamentation in pursuit of a ‘less is more’ aesthetic. There are, in fact, solid, scientifically grounded reasons to pursue such a direction.

Two main facts support this endeavour, both grounded in biological and evolutionary evidence: (i) human language is a relatively recent biological innovation, which did not emerge gradually but quite suddenly, and (ii) its appearance predates early humans’ migration out of Africa and has remained essentially unchanged since (see a.o. Givón 2002; Hauser et al. 2002; Di Sciullo and Boeckx 2011; Berwick and Chomsky 2016; Chomsky 2016, 2017; Huybregts 2017 for discussion).

Let me once again draw from Hornstein’s wording to clarify why this supports the quest for ‘simplicity’ (Hornstein 2018: 53):

Both facts point to the same conclusion: that whatever prompted the emergence of FL is recent and very minor—a small, recent, simple addition to the prelinguistic cognitive apparatus. A simple or minor change could occur all at once. A simple or minor recent change would be similar across the species. In other words, if FL is composed largely of domain-general computational operations with a recent minor or simple domain-specific add-on, then the two facts above could be explained. To put this yet another way: The less UG-ish FL is, the easier it is to explain how it arose in the species so recently (fact 1) and why it has remained largely unchanged (fact 2).

This passage neatly underscores one of the core tenets of minimalism: the goal of positing as few language-specific constraints as possible while deriving linguistic phenomena from independent, domain-general principles of cognition. The aim is thus to reduce the language faculty to a minimal, language-specific UG component and a broader, computationally generic component (Hornstein 2018: 53). From this perspective, it is entirely expected that UG principles should interact with general cognitive mechanisms. It is a natural consequence of the generative view that “language (in the narrow sense [...]) involves a computational device embedded within an array of human cognitive systems, and interacting with them” (Chomsky et al. 2023: 4–5).

Let me clarify this further. The language faculty is conceived as an “organ of the body” (Chomsky 2005: 1), forming a core component of human cognition. The system of rules and principles that underlie grammatical competence constitutes what is referred to as the I-language: the “internal-individual language system [...] that permits the composition of infinitely many meaningful expressions from a finite stock of discrete units” (Chomsky et al. 2019: 230). A human child is not born with a fully developed I-language, but is biologically equipped with the capacity to acquire one. This capacity is shaped by what are often called the *three factors in language design* (Chomsky 2005): (1) the genetic endowment, which includes the innate principles of UG; (2) experience and interaction with the environment, i.e. linguistic input; and (3) general cognitive principles that are not specific to language (and perhaps not even to our species) but instead reflect broader laws of nature or computational constraints.

Now, advancing the scientific inquiry into human language essentially means hypothesising and evaluating which properties of the language faculty are attributable to which of the Three Factors. Ideally, most such properties should be assigned to the third factor, making it possible to reduce seemingly idiosyncratic traits of the language faculty to more general, independently motivated ‘laws of nature.’ The first factor (i.e., UG) should instead be restricted to genuinely irreducible linguistic properties. The smaller this set, the better. Only then can we meaningfully address the questions of *learnability* and *evolvability*; that is, work towards a “theory that must account for how a child can, through the interaction of the innate faculty and the environment, grow an I-language”, while also recognising that “the innate computational system must have emerged in accord with the conditions on the evolution of *Homo sapiens*” (Chomsky et al. 2023: 11).

It is on the basis of these very premises that Minimalism, initially conceived more as a research program (Chomsky 1995), soon gave rise to what is known as the *Strong Minimalist Thesis* (SMT). The SMT holds that language is an optimal (and minimal) solution to interface conditions (Chomsky 2001).⁶

According to this view, the set of UG-specific principles must be reduced to its bare minimum—ideally, to just one core operation: Merge. I will not, in this context, elaborate on the formal properties of Merge; some of its relevant features will be discussed in Section 6.2.3.⁷ For now, let me just highlight the following. If Merge is indeed as-

⁶Somewhat simplifying, we can say that the SMT shifts Minimalism from a methodological stance (how simple can the theory get?) to a hypothesis about the biological design of language. In plain words, language exists the way it does because it is the most efficient and minimal way to bridge thought and sound. For discussion of SMT, see a.o. Chomsky (2000, 2004a, 2005, 2017, 2020, 2021), Boeckx (2006a) and Epstein et al. (2017); and especially Freidin (2021) for notes on its historical development.

⁷For discussion, see Chomsky (1995, 2000, 2004a, 2013, 2015), Collins (1997, 2002), Citko (2005, 2011), Seely (2006), Fortuny (2008), Collins and Stabler (2016), Collins and Postal (2017), Collins and Seely (2020), Epstein et al. (2021) and Marcolli et al. (2025).

sumed to be the sole innate, language-specific generative principle, then all dependencies between syntactic objects should be established via Merge.⁸ If this assumption is on the right track, then the emergence of Merge marks the evolutionary advent of the human language faculty: the capacity to recursively generate sets is minimal enough to plausibly result from a single, sudden (and conceivably random) rewiring of the brain, yet powerful enough to yield an enormous evolutionary advantage. On this view, Merge lies at the foundation of all the central properties of human grammar. Note that this insight nicely connects the SMT with the original methodological thrust of the Minimalist Program. Indeed, the SMT “also suggests a program,” essentially challenging us to “[s]how that all G[rammatical] dependencies (all those GB described in its various modules) are actually mediated by Merge” (Hornstein 2018: 58).

Before summarising and moving on to the next section, let me stress one final, crucial aspect of the SMT. Because language is embedded within the broader cognitive system, each and every one of its properties is expected to interact with general cognitive constraints. Even Merge (the most UG-specific component of the language faculty) is arguably shaped by third-factor principles.

Without going into detail, Merge is conceived as the simplest possible structure-building operation: one that takes elements from a given workspace and forms a new object from them. Notably, there is no a priori reason why Merge should be limited to combining just two elements. For Merge to qualify as *the* simplest possible combinatorial operation, it must be defined in the most general way: take n objects and create a new object from them. In other words, the binarity of Merge must be justified, not simply assumed.

It is therefore reasonable to view the binary nature of Merge as the result of its interaction with more general constraints on computational economy and efficiency (Collins 1997). Constraints, these, that are not specific to the language faculty, but apply across cognitive domains. Put differently, Merge, like any other property of human language, is shaped by third-factor principles (i.e., general laws of nature) that favour minimal computational complexity. Why two arguments? Because two is the smallest number that allows the creation of relations between elements (i.e., syntactic dependencies). That is, we need *at least* two arguments for Merge to yield meaningful structure, but any more would unnecessarily increase computational cost without a corresponding gain. The following excerpt from Chomsky et al. (2023: 17) nicely articulates this reasoning:

The simplest form of structure building, with Merge characterized as just ‘form set,’ could operate on any number of WS [workspace] objects. [...] But there are external

⁸In a footnote, Chomsky et al. (2023) comment that “[t]he strongest view is that Merge is not just the only structure-building operation, but the only operation at all, relegating Agree to externalization” (Chomsky et al. 2023: fn. 109).

forces that constrain selection of input objects. One third-factor principle associated with computation involves a form of simplicity: less computation is simpler than more. Of course this is not unique to linguistics; it's a scientific truism representing the basic methodology of science. Simplicity-by-least-effort expresses itself in various ways. One involves the number of elements in the WS that Merge can target. The simplest choice for natural language syntax is two elements at a time because syntax involves relations between distinct objects. To build any relevant syntactic structure at all, Merge must target a minimum of *two* objects. By parsimony, then, the simplest form of Merge is binary; Merge targets two and only two elements of the WS on any given application. [...] Two is necessary; by parsimony, we seek a system where two is sufficient.

So, Merge is an innate operation constrained by third-factor requirements. But this is not the end of the story. The ultimate property of human language is, of course, (efficient) communication, which requires the generation of meaningful and interpretable representations. In particular, the output of Merge must be compatible with the demands of the systems tasked with interpreting it. That is, the structures built by Merge must be legible to the so-called *Conceptual-Intentional* (CI) and *Sensory-Motor* (SM) interfaces. These systems are responsible for analysing the syntactic objects they are fed and extracting from them the relevant information: specifically, how to assign semantic interpretations (CI) and how to linearise hierarchical structures for externalisation (SM).⁹

Now, CI and SM are not passive processors that simply accept whatever structures the generative engine produces. Rather, they impose their own interpretive constraints—meaning they will reject or fail to interpret representations that do not meet their requirements. For instance, CI demands that syntactic representations yield coherent predicate–argument structures, compliant with traditional θ -theory (or TT, in Chomsky et al.'s 2023 terminology). As a result, Merge cannot blindly generate arbitrary structures: it must build representations that are well-formed with respect to both CI and SM interface conditions.

Crucially, these interface requirements are considered “language-specific conditions on interpretation” (Chomsky et al. 2023: 30), and thus pertain to what Chomsky identifies as the first factor. To summarise: while the third-factor-compliant form of Merge is binary, “[c]ritically, Merge is also subject to the first factor; that is, to language-specific conditions like TT. Theta Theory is not a general efficiency condition constraining formal systems; it is unique (as far as we know) to language” (Chomsky et al. 2023: 30). Figure 3.1 schematically represents these concepts (in a deliberately simplified form).

⁹CI and SM replace the traditional LF and PF levels of representation. I will elaborate on the notion of interface(s) in the next section.

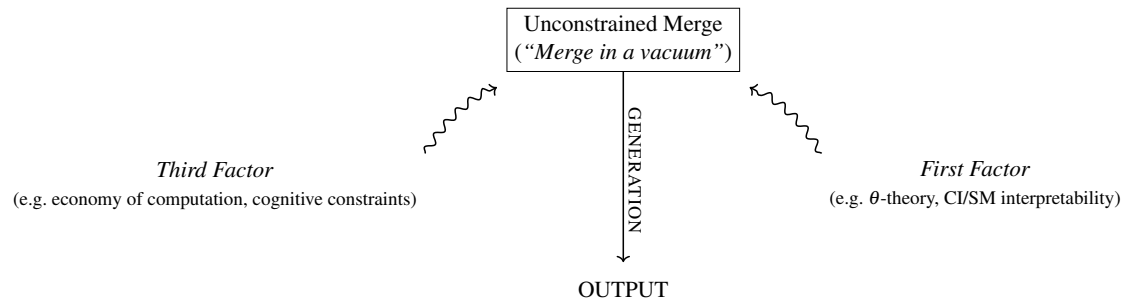


Figure 3.1: Schematic interaction between Merge and interface constraints

Let me conclude by briefly clarifying what is arguably the fundamental purpose of all ‘minimalist inquiries.’ The core aim of the Minimalist Program is *not* to describe the idiosyncrasies of particular constructions in particular languages (see Hornstein 2018: 64).¹⁰ Rather, it aspires to uncover and explain the properties of the language faculty itself; in particular, to determine which of those properties are truly language-specific. Crucially, then, Minimalism should not be seen as a fixed, bona fide theory, but rather more as a methodological and epistemological stance. “As Chomsky has been fond of saying, Minimalism is not a theory but a program. Programs are not right or wrong but rather fecund or sterile” (Hornstein 2018: 65).

I trust it is clear that these pages were not intended as a comprehensive survey of the Minimalist Program or the SMT.¹¹ Instead, my aim has been to foreground a few core tenets of Minimalism that directly support the claims developed in the chapters that follow. These claims, I believe, can be coherently framed within the set of assumptions outlined here.

To summarise, these are the core minimalist principles that will guide the argumentation in what follows:

1. The theory should be simplified as much as possible. Wherever feasible, explanations grounded in third-factor principles should be preferred over stipulative mechanisms.
2. Ultimately, the only genuine UG operation is Merge, which is shaped by both third-factor (general cognitive) and first-factor (language-specific) constraints.

Note: Any property attributed to UG should, insofar as possible, be derivable from or reducible to the workings of Merge.

¹⁰As Hornstein put it, we should not think of Minimalism as a “slimmed-down version of GB.”

¹¹For further reading, see, a.o., Hornstein et al. (2005), Boeckx (2006a,b, 2008b), Freidin and Vergnaud (2013) and Chomsky (2021), as well as the other works cited throughout this section.

3. It is important to recall the purpose of the Minimalist Program, conceived primarily as an epistemological project rather than a dispositive theory of linguistic phenomena.

3.2 The Interface

Back to the hypothesis, I argue that locality effects originate at the *Interface*, intended here as the abstract space of interaction between the grammar (in a broad sense) and the cognitive systems it is embedded in. As reviewed in the previous section, it is a standard minimalist assumption that third-factor principles actively shape the grammatical system. In fact, a central goal of the minimalist enterprise is to derive as many linguistic properties as possible from general cognitive constraints on computation. Even the binary nature of Merge is argued to follow from general principles of computational economy (see above). So, when we recall the SMT claim that language is an *optimal solution to interface conditions*, we can interpret it in two complementary ways. In a narrower (and perhaps more literal) sense, this means that the language faculty is shaped by the requirements of the Conceptual-Intentional (CI/LF) and Sensory-Motor (SM/PF) systems. For instance, the condition that verbs must discharge their θ -roles can be viewed as an interpretability requirement imposed by the CI/LF interface. In a broader sense, however, the notion of interface could be understood to encompass more than just CI/LF and SM/PF. The language faculty has reasonably evolved as an optimal solution not only to these linguistic interpretive systems, but also to the general-purpose cognitive mechanisms responsible for processing linguistic information. Put differently, we could say that linguistic derivations must be *interpretable* (CI/LF), *linearisable* (SM/PF), and *processable*. By *processable*, I mean that the output of Merge must be optimally structured for the computational system tasked with parsing and interpreting language in real time.¹²

The existence of such a processing system is self-evident, and the idea that it exerts pressure on the form of grammar is both plausible and consistent with third-factor reasoning. Thus, to reiterate, it seems entirely natural to interpret the SMT claim that language is an optimal solution to interface conditions as referring not only to CI/LF and SM/PF, but also to the domain-general processing systems that interact with and contribute to shaping the language faculty.

At this point, one might wonder how the complex interaction among Grammar, LF, PF, and what I broadly refer to as ‘processing’ could be represented within a coherent model of the language faculty. Although this is a much larger issue than I can address here,

¹²This was similarly articulated in Phillips (1996), who introduced the notion of *parsability*. In his account, a sound–meaning pair is parsable if the system can compute the relevant correspondence within its resource constraints (Phillips 1996: 255).

it is worth briefly situating my proposal within the broader architectural debate. To my knowledge, generative linguists have long expressed scepticism towards the traditional Y-model of grammar, calling for its reconsideration. Indeed, while it has served as a useful heuristic, there is broad agreement that it oversimplifies (and perhaps even misrepresents) the architecture of the language faculty (see, e.g., Jackendoff 1997; Richards 2016, 2017, 2023 for discussion). Yet despite this general dissatisfaction, relatively little effort has been made, at least within the generative tradition, to propose a viable alternative that commands broad consensus (but see Sauerland and Alexiadou 2020; and see, a.o., Phillips 1996; Momma and Phillips 2018; Parker 2019 for parser-oriented perspectives).¹³ I do not attempt to resolve these debates here. My aim is simply to offer a modest reorganisation of some fairly established assumptions, as a way of clarifying the architectural background to the hypothesis defended in this thesis.

Let me begin by noting that, despite said scepticism towards the Y-model, many continue to emphasise the importance of maintaining a modular distinction between LF/CI and PF/SM. More recently, Chomsky has even suggested that the interfaces are not only distinct, but also conceivably hierarchically organised, with PF considered somehow ‘less central’ than LF.¹⁴ Now, if we adopt the standard assumptions that (i) LF and PF hinge upon language-specific interpretive factors (such as θ -theory and the linearisation algorithm), and (ii) the processor operates according to general, domain-independent principles of computation (i.e., third-factor constraints), then it follows quite naturally that LF and PF are structurally ‘closer’ to the grammatical core than the broader cognitive systems involved in processing. Put simply: the more language-specific a module is, the more tightly it is connected to the core of the grammar component—that is, ultimately, to Merge. For clarity, what follows is not the proposal of a new architectural model, but a re-articulation of widely accepted premises, made explicit to spell out the assumptions on which my hypothesis rests.

Continuing this reasoning, if LF is indeed ‘closer’ to the core grammar than PF, and

¹³For a lively take on Sauerland and Alexiadou’s (2020) *Meaning First Approach*, see a blog post by Omer Preminger (Preminger 2021). A further insightful discussion can be found in Thomas Graf’s post on the T/Y-model, which explores parallels with computational theory (Graf 2019).

¹⁴The claim is that “the CI interface is primary [and] SM is secondary” (Chomsky et al. 2023: 12), meaning that “[o]f the two interfaces, Merge primarily serves CI” (Chomsky et al. 2023: 30). Admittedly, this does not explicitly posit a hierarchical organisation of the interfaces. It does, however, highlight a clear asymmetry between them. As I understand it, the authors argue that Merge, and more generally the language faculty, has been shaped for the most part by the demands of CI, with SM imposing only fewer, more ‘peripheral’ constraints related to linearisation. My reasoning in the following lines does not hinge on whether the inverse holds (i.e., SM is more prominent than CI), or whether there is no hierarchical/asymmetric relation at all. It simply builds on the (quite natural) assumption that Merge interacts more closely with the more language-specific interfaces, while the ‘processing interface’ occupies a more peripheral position. This seems rather self-evident: the processor must operate on actual linguistic input, which has already been ‘sifted through’ CI/LF and SM/PF. For a discussion of the independent status of the interfaces and their interaction with Merge in a formal algebraic framework, see Marcolli et al. (2025: 30ff).

PF in turn closer than the general processing architecture, we might sketch a *highly simplified* model of the interface as shown in Figure 3.2, with LF more tightly connected to Merge than PF, and both in turn interfaced with what I will refer to as *CF*. A brief clarification: I use the terms LF and PF here instead of CI and SM because I am referring to their representational rather than functional roles. The label CF stands for *Computational Form*. I am not proposing that this be introduced as a formal theoretical primitive; it is merely a descriptive term intended to capture the third-factor-driven mechanisms that underlie the online processing of linguistic structures. Finally, I stress once again that Figure 3.2 is meant only as a tentative illustration of how the interface system might be representationally organised from a *purely theoretical perspective*. It is *not* a parser model, it does not carry any implications regarding the actual procedures involved in real-time sentence processing. Like the Y-model, it is a conceptual construct: an attempt to represent an idealised architecture that (a) is consistent with a structured formal hypothesis and (b) yields testable predictions.¹⁵ What is crucial for the present discussion is the distinction between CF and LF/PF: while the former constitutes an outer layer representing the interface between the grammar and the performance system (that is, the locus of third-factor pressures) the latter represent the domains of application for most grammatical, language-specific operations, which interact more directly with the hierarchical output of Merge.

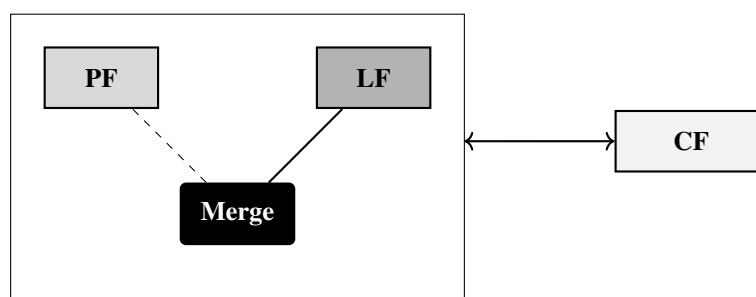


Figure 3.2: Simplified schematic representation of interface organisation: LF is tightly connected to Merge, with PF and LF forming a bounded language-specific domain. This domain interfaces with CF, which represents the third-factor-driven mechanisms responsible for processing.

¹⁵I recall a particularly insightful parallel drawn by Andrea Moro during a presentation, in which he likened syntactic trees to Bohr's model of the atom. Just as Bohr's model was never meant as a literal depiction of atomic structure, but rather as a theoretical construct that offered better predictions than its predecessor (Thomson's plum pudding model), syntactic trees too are best understood as abstract representations that are designed to capture structural relations, not to reflect the actual computational procedures involved in linguistic processing. This, after all, is what theoretical models are for: structuring a hypothesis and generating predictions. Bohr's model was eventually replaced by the quantum atomic model in the 1920s. In the same spirit, the model illustrated in Figure 3.2 can be intended as an (infinitely more modest) attempt at representing the interface system. This should be readily revised or abandoned the moment a more theoretically coherent and empirically predictive model becomes available (hopefully not only in the domain of locality).

That said, I believe we need not draw an overly rigid line between abstract, competence-centred models and psychologically plausible architectures. Indeed, a model of this kind is not only compatible with core minimalist assumptions, but also largely aligned with claims from psycholinguistic literature suggesting the existence of a unified sentence analyser responsible for language computation. That is, a single cognitive workspace in which computational mechanisms interact with grammar-specific representations and principles (Phillips 1996; Parker 2019; Lidz 2022). As Parker (2019: 5) puts it:

This [...] conception of the grammar as a structure-building system leads to a single-analyzer view of the cognitive architecture [...], in which both online and offline tasks rely on the same properties, namely the lexicon, the grammar, and limited general-purpose resources. On this view, the traditional notions of the “parser” and “grammar” simply reflect different descriptions of the same system: the grammar is just an abstraction from the processes involved in real-time sentence comprehension under the idealization of unbounded resources (Phillips 1996).

Crucially, however, this view stands in contrast to earlier two-module architectures, which posit that sentence comprehension involves two distinct components, one of which is entirely independent from the grammatical system (see, e.g., Bever 1970; Fodor et al. 1974; Townsend and Bever 2001). A similar division underlies also more recent accounts that distinguish between a slower, more accurate grammatical analysis and a “quick and dirty” parsing strategy. On such accounts, sentence comprehension unfolds in two stages. First, a “shallow” processing heuristic is engaged, relying on surface-level cues and structural templates to construct “good-enough” representations (Ferreira et al. 2002; Ferreira and Patson 2007; Karimi and Ferreira 2016). These representations may be error-prone, and if comprehension fails or ambiguity arises, a second, grammar-based process is triggered to fill in details that were missed during the initial parse (Parker 2019: 2–5). Albeit developed in the context of a processing model, this distinction is highly relevant here, as it suggests that grammatical analysis may be distributed across two cognitive systems with distinct properties and functions. Specifically, in this type of framework the parser is taken to be independent from the grammatical system.¹⁶ In a way, then, we could say that ‘non-principled’ processing-based approaches to locality (i.e., those that escape explicit reference to grammatical representations, see fn. 3 and the discussion in Section 4.3.7) ultimately relegate locality effects to a processing component of the kind just described: a largely grammar-independent cognitive module that exploits domain-general computational heuristics to process linguistic input. Such a component

¹⁶As Parker (2019: 2) put it: “the parser and grammar reflect *separate cognitive systems* because they have independent functions (rapid communication vs. knowledge representation), operate over representations of a distinct kind (noisy ‘good-enough’ templates vs. detailed hierarchical structure), and use a distinct set of rules (fallible heuristics vs. grammatical constraints)” (emphasis mine).

is not represented in Figure 3.2: the only plausible candidate to contain it would be CF, which, however, describes the interface layer of dense interaction between ‘processing’ and ‘grammar’, and not a grammar-independent module. As a matter of fact, a fully grammar-independent parsing component necessarily falls outside the interface system as I have defined it, for the reasons articulated above. This brings me to the theoretical consequences of the hypothesis proposed here.

3.2.1 Consequences and predictions of the Interface Hypothesis

The first theoretical consequence of the hypothesis advanced here is the following: any account that explains locality by appealing to a parsing system cognitively independent of grammar is ruled out. Once again, since the central claim is that locality effects arise *at the Interface*, any explanation that locates them outside the theoretical space defined by that interface cannot qualify as a valid account of locality phenomena.¹⁷

This idea was already introduced in the Introduction, where I represented the Interface as in Figure 1.1 (reproduced here as Figure 3.3, for convenience).



Figure 3.3: A conceptual illustration of the Interface-spectrum and its interactions between language-specific and non-language-specific factors.

Now that the notion of Interface has been further clarified, however, a more detailed representation may offer a clearer perspective. In particular, Figure 3.4 helps visualise how the interaction between first- and third-factor pressures is confined to a specific theoretical space—that is, one with defined boundaries. Outside of this space lie components that fall beyond the Interface proper: to the left, the derivational module prior to LF/PF evaluation; to the right, grammar-independent processing mechanisms (as discussed in, e.g., Bever 1970; Fodor et al. 1974).

Essentially, I propose that locality emerges from the interaction of pressures *within* the Interface. As discussed above, this entails that accounts based on grammar-independent systems are excluded, since these fall beyond the right boundary of the Interface.

But what about the left boundary? On the view adopted here, anything beyond that edge corresponds to what we might call ‘just grammar’: structural operations internal to

¹⁷Naturally, this not only excludes a class of processing-based reductionist accounts, but also broader views of language as “mere cultural objects potentially subject to arbitrary variation (Evans and Levinson 2009)”, as discussed in Rizzi (2013a: 183).

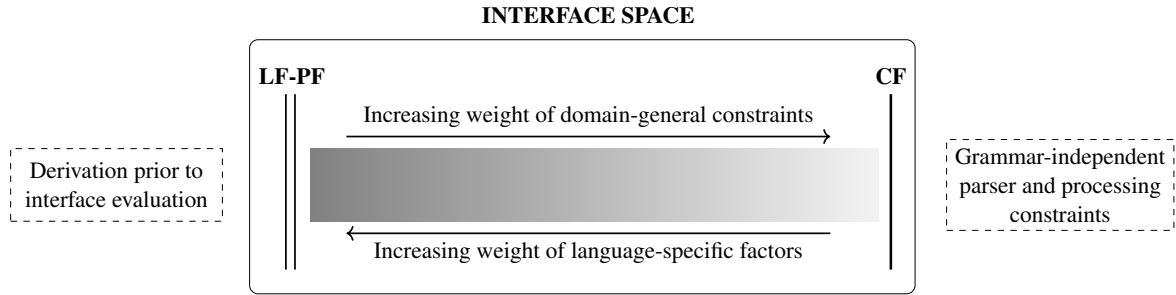


Figure 3.4: A conceptual illustration of the Interface space, flanked by grammar-internal derivational processes on the left and grammar-independent processing systems on the right.

the derivation, prior to evaluation at the CI/LF and SM/PF interfaces. Put simply, such operations are the restricted set of principles that directly affect the workings of Merge, before its output is sent to Spell-Out. From this follows the second theoretical consequence of the Interface Hypothesis: locality effects do not originate within the derivation itself. I therefore set aside approaches in which locality is determined by constraints on structure-building operations that block movement/Internal Merge at the source. In other words, locality does not originate within narrow syntax.¹⁸

In sum, purely derivational accounts are excluded, just like purely processing-based ones. I once again emphasise, however, that the distinction between derivational and representational approaches is a delicate one (see Section 2.1.1). I return to this in Section 3.2.3, where I also present further theoretical arguments against derivational accounts. For now, however, the basic prediction of the Interface Hypothesis should be clear: both purely derivational and purely processing-based accounts are ruled out. Locality effects originate within the Interface, not outside it.

3.2.1.1 A gradient of pressures

Now, I have so far described how the Interface Hypothesis may *exclude* certain accounts of locality—what one might grandly call a *pars destruens*. It seems only fair, then, to attempt a *pars construens* as well. This, I believe, begins with understanding how the relevant factors within the Interface interact.

As outlined in the Introduction, the Interface can be understood as a space of graded interactions, where different constraints carry different weights. In this view, first- and third-factor pressures exert influence on syntactic representations in a continuous, rather

¹⁸It is quite possible, however, that at least some strong islands arise from the workings of Merge itself, and in particular from its operation across parallel, independent workspaces. In essence, once a workspace is closed, it is no longer accessible from the main workspace. Whether this amounts to locating the origin of such island effects within narrow syntax depends largely on whether one adopts a derivational or a representational approach to movement. I return to these issues in Chapter 6.

than stepwise, fashion. Even though it is heuristically useful to represent LF, PF, and CF as discrete regions (see Figures 3.2 and 6.2), we should be cautious about interpreting them as literal *stages* in a generative algorithm. Instead, I propose that each of these (sub)interfaces constitutes a ‘sphere of influence’, whose pressure diffuses across a continuous representational space. By hypothesis, the closer we move to the right edge of this space, the more dominant domain-general computational constraints become (i.e., third-factor effects). Conversely, the closer we move to the left edge, the stronger the influence of language-specific principles. That is, those encoded at LF and PF.¹⁹

This framing leads to the third major theoretical consequence of the Interface Hypothesis: locality effects do not stem from a single, unified source, but span the full range of interactions across the Interface. This perspective naturally accommodates the multifaceted nature of locality phenomena, which (as repeatedly emphasised) are shaped by diverse and often overlapping syntactic, semantic, and processing-based influences. If locality effects (and their amelioration) arise at different points along the Interface spectrum, then their sensitivity to both grammatical and processing-based factors is not anomalous, but fully expected. It also follows that the apparent success of multiple, seemingly incompatible accounts should not be seen as a theoretical failure, but rather as a reflection of the system’s architecture. By locating locality within the Interface interaction space, we thus obtain a principled explanation of why different perspectives (syntactic, semantic, pragmatic, and processing-oriented) can each succeed in capturing distinct aspects of the phenomenon. Moreover, this view not only aligns with core minimalist assumptions and recent single-analyser models of the parser (see above), but also offers a principled rationale for the theoretical flexibility that many now consider essential for modelling locality (see especially the contributions in Sprouse and Hornstein 2013).

That said, I would like to take this line of reasoning one step further. Specifically, I propose that variation in the position of locality effects along the Interface spectrum may help clarify the long-standing divide between impenetrability and intervention effects. I suggest, in particular, that impenetrability effects cluster towards the left edge of the Interface, closer to core grammatical operations. By contrast, intervention effects arise further to the right, where general computational constraints exert more influence. This proposal yields two clear, testable predictions. First, intervention effects should be more susceptible to amelioration via contextual and/or morphosyntactic disambiguation: their

¹⁹It seems to me that the Interface space exhibits a kind of duality, depending on how we approach it analytically. From one angle, it can be seen as a sequence of discrete interfaces: a sort of ‘pipeline,’ where Merge feeds LF and PF, and their output is passed on to CF. From a broader systems perspective, however, the architecture appears less modular and more gradient: a continuous space in which different pressures overlap and blur into one another. This is, of course, a mere metaphor, but it is meant to capture the idea that different levels of interpretation may well yield different (and equally valid) representations of the same underlying system—a situation not unfamiliar in the natural sciences.

position within the Interface makes them more sensitive to processing pressures and parser heuristics. Facilitating the parser's task of correctly identifying and reconstructing the relevant filler–gap dependency should thus result in measurable improvement. Second, strong islands (typically associated with impenetrability) should be less amenable to such amelioration, as they are governed by stricter, grammar-internal constraints. Crucially, the experimental results presented in Chapter 4 and Chapter 5 point precisely in this direction.

Another welcome consequence of this interface architecture is that it allows for a more nuanced classification of islands, one that need not rely on a rigid binary distinction between strong and weak. Indeed, if locality effects are distributed across the Interface space, then we should expect their 'strength' to vary as a function of their position: the further to the right, the less 'grammar-driven' the constraint, and thus the more amenable it is to being accommodated.²⁰

By contrast, constraints that are more structurally determined (e.g., the Subject Condition or the CNPC) arguably lie towards the left edge of the Interface, and are correspondingly more resistant to amelioration. In this view, island strength can be conceptualised not only in categorical terms, but also as a gradient in the degree of degradation they produce. A perspective, this, that aligns neatly with recent large-scale experimental studies by Jon Sprouse and Sandra Villata (see e.g. Sprouse 2023),²¹ which show that different island types, when tested through acceptability judgement tasks, yield results distributed along a gradient.²²

At the same time, however, the Interface Hypothesis keeps alive what Boeckx (2012) calls "the dream of a unified theory of islands." Indeed, even though the distinction between intervention and impenetrability is recognised, and as we have seen, even predicted, it is nonetheless understood as emerging from a common source. On this view, different locality effects reflect their relative position within a shared interactional space, rather than stemming from fundamentally distinct explanatory systems. So while strong and weak islands may differ in how and where they arise along the Interface spectrum, this framework still holds promise for capturing them within a single, coherent theoretical model: a goal that many have long regarded as a key desideratum in the theory of locality.

²⁰Come to think of it, this might also shed light on other long-standing issues in the debate on islandhood. For instance, if the problem with weak(er) islands is indeed one of ambiguity and/or gap misidentification at the level of parsing, it is not too surprising that these constraints would be more sensitive to individual variation and/or satiation effects (see, e.g., Phillips 2013b for discussion). Some individuals may resolve ambiguity more efficiently or access the correct underlying representation more quickly; likewise, repeated exposure may facilitate such access. I do not pursue these points further here, but leave them for future consideration.

²¹These studies have been presented at several conferences and are expected to form the basis of a forthcoming publication. I thank Jon Sprouse for sharing the slides with me.

²²As the authors note, this finding lends empirical support to the long-held intuition that the distinction between strong and weak islands reflects not only a difference in kind (e.g., selectivity to certain dependencies), but also a difference in degree (i.e., the magnitude of the effects).

3.2.2 A reflective summary

Taking stock, my central claim is that locality effects originate at the Interface: a theoretical domain representing the interaction space between grammatical constraints and domain-general computational processes. Within this space, the influence of first- and third-factor constraints varies: grammatical factors dominate on the left, while domain-general constraints gain prominence towards the right. Locality effects can be mapped along this spectrum: the closer to the grammatical side, the more they resemble strong island effects (impenetrability); the closer to the cognitive side, the more they align with weak islands or intervention.

This architecture brings several welcome consequences:

- It integrates syntactic, semantic, pragmatic, and processing factors into a unified explanatory model, incorporating key insights from earlier accounts (see Chapter 2);
- It explains the apparent success of multiple, seemingly incompatible theories as a reflection of graded pressures across a shared interface spectrum;
- It qualifies as a genuine theoretical framework in its own right, with clear empirical consequences.²³ Specifically, it rules out two major explanatory classes: (i) purely grammar-independent processing accounts, and (ii) purely derivational grammatical accounts. Both of which lie outside the Interface space.

Of course, the Interface Hypothesis is not presented here as a conclusive theory of locality. Yet I believe the arguments laid out point in a promising direction for advancing our understanding of the phenomenon. In this thesis, I support the hypothesis in two main ways:

- **Theoretically**, I have shown that it follows naturally from core assumptions of the Minimalist Program and the SMT, and it aligns with single-analyser models in psycholinguistics. Moreover, the emphasis on interface interactions is neither surprising nor exotic: work on locality has long recognised the need to model multi-layered interactions. Already over a decade ago, Boeckx (2012) argued that any valid theory of locality must give “priority to the interfaces.” And more recently, Chomsky et al. (2023: 65) claim that “[f]or islands, a range of different extra-syntactic principles could conceivably be relevant,” suggesting that “progress depends on identifying the precise contribution of independent syntactic and extra-syntactic mechanisms.”

²³That is, the Interface Hypothesis does not constitute a mere restatement of prior views under a new umbrella, but a distinct framework that yields original predictions.

- **Empirically**, it is supported by the results of Experiments 1–6 (Chapters 4 and 5), which show that:
 - Intervention effects are ameliorated through disambiguation strategies that ease processing (Experiments 1–2), pointing to a (principled) processing-based account.
 - Such strategies must rely on grammatical representations (Experiments 2–3), excluding grammar-independent accounts based on statistical heuristics.
 - Weak-, but not strong-islands can be improved via contextually legitimising the internal gap (Experiments 4–6), indicating a different status of impenetrability and intervention.

Taken together, this evidence refines the proposal: intervention and impenetrability effects share a common source (the Interface) yet differ in their ontological grounding. Intervention effects are more responsive to processing-based disambiguation, while impenetrability effects are more rigidly tied to grammatical constraints and thus less amenable, if at all, to amelioration. Once again, these patterns reflect their origin at different points along the Interface spectrum, where language-specific and language-independent pressures exert varying influence. This may also explain the independently observed gradient in the severity of island-induced degradation (see fn. 22).

Let me now try to do justice to the title of this section by adding a more reflective note. Clearly, the empirical findings presented in this thesis can only go so far in substantiating the Interface Hypothesis, and much more work is needed to develop and test it further. I therefore prefer to treat both the data and the ideas advanced here not as an endpoint, but as a starting point, a step towards a novel path for investigating locality.

Moreover, locality is such a broad and intricate topic that any attempt to frame it is bound to leave important questions unresolved. This thesis is no exception. Many issues remain open, and perhaps just as many new ones are raised. One, in particular, stands out: why is it that only certain specific syntactic environments give rise to impenetrability effects? In other words, there appear to be only a few *genuinely* strong islands: a restricted set of syntactic configurations that consistently block (sub)extraction. What determines their status? The practised syntactician will recognise this as the classic problem of the Condition on Extraction Domains (CED), which has been at the centre of a particularly fertile debate. Although I do not claim to solve it, I will offer a speculative attempt to capture it within the interface-based approach defended here.

3.2.3 Further reflections and the (open) issue of the CED

The CED was already introduced in Section 2.1.3. Here is how it was originally proposed by Huang (1982: 505):

(1) **Condition on Extraction Domain**

A phrase *A* may be extracted out of a domain *B* only if *B* is properly governed.

Traditionally, complements of *V* are taken to be properly governed, unlike specifiers and adjuncts. Essentially, then, the CED bans extraction from *non-complements*. The relevant English examples were presented in Section 2.1.3 and are repeated in (2) for convenience.

- (2) a. Who did you see [pictures of ___]?
 b. *Who did [pictures of ___] cause Bill to leave the room?
 c. *Who did Bill arrive [after seeing (pictures of) ___]?

Whereas *wh*-extraction from *Comp,VP* is acceptable (2a), extraction from subjects (2b) or adjuncts (2c) is ungrammatical. The picture is actually more nuanced, with notable exceptions both within and across languages. But a useful (and reasonably accurate) approximation is that strong islandhood essentially reduces to extraction from adjoined domains (Boeckx 2012: 55). I will return to the syntactic implications of this generalisation in Section 6.2. For now, let us simply ask: what is it about these specific domains (non-complements) that gives rise to impenetrability effects?

A minimalist approach, committed to principled explanation, cannot be satisfied with the original appeal to government. Ideally, the CED should be derived from independent properties of the language faculty, such as phasehood (e.g., Müller 2010, 2011), PF-linearisation constraints (e.g., Uriagereka 1999a; Richards 2001; Fox and Pesetsky 2005; Hornstein et al. 2007; Sheehan 2013), the well-formedness of chains (e.g., Rizzi 2006; Rizzi and Shlonsky 2008; Boeckx 2008a), or the functioning of (re-)Merge (e.g., Takahashi 1994; De Belder and van Craenenbroeck 2015) and Agree (Boeckx 2003). I will consider representatives of these proposals in due course. At this stage, I would just note that many of these accounts have been already questioned in Boeckx (2012), which offers a detailed critique of some of the most influential proposals on the market. I cannot hope to match the clarity of Boeckx's discussion, so even though I will briefly highlight the central issues with these earlier minimalist accounts of the CED (see Section 6.2), I encourage the interested reader to consult Boeckx directly. My focus, instead, will be on the alternative proposal he himself advances as a promising way forward.

Boeckx strongly rejects derivational accounts of the CED and instead advocates for an interface-based approach, which he sees as something of a minimalist reinterpretation

of Ross’s original filter-based explanations. Extraction from islands, he argues, is not ‘non-derivable’ per se, but rather results in representations that challenge the interpretive systems. Like many of the accounts above, Boeckx (2012) thus treats the CED as an interface condition on the legibility of syntactic structures. The core idea is this: both adjuncts and subjects involve particularly complex derivations, which trigger early Spell-Out in the syntax. Once a domain has been Spelled-Out, it becomes opaque, meaning that its internal structure is no longer accessible for sub-extraction. This is what blocks movement from within these domains. But what makes adjuncts and subjects so complex as to require early Spell-Out? For adjuncts, Boeckx adopts the widely held view that they are constructed in separate workspaces and adjoined as syntactically atomic (and thus inaccessible) units. More innovatively, he claims that subject specifiers undergo early Spell-Out as a consequence of subjects’ A-movement (i.e., EPP-driven movement to Spec,TP). The idiosyncratic complexity of A-chains (Boeckx 2008a; Chomsky 2008), he argues, imposes extra interpretive demands, forcing early Spell-Out and preventing later extraction from within the subject. Moreover, if sub-extraction via, say, wh-movement applied to such a chain, the interface would need to interpret both an A- *and* an A’-chain over partially overlapping material. This, in turn, could yield a potentially ambiguous configuration, reminiscent of classic garden-path effects (Boeckx 2012: 115). Once again, then, ambiguity appears central to locality effects: a perspective that aligns with, and in fact reinforces, the Interface Hypothesis defended in this thesis.

I will not go into technical detail here, as this is merely a preliminary sketch of the problem. I do note, however, that Boeckx’s account does not fully resolve the puzzle. In fact, I will suggest that it faces two main challenges. Theoretically, it still relies on several assumptions that lack grounding in independent principles. Moreover, the appeal to the A/A’ distinction may not be ideal, as recent minimalist work encourages us to eschew the postulation of such primitives (see Obata and Epstein 2011; Urk 2015; Fong 2022; Chomsky et al. 2023 for discussion). Empirically, I will present examples that run counter to the account’s predictions.

In light of these concerns, I will explore two alternatives. The first draws on De Belder and van Craenenbroeck’s (2015) proposal, which derives the opacity of subjects from the mechanics of the Merge algorithm. The second develops, instead, my own account. In pursuing this alternative, I follow Boeckx’s own spirit of critique, which calls for identifying weaknesses in existing proposals and venturing a new account grounded in simpler, independently motivated principles. Indeed, he himself “urge[s] the readers to critically examine [his] proposal [and] judge for themselves if [his] claim concerning the robustness of CED-effects [...] is valid, and if they believe that [his] proposal has a fighting chance of making the right cut” (Boeckx 2012: 134). Specifically, I will retain

his (and others') insights concerning adjunct opacity, as well as the broader interface-based perspective. But I will pursue a different path regarding the opacity of subjects, one that ties strong island effects to independently motivated properties of the interfaces. In particular, I argue that subject opacity arises when two otherwise tolerable interface costs (at PF and LF, respectively) co-occur, yielding a *superadditive* interaction.

Asymmetries in Spanish Subject and Object Relative Clauses

This chapter starts from a well-attested observation: Subject Relative Clauses (SRCs) are easier to interpret than Object Relative Clauses (ORCs). This asymmetry has been documented across numerous languages and populations and is supported by a substantial body of experimental evidence. Theoretically, the difficulty associated with ORCs is typically attributed to intervention locality. Here, I investigate RCs in Spanish from both syntactic and experimental perspectives, showing how they can shed new light on our understanding of intervention effects more generally.

The chapter is structured as follows. I first introduce the relevant morphosyntactic properties of Spanish RCs, with particular focus on how Differential Object Marking (DOM) is realised across different RC types. I outline their derivational structure and propose a syntactic analysis building on previous work by Poletto and Sanfelici (2018, 2019) and Svenonius (2012, 2016, 2020). I then present two studies examining how adult native speakers of Peninsular Spanish process and comprehend SRCs and ORCs. This is followed by two further studies examining children's comprehension of the same materials. The chapter concludes with a summary.¹

¹Parts of this chapter are based on and expand upon material published in Presotto (2024a).

4.1 The empirical domain: Spanish relative clauses

Linguistic theory has long been concerned with RCs. Over the years, a remarkably large number of contributions have proposed a wide array of analyses for these structures. What is probably at the root of such proliferation is the fact that RCs come in disparate types, both within and across languages. Research on this topic has thus traditionally focused on defining the syntax underlying different relativisation strategies, with the aim of unifying their intricate variation under a single analysis. Central to this effort is the question of the categorial status of relativisers, or more generally, of the elements that introduce RCs. In particular, debate has centred on whether, and which of, these elements should be treated as complementisers or relative pronouns.

In this section, I contribute to this debate by examining the specific case of Spanish subject and object RCs. I propose a unified analysis of *que* as a wh-determiner, extending for the first time to Spanish a prominent analysis that has been proposed for several other Romance languages (cf. Section 4.1.3). In doing so, I also review other salient morpho-syntactic properties of Spanish RCs and show how they can all be derived in a uniform fashion based on current syntactic assumptions. These properties include: (i) variation in the position of the subject in ORCs, and (ii) the alternation between bare *que* and *al/a la que*. As far as the latter is concerned, I argue that a single *que* exists that can realise different portions of the same underlying structure, drawing on Spanning theory (Svenonius 2012, 2016, 2020).

I proceed as follows: Section 4.1.1 reviews the main analyses of RCs in syntactic theory, with a focus on their derivations; Section 4.1.2 illustrates the word order properties of Spanish RCs and their syntactic interpretation; Section 4.1.3 discusses the morphosyntactic variation of *que*; and Section 4.1.5 presents the proposed analysis.

4.1.1 Relative clauses in syntactic theory

A relative clause is, at its simplest, a clause that modifies a noun. Traditionally, two major types have been distinguished: *restrictive* RCs and *non-restrictive* RCs. The former limit the meaning of the noun they modify, *restricting* the identification of its referent to the intersection of the denotations of the RC and the head noun. For instance, in “*The tree that John planted is a spruce*” the RC specifies the referent of the noun as the one tree that John planted. On the other hand, non-restrictive, or *appositive* RCs do not intersect with the denotation of the noun, but rather provide additional independent information, as in “*The tree, which is a spruce, was planted by John.*” Semantically, restrictive RCs behave like other intersective modifiers such as adjectives and predicates, and are described as being of type $\langle e, t \rangle$ (Heim and Kratzer 1998; Zimmermann and Sternefeld 2013). In

contrast, non-restrictive RCs are seemingly closer to an independent sentence of type *t* that follows the noun, as in, e.g., “*The tree was planted by John. The tree is a spruce*” (Ross 1967; Demirdache 1991; Trabant 2016). In what follows, I will be concerned only with restrictive RCs, and among these, exclusively with SRCs and ORCs. I will thus leave aside oblique and non-restrictive RCs altogether.

At the syntactic level, RCs have been at the centre of a fertile debate, giving rise to numerous analyses over the years. Despite some inevitable idiosyncrasies that distinguish individual proposals, they can broadly be grouped into three main theoretical families: so-called *head-external analyses*, *matching analyses*, and *raising analyses*. These offer (partially) different solutions to what Salzmann (2017) terms the *connectivity problem* and the *modification problem*. Concretely, any analysis of RCs must address two core issues. First, how to grammatically define the link between the surface position of the head noun and the RC-internal position with which it is related (connectivity problem). Second, where the RC attaches to the noun it modifies, and what kind of syntactic relation is established between the two (modification problem). The next subsection briefly outlines the main characteristics of the three approaches and how they address these challenges.

4.1.1.1 Head external analysis, matching analysis, and raising analysis

In the head external analysis, the head noun is not represented inside the RC, and a relative operator or a (c)overt pronoun is A'-moved to the C-domain, where it is co-indexed with the external head noun (see Plann 1975; Montague 1973; Chomsky 1977; Jackendoff 1977; Boef 2012), as in (1).

- (1) I like the woman [_{CP} whom₁/Op₁ C_{rel} John kissed ____₁].

(Salzmann 2017: 39)

This type of analysis was extensively adopted throughout the Government and Binding era but later gave way to accounts that posit a representation of the head noun *internal* to the RC. Indeed, many have regarded such an alternative as superior in many respects. First and foremost, it seems better equipped to analyse the rich typology of relativisation strategies, which includes head-internal RCs (see Alexiadou et al. 2000; de Vries 2002; Salzmann 2017; Cinque 2020). Moreover, assuming a RC-internal representation of the head noun offered a more natural interpretation to a number of properties largely (but not only) related to reconstruction effects. For instance, the fact that certain expressions keep their idiomatic meaning when split between the RC and the matrix clause suggests that the head noun forms a constituent with the RC-verb at some stage of the derivation. One such classic example is the one presented by Schachter (1973) and Vergnaud (1974) for French, where the interpretation of the idiom chunk *tirer parti* “to take advantage” keeps

intact when *parti* is the head of the RC:

- (2) Peu de gens ont parlé du parti qu'il a tiré des difficultés économiques.
 few of people have spoken of-the advantage that-he has taken of-the difficulties economic
 'Few people have talked about the advantage that he took from the economic crisis.'

(From Perpiñán 2010: 37)

Again, under the assumption that the idiomatic meaning can only be assigned when the verb and its argument (here *tirer* and *parti*) form a constituent, it follows that the head noun in (2) must have occupied the complement position within the RC-predicate structure at some point. A somewhat similar case is that of binding effects, and notably Principle A effects (Schachter 1973; Kayne 1994). Consider the sentence in (3). If the relative head *picture of himself* were generated in a position external to the RC, at no point in the derivation could the anaphor *himself* be c-commanded by its antecedent (*John*).

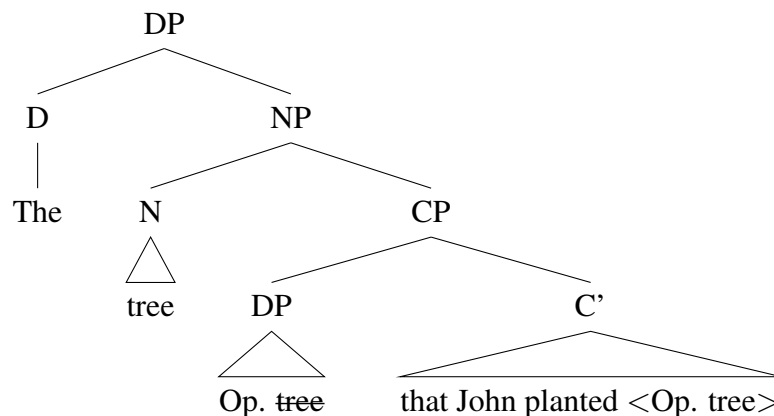
- (3) the picture of himself_i that John_i likes___ best

(Salzmann 2017: 2)

The actual interpretation of (3) is thus only obtained if *picture of himself* is reconstructed within the RC as the internal argument of the verb *like*. After a fashion, both (3) and (2) imply that the head noun can be interpreted as part of the RC-predicate. To secure this scenario, the head noun must have a RC-internal representation, which is exactly what constitutes the premises of both matching and raising analyses.

While its original formulations can be traced back to the early 60s (Lees 1960, 1961; Chomsky 1965), the matching analysis has only garnered a renewed consensus in recent years (Sauerland 1998, 2003, 2004; Citko 2001; Salzmann 2006 a.o.). At its core, the matching approach posits that the head noun is represented twice: once in its surface position (i.e., the matrix clause), and once within the RC, as illustrated in (4).

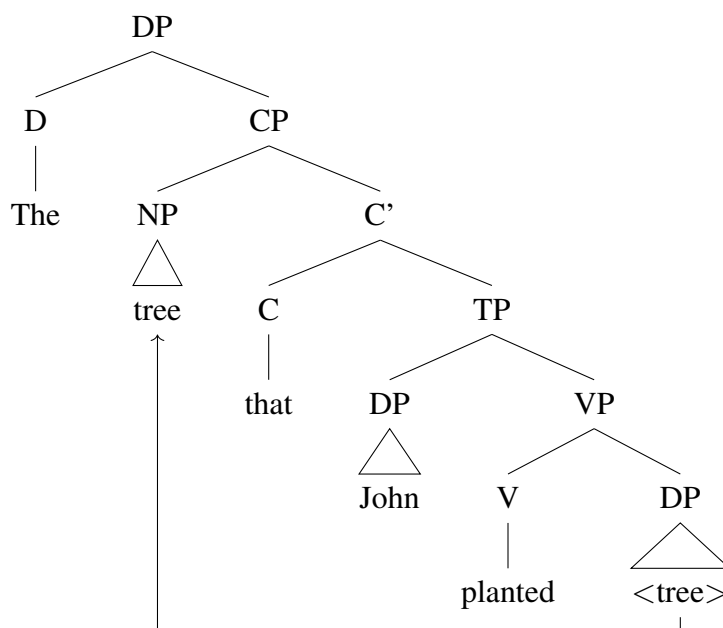
- (4) **Matching derivation**



In this derivation, the RC-internal instance of the head noun moves along with a relative operator to the left periphery of the RC. From there, it gets deleted by its non-distinct, c-commanding copy in the matrix clause, so that only the latter is phonetically realised (Sauerland 2003, 2004).

Conversely, in raising analyses there is a single instance of the head noun. This is base-generated within the RC and then moved (or *raised*) to the CP-layer of the RC (see e.g., Kayne 1994; Cinque 2008, 2020) or, in some versions, up to a higher projection in the complement position of the external DP (e.g., Bhatt 2002; Donati and Cecchetto 2011; Cecchetto and Donati 2015). This approach finds its roots in works by Brame (1968), Schachter (1973), and Vergnaud (1974), though likely owes its popularity to Kayne's (1994) influential re-elaborated version, upon which all recent analyses are based (Bianchi 1999, 2000, 2002; Cinque 2008, 2020; Poletto and Sanfelici 2018 a.o.). The tree in (5) illustrates a typical raising derivation.

(5) **Raising derivation**

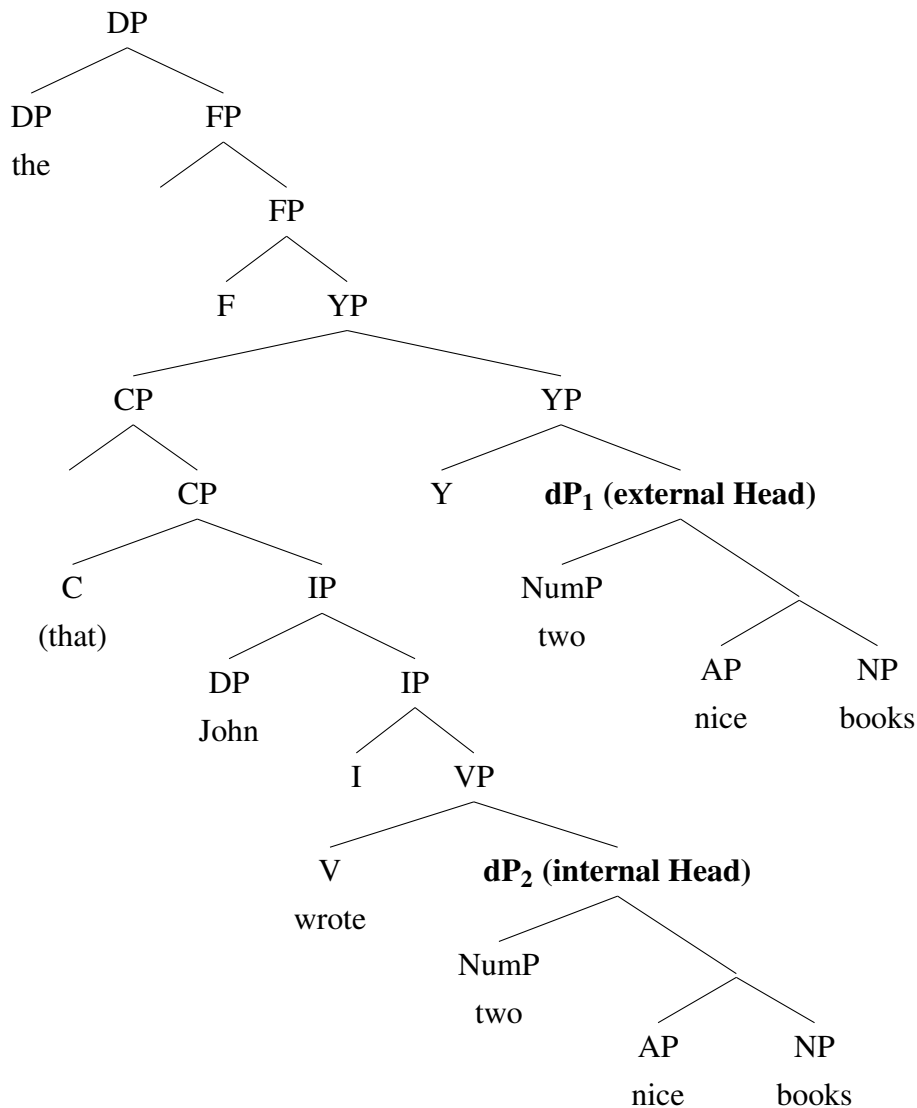


On closer inspection, however, neither the matching nor the raising derivation is completely immune to criticism. One major aspect that raising accounts tend to leave somewhat unaddressed is the fact that the head noun potentially receives two conflicting theta roles and cases.² At the same time, a genuine matching approach is not always fully equipped to accommodate cases like (2), where idioms are seemingly split between the matrix clause and the RC. It is not my direct purpose to delve into all the arguments and counterarguments of either approach, for which I refer to the comprehensive review of the topic in Salzmann (2017). I will instead limit myself to acknowledging that many scholars have proposed that both raising and matching derivations are supported by substantial

²See Kayne (1994: ch:8) and Bianchi (1999) for possible solutions.

evidence, suggesting that they are in principle equally available to UG (Áfarli 1994; Aoun and Li 2003; Sauerland 2003, 2004; Hulsey and Sauerland 2006; Szczegielniak 2004; Cinque 2008, 2020; Mursell and Hartmann 2025). This is why, when confronted with the vast array of options for analysing Spanish RCs, I opted for a syntactic structure that accommodates both types of derivations, following Cinque (2008, 2011, 2020). Cinque pursues the idea that matching and raising approaches are not mutually exclusive but, rather, structural variants that emerge from one and the same “double-headed” underlying structure. A representation of the latter is given in (6), where the RC is, much as other pre-nominal modifiers, attached to the functional spine of the external head noun, in accord with standard cartographic assumptions (Cinque and Rizzi 2009; Rizzi 2013b) and Kayne’s LCA (Kayne 1994).

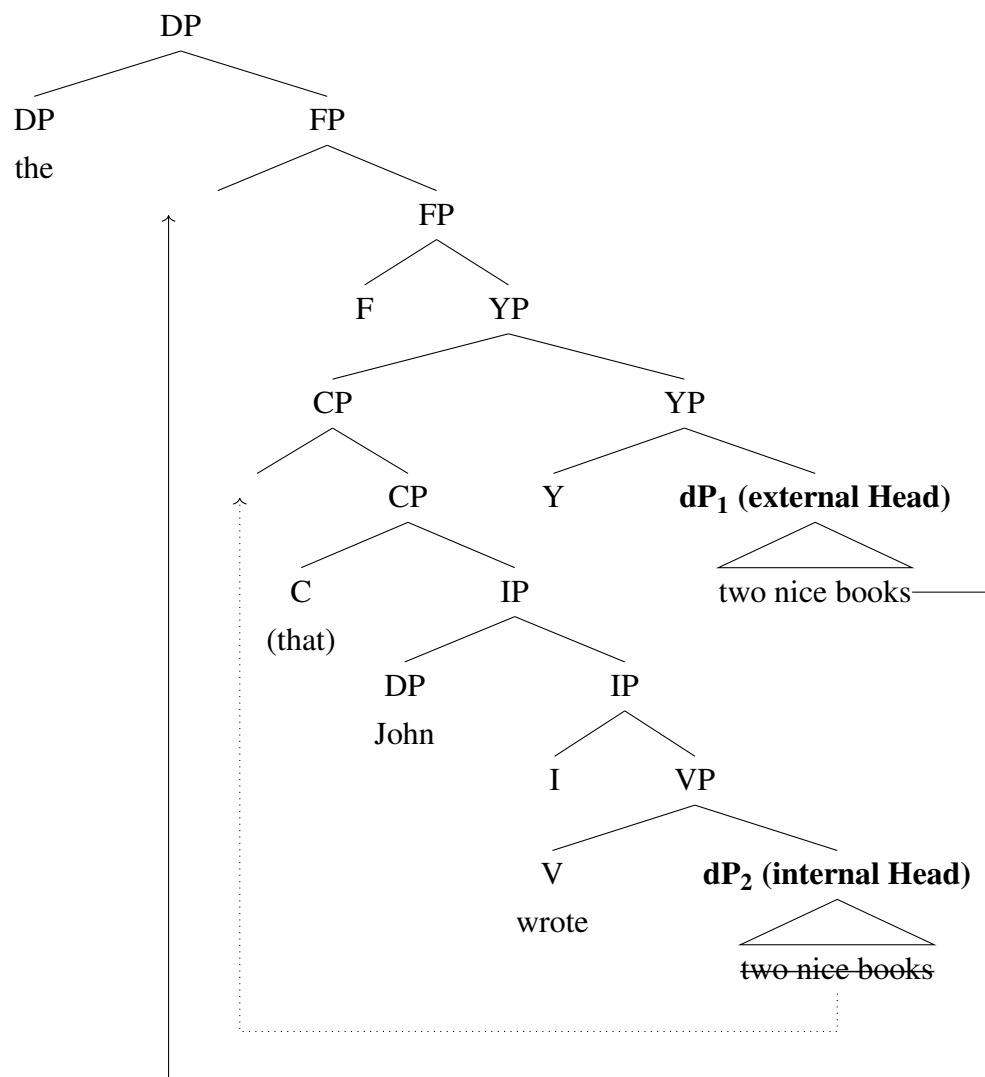
(6) **Double-headed RC structure** (Adapted from Cinque 2020)

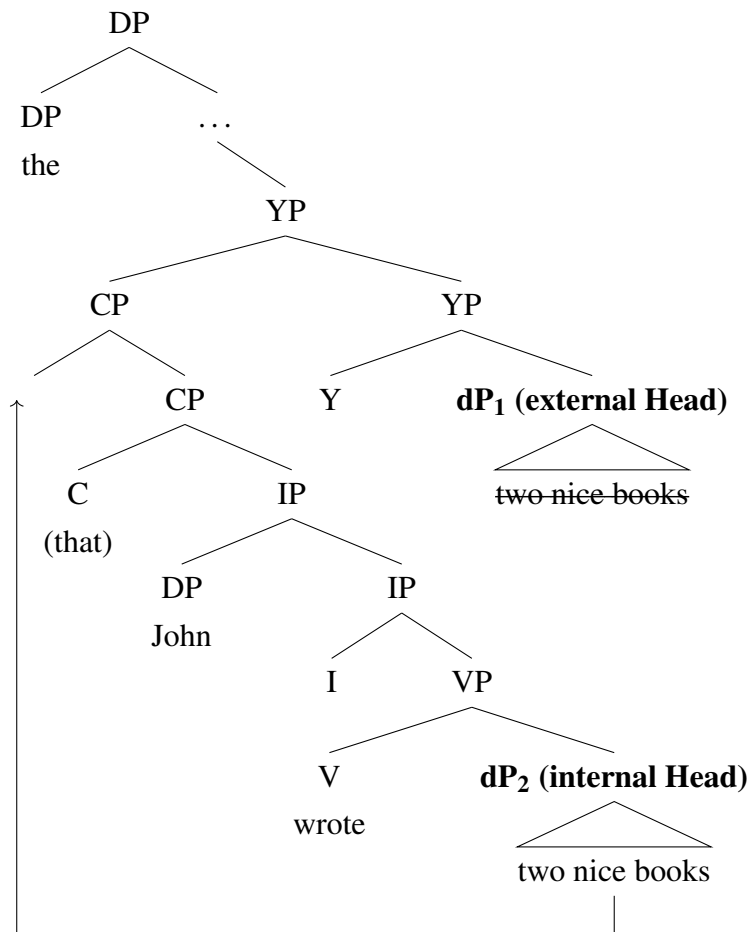


Cinque’s (2020) book provides an extensive discussion of how this base structure

adapts to the needs of typologically diverse RC configurations. In particular, it shows that, depending on the properties of the language or the RC in question, either type of analysis may be available. In a matching derivation, the external head moves above the attachment site of the internal head and deletes it (Cinque 2020: 18, see tree in 7). In a raising derivation, the external head remains in situ, while “it is the Head internal to the relative clause that ends up being the *overt* Head by raising to Spec,CP and licensing the deletion of the external Head” (Cinque 2020: 16), as shown in (8).

(7) **Double-headed RC structure - Matching** (Adapted from Cinque 2020)



(8) **Double-headed RC structure - Raising** (adapted from Cinque 2020)

In the next sections I will take into account the specific case of Spanish subject and object RCs and, after discussing some relevant properties of the relative *que*, show how these can be analysed based on a (slightly modified version of) double-headed structure à la Cinque.

4.1.2 Spanish RCs - word order

Like all Romance languages, Spanish features head-external, post-nominal RCs.³ The sentences in (9) and (10) show an example of, respectively, SRC and ORC.

- (9) La chica que mira a la profesora es Isabel.
 the girl that watches DOM the professor is Isabel
 ‘The girl looking at the teacher is Isabel.’
- (10) La chica que la profesora mira es Isabel.
 the girl that the professor watches is Isabel
 ‘The girl the teacher is looking at is Isabel.’

³See Bianchi 2002; Salzmann 2017; de Vries 2018; Cinque 2020 for a complete review of RC types.

Moreover, ORCs can vary with respect to the position of the subject within the RC. Subjects can appear pre-verbally, as seen in (10), but also post-verbally, like in (11).⁴

- (11) La chica que mira la profesora es Isabel.
 the girl that watches the professor is Isabel
 ‘The girl the teacher is looking at is Isabel.’

Gutiérrez-Bravo (2005) proposes that, unlike in matrix clauses, the unmarked word order in Spanish ORCs requires a post-verbal subject, yielding OVS order. He further argues that subjects in post-verbal position function as foci conveying new information, while pre-verbal subjects rather behave as topics linked to old information.⁵ Moreover, he observes that the alternation between OVS and OSV orders is not governed solely by information structure. Rather, it interacts with prosodic factors and, specifically, the prosodic weight and intonational prominence of the phonological phrases involved. In essence, given that the nuclear accent in Spanish is always clause-final (Contreras 1976; Zubizarreta 1998) and that subjects, as stressed lexical heads, constitute the head of the phonological phrase in which they occur (here the RC), these must follow the verb (i.e., a lighter phonological phrase) as in (11).⁶ When this prosodic configuration is altered, however, the subject surfaces in its canonical pre-verbal position, as is the case in RCs with two complements, about which “speakers have clear intuitions that the unmarked position of the subject is not the post-verbal position, but rather the canonical pre-verbal position” (Gutiérrez-Bravo 2005: 160):

- (12) *Qué pasó?* ‘What happened?’
- a. # Estoy leyendo la carta que le mandó la maestra a Pedro.
 am reading the letter that DAT-CL sent the teacher to Pedro
 ‘I am reading the letter that the teacher sent to Pedro.’
- b. Estoy leyendo la carta que la maestra le mandó a Pedro.
 am reading the letter that the teacher DAT-CL sent to Pedro
 ‘I am reading the letter that the teacher sent to Pedro.’

(Gutiérrez-Bravo 2005: 160)

⁴Note that this sentence is not ambiguous with an SRC reading. Had (11) been an SRC, the RC-internal DP (*la profesora*) should have been introduced by DOM. Because DOM is missing here, the only available reading is the ORC one. I will discuss this more in depth in Section 4.2.1.

⁵See Hartmann and Zimmermann (2008) for a discussion of the distinction between focus, as an information-structural category, and *focus marking* (the linguistic means used to signal it) as well as an analysis of crosslinguistic variation in this respect.

⁶One crucial assumption here is that, just as there are light and heavy syllables at the word level (Prince 1990; Hayes 1995; Ryan 2016), there are also light and heavy prosodic phrases at the level of sentence prosody (Gutiérrez-Bravo 2005: 156-157).

In the above examples, since the nuclear accent falls on the indirect object (a heavy phonological phrase), the subject can occur pre-verbally without interfering with conflicting prosodic requirements.⁷ As the author puts it, Spanish can thus be characterised as “a language that prioritises intonational considerations over canonical subject position,” which is, he continues, “not surprising” (Gutiérrez-Bravo 2005: 157), given that similar contingencies arise elsewhere in the language—for instance, with narrow-focused subjects in matrix clauses (see Büring and Gutiérrez-Bravo 2001). As Katharina Hartmann (p.c.) has pointed out to me, since any material following the verb will tend to place the subject in a pre-verbal position, this word order is more plausibly derived post-syntactically than driven by information-structural features in the syntax.⁸ In this view, post-verbal subjects and pre-verbal subjects in RCs where the verb is followed by additional phonological material are equally new-information foci. By contrast, pre-verbal subjects in RCs whose verb is not followed by further material represent a marked word order, plausibly associated with an old-information topic (see above).

These observations appear to find further support in languages with similar prosodic properties.⁹ In Italian, for instance, subjects in ORCs are typically realised post-verbally in out-of-the-blue contexts (13a). Much as in Spanish, however, a pre-verbal subject is preferred when additional (heavy) phonological material follows the verb. This may be an argument (as with the indirect object in 13b), an adjunct (13c), or even a single adverb (13d).

(13) *Che fai?* ‘What are you doing?’

- a. Sto leggendo la lettera che ha mandato la maestra.
am reading the letter that has sent the teacher
‘I am reading the letter that the teacher sent.’
- b. Sto leggendo la lettera che la maestra ha mandato a Pietro.
am reading the letter that the teacher has sent to Pietro
‘I am reading the letter that the teacher sent to Peter.’
- c. Ho incontrato la maestra che Pietro ha disegnato nel suo diario.
have.1SG met the teacher that Pietro has drawn in.the his diary
‘I met the teacher who Peter has drawn in his journal.’

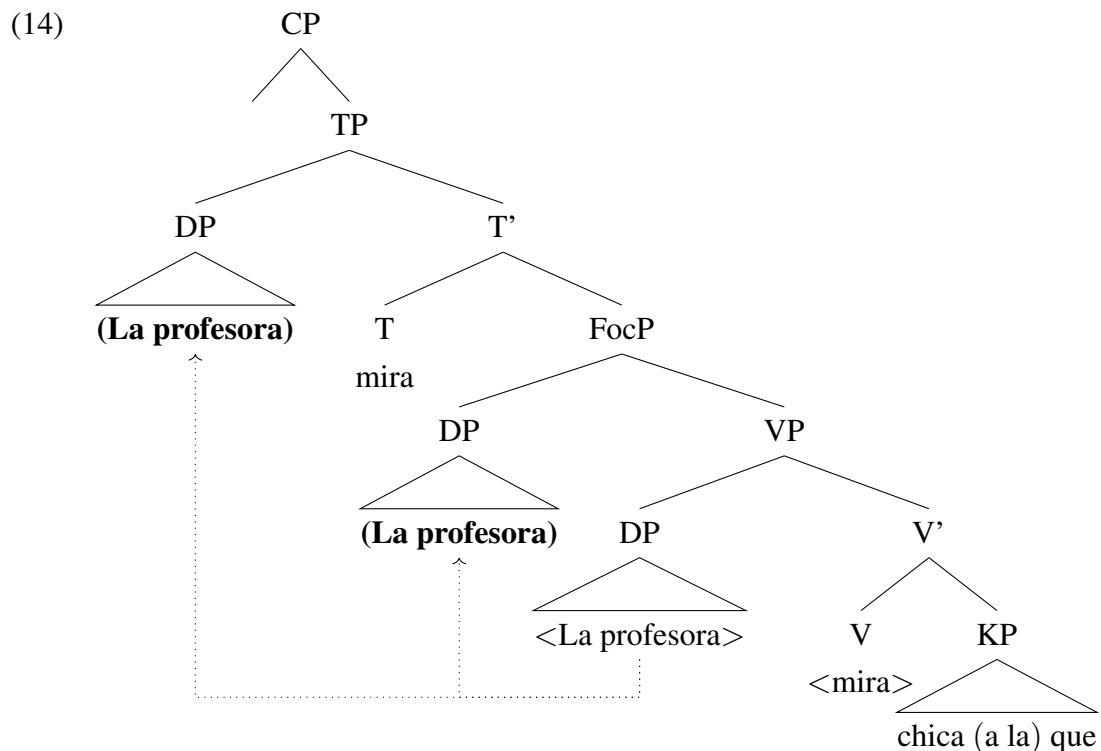
⁷The conflict between syntactic and prosodic requirements is addressed by Gutiérrez-Bravo 2005 within the Optimality Theory framework (cf. Prince and Smolensky 2004; Gutiérrez-Bravo 2005: 157–164).

⁸Empirically, this is consistent with the Italian data presented below; theoretically, it aligns well with proposals drawing a direct link between syntactic and prosodic structure (Selkirk 2011; Elfner 2012; Bennett et al. 2016; Richards 2016, 2017). I am grateful to her for this insight.

⁹Italian, in particular, closely resembles Spanish in this respect. In addition to assigning the nuclear accent to the rightmost constituent (Calabrese 1982; Bocci and Cruschina 2018, a.o.), Italian is, like Spanish, a so-called *stress accent* language, which modulates metrical prominence through spectral cues such as fundamental frequency (f0), intensity, and duration (D’Imperio 2002).

- d. Ho incontrato la maestra che Pietro ha disegnato ieri.
 have.1SG met the teacher that Pietro has drawn yesterday
 'I met the teacher who Peter has drawn yesterday.'

Steady though these contrasts are in Italian, I would not go as far as suggesting that acceptability is ever deeply compromised based solely on subject position. In all displayed cases, both pre- and post-verbal subjects are in fact fully grammatical. This apparently holds true also for Spanish, where the position of the subject does not alter the interpretation of the sentence, nor does either order characterise as non acceptable.¹⁰ In sum, barring said prosodic restrictions, pre- and post-verbal subjects in Spanish ORCs are largely interchangeable and, as far as their syntax is concerned, can be reasonably analysed in a fairly standard fashion. Specifically, I argue that pre-verbal subjects occupy the Spec,TP position, and post-verbal subjects a low Spec,FocP *à la* Belletti (2004), as illustrated in the following tree.¹¹



4.1.3 Spanish RCs - DOM and the status of *que*

A second source of variation in Spanish ORCs concerns the (non-)realisation of DOM in front of the relative particle *que*. As is well known, Spanish marks a specific class of direct objects with the particle *a*: a case-like marker homophonous with the dative preposition,

¹⁰See, in this regard, Betancort et al. (2009: 1918), Brown and Rivas (2011: 23), del Río et al. (2012: 2111), Ezeizabarrena (2012: 2–3), and Sagarra et al. (2019: 123–124).

¹¹I discuss in greater detail the other relevant steps of the derivation in the following section.

commonly referred to as A-marker (Rodríguez-Mondoñedo 2007). In the main clause, at least [human, specific, definite] direct objects are obligatorily A-marked, and the absence of DOM results in ungrammaticality:¹²

- (15) La profesora mira *(a) la chica.
 the teacher looks DOM the girl
 ‘The teacher is looking at the girl.’

In ORCs, however, the marking of the head noun is not as straightforward. In fact, even when the head noun is [human, specific, definite], the realisation of DOM (in conjunction with the definite article *ella*) is optional (see, e.g., Sagarra et al. 2019: 123):

- (16) a. la chica a la que la profesora mira
 the girl DOM the that the teacher looks
 ‘the girl that the teacher is looking at’
 b. la chica que la profesora mira
 the girl that the teacher looks
 ‘the girl that the teacher is looking at’

To better understand why ORCs appear to permit different conditions on the morphological realisation of DOM, it is worth taking a closer look at the relative particle *que*.

Defining the nature of *que* is not an easy task. Ever since the seminal work of Klima (1964), Kayne (1975), and Radford (1981), a prolific strand of research has focused on establishing the categorial status of relative elements. Central to this debate is the question of whether these elements are subordinating heads, like clausal complementisers, or rather relative pronouns. A well-established set of diagnostics helps us distinguish between the two. As illustrated in (17), complementisers and relative pronouns/wh-words differ in three main respects.

- (17) a. **Case marking:** complementisers do not inflect for case, relative pronouns can
 b. **Sensitivity to animacy:** complementisers do not encode animacy features, relative pronouns can
 c. **Compatibility with prepositions:** complementisers cannot be combined with prepositions, relative pronouns can

If we take English, for instance, we can distinguish *that* from *who* and *which* based on these properties. *That* does not inflect for case (18a), *who* can (18b). Similarly, *that* cannot pied-pipe a preposition (19a), while both *whom* and *which* can (19b, c). Finally,

¹²For a comprehensive review of the topic, see Bossong (1991), Torrego (1998), Aissen (2003), Leonetti (2008), Rodríguez-Mondoñedo (2008), López (2012) and Fábregas (2013).

whereas *that* is not sensitive to animacy, *who* can only be used with animate head nouns (20).

- (18) a. the boy that you like/likes you
- b. the boy who/whom you like
- (19) a. * the girl about that I told you
- b. the girl about whom I told you
- c. the car about which I told you
- (20) the car that/*who Jenna bought

For these reasons, *that* has been traditionally assigned the status of relative complementiser (or relativiser) and *who/which* that of wh-words.

But what about Spanish? Like all Romance languages, Spanish exhibits a considerably broader paradigm of relative elements. This includes full-fledged relative pronouns like, e.g., *el/la cual*; *los/las cuales*,¹³ as well as elements like *que*, whose nature is less straightforward to classify. A good starting point to determine its status is, once again, observing how *que* behaves with respect to the properties in (17). Let us start with the third one—namely, the possibility to occur with prepositions in pied-piping configurations. The sentence in (21) illustrates that *que* can be preceded by, for instance, the preposition *en*, indicating that in this respect *que* patterns with wh-words rather than relativisers.

- (21) la casa en que Juan vivió
- the house in that Juan lived
- ‘the house where Juan lived’

The other two properties concern case marking and sensitivity to animacy. In a way, the example presented in (16a) already provides evidence suggesting that *que* is responsive to both. By definition, the A-marker encodes animacy and case features. More precisely, it has been interpreted as an accusative marker that applies to animate (and definite, specific, etc.) direct objects (Brugè and Brugger 1996; Aissen 2003; Leonetti 2004; Rodríguez-Mondoñedo 2007; Fábregas 2013). Hence, the sheer fact that *que* can be preceded by DOM is sufficient to suggest that, even with respect to case and animacy, *que* exhibits properties typically associated with wh-words. Further support in this direction comes from oblique RCs such as those in (22). Under the assumption that *cuyas* corresponds to the genitive form of *que*, and *quien* to its animate variant, these patterns reinforce the idea that *que* displays behaviours characteristic of *bona fide* relative pronouns.

¹³See Zagana (2001: 58-60) for a review.

- (22) a. la casa cuyas ventanas son verdes
 the house whose windows are green
 ‘the house whose windows are green’
- b. el profesor/*libro con quien he estudiado
 the professor/book with that(animate) AUX-have.1SG studied
 ‘the book with which I studied’

If we take the Klima tests seriously (see the list in 17), these few examples are already suggestive: they indicate that *que* functions as a wh-word rather than a relativiser.¹⁴ Nonetheless, the literature on this topic remains divided, and several analyses have reached divergent conclusions. I briefly review in the next paragraph the two most prominent analyses of Spanish *que*. After that, I will present my own proposal.

4.1.4 The ‘bare *que* hypothesis’ and the ‘unique *que* hypothesis’

Suñer (2000) proposes that *que* is a wh-pronoun; except when it introduces an SRC or an ORC, in which cases it rather characterises as a complementiser. This analysis is known as the *bare que hypothesis*. An alternative view is offered by the so-called *unique que hypothesis*, proposed by Brucart (1992), which holds that *que* is always a complementiser, regardless of the type of RC in which it occurs.

The main piece of evidence in support of Brucart’s approach comes from some Canarian and Puerto Rican sentences such as “*La amiga con la [más confianza] que tengo*” ‘The friend I trust the most’, where *que* does not seem to form a constituent with the preposition and the determiner following the head noun. This is readily accounted for if *que* is in the C head, while it becomes more problematic if *que* is analysed as a wh-word. Note, however, that sentences of this type are not allowed in standard Peninsular Spanish. Moreover, Suñer (2000) does not consider these cases to be genuine counter-evidence against a relative pronoun analysis of *que*, as they are probably instances of misanalysis by analogy with other similar structures.¹⁵

The *unique que hypothesis* has the obvious advantage of positing a simpler, more economical system than the one ensuing from the *bare que hypothesis*, as it posits a single *que* in the lexicon. However, treating *que* as a C head leaves open the question of why it exhibits all the relevant properties of relative pronouns (see above). On the whole, then,

¹⁴As a matter of fact, Poletto and Sanfelici (2018) claim that there are reasons to believe that the dichotomy between relativisers and relative pronouns is not as clear-cut as originally thought. I will return to this below and show that my proposal aligns with Poletto and Sanfelici (2018).

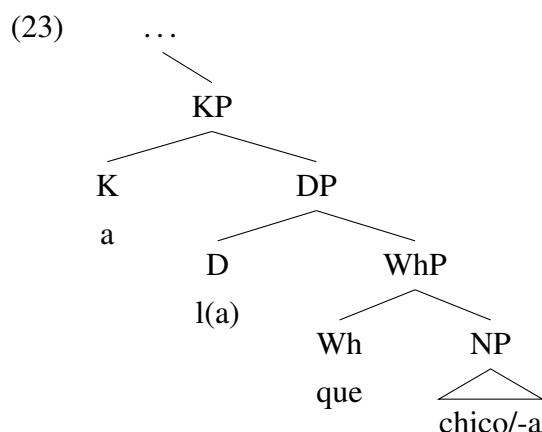
¹⁵A full discussion of this (counter-)argument lies beyond the scope of this chapter, as does a detailed examination of Brucart’s and Suñer’s proposals. For further discussion, see the cited works as well as Perpiñán’s (2010: 49–56) review of the topic.

neither account is fully exempt from criticism. Yet on closer inspection, a third possibility remains: we could assume that *que* is always a wh-word. This preserves the simplicity of Brucart’s proposal, while offering a more natural account of the similarities between *que* and canonical relative pronouns. This is the core of my proposal, which I will develop in the next section.

4.1.5 The proposal: *que* as a wh-word

In essence, I propose that Spanish *que* should always be analysed as a wh-word, and not as a complementiser. This aligns with a number of well-established analyses of *que*’s counterparts in other Romance languages. Indeed, the view that RCs are always introduced by a wh-word has been advanced for Italian (Manzini and Savoia 2003, 2011; Poletto and Sanfelici 2018; Rugna 2023), old and modern Italo-Romance dialects (Poletto and Sanfelici 2018, 2019), French (Rooryck 2000; Sportiche 2011), European Portuguese (Rinke and Aßmann 2017), Brazilian Portuguese (Kato and Nunes 2009).

All these accounts can be somehow connected with Kayne’s (2008; 2014) claim that there is no real distinction between complementisers and relative pronouns, as the former do not exist and all relative particles are, in fact, determiners. Building on this theoretical stance, Poletto and Sanfelici (2018) argue that *che* (and its variants) in the Italo-Romance system is always a determiner-like element that bears (at least) a [Wh] feature and has the lexical property of requiring a nominal restriction. I here extend this proposal to Spanish and contend that *que* is a wh-item that merges as a specifier of the internal head’s NP. In more detail, I propose that *que* realises a [Wh] feature and is topped by a DP and a KP, whose heads are realised by, respectively, the definite article *el/la* and the A-marker *a* (I henceforth refer to this configuration as *A + L que*), as represented in (23).



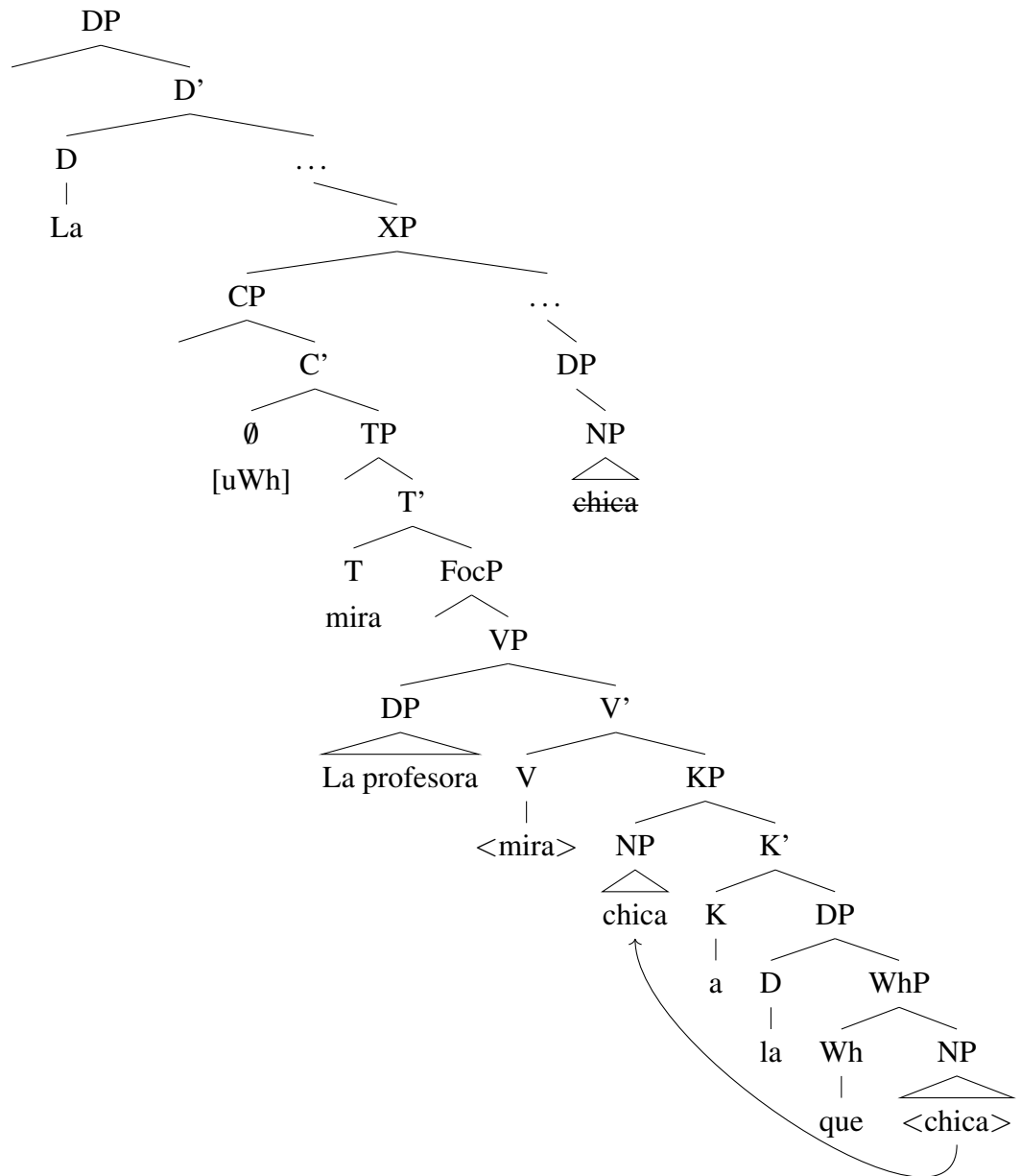
As far as the derivation of the entire RC is concerned, I propose that it proceeds in a raising fashion, with the internal head of a Cinque-style double-headed structure raising to Spec,CP (see 6). However, I adopt an analysis closer to the one proposed by

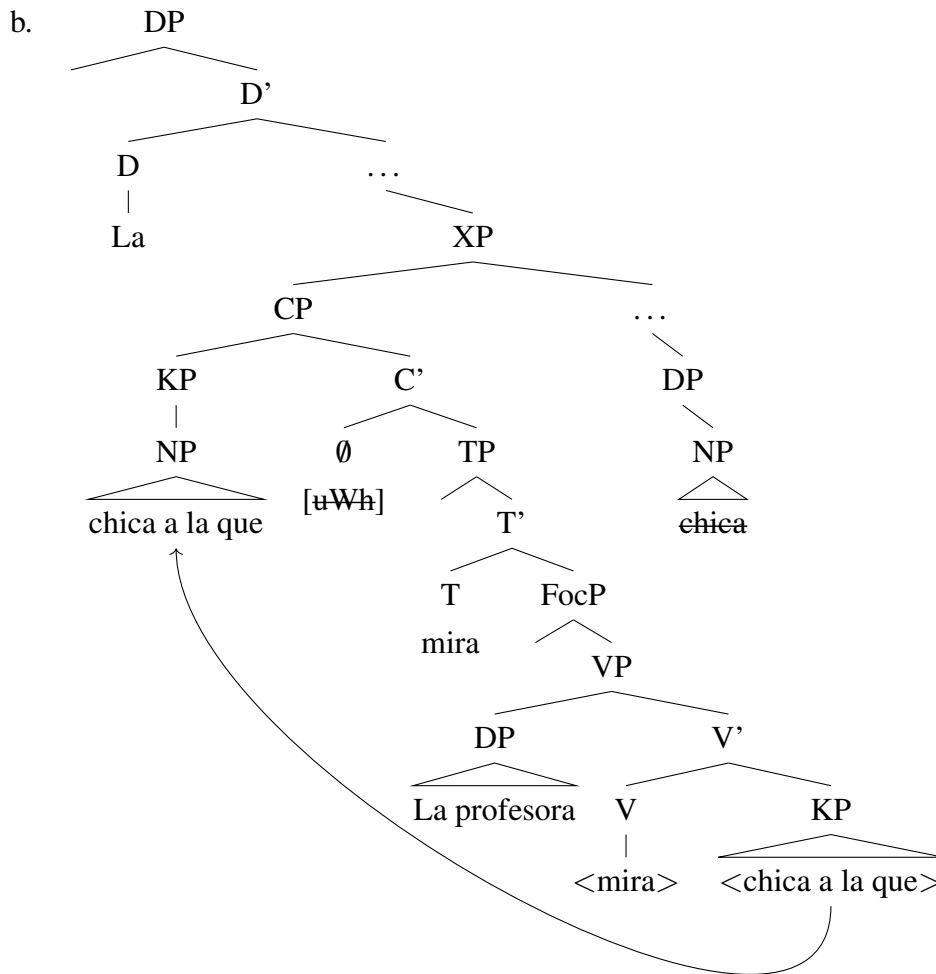
Poletto and Sanfelici (2018), which differs from Cinque's in a few aspects. First, their RC is a CP and not an IP (like in Rizzi 1997 and Bianchi 1999). As already mentioned, moreover, they claim that a *wh*-element (here *que*) modifies the internal head, similarly to what Kayne (1994) originally proposed for *which*-relatives. This way, the derivations of *which*-relatives and *that*-relatives are unified. Last, the initial step of Poletto and Sanfelici's (2018) derivation involves movement of the internal-head's NP to the specifier of its dominating DP (here KP), in order to escape binding of the *wh*-variable.¹⁶ In the case of Spanish ORCs like (16), then, the derivation proceeds as in (24): the internal-head's NP *chica* moves to Spec,KP to c-command the WhP (24a). Then the entire KP is attracted to Spec,CP to check against a [*uWh*] feature in C (24b). Finally, the external-head is deleted under (kaynean) c-command and non-distinctness, *à la* Sauerland (2003, 2004).¹⁷

¹⁶The rationale for this movement is rooted in the differing semantics of nominal expressions in *wh*-questions and RCs. In the former, the NP introduces a variable in an open set, while in RC-contexts the NP introduces a fixed referent. As such, the nominal expression in RCs needs to reach a position that is generally occupied by (in)definite articles and encodes the presupposition of existence (Poletto and Sanfelici 2018: 287).

¹⁷Under Kayne's antisymmetric c-command (defined over categories rather than segments) the NP in question can c-command the lower external-head as the former is in the specifier of the phrase which is, in turn, the specifier of the CP.

(24) a.





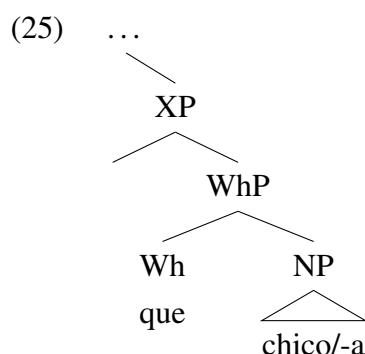
The decision to adopt a raising analysis for Spanish ORCs is primarily motivated by the observation that this type of derivation best accounts for intervention effects. In (24), the copy of *chica* that is ultimately realised is the one that directly crosses over the subject of the RC, thereby triggering the disruption predicted by Relativised Minimality constraints (see Chapter 2).¹⁸ If we aim to construe these effects syntactically, this seems to be the most plausible analysis. Still, in principle, a matching approach could also accommodate the relevant facts.¹⁹

¹⁸In brief, the fact that the object crosses over the subject is what should make ORCs more costly than SRCs—an asymmetry that has long been attested across languages and populations (see below for discussion). In Sections 4.2 and 4.3, I will show that this pattern is confirmed for Spanish in both child and adult populations.

¹⁹As discussed above, there is now broad consensus that matching and raising need not be mutually exclusive. In the Spanish RCs considered here, a matching analysis could be adopted without major drawbacks. After all, even if the head that gets realised in a matching derivation is the external one, the movement of the object over the subject within the RC would still be part of the derivation (see 4, and Cinque 2020: fn.21). I do not consider this issue crucial for the purposes of the present chapter, as none of the central

4.1.6 The case of covert DOM

It remains to be explained what happens in cases like (16b), where DOM is not realised (at least not overtly). There are three ways I can think of to account for this. The first is to argue that when DOM is absent, it is simply because the functional heads required for its realisation have not been merged. This is represented in (25), where the internal-head is composed of an NP topped by the WhP hosting *que* and at least an XP (evidently different from DP, as the definite article is missing, too) to whose specifier the NP moves (see fn. 16).



Alternatively, one could argue that KP and DP are always realised as in (23) above, and that they are occupied by null elements when *que* surfaces in its bare form (see, e.g., Collins and Kayne 2023 for a way to implement this formally).

A third possibility, which I consider more elegant, is that *que* can Spell-Out either its own head or the entire K-D-Wh sequence. Put differently, on this view, there is just one type of *que* in the lexicon that is always merged as a modifier of the NP and is capable of Spelling-Out different portions of the structure. In one case, it Spells-Out only the Wh head, while D and K are realised independently by their respective lexical items, i.e. *el/la* (the definite article) and *a* (the A-marker). In the other case, *que* also Spell-Out K and D. This type of variation can be handled straightforwardly within any framework that permits a one-to-many mapping between lexical items and syntactic heads, such as Nanosyntax (Starke 2009; Caha 2009b,a; Baunaz et al. 2018) or Distributed Morphology (Halle and Marantz 1993, 1994; Harley and Noyer 2000). In what follows, I adopt the formal tools provided by Svenonius's theory of Spanning (Svenonius 2012, 2016, 2020), which I briefly summarise in the next paragraph.

arguments hinge on the choice between these alternatives. As a final remark, the only analysis that would likely prove inadequate for explaining Relativised Minimality effects is the head-external analysis represented in (1). In such a derivation, the object crosses over an empty category which, lacking relevant features, would fail to trigger intervention effects (see 4.2.1 for the technical aspects).

4.1.6.1 Spanning theory

According to Svenonius, the operation of Spell-Out targets *spans*. A span is defined as a syntactic head (a *trivial span*) or a sequence of syntactic heads (a *non-trivial span*) in a functional projection. That is, given an extended projection (e.g., N and all its functional heads), a span characterises as a subpart of a *complement line* that links an unbroken sequence of heads in a complement-relation. This is formally defined in (26):²⁰

(26) *Definition of Span* (from Svenonius 2020)

- a. A span is a head (a minimal X^0) or a sequence of heads $\langle h_1, h_2 \dots \rangle$ such that for each $n > 1$, $R_C(h_{n-1}, h_n)$ (Svenonius 2012; Merchant 2015)
- b. $R_C(h_m, h_n) = h_m$ is the head of the complement h_n

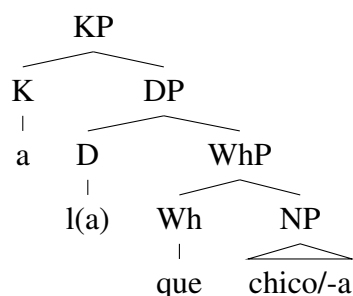
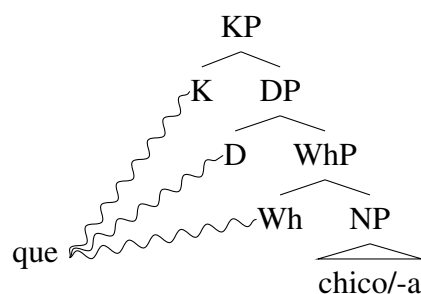
Each morphological exponent is associated with a span, be this trivial or non-trivial. This association takes place during Spell-Out, defined as the operation that maps syntactic structures onto representations that serve as input to the phonological component (see Chomsky 1995 and the discussion in Section 3.1.1). According to Svenonius, Spell-Out operates cyclically (Chomsky 2000, 2001), and in a two-step fashion.²¹ In a first step, called *L(exical)-Match*, the syntactic structure is associated with lexical entries. These, in a way, mediate the pairing between (portions of) syntactic representations and their phonological realisation. Crucially, the L-Match operation is exclusively concerned with features “on the syntactic side of the pairing” (Svenonius 2012: 3), as it is only in the second step of Spell-out, called *Insert* (Bye and Svenonius 2010, 2012), that the phonological properties of the matched exponents are interpreted.

In the case of the Spanish ORCs at issue here, L-Match has two options. As hinted at above, the lexical entry *que* can L-match either a trivial span represented by the single Wh-head (27a), or the non-trivial span including the sequence K-D-Wh (27b).²²

²⁰The definition assumes that there exists a complement relation R_C whereby each head in a functional sequence takes the immediately lower head in the sequence as a complement (see Grimshaw 2005a). Crucially, this relation must be distinct from the relations that apply to specifiers and to adjuncts (see Svenonius 2020).

²¹This would be motivated by the strict modular disjunction that holds between phonology and syntax (Zwicky 1969, 1987; Zwicky and Pullum 1986). Although see Bruening (2018, 2019) and Richards (2010, 2016) for perspectives rejecting the hypothesis of a phonology-free-syntax.

²²The representations in (27) are probably a simplification, as it might be more accurate to distinguish two different lexical entries for *que*: one endowed with [K-D-Wh] features and the other only with [Wh]. This does not affect the core of my analysis and, in fact, the two alternatives are not necessarily mutually exclusive (Svenonius p.c.).

(27) a. **Trivial Span L-Match**b. **Non-trivial Span L-Match**

4.1.7 Why Spanning?

At this point, however, a crucial question arises: is there evidence for postulating the full K–D–Wh sequence even in cases where only bare *que* surfaces? Why not assume, instead, that in such cases KP and DP are simply absent from the underlying structure (as discussed above)? One way to approach this problem is by considering the parallelism with the main clause. As mentioned earlier, it is surprising that the same class of direct objects requires overt A-marking in matrix clauses but not in RC-contexts. Yet this asymmetry becomes less puzzling if we assume that *que* can realise the entire set of projections associated with DOM. On this view, the only difference between matrix and relative clauses would be restricted to independent lexicalisation properties of *que*: DOM does not overtly surface when *que* realises the whole K-D-Wh sequence, but the underlying syntax is always the same.

This line of reasoning, however, may not provide sufficient support for the span-based analysis illustrated above. One could still argue that the difference between bare *que* and A + L *que* reflects an actual structural contrast: in the former case, the direct object in the RC may simply fail to meet the conditions that trigger DOM. On this view, bare *que* occurs when the direct object in the RC is not specific (or perhaps even definite), and thus does not require DOM. If that were the case, the projections responsible for A-marking (KP) and specificity/definiteness (DP) would not be merged. In contrast, when (animate) specific, definite direct objects are selected at the RC-level, the conditions for DOM are met, requiring both KP and DP (and the ensuing emergence of A + L *que*).

Now, as minimal as this alternative analysis may seem, it still leaves certain issues unaddressed. Most notably, it fails to explain why there are no interpretative differences between ORCs with bare *que* and those with A + L *que*. Again, sentences like (16a) and (16b), repeated in (28), are semantically equivalent.

- a(28). a. la chica a la que la profesora mira
 the girl DOM the that the teacher looks
 ‘the girl that the teacher is looking at’

- b. la chica que la profesora mira
 the girl that the teacher looks
 ‘the girl that the teacher is looking at’

One might argue that in (28b), the interpretation of the head noun perfectly matches that of (28a) not because of an identical underlying structure, but rather because the ultimate semantics is assigned, in both cases, by the external DP, whose features override those assigned RC-internally. However, I do not find this analysis to be either superior or conclusive. First, it is not the null hypothesis, but simply one possible hypothesis, just like the span-based account proposed above. More importantly, there appear to be contexts in which positing the absence of DOM-relevant projections in the syntactic representation becomes problematic, regardless of whether *que* appears in its bare form or is preceded by A + L. I discuss a few such examples below.

In his seminal monograph, López (2012) discusses cases where DOM must be obligatorily realised.²³ In particular, beside reviewing some well-described semantic requisites of the direct object, he illustrates that there are specific syntactic configurations that impose the realisation of DOM. These include small clauses, clause union, and object control structures (López 2012: 23–25). For instance, (29) shows that the nominal phrase in a small clause must be A-marked:

- (29) El profesor consideró a/* $\emptyset$ un estudiante inteligente
 The professor considered DOM a student intelligent
 ‘The professor considered a student intelligent’ (López 2012: (56))

Similarly, affected direct objects in clause union also require DOM. This can be seen with causative constructions like in (30), as “[t]he causee of an intransitive predicate is accusative [and] obligatorily introduced by accusative A” (López 2012: 23), but also with complements of perception verbs, as in (31).

- (30) María hizo llegar tarde a/* $\emptyset$ un niño
 María made arrive late DOM a boy
 ‘María made a boy be late’ (López 2012: (58))
- (31) María vio caer a/* $\emptyset$ un niño
 María saw fall DOM a boy
 ‘María saw a boy fall’ (López 2012: (64))

In all these (and similar) contexts, DOM cannot be omitted. What makes these cases particularly interesting for the present discussion is the fact that the conditions imposing the A-marker are strictly syntactic and have nothing to do with the specificity of

²³López (2012) focuses on indefinite objects, whereas the current discussion involves definites. Nonetheless, whenever DOM is obligatory with indefinites, it is *a fortiori* obligatory with definites, which occupy a higher position on the definiteness scale (Aissen 2003).

the marked objects, as demonstrated by the fact that these can be naturally interpreted as non-specific (López 2012: 25). Thus, if we embed the above examples into a RC-structure, we expect the very same conditions to apply, independently of the semantic contribution of the external D-head. That is, because the A-marking is imposed by the argument-structure of the RC-internal predicate, potential interactions with the definiteness/specificity features of the external DP will not affect the realisation of DOM. This means that the projections relevant for DOM (here described as the K-D sequence) must be there. How is this relevant? Well, if *que* can introduce these types of RCs, then we do have some evidence in support of the span approach advanced above.

Let us see the relevant examples. The sentences in (32) reproduce the case of small-clause complements illustrated in (29). A + L *que* is perfectly fine and, in fact, seems to be the more natural form to use. Albeit probably more marginal, also bare *que* can fit this context.

- (32) a. Un estudiante al que considero inteligente fue admitido en
 a student DOM.the that consider.1sg intelligent was admitted in
 Oxford.
 Oxford
 ‘A student I consider intelligent was admitted to Oxford.’
 b. ? Un estudiante que considero inteligente fue admitido en Oxford.
 a student that consider.1sg intelligent was admitted in Oxford
 ‘A student I consider intelligent was admitted to Oxford.’

The same holds true for (33) and (34), which reflect their matrix-counterparts (reported in 30 and 31). Again, even though A + L *que* is preferred here, too, bare *que* remains a viable option, at least in some speakers’ grammar.²⁴

- (33) a. El chico al que María hizo llegar tarde se llama Juan.
 the boy DOM.the that María made arrive late REFL call Juan
 ‘The boy Maria made late is called Juan.’
 b. ? El chico que María hizo llegar tarde se llama Juan.
 the boy that María made arrive late REFL call Juan
 ‘The boy Maria made late is called Juan.’
 (34) a. El niño al que María vio caer se hizo daño.
 the child DOM.the that María saw fall REFL made damage
 ‘The child that Mary saw fall hurt himself.’

²⁴It would be interesting to run a controlled study to gauge the extent to which this type of sentences are accepted across participants and in different contexts. Due to space and time restrictions, I had to limit myself to collecting judgements through an online form. Twenty native speakers (most of whom were linguists) completed the forms and gave various answers. In the sentences displaying bare *que* reported here, at least 50% of my informants considered the sentences acceptable. As is often the case with DOM, I suspect there is considerable variability depending on (sub-)regional varieties.

- b. ? El niño que María vio caer se hizo daño.
 the child that María saw fall REFL made damage
 ‘The child that Mary saw fall hurt himself.’

If we were to believe that bare *que* can only realise (the head of) WhP, the acceptability of (32b, 33b, 34b) would escape an immediate explanation. On the contrary, if the span account defended here is on the right track, these facts are actually predicted: DOM is always represented at the syntactic level; it simply does not surface with its typical *a* morpheme whenever *que* realises the entire K-D-Wh span. This is not to say that this is necessarily the case. I am only suggesting that *que* (in its bare form) *can* realise DOM, but need not. It is thus possible that bare *que* also serves all cases where the direct object internal to the RC need not or must not be introduced by DOM.

That an alternation between different exponents in syntactically equivalent environments exists should not sound too surprising. Similar cases can be found elsewhere in the Romance domain, confirming that this is indeed an option available to grammars. Take the case of Italian *cui* ‘to whom.’ *Cui* is a relative element introducing indirect object RCs. Interestingly, it can occur either alone or preceded by the dative preposition *a*.

- (35) a. La professoressa cui devo consegnare la tesi è in ferie.
 the professor that-DAT must.1SG submit the thesis is in holiday
 ‘The professor to whom I have to hand in my thesis is on holiday.’
 b. La professoressa a cui devo consegnare la tesi è in ferie.
 the professor to that-DAT must.1SG submit the thesis is in holiday
 ‘The professor to whom I have to hand in my thesis is on holiday.’

The use of either form has no reflexes in the interpretation, as both *cui* and *a cui* unquestionably bear dative case in the given examples.²⁵ Once again then, we are looking at an example in which two different morphological exponents presuppose the same underlying syntax, and one of these exponents is morphologically richer. Although there is probably more than one way to modulate this alternation, spans offer a straightforward account. In (35a), *cui* realises a K-Wh span, while in (35b) K is realised by the dative marker *a* and Wh by *cui*, in the exact same fashion discussed for Spanish (A + L) *que*.

4.1.8 Summing up on Spanish RCs

The literature on the syntax of restrictive SRCs and ORCs offers a host of different analyses, sometimes differing only in a few marginal aspects. After evaluating the alternat-

²⁵The only difference between the two forms can be related to their register. *Cui* is more likely used in a higher diaphasic context, while *a cui* is preferable in every-day speech. Interestingly, some Spanish speakers have very similar intuitions and reported that A + L *que* might sound more formal than bare *que*.

ives, I decided to implement a raising approach based on Cinque's double-headed RC-structure, whereby the internal head raises to a position c-commanding the external head and deletes it. As mentioned, however, I do not exclude that other analyses (notably a matching derivation) could equally serve the purpose.

A second important aspect considered is the categorial status of *que*. Again, there is a vast literature discussing the nature of relative elements, with the core of the debate revolving around whether and when these should be analysed as complementisers or wh-words. I argued that Spanish *que* is always a wh-word that merges as a modifier of the NP in the internal-head position, extending to Spanish the (by now standard) analysis adopted for most Romance languages.

Lastly, I considered the surface variability of Spanish ORCs. Without affecting interpretation, these can vary along two dimensions: the position of the subject within the RC and the realisation of overt DOM in front of the relative element *que*. As far as word order is concerned, I followed work by Gutiérrez-Bravo (2005) and argued that the preference for a pre- or post-verbal subject is likely determined by independent phonological conditions. At the syntactic level, I represent pre-verbal subjects in the specifier of TP, and post-verbal subjects in the specifier of a low FocusP, as is standardly assumed.

Accounting for the alternation between bare *que* and A + L *que* is less straightforward. Again, interpretation is unaffected, but the question of what constitutes the underlying syntax of the two forms remains complex. I propose that the syntactic representation need not differ, and that both *que* and A + L *que* can realise the same structural sequence, crucially including the projections relevant for DOM. I developed this account building on Svenonius' theory of spanning, whereby single lexical entries can realise sequences of heads of different sizes. In the case at hand, *que* solely realises the Wh head when it is preceded by *al/a la* (which realise, respectively, a K head and a D head), and the full K–D–Wh sequence when it occurs in its bare form.

4.2 Experiments 1–2: comprehension and processing of SRCs and ORCs by adults

It is a well-established observation that SRCs and ORCs pattern asymmetrically. SRCs are typically easier for children to comprehend (de Villiers et al. 1979; Tavakolian 1981; Goodluck and Tavakolian 1982; Corrêa 1995; Friedmann et al. 2009), and they also pose fewer challenges for adults during processing (Frauenfelder et al. 1980; Ford 1983; Clifton and Frazier 1989; Schriefers et al. 1995; Mak et al. 2002). A comprehensive review of the relevant data and theoretical proposals can be found in Lau and Tanaka (2021).

In this and the following sections, I focus on Spanish RCs, and in particular on how they are processed and comprehended by adult and child native speakers, respectively. The rationale for choosing Spanish is twofold. First, even though the literature on subject–object asymmetries in RCs is quite extensive, experimental work on Spanish remains relatively scarce. To the best of my knowledge, only a handful of studies have addressed adult populations (see, e.g., Betancort et al. 2009; Sagarra et al. 2019); and as far as children are concerned, the study presented in Section 4.3 constitutes the first experimental investigation, published in Presotto and Torregrossa (2024).

Second, as I will illustrate in the next section, Spanish constitutes an ideal testing ground for evaluating the role of disambiguating cues in the comprehension and processing of RCs. This relates to another crucial aspect in the literature: although ORCs are generally rather demanding, it has long been observed that they can be facilitated under specific conditions. Different frameworks attribute these effects to different sources (see discussion below). Nonetheless, there appears to be a consistent regularity in the kinds of sentences that show improvement. In Presotto and Torregrossa (2024), we argued that ORCs can be facilitated by language-specific cues that satisfy the following descriptive generalisation:

(36) **Descriptive generalisation of ‘morphosyntactic cue’**

A morphosyntactic cue is a grammatical property whose presence or absence implies that a given constituent is either the subject or the object of a given clause.

The main support for this generalisation comes from studies on children’s comprehension of ORCs, which consistently show that facilitation effects arise when such cues are involved (see Section 4.3 and Presotto and Torregrossa 2024 for detailed discussion). These considerations are directly relevant to the main research questions addressed in

this thesis. As emphasised in Section 3.1, this work investigates the source of locality effects by examining the conditions under which they can be facilitated. Understanding how and why locality effects can be eased (and through which cues) provides insight into their underlying cause. As illustrated in the following sections, Spanish RC data prove particularly revealing in this respect.

4.2.1 The relative penalty of ORCs and their amelioration: reductionist vs. grammatical theories

I have already mentioned that the literature on the degradation of ORCs is vast and hosts a variety of competing accounts. In Section 2.2.2, for instance, I reviewed two major reductionist proposals—namely, the Increased Referential processing approach (Warren and Gibson 2002, 2005) and the Similarity-based approach (Gordon et al. 2001, 2004). I will not repeat their details here, but it is important to recall that both identify independent, domain-general pressures as the source of ORC degradation (and more generally, of object A-bar dependency effects). Specifically, ORCs involve a filler–gap dependency that crosses over an intervening subject. This increases the processing cost, as the filler must be maintained in working memory while another referent (the subject) is processed. By contrast, in SRCs the dependency is not disrupted by any such intervener. Crucially, ORCs are predicted to be easier when the intervening subject is referentially “lighter” (Increased Referentiality account) or sufficiently distinct from the object (Similarity-based account). Details aside, what is key here is the underlying logic shared by these approaches: ORCs are harder because they tax domain-general processing resources, yet this difficulty can be mitigated by manipulating factors that reduce processing load. This places such accounts on the ‘pure reductionist’ end of the spectrum, firmly outside the Interface space as defined in this thesis (see Figure 6.2 and Section 3.2).

A similar group of extra-grammatical accounts includes so-called *experience-based theories* (e.g. Bever 1970; MacDonald and Christiansen 2002; Wu et al. 2018). These propose that comprehenders are sensitive to the statistical regularities of linguistic input and use prior experience to predict upcoming structures. As suggested by Wu et al. (2018: 24), “[...] comprehenders are sensitive to the statistical regularities in lexical and structural input, and can use their prior experience with linguistic signals to predict the upcoming structure.” Although these theories differ in scope and formulation, they share the assumption that RC processing is governed by general learning and predictive mechanisms, thus placing them, again, outside the language-specific component and outside the Interface space.

By contrast, there exists a prolific strand of research attributing both the difficulty and the conditions for amelioration of ORCs to purely grammatical constraints—namely,

those regulating intervention locality (see Chapter 2). Chief among these is Relativised Minimality (RM), reviewed in Section 2.1.1. According to RM, the integration difficulty in ORCs (and in object A-bar dependencies more broadly) is modulated by the featural specification of the object and the subject. As already discussed, RM assigns to the object (target) and subject (intervener) a set of syntactically active or “relevant” features (Rizzi 2018). A set-theoretic computation then evaluates the similarity between these feature sets. This computation is not binary, but gradient, as regulated by the Distinctness Hierarchy (see 46 in Section 2.1.1). The following example illustrates possible ORC configurations, where [A] and [B] represent RM-relevant features:

- | | | | | |
|------|----|----|--------------------------------------|--------------------|
| (37) | a. | * | The [object] that the [subject] V __ | Identity |
| | | | [A] [A] | |
| | b. | ?? | The [object] that the [subject] V __ | Inclusion |
| | | | [A; B] [A] | |
| | c. | | The [object] that the [subject] V __ | Disjunction |
| | | | [A] [B] | |

Under this view, an evaluation procedure at the interface assigns a different penalty to each of these relations. Configurations where the target and the intervener are identical are assigned the strongest penalty. A weaker penalty is associated with inclusion,²⁶ and no penalty is assigned with disjunction. ORCs are degraded when the object and subject bear identical, or in certain circumstances overlapping, features (identity or inclusion), while they are ameliorated when their feature sets are disjoint (disjunction). Thus, to reiterate, RM represents a strictly grammatical approach: both degradation and amelioration effects are driven by syntactic principles internal to the grammar.

This brings us to the core empirical question addressed in this chapter: can Spanish help adjudicate between these competing accounts of ORC degradation and amelioration? Here is how I proceed. Experiment 1 tests whether an SRC–ORC asymmetry is actually observable in Spanish adult comprehension and processing. Since the results will confirm the asymmetry, Experiment 2 investigates four types of ORCs that differ in the presence of morphosyntactic cues matching the definition in (36), in order to determine whether such cues yield any amelioration effects. Specifically, I examine two properties of Spanish ORCs: the position of the subject within the RC, and the presence or absence of DOM before the relative particle *que* (see Section 4.1.3). The full paradigm is shown in (38), where the subject occurs either after (38a, b) or before (38c, d) the inflected verb, and the

²⁶Crucially, inclusion is argued to mark the divide between the child and adult systems. While children generally fail to comprehend structures involving inclusion, adults typically succeed, though they may still experience processing difficulty (Belletti et al. 2012; Belletti and Rizzi 2013).

DOM-marker *al* between the relative head and *que* is either realised (38b, d) or not (38a, c):

- (38) a. el niño que saluda el futbolista
the boy that greet.3SG the footballer
'the boy that the footballer greets'
- b. el niño al que saluda el futbolista
the boy DOM.the that greet.3SG the footballer
'the boy that the footballer greets'
- c. el niño que el futbolista saluda
the boy that the footballer greet.3SG
'the boy that the footballer greets'
- d. el niño al que el futbolista saluda
the boy DOM.the that the footballer greet.3SG
'the boy that the footballer greets'

Leaving aside possible information-structural discrepancies (see discussion above), the sentences in (38) all convey the same meaning. That is, they are equally unambiguous ORCs, since each features at least one disambiguating morphosyntactic cue as defined in (36).

In particular, (38a) is disambiguated by the *absence* of DOM: the NP within the RC is not introduced by DOM, which forces a subject interpretation. Had *el futbolista* been the object, DOM would have been required, as in (40). Because its relevance lies in the absence of a morphological marker, I label this an *in-absentia* cue. Conversely, (38b) provides a case of an *in-praesentia* cue: the overt DOM marker *al* on the relative head explicitly signals its object status.²⁷ (38c) is disambiguated by word order: the NP within the RC precedes the inflected verb, a position dedicated to subjects in Spanish relative clauses. Finally, (38d) contains both a preverbal subject and an overt DOM marker.

To sum up, (38) illustrates ORCs disambiguated by a single cue (38a–c) and one condition involving two simultaneous cues (38d). Importantly, (38a) involves an *in-absentia* cue, while the others rely on *in-praesentia* cues or a combination thereof (38b–d).

In what follows, I will illustrate that such cues have different effects on the interpretation (and processing) of ORCs, with crucial consequences for traditional accounts. In the interim discussion in Section 4.2.3.4, I will argue that the results of Experiment 2 do not support a grammatical account like RM but rather point towards a more processing-oriented explanation. Not of the kind represented by Increased Referentiality or Similarity-

²⁷The distinction between *in-praesentia* and *in-absentia* cues is purely descriptive. It refers to whether the cue consists of an overt morphological marker that is present and easily accessible to the parser (*in-praesentia*), or instead arises from the absence of a morphological element whose omission has grammatical consequences (*in-absentia*). I return to this distinction in Section 4.3.7.

based approaches, however. Instead, I propose that intervention effects are ameliorated via disambiguation strategies that ease processing depending on the quality, number, and linear occurrence of surface disambiguating cues. Cues that, crucially, presuppose access to abstract grammatical representations.

4.2.2 Experiment 1

Experiment 1 addresses the following research question:

- (39) [RQ1] Is there a subject–object asymmetry in Spanish adults’ comprehension and processing of relative clauses?

4.2.2.1 Materials and design

The stimuli consisted of 24 sentences arranged in a 2×1 design, with SENTENCE TYPE (SRC vs. ORC) as the independent factor. All experimental items featured two [human; definite; specific] DPs, serving as the agent and the patient of transitive constructions (see Appendix for the full list). The verbs and their participants were selected so that all sentences were semantically reversible. SRCs were like (40a), and ORCs like (40b).

- (40) a. el niño que saluda al futbolista
the boy that greet.3SG DOM.the footballer
‘the boy that greets the footballer’
b. el niño que saluda el futbolista
the boy that greet.3SG the footballer
‘the boy that the footballer greets’

Note, again, that (40b) is *not* ambiguous: the absence of the DOM-marker (obligatory with [human; definite] objects) in front of *futbolista* imposes that the latter is the subject (see the discussion above as well as fn. 4). To put it in other words, lack of DOM in (40b) results in ungrammaticality under an SRC-reading. The only available interpretation would be, in fact, one where *futbolista* is the subject of the RC, effectively turning the structure into an ORC.²⁸

Experimental items were divided into two lists and integrated with 12 filler sentences. Each list featured 6 items per condition, arranged so that participants would see each experimental item only once. Fillers were the same across lists. All sentences were recorded by a native speaker of Spanish and presented as audio stimuli. The test was run online and used a sentence–picture matching task in the style of Friedmann et al. (2009),

²⁸On the disambiguation between SRCs and ORCs based on (absence of) DOM, see also Betancort et al. (2009: 1917–1918), Ezeizabarrena (2012: 165–166), and Perpiñán (2010: fn 37)

where participants listened to a sentence and selected the corresponding picture from two options: one depicting the event described by the item, and the other depicting the same event with the agent and patient roles reversed (see Figure 4.1).



Figure 4.1: Sample picture used in the experiments (© alicelipparini.com).

Participants were required to wear headphones, listen to the stimulus, and select the picture they believed matched the sentence by pressing either the ‘F’ or ‘J’ key. The introductory question for the experimental items was always as in (41):

- (41) ¿Dónde está el niño que saluda el futbolista?
 where is the.M.sg boy that greet.3SG the footballer
 ‘Where is the boy that the footballer greets?’

In this way, the relative head was always the subject of the main clause and introduced by the determiner *el*. This was done to avoid potential confounds or interactions between DOM assigned by the verb of the RC and that assigned by the main clause.²⁹ Filler sentences were introduced by the Spanish translation of “Show me where the NP V the NP.”

Both accuracy and reaction times (RTs) were recorded. RTs were measured as the time between the end of the audio stimulus and participants’ keypress. The task was built using PciBex (Zehr and Schwarz 2018).

4.2.2.2 Participants

Participants were recruited via Prolific, an online platform that allows targeting based on demographic filters. For this study, only adult native speakers of Peninsular Spanish with no reported history of language-related disorders were considered. In total, 24 participants (mean age: 30.5; SD = 10.32) took part in the study. Each was paid £2 for participation. An accuracy threshold of 85% on filler items was set for inclusion in the analysis. All participants exceeded this threshold.

²⁹This becomes especially relevant for Experiment 2. An introductory question like “Show me the X that the Y VERB”, as used in e.g. Friedmann et al. 2009, would result in “*Muéstrame al niño al que (el futbolista) saluda (el futbolista)*” in sentences like (46b) and (46d) (see below).

4.2.2.3 Results and analysis

The bar graph in Figure (4.2) shows the proportion of correct answers in association with SRCs and ORCs, respectively. The mean proportion of accurate answers was .97 (SD = .18) for SRCs and .58 (SD = .49) for ORCs.

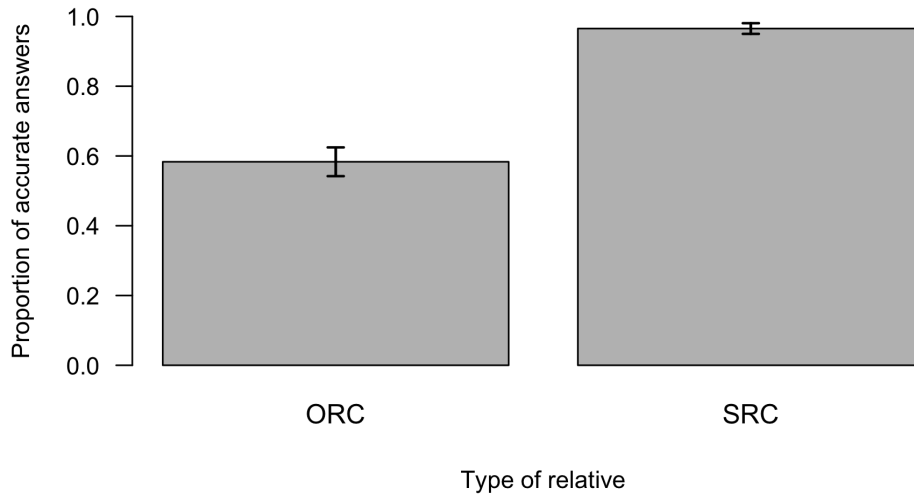


Figure 4.2: Bar graph of the proportion and standard error of accurate answers with SRCs and ORCs (adults).

R (R Core Team 2021) and the lme4 package (Bates et al. 2015) were used to fit a generalised linear mixed-effects model with response accuracy as the dependent variable and relative clause type as the fixed effect. The model included a random intercept and slope for relative clause type by participant.³⁰ SRCs were set as the reference level. The results, summarised in Table 4.1, revealed a significant effect of condition, with participants performing significantly worse with ORCs than with SRCs ($\beta = -3.88$, $p = .006$).

Table 4.1: Parameters of the generalised linear mixed-effects analysis with response accuracy as dependent variable and type of relative clause as fixed effect.

Fixed effects	<i>b</i>	SE	<i>z</i>	<i>p</i>
Intercept	4.35	1.35	3.22	.001
Condition (ORC)	−3.88	1.41	−2.75	.006

³⁰The model was fitted using the following formula: `m1 <- glmer(Accuracy ~ 1 + Condition + (1 + Condition | Participant), data = data, family = binomial(link = "logit"), glmerControl(optimizer = "bobyqa"))`.

Turning to RTs, only correct responses were included in the analysis. The average RT was 715 ms (SE = 62 ms) for SRCs and 2112 ms (SE = 218 ms) for ORCs. As shown in Figure 4.3, participants were slower to respond to ORCs than to SRCs.

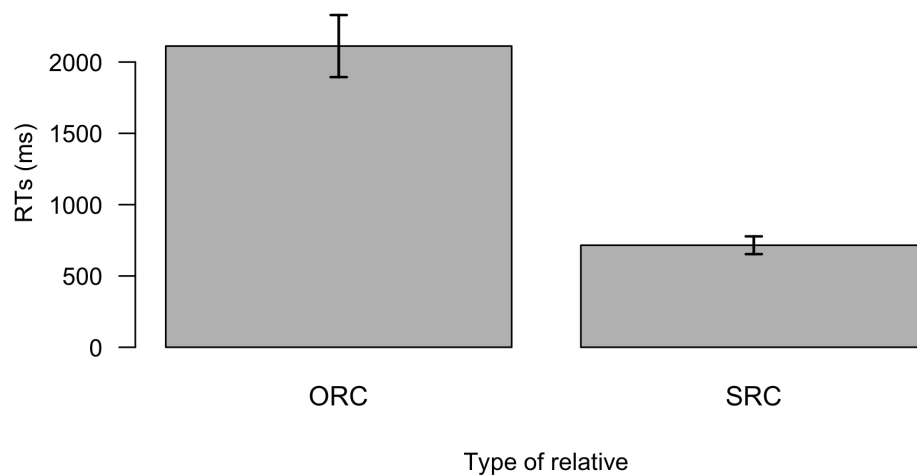


Figure 4.3: Bar graph of the average RTs (ms) for correct responses with SRCs and ORCs (adults). Error bars represent standard error.

A linear mixed-effects model with RTs as the dependent variable and clause type as the fixed effect (with random intercept by participant)³¹ revealed a significant main effect of clause type: participants were significantly slower with ORCs compared to SRCs ($\beta = 936.82$, $p < .001$), as summarised in Table 4.2.

Table 4.2: Parameters of the linear mixed-effects model with RTs as dependent variable and type of relative clause as fixed effect.

Fixed effects	<i>b</i>	SE	<i>df</i>	<i>t</i>	<i>p</i>
Intercept	712.17	158.04	36.53	4.51	<.001
Condition (ORC)	936.82	144.46	263.00	6.49	<.001

4.2.2.4 Interim discussion

To summarise, the results of Experiment 1 reveal a clear subject–object asymmetry in Spanish adult speakers’ comprehension and processing of RCs. ORCs brought about

³¹Due to convergence issues, random slopes were not included. The model was fitted as follows: `m2 <- lmer(RTs ~ 1 + Condition + (1 | Participant), data = data0.1)`

significantly lower accuracy rates as well as longer RTs compared to SRCs.

On the one hand, these findings align with a large body of literature confirming robust asymmetries between RC types. In particular, our results show ceiling-level comprehension for SRCs and, as in Betancort et al. (2009), faster RTs relative to ORCs. On the other hand, the results are somewhat unexpected: while adults are well known to experience difficulty processing ORCs, comprehension is typically spared.

Crucially, none of the accounts discussed in Section 4.2.1 predict this pattern. Frequency-based, referentiality-based, and similarity-based accounts all predict increased processing cost, not impaired comprehension. The same goes for RM, unless the configuration at issue involves feature identity. Yet this cannot be the case here for two reasons. First, RM assumes that the relative head carries a [R(eltative)] feature in its syntactic bundle, which cannot be shared by the subject DP in sentences like (40b), as the subject is not itself relativised. Whatever the full feature specification of the two DPs in (40b), they must differ at least in the presence of [R], ruling out identity. Second, and more intuitively, feature identity between target and intervener should yield full ungrammaticality under RM. However, the ORCs tested in Experiment 1 are unquestionably grammatical in Spanish; they are ‘just’ extremely difficult to process and comprehend.

What, then, makes these ORCs so demanding? And how is it possible that entirely grammatical sentences prove so challenging for unimpaired adult native speakers? I argue that these sentences are, in fact, not particularly exceptional, but rather fall into a broader class of constructions long recognised in the literature: garden-path sentences. These are syntactically well-formed structures that cause severe processing disruptions, typically leading the parser to commit to an incorrect structure that must later be revised. As noted by Siloni (2014: 276), garden-path sentences are “very difficult for humans to process,” and their correct interpretation often involves “a perceptible break-down and conscious reanalysis.” In short, the key feature of such sentences is that they temporarily induce an illusion of ungrammaticality, which disappears once the correct structure is recovered. Consider the classic example in (42), taken from Pritchett (1991):

(42) After Susan drank the water evaporated.

In (42), *the water* is first analysed as the direct object of *drank*. Upon encountering *evaporated*, however, the parser must reanalyse *the water* as the subject of the second predicate, and reinterpret *drank* as intransitive to avoid ungrammaticality. This sharply contrasts with structurally ambiguous sentences like (43), in which multiple grammatical parses remain simultaneously available:

(43) Max saw the suspect with the binoculars yesterday.

(Siloni 2014: 277)

In other words, whereas ambiguous sentences permit multiple grammatical interpretations, garden-path sentences momentarily lead the parser into a syntactic dead end (what Siloni terms a case of “transitory ungrammaticality”) which must be reanalysed to yield a licit parse. I contend that the ORCs tested in Experiment 1 constitute precisely this type of structure.³² Disambiguation between an SRC and ORC reading only becomes possible once the final DP is encountered. That is, participants hearing the sentence initially assign an SRC interpretation (generally regarded as the default and statistically more prominent reading, see Lau and Tanaka 2021) and treat the first DP, *el niño*, as the subject. This bias is reinforced upon encountering the transitive verb *saluda*, prompting early assignment of the subject θ -role, consistent with the general principle that θ -roles are discharged as early as possible during processing (Siloni 2014: 280). It is only when the second DP, *el futbolista*, is reached that reanalysis becomes possible and, in fact, necessary.

In a time-constrained experimental setting like the one at hand, this reanalysis becomes particularly effortful. Moreover, the difficulty is likely compounded by the arduousness of the disambiguation process, which requires recognising that *el futbolista* is animate, definite, and in object position, yet introduced by *el* rather than *al* (i.e., the only grammatical option in such contexts). In other words, what truly disambiguates (40b) from (40a) is the *absence* of the morphological exponent realising DOM. A fact that arguably renders the disambiguation process particularly demanding. This explains both the low comprehension accuracy and the higher RTs observed for ORCs in the study.

Let me now emphasise a crucial point of this explanation: processing these structures relies on grammatical representations. Specifically, it presupposes that θ -role assignment and the grammar-specific rules constraining the distribution of DOM are accessed during processing. In other words, the parser is not merely guided by surface cues or plausibility; it actively engages with syntactic knowledge to resolve temporary ambiguities or reanalyse misleading input.³³

Crucially, this type of interaction is exactly what the Interface Hypothesis laid out in Section 3.2 predicts: a bias to assign θ -roles early interacts with grammar-specific representations and constraints, such as those governing DOM. This bias is reasonably driven by domain-general pressures on processing, as discharging semantic information such as θ -relations reduces the computational load, facilitating the processing of structures within the limited resources (e.g., working memory) available to the human parser (Siloni 2014:

³²Garden-path sentences are typically restricted to written stimuli that, crucially, lack punctuation. Often, the addition of punctuation or the use of appropriate intonation resolves the ambiguity, which is why garden-path effects are largely absent in natural speech. In the task used in this experiment, however, intonation was purposely kept neutral, and no contextual cues were available, as both SRC and ORC interpretations were equally plausible.

³³Note that, despite the low accuracy, participants still identified the correct interpretation of ORCs above chance, suggesting that reanalysis did occur in the majority of cases.

280). Interestingly, Siloni’s work points in a similar direction, suggesting that the operations underlying sentence processing mirror those of syntactic derivation, particularly Merge (Siloni 2014: 277ff.), and that reanalysis is constrained in ways similar to syntactic movement: a misanalysed constituent can only be ‘moved’ to a higher, c-commanding position. A full engagement with Siloni’s proposal would take me too far afield, but it is worth noting that analysing ORCs like (40b) as garden-path structures fits her model well: the misanalysed object DP is reinterpreted as occupying a higher, c-commanding subject position. Additionally, Siloni builds on insights from Pritchett (1991), who argues that parsing and grammatical derivation are deeply interconnected, once again reinforcing the view defended here: that both domain-general and grammar-specific factors are jointly responsible for effects like the ones emerged in the present results.

In sum: Experiment 1 confirms the presence of a subject–object asymmetry in Spanish RCs. ORCs like (40b) are particularly taxing because they instantiate garden-path effects, in line with Siloni’s theory. Crucially, this account fares better than a purely grammatical model like RM, which fails to predict such a dramatic drop in comprehension. If this interpretation is correct, then inserting an earlier cue which either prevents the garden-path from forming or enables earlier reanalysis should facilitate comprehension. This is precisely what Experiment 2 is designed to test.

4.2.3 Experiment 2

At the start of Section 4.2, I emphasised that ORCs, albeit overall more taxing than SRCs, can be ameliorated, arguably when morphosyntactic cues are available that support assignment of θ -roles (see the generalisation in 36). I identified four such cues in Spanish ORCs: *in absentia* DOM, *in praesentia* DOM, and word order (henceforth WO, i.e. the position of the subject within the RC).

Experiment 1 showed that *in absentia* DOM cues are either too complex or occur too late in the linear string for the parser to benefit from them, leading to garden-path effects. On these grounds, Experiment 2 is designed to address the following two research questions:

- (44) **[RQ2]** Both *in praesentia* DOM and WO are overt and linearly earlier than *in absentia* DOM. Do they ameliorate the comprehension and processing of Spanish ORCs?

The second research question situates the findings within the broader theoretical framework of this thesis, by asking whether and how such amelioration effects can inform our understanding of locality:

- (45) **[RQ2.1]** What implications do the results have for locality theory?

4.2.3.1 Materials and design

Experiment 2 followed the exact same procedure as Experiment 1. Stimuli consisted of 16 ORCs of the types illustrated in (46), where the subject within the RC appeared in either pre-verbal (46c, d) or post-verbal (46a, b) position, and the relative head was either unmarked (46a, c) or marked (46b, d) with an overt DOM-marker.³⁴

- (46) a. el niño que saluda el futbolista (-DOM ; post-verbal)
 the boy that greet.3SG the footballer
 ‘the boy that the footballer greets’
- b. el niño al que saluda el futbolista (+DOM ; post-verbal)
 the boy DOM.the that greet.3SG the footballer
 ‘the boy that the footballer greets’
- c. el niño que el futbolista saluda (-DOM ; pre-verbal)
 the boy that the footballer greet.3SG
 ‘the boy that the footballer greets’
- d. el niño al que el futbolista saluda (+DOM ; pre-verbal)
 the boy DOM.the that the footballer greet.3SG
 ‘the boy that the footballer greets’

The items were arranged in four experimental conditions in a 2×2 design crossing SUBJECT POSITION (pre-verbal vs. post-verbal) and DOM PRESENCE (–DOM vs. +DOM) as independent variables. They were integrated with 16 fillers. Four experimental lists were created using a Latin Square design, with each list featuring a different item per condition. As in Experiment 1, fillers were held constant across lists.

4.2.3.2 Participants

A total of 48 adult native speakers of Spanish (mean age = 31.38, SD = 10.99) were recruited via Prolific using the same demographic criteria as in Experiment 1. Each participant received £2 for completing the task. One participant was excluded for failing to meet the 85% accuracy threshold on fillers. An additional participant was recruited to ensure that the final sample was fully balanced across the four lists (12 participants per list).

4.2.3.3 Results and analysis

Figure 4.4 shows the proportion of accurate responses across the four conditions. Accuracy was near ceiling in all conditions except for –DOM, POST-VERBAL, which yiel-

³⁴For simplicity, the analysis below uses the labels “-DOM” and “+DOM” instead of *in-absentia* and *in-praesentia* DOM, respectively.

ded substantially lower performance ($M = .59$, $SD = .49$). The remaining conditions all showed high accuracy: DOM, POST-VERBAL $M = .97$ ($SD = .17$); DOM, PRE-VERBAL $M = .96$ ($SD = .20$); and $-DOM$, PRE-VERBAL $M = .99$ ($SD = .07$).

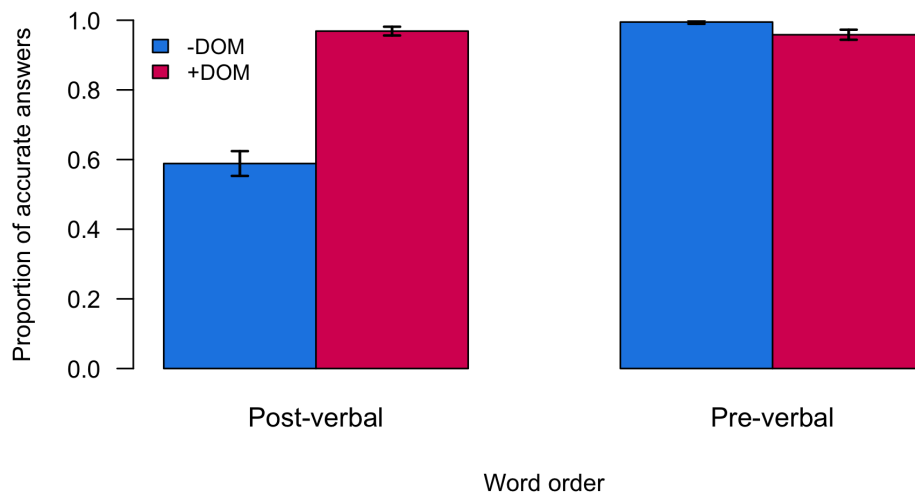


Figure 4.4: Bar graph of the proportion and standard error of accurate answers across conditions (adults).

A generalised linear mixed-effects model was fitted using R and the `lme4` package, with response accuracy as the dependent variable and the interaction between DOM and SUBJECT POSITION as predictors. The model included random intercepts for items and both random intercepts and by-participant slopes for position.³⁵ Results are summarised in Table 4.3. The main effect of SUBJECT POSITION ($\beta = 5.66$, $p < .001$) indicates that ORCs with pre-verbal subjects were comprehended significantly better than those with post-verbal subjects. The main effect of DOM ($\beta = 3.48$, $p < .001$) shows that ORCs with overt DOM were comprehended better than those with in-absentia DOM. Crucially, though, a significant interaction ($\beta = -5.67$, $p < .001$) reveals that among the ORCs with pre-verbal subjects, response accuracy actually decreased when the ORC exhibited overt DOM.

Pairwise comparisons were calculated with the `emmeans` package (Lenth 2025) and showed that accuracy in the $-DOM$, POST-VERBAL condition was significantly lower than in all other conditions ($ps < .0001$). The three remaining conditions ($+DOM$, POST-VERBAL; $-DOM$, PRE-VERBAL; and $+DOM$, PRE-VERBAL) did not differ significantly from one another, reflecting the near-ceiling proportions in Figure 4.4.

³⁵Model formula: `m1 <- glmer(Accuracy ~ Position * Dom + (1 + Position |`

Table 4.3: Fixed-effect estimates from the generalised linear mixed-effects model with response accuracy as the dependent variable.

Fixed effects	<i>b</i>	SE	<i>z</i>	<i>p</i>
Intercept (−DOM, post-verbal)	0.43	0.26	1.62	.104
Position (pre-verbal)	5.66	1.19	4.76	<.001
DOM (+DOM)	3.48	0.49	7.12	<.001
Position * DOM	−5.67	1.19	−4.77	<.001

Turning to RTs, only correct responses were retained. As shown in Figure 4.5, participants were fastest in the +DOM, PRE-VERBAL condition ($M = 1569$ ms, $SE = 178$), followed by −DOM, PRE-VERBAL ($M = 1878$ ms, $SE = 164$) and +DOM, POST-VERBAL ($M = 1894$ ms, $SE = 182$). Responses in the −DOM, POST-VERBAL condition were by far the slowest ($M = 3280$ ms, $SE = 333$).

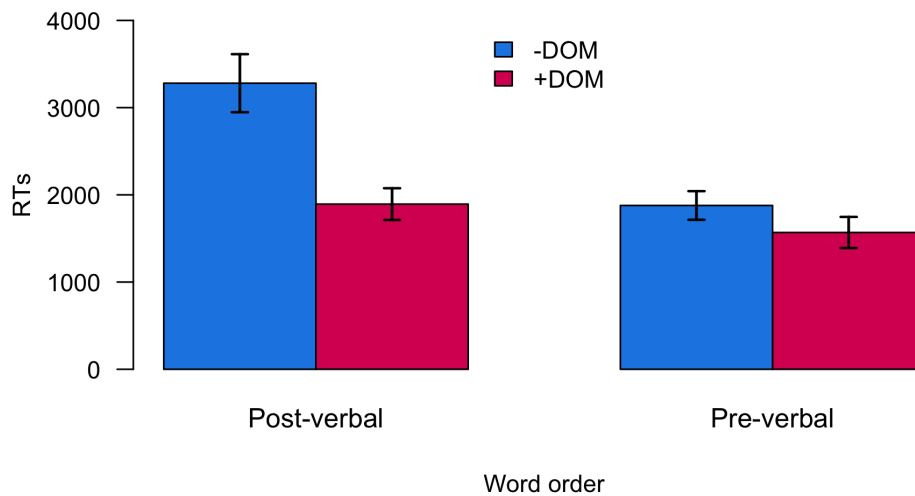


Figure 4.5: Bar graph of the average RTs (ms) for correct responses across conditions. Error bars represent standard error.

A linear mixed-effects model was fitted with RTs as the dependent variable and the interaction between DOM and SUBJECT POSITION as predictors. Random intercepts for items and participants were included, along with separate by-participant random slopes for DOM and SUBJECT POSITION.³⁶ Table 4.4 summarises the results, which revealed a

Participant) + (1 | Item), family = binomial(link = "logit")).

³⁶Model formula: `m2 <- lmer(RTs ~ 1 + Dom * Position + (1 + Dom | Participant) + (1 + Position | Participant) + (1 | Item))`.

pattern reflecting accuracy rates. The main effect of DOM ($\beta = -1157$, $p = .001$) shows that ORCs with overt DOM yielded significantly faster responses compared to those with in-absentia DOM. The main effect of POSITION ($\beta = -1198$, $p < .001$) indicates that ORCs featuring a pre-verbal subject brought about significantly faster RTs compared to those with post-verbal subjects. Yet the significant interaction ($\beta = +841$, $p = .006$) reveals that the RTs in ORCs with pre-verbal subjects increased when these displayed overt DOM.

Pairwise comparisons also reflected the pattern observed for accuracy: RTs in the –DOM, POST-VERBAL SUBJECT condition were significantly slower than in all other conditions ($ps < .01$). However, no significant differences were found among the remaining three conditions (+DOM, POST-VERBAL, –DOM; PRE-VERBAL, +DOM; PRE-VERBAL; all $ps > .4$).

Table 4.4: Fixed-effect estimates from the linear mixed-effects model with reaction times as the dependent variable.

Fixed effects	<i>b</i>	SE	<i>df</i>	<i>t</i>	<i>p</i>
Intercept (–DOM, post-verbal)	3074.43	354.17	87.93	8.68	<.001
DOM (+DOM)	–1156.91	342.30	89.22	–3.38	.001
Position (pre-verbal)	–1198.48	261.80	150.06	–4.58	<.001
Position * DOM	840.91	306.67	541.06	2.74	.006

4.2.3.4 Interim discussion

The findings of Experiment 2 confirm that ORCs with a post-verbal subject and no overt DOM marker are particularly troublesome for native speakers. As in Experiment 1, this is reflected in both comprehension and processing. In Section 4.2.2.4, I argued that these effects can be understood as instances of garden-path: a late and complex cue such as in-absentia DOM may not suffice for successful reanalysis, especially in a time-constrained task. If this interpretation is on the right track, then an earlier cue should either allow for earlier reanalysis or prevent the garden-path altogether.

The results of Experiment 2 support this prediction. Both the overt DOM marker preceding the relative *que* and the presence of a pre-verbal subject yield near-ceiling accuracy and significantly lower RTs. This is, again, expected: in sentences like (46b, d), a subject-reading can be ruled out as early as *al que*, which, given the grammatical constraints discussed in Section 4.1.3, can only follow a relative head that satisfies the conditions for DOM, and thus never a subject. Similarly, in sentences like (46c, d), the presence of a second DP *before* the inflected verb clearly identifies it as the subject of the RC, which by

necessity forces the first DP to represent the object of the predicate (see Section 4.1.2). This answers the first research question:

- (47) **[RQ2]** *Both in praesentia DOM and WO are overt and linearly earlier than in absentia DOM. Do they ameliorate the comprehension and processing of Spanish ORCs?*

In-praesentia DOM and pre-verbal subjects have a clear ameliorating effect in the comprehension and processing of Spanish ORCs.

The second research question concerns the implications of these results for our understanding of locality theory. As already discussed, a grammatical account such as RM explains the difficulty associated with ORCs as a classic case of intervention locality, where the movement of the object from its argument position to the relative head position crosses over an intervening subject DP. I have already noted, however, that the comprehension difficulties observed in –DOM, POST-VERBAL ORCs are not straightforwardly predicted under RM (see Section 4.2.2.4). Suppose, nevertheless, that even in non-identity configurations, some ORCs may still pose substantial difficulties at the level of comprehension. Can RM account for the ameliorating effects observed in Experiment 2? The challenge lies in the nature of the factors that boost performance—namely, subject position and DOM. Under RM, these would need to be encoded as syntactically active features capable of modulating intervention configurations along the Distinctness Hierarchy.

Let us begin with subject position. One might propose that pre-verbal subjects bear a [Topic] feature, following Gutiérrez-Bravo (2005). But this would immediately run into trouble: RM-based models assume that [Topic] belongs to a distinct featural class that does not interact with A-bar dependencies (Rizzi 2004).³⁷ A possible workaround would be to instead treat pre-verbal subjects as bearing a [Focus] feature. As illustrated in (48), this would yield an Intersection configuration, which RM predicts to be unproblematic (Belletti et al. 2012).³⁸

- (48) El niño que el futbolista saluda **Intersection**
[R; NP] **[Foc; NP]**

Under this interpretation, RM would align with the findings presented above. As said, however, this analysis stands in contradiction with the standard interpretation of pre-verbal subjects in Spanish ORCs. Moreover, if, as Gutiérrez-Bravo (2005) suggests, *post-verbal* subjects are the ones bearing focus, then the RM-based prediction is that ORCs

³⁷Specifically, topic features form “a class of its own [...], so that [they] do not trigger minimality effects on operator chains nor are they affected by intervening operators” (Rizzi 2004: 344).

³⁸I here adopt the standard representation of featural bundles within the fRM-framework, see also fn. 41.

with post-verbal subjects should be easier than those with pre-verbal ones. A prediction that, however, runs counter to the actual results.

Turning to DOM, a similar problem emerges. In order to factor DOM into an RM-based account, one would need to treat it as a syntactically active feature. But if DOM is a case marker (as argued by, a.o., Bossong 1985, Brugè and Brugger 1996, Aissen 2003, and López 2012) then it does not count as an intervening feature under RM and should be irrelevant to the computation (see Friedmann et al. 2017 for a discussion of why case should be excluded from RM-based computations). This is, for that matter, exactly what Bentea (2012, 2016) and Sevcenco and Avram (2012) conclude for Romanian, where DOM seems to not trigger a clear amelioration effect in children's comprehension of ORCs.³⁹ The fact that Spanish DOM *does* facilitate ORC comprehension is therefore problematic for RM. But let us nonetheless grant the possibility that DOM can be treated as syntactically active and enters the RM computation. Even under this assumption, its presence does not alter the relevant featural configuration in a way that RM could exploit. Consider the following illustrative cases:

- | | | |
|------|--|------------------|
| (49) | a. El niño que saluda el futbolista | Inclusion |
| | [R; NP] [NP] | |
| | b. El niño al que saluda el futbolista | Inclusion |
| | [R; NP; DOM] [NP] | |

In (49a), the configuration is one of featural inclusion. RM thus predicts that adults should not encounter significant difficulty, contrary to the empirical findings. Crucially, adding DOM to the featural bundle of the object in (49b) does not alter the inclusion configuration. RM would therefore predict that (49a) and (49b) should yield similar results, yet Experiment 2 reveals significantly better comprehension and faster RTs in the latter. Even with flexible assumptions about feature assignment, then, RM fails to derive the observed contrasts.

This, I believe, represents an important example in support of the perspective adopted in this thesis: that amelioration effects offer a valuable diagnostic for assessing the limits of current locality theories. In the specific case of Spanish ORCs, the findings indicate that a theory such as RM, based exclusively on grammar-internal constraints, is insufficient to account for the observed data.

On the other side, purely reductionist accounts like the Increased Referentiality and Similarity-based approaches also fall short. These models predict variation based on the

³⁹In this regard, I must underline that Romanian DOM and Spanish DOM are not entirely comparable. Albeit sharing similar syntactic behaviours, especially in the context of ORCs, a crucial difference between the two is that Romanian DOM is not strictly obligatory. In spoken language, the DOM particle *pe* is actually quite consistently omitted (Guțu Romalo 2000; Sevcenco and Avram 2012; Bentea 2016).

referential quality or similarity of the noun phrases involved (see above). But in both experiments reported here, the relevant nominals were held constant: all were human, definite DPs. As such, no variation in accuracy or RTs should be expected. Again, this prediction is falsified by the empirical data.

Hence, taken together, the results of Experiments 1 and 2 challenge both narrowly grammatical and pure reductionist explanations. Under the Interface Hypothesis defended in this thesis, however, these findings actually align with theoretical expectations. To reiterate, I argue that intervention effects of this type emerge within the Interface system, where domain-general pressures on processing interact systematically with grammar-specific constraints. The observed pattern reflects precisely this kind of interaction: a processing bias towards early θ -role assignment is modulated by grammar-specific constraints governing DOM distribution, determining the amelioration effects documented in these experiments. Now, let me emphasise a crucial theoretical point here. Under this view, the parser actively accesses abstract grammatical representations rather than being driven solely by surface cues or plausibility heuristics. That is, it draws on syntactic knowledge to resolve temporary ambiguities and reanalyse misleading input. Note that this wholly distinguishes the present account from frequency-based alternatives, which, in principle, could also provide an explanation for the observed results. Specifically, they could potentially explain DOM's ameliorating effect through statistical associations between its surface form and ORC interpretations. Importantly, however, these two approaches make testable predictions. If DOM's facilitative effect stems from its grammatical role, only participants who have acquired its syntactic distribution should benefit from its presence. Conversely, if the parser relies primarily on surface-level associations, DOM should aid interpretation regardless of whether its underlying grammar is fully mastered. Experiments 3 and 4 explore this distinction by replicating the present design with child participants, whose grammatical knowledge (and, in particular, that underlying DOM) is still developing.

4.3 Experiments 3–4: comprehension of SRCs and ORCs by children

This section investigates how children comprehend the RCs tested in Experiments 1 and 2. I begin with a brief overview of key acquisitional findings in this domain to contextualise the generalisation formulated in (36) and clarify the rationale behind it, as developed in Presotto and Torregrossa (2024). I then present the two studies. After discussing the results, Section 4.4 provides a summary of the whole chapter.

4.3.1 Amelioration effects in children’s comprehension of ORCs

Much of the literature on SRC–ORC asymmetries (particularly that within the RM framework) has drawn on data from children’s comprehension of these structures. As already mentioned, children consistently struggle more with ORCs than SRCs, yet a growing body of evidence shows that certain grammatical configurations can ease this difficulty. For instance, Friedmann et al. (2009) demonstrated that Hebrew-speaking children comprehend ORCs more accurately when the object, but not the subject, is lexically restricted. A comparable effect has been observed in Italian, where mismatches in number between the subject and the relative head improve ORC comprehension (Adani et al. 2010). Similarly, German-speaking children show improved performance when ORCs are disambiguated through morphological case marking (Arosio et al. 2012).

To reiterate, these improvements are known as *amelioration effects*, and the grammatical property responsible for them is referred to as the *ameliorating cue*. What is particularly interesting is that these cues are not cross-linguistically uniform: speakers of different languages appear to rely on different morphosyntactic features when processing otherwise difficult structures like ORCs.⁴⁰

In a particularly influential study, Belletti et al. (2012) tested Hebrew- and Italian-speaking children with gender-matched and gender-mismatched ORCs. They found that gender mismatch facilitated comprehension in Hebrew but not in Italian. Hebrew stimuli were structured as in (50a) for the gender-match condition, and as in (50b) for the mismatch condition.

- (50) a. Tare li et ha-yalda she-ha-isha mecayeret
show to-me ACC the-girl.F that-the-woman.F draw.3SG.F

⁴⁰It is of course possible that broader factors, such as semantic plausibility, can ease ORC processing across languages. A sentence like *The mouse that the cat chases* is intuitively more accessible than *The cat that the mouse chases*. However, plausibility effects are not specific to relative clauses but apply broadly to filler–gap dependencies (Traxler and Pickering 1996). For this reason, studies on amelioration typically control for these variables in order to isolate the role of morphosyntactic structure. The studies reported here follow this approach.

‘Show me the girl that the woman draws.’

- b. Tare li et ha-yalda she-ha-rofe mecayer
 show to-me ACC the-girl.F that-the-doctor.M draw.3SG.M
 ‘Show me the girl that the (male) doctor draws.’

(Adapted from Belletti et al. 2012)

Likewise, the subject and the object in the tested Italian ORCs were specified for either the same (51a), or different (51b) gender features.

- (51) a. Mostrami la bambina che la nonna bacia
 show-to-me the.SG.F girl.SG.F that the.SG.F grandma.SG.F kiss.3SG
 ‘Show me the girl that the grandma kisses.’
 b. Mostrami il dottore che la bambina disegna
 show-to-me the.SG.M doctor.SG.M that the.SG.F girl.SG.F draw.3SG
 ‘Show me the (male) doctor that the girl draws.’

(Adapted from Belletti et al. 2012)

A central difference between the two languages is that, differently from Italian, Hebrew’s main verb inflects for gender in undergoing agreement with the subject (Shlonsky 1997). For this reason, Belletti et al. (2012) interpreted their results as supporting the RM approach: since gender qualifies as a movement-triggering, *active* feature in Hebrew but not in Italian, the intervention configurations in Hebrew and Italian ORCs will differ in whether this feature is involved. That is, gender enters into the locality computation in Hebrew, but not in Italian, as exemplified below.⁴¹

(52) Hebrew

- a. Tare li et ha-yalda she-ha-isha mecayeret
 show to-me ACC the-girl.F that-the-woman.F draw.3SG.F
 [+R,+NP,+fem] [+NP,+fem]
 b. Tare li et ha-yalda she-ha-rofe mecayer
 show to-me ACC the-girl.F that-the-doctor.M draw.3SG.M
 [+R,+NP,+fem] [+NP,+masc]

(53) Italian

- a. Mostrami la bambina che la nonna bacia
 show-to-me the.SG.F girl.SG.F that the.SG.F grandma.SG.F kiss.3SG
 [+R,+NP] [+NP]

⁴¹I report the relevant featural bundles as originally illustrated by the authors, where [+R] stands for the relative feature and [+NP] for lexical restriction. Both, according to this approach, qualify as *bona fide* triggers for movement.

- b. Mostrami il dottore che la bambina disegna
 show-to-me the.SG.M doctor.SG.M that the.SG.F girl.SG.F draw.3SG
 [+R,+NP] [+NP]

The fact that gender is represented in (52) determines two distinct set-theoretic relations between the relative head and the intervening subject. In (52a), the feature bundle of the object properly includes that of the subject, whereas in (52b) the two bundles intersect. By contrast, in Italian, gender is never involved in the locality computation, hence the set-theoretic relation remains one of inclusion in both (53a) and (53b). This distinction is crucial, as it is precisely the type of feature relation that modulates the disruptiveness of ORCs, with inclusion yielding significantly poorer comprehension than intersection.⁴² This, then, explains why only Hebrew-speaking children benefit from gender-mismatched ORCs. This line of analysis has been extended to other languages as well. For instance, in French, number mismatches between the subject and the object improve children's comprehension of ORCs, relative to number-matched counterparts (Bentea and Durrleman 2017; Durrleman and Bentea 2021). Such results follow naturally from RM, since number in French is a movement-triggering feature and thus enters the computation of intervention effects.

The RM framework has undoubtedly shed light on crucial aspects of children's comprehension of ORCs and their amelioration, while also providing a rigorous set of tools for cross-linguistic comparison. Yet despite its theoretical precision, RM is not always supported unambiguously by empirical data. I have already discussed this in relation to the adult findings from Experiments 1 and 2, but similar issues emerge in acquisitional work. For instance, Guasti et al. (2012b) tested ORC comprehension in Greek- and Italian-speaking children and found that the former performed better when overt case marking disambiguated the arguments (as also shown for German by Arosio et al. 2012). Even though the authors invoke RM to interpret their results, it is not clear how case could act as an ameliorating feature within that framework. As noted in Section 4.2.3.4, case is predicted not to count as a syntactically active feature, since it does not trigger movement (Friedmann et al. 2017).

Another central assumption of RM-based accounts is that lexical restriction, encoded as a [+NP] feature, modulates intervention effects *qua* licenser of syntactic positions (Rizzi 2018: 349–355). The prediction for ORCs is that anytime the object and the (intervening) subject share a [+NP] feature specified with the same phi-features (see Belletti et al. 2012 et seq.), intervention effects arise resulting in poorer comprehension of the

⁴²As Belletti et al. (2012: 1064) put it: “any configuration in which the intervener would differ from the target in at least one relevant feature specification would create a structure which could be computed by the child system. Thus, cases of inclusion are problematic, but cases of [...] intersection (at least one feature in the intervener differs from the target) are fine for children”.

sentence by children. Conversely, when either the subject or the object is not lexically restricted (as is the case with pronouns), an amelioration effect is expected.

This prediction has been confirmed in many studies. However, Sevcenco and Avram (2012) report findings from Romanian that point in the opposite direction. In one of their tasks, children were tested on sentences like those in (54). In (54a), the subject DP (*piticul*) occurs in post-verbal position, whereas in (54b) and (54c), the subject is expressed by a null pronoun. As the authors note, sentences like (54c) are ambiguous between SRC and ORC readings,⁴³ while those like the one in (54b) are unambiguous due to a plural verb combined with a singular relative head.

(54) Adapted from Sevcenco and Avram 2012

- a. *prințesa* *care o* *fotografiază* *piticul*
 princess-the.F.SG who her.ACC.3.F.SG photograph.3SG dwarf-the.M.SG
 ‘the princess whom the dwarf is photographing’
- b. *fata* *care o* *stropese*
 girl-the.F.SG who her.ACC.3SG.F splash.3PL
 ‘the girl whom they are splashing’
- c. *broaștele* *care le* *salută*
 frog-the.F.PL who them.ACC.3.F.PL greet.3SG/PL
 1: ‘the frogs whom they/he is/are greeting’
 2: ‘the frogs that are greeting them’

Participants (monolingual children aged 4;0–7;5) correctly interpreted sentences like (54a) in 80.29% of cases, but were only 67.9% accurate with items like (54b). Moreover, they assigned an ORC interpretation to ambiguous sentences like (54c) only 51.2% of the time. Setting aside the results for (54c), whose interpretation is confounded by their inherent ambiguity, children performed more accurately with (54a) than with (54b). That is, the presence of a lexical restriction (i.e. a [+NP] feature) on the subject appeared to *facilitate* comprehension, despite the relative head being lexically restricted as well.⁴⁴ Conversely, the [+NP] mismatch between the object and the (null) subject in (54b) resulted in significantly lower accuracy.

⁴³The verbs used in these items came from a class in which the 3rd person present-tense form of the indicative is syncretic for number.

⁴⁴In this example, the relative head bears masculine gender while the subject bears feminine gender. One could argue that this mismatch alone weakens or nullifies intervention effects, as observed in Hebrew (Belletti et al. 2012). If so, two issues arise. First, it is unclear why gender should qualify as a syntactically active feature in Romanian but not in Italian, given that neither language displays gender agreement on the main verb. Second, if phi-features in Romanian are capable of weakening intervention effects regardless of their realisation on verbal morphology, then number mismatches, such as that in (54c), should yield similar (if not better) accuracy rates, especially given the absence of lexical restriction on the intervener. Yet a substantial difference emerges between the two conditions.

On this basis, we argued in Presotto and Torregrossa (2024) that children’s comprehension is enhanced when morphosyntactic cues enable disambiguation between subjects and objects, allowing more effective parsing of argument–verb relations.⁴⁵ This is the idea I already presented in the generalisation introduced in (36), repeated here in (55).

(55) **Descriptive generalisation of ‘morphosyntactic cue’**

A morphosyntactic cue is a grammatical property whose presence or absence implies that a given constituent is either the subject or the object of a given clause.

All the amelioration effects reviewed so far conform to this generalisation. For instance, the fact that a gender mismatch between the subject and the object improves Hebrew-speaking children’s comprehension of ORCs follows directly from it: in such cases, gender morphology on the verb unambiguously identifies the subject DP. Of course, verbal inflection is not the only expected ameliorating cue. We deliberately adopted the general term “grammatical property” to ensure that other relevant factors are included. Among these are case marking, word order, and, in fact, any property that signals whether a nominal is the subject or the object of the clause. On this view, the ameliorations triggered by (unambiguous) case morphology in, e.g., German (Arosio et al. 2012) or Greek (Guasti et al. 2012b) naturally fall out. In fact, word order is also predicted to play a role to the extent that it disambiguates argument roles, and the results of Experiment 2 confirm this expectation.

Another welcome outcome of this generalisation is that it circumvents the issues posed by certain ameliorating cues for RM. Whereas case is not expected to ease intervention effects in that framework, it is inherently predicted to do so under the present approach. Moreover, the Romanian data in (54) also align naturally with (55). As discussed by Sevcenco and Avram (2012), the mere presence of an overt NP subject is sufficient to disambiguate between an ORC- and an SRC-reading. While sentences like (56a) are only unambiguous if the direct object clitic (in this case *îl*) and the relative head differ in phi-features, no such proviso is required in (56b), “since an overtly realised pre-verbal or post-verbal subject [...] in the embedded clause bans the [SRC-] reading” (Sevcenco and Avram 2012: 221).⁴⁶

(56) Adapted from Sevcenco and Avram 2012

- a. fata care îl vede
 girl-the.F.SG that him.ACC.3SG.M see.3SG
 ‘the girl who sees him’

⁴⁵A similar argument is made by Polinsky et al. (2012) in a study on Avar, an ergative–absolute language. There, the authors propose that ergative case-marking facilitates the comprehension of relative clauses with absolute object gaps, as the ergative marker helps the parser anticipate the presence of a gap (see also Bornkessel and Schlesewsky 2006).

⁴⁶For more detailed information on this contrast, I refer the reader to the original source.

- b. băiatul care (tata) îl vede (tata)
 boy-the.M.SG that (father.SG.M) him.ACC.3SG.M see.3SG (father.SG.M)
 ‘the boy whom the father sees’

To summarise, I propose that the available evidence on children’s comprehension of ORCs supports the following conclusion: ORCs can only be facilitated when a disambiguation between subject and object is possible. This insight is captured in the descriptive generalisation in (55), which characterises the properties of morphosyntactic cues. These cues, I argue, represent a necessary (though not sufficient) condition for amelioration effects to emerge (see Presotto and Torregrossa 2024 for further discussion). As illustrated in the previous sections, subject position and DOM constitute two cues that conform to this generalisation. The study presented in the next section investigates their impact on children’s comprehension.

4.3.2 The study

Experiment 1 revealed a robust SRC–ORC asymmetry in Spanish adults’ comprehension and processing. A natural first question is whether, and to what extent, this asymmetry is reflected in children:

- (57) **[RQ3]** Is there a subject-object asymmetry in Spanish children’s comprehension of RCs?

In addition, replicating the design of Experiment 2 with children allows us to investigate whether and how WO cues (i.e. subject position) and morphological cues (i.e. DOM) affect their comprehension:

- (58) **[RQ4]** Do DOM and WO cues ameliorate the comprehension of ORCs by Spanish-speaking children? Do these cues interact? Is there a difference between in-absentia and in-praesentia DOM cues?

Difficulties in adult online processing have been argued to mirror children’s challenges in offline comprehension (Belletti and Rizzi 2013; Phillips and Ehrenhofer 2015, and references therein). It is therefore reasonable to predict that children, like adults, will perform more poorly on ORCs overall. And particularly on those ORCs featuring a single in-absentia cue, which proved to yield the lowest accuracy rates with adults.

However, as discussed in Section 4.2.3.4, I proposed that both intervention effects and their amelioration are determined at the Interface through the interaction of computational limits and grammatical constraints. Crucially, this view assumes that the parser accesses the grammatical representations underlying morphosyntactic cues. This makes the child data especially informative: the extent to which the grammar underlying a cue

has been acquired may directly affect its role in comprehension. In particular, we can anticipate potential discontinuities between child offline responses and adult online processing. For instance, whereas the adult parser likely relies upon a number of substantially automatised procedures (Baddeley and Hitch 1974; Ferreira et al. 2002; Givón 2002; Jeffries et al. 2004; Sturt et al. 2004; Ferreira and Patson 2007; Ferreira and Lowder 2016, a.o.), children must sometimes devote substantial resources to newly acquired grammatical structures.

Following Biberauer's *Maximise Minimal Means* model (Biberauer 2019a,b), children are predicted to maximise the cognitive resources at their disposal by “consistently focus[ing] on input that is ‘just right’, [...] minimising attention to any [input] which has become ‘too simple’ (as this has already been accounted for and is therefore ‘uninteresting’ and not worthy of cognitive resources)” (Biberauer 2019a: 216). In other words, children actively allocate greater processing effort to aspects of the target grammar that are currently being acquired or have only recently been mastered, while devoting less attention to those that are already consolidated. This perspective suggests that the parser is particularly committed when processing linguistic material that is still ‘under construction’ in the grammar. Cues that engage these developing aspects of grammatical knowledge will likely draw more cognitive resources, while cues tied to long-established representations will be processed more automatically. We may therefore expect the computational cost associated with a cue to vary depending on the acquisition status of the relevant grammatical structure. As a result, the processing load imposed by a cue is not fixed but modulated by its developmental status within the learner's grammar.⁴⁷

If this hypothesis is on the right track, one would expect WO cues to be processed more automatically and in a more adult-like fashion, as word order is acquired relatively early in Spanish (Grinstead 1998, 2000, 2004; Bel 2001, 2003).⁴⁸

DOM, by contrast, has been shown to be acquired only later (Rodríguez-Mondoñedo 2008).⁴⁹ If children have not yet acquired the grammatical representation associated with

⁴⁷This view aligns with the widely accepted distinction between *input* (all linguistic stimuli) and *intake* (what is selectively attended to during acquisition) (Evers and van Kampen 2008; Gagliardi 2012; Lidz and Gagliardi 2015). If, as argued by Lidz and Gagliardi (2015: 349), children “filter their input to identify critical information to learn from”, it follows that they will concentrate resources on input deemed relevant for acquisition.

⁴⁸More generally, knowledge of language-specific constraints on constituent ordering is known to be acquired early in development, often during the second year of life (Bloom 1970; Brown 1973; Abbot-Smith et al. 2001; Gavarró et al. 2015, among others). For a cross-linguistic overview, see Guasti (2017: ch.4), and for Spanish in particular, Montrul (2004: 190–205; 262–279).

⁴⁹Rodríguez-Mondoñedo found that the differential marker *a* is used in a target-like way by Spanish children around age 3. While this is relatively early, it is still later than the acquisition of word order constraints. Yet full mastery of DOM remains difficult to determine, given its complexity and variation across contexts. In light of this, Presotto and Torregrossa (2024) included an independent test to assess participants' knowledge of DOM (see Section 4.3.5.1).

DOM, this might influence how DOM cues affect comprehension. In Presotto and Torregrossa (2024) we therefore hypothesised that DOM cues would be more demanding than WO cues for children who had acquired DOM, but would nonetheless lead to amelioration effects thanks to their disambiguating function. For those who had not yet acquired DOM, two possibilities were considered: (i) the cue would be disregarded, resulting in no amelioration effect; or (ii) amelioration would still occur due to probabilistic expectations associating the presence of the sound *a/al* with an ORC structure, consistent with experience-based models of RC processing (see the discussion above).

Experiments 3 and 4 therefore serve as a testing ground for the Interface Hypothesis defended here. If DOM cues improve comprehension even when DOM has not been acquired, then cue effectiveness cannot depend on grammatical representations, favouring a reductionist, experience-based explanation. If, conversely, amelioration only emerges when DOM has been acquired, then grammar must be assigned a central role in shaping locality effects, ruling out a purely extra-grammatical account.

This leads to the final research question:

- (59) **[RQ4.1]** Does the acquisition of the grammar underlying a cue affect its effectiveness?

In the following section, I describe how participants' knowledge of DOM was assessed and detail the overall methodology followed in the experiments.

4.3.3 Materials and methods

This study comprised two main parts. The first tested the comprehension of SRCs and ORCs by Spanish-speaking children, replicating Experiment 1. The second examined whether, and to what extent, different types of disambiguating cues (and their combinations) impact the comprehension of ORCs specifically, replicating Experiment 2. Additionally, children's mastery of DOM grammar was assessed to evaluate its effect on their ability to comprehend ORCs disambiguated by DOM cues alone. In total, the study consisted of three experiments: two sentence-picture-matching tasks (Experiments 3 and 4), and one sentence repetition task (SRT).

4.3.4 Participants

Participants were recruited in a school in the vicinity of Granada (Southern Spain), where a total of 57 children were tested. Of these, 47 monolingual native-speakers of Spanish were eventually considered in the analysis (age range: 4;5–7;1, $M = 5;9$, $SD = 0;8$).⁵⁰

⁵⁰Specifically, speakers of Peninsular Spanish born and raised in the region. This was done to avoid possible interactions between conflicting DOM systems.

Participants were arranged into three age groups, as shown in Table 4.5. None of the children included in the final sample had a reported history of speech, hearing, or language disorders.

Table 4.5: Participants' demographics per age group

Age group	No. of participants	Age range	Mean age (<i>sd</i>)
4 y.o.	14	4;5 - 5;3	4;11 (0;3)
5 y.o.	17	5;3 - 6;3	5;9 (0;4)
6 y.o.	16	6;4 - 7;1	6;9 (0;3)

4.3.5 Experiments design

All participants completed all experiments in two separate sessions held approximately one week apart. Each session included one of the two sentence-picture-matching tasks, presented in randomised order. The sentence repetition task was administered either in the first or second session, also in randomised order, and we additionally randomised whether the SRT or the comprehension task appeared first. Each session lasted no more than 30 minutes, with breaks offered between tasks. Children could pause or stop the session at any point.

Experiment 3 tested comprehension of SRCs and ORCs. Experiment 4 focused on ORCs featuring various types of disambiguating cues. The sentence repetition task evaluated participants' overall grammatical competence, with special attention to DOM. Experiments 3 and 4 followed the same methodology as Experiments 1 and 2, except that testing was conducted in person rather than online.

During each session, children wore headphones and followed pre-recorded instructions. The task was framed as a game: they were to help a small teddy bear by pointing to the picture that matched the sentence they heard. An experimenter recorded their responses by pressing the left or right arrow key, and results were automatically uploaded to the server at the end of the session.

4.3.5.1 Sentence Repetition Task

We used a sentence repetition task (SRT) to assess participants' syntactic knowledge. Prior research shows that SRTs are an effective tool for evaluating children's overall grammatical competence (Armon-Lotem et al. 2015; Klem et al. 2015; Spada et al. 2015; Andreou et al. 2021; Torregrossa et al. 2022). At the comprehension level, SRTs tap into participants' ability to decode and interpret stimulus sentences. At the production level,

they involve lexical retrieval, grammatical analysis, sentence encoding, and phonological realisation.

We built our SRT on the basis of Bravo et al. (2020). The task included 35 sentences targeting 43 different structures, some of which appeared more than once, yielding a total of 71 targets. Among these were 9 instances of DOM, embedded in contexts of increasing syntactic complexity and involving both obligatory and optional cases. Specifically, DOM occurred in simple SVO sentences, in wh-object questions, in SRCs where the head functions as the matrix object, and in clitic-left-dislocated structures. The direct objects featured a variety of nominal types: proper names, [+human; +specific; +definite] DPs, [+human; +specific; –definite] DPs, and strong quantifiers (e.g. *Llamó a toda chica con sombrero* ‘(S)he called every girl in a hat’).⁵¹

Participants’ responses were transcribed and coded for the presence or absence of each target structure. Following the methodology in Armon-Lotem et al. 2015 and Torregrossa et al. 2022, we assigned 1 point for each accurately repeated target structure, regardless of whether the rest of the sentence was reproduced verbatim. 0 points were assigned for omissions or substitutions. For example, consider the sentence *A Sofía la vi ayer antes de ir a trabajar* (‘Sofía, I saw her yesterday before going to work’), which targeted four structures: clitic left dislocation (CLLD), the use of DOM with a proper name, temporal adverbials, and non-finite complement clauses introduced by *de*. If a child omitted the verb *ir* but produced, e.g. *A Sofía la vi ayer antes de trabajar*, we would assign 1 point each for the CLLD, DOM, and temporal adverbial targets, and 0 points for the non-finite complement clause.

Although various scoring methods exist for SRTs, we focused exclusively on the accurate reproduction of target structures. This approach has been argued to reflect syntactic knowledge more directly than alternatives based on verbatim repetition, overall grammaticality, or the omission of functional/content words, which are more sensitive to vocabulary size and working memory limitations (Hamann and Abed Ibrahim 2017).

We extracted two scores. The first, ranging from 0 to 71, reflects the number of accurately reproduced target structures (henceforth SRT-score). The second, ranging from 0 to 9, captures only the accurate reproduction of the DOM-related targets (henceforth DOM-score). Our aim was to obtain a reliable measure of participants’ mastery of the grammar underlying DOM at the time of testing. In Section 4.3.7, I discuss why the DOM-score in particular is crucial for interpreting the results of Experiment 4.

⁵¹See the supplementary materials in the Appendix for the complete list of SRT stimuli. The items targeting DOM are: 2, 22, 23, 24, 25, 26, 29, 31, and 35.

4.3.6 Results and analysis

Before presenting the results of Experiment 1 and Experiment 2, I report the descriptives for participants' scores in the SRT and the subpart of the SRT related to the mastery of DOM. Participants' SRT-scores exhibit a range of correct responses between 27 and 70 (M: 58.4, SD = 8.33). As for DOM-scores, the number of correct answers ranges between 0 and 9 (M: 5.81, SD = 1.94). The two scores (SRT-score and DOM-score) are highly correlated with each other ($r(45) = .88, p < .001$). Participants' age correlated with both their SRT-scores and DOM-scores ($r(45) = .55, p < .001$ and $r(45) = .45, p = .002$).

4.3.6.1 Experiment 3

The bar graph in Figure 4.6 shows the proportion of correct answers in association with SRCs and ORCs, respectively. The mean proportion of accurate answers was .92 (SD = .27) for SRCs and .11 (SD = .31) for ORCs.

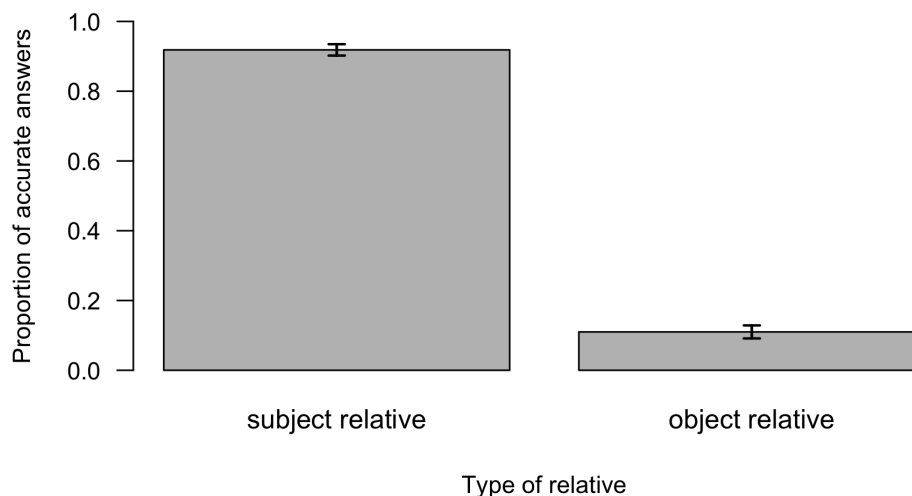


Figure 4.6: Bar graph of the proportion and standard error of accurate answers with SRCs and ORCs (children).

A generalised linear mixed-effects model was run with R and lme4 to investigate the extent to which participants' accuracy was dependent on the type of relative (SRC vs. ORC), their age, and their syntactic abilities as measured through the SRT. In particular, we considered response accuracy as dependent variable and the interaction between type of RC and age, and SRT-score as predictors. For the type of RC, we considered SRCs as the reference level. We mean-centred the values related to age and SRT-score. We fitted

the model with random slopes for type of relative by participant.⁵² The results reported in Table 4.6 revealed that there was a significant lower-order effect of type of RC, indicating that participants were less accurate with ORCs (when age was at its mean value). We also found a significant lower-order effect of age, suggesting that accuracy with SRCs (taken as the reference level in the intercept) increased with age. The significant interaction between type of RC and age indicated that contrary to the pattern observed with SRCs, accuracy with ORCs decreases with age (Figure 4.7). Finally, we observed a main effect of the SRT-score, whereby participants with higher syntactic abilities were more accurate in the comprehension of both SRCs and ORCs.

Table 4.6: Parameters of the generalised linear mixed-effects analysis with response accuracy as dependent variable and type of relative clause in interaction with age, and SRT-score as fixed effects.

Fixed effects	<i>b</i>	SE	<i>z</i>	<i>p</i>
Intercept	2.91	0.42	7.01	<.001
Type of relative clause (object)	−5.40	0.54	−9.96	<.001
Age	0.73	0.32	2.25	.02
SRT score	0.34	0.17	2.01	.04
Type of relative clause (object) * Age	−1.54	0.43	−3.54	<.001

4.3.6.2 Experiment 4

The bar graph in Figure 4.8 shows the proportion of correct answers in association with all types of ORCs, as a function of the position of the subject (post-verbal vs. pre-verbal) and the presence (vs. absence) of overt DOM. The mean proportion of accurate answers with ORCs featuring a post-verbal subject and no overt DOM was .08 (SD = .27), similarly to the results shown in Experiment 3, in which the same type of ORCs was used. The mean proportion of accurate answers with ORCs featuring a post-verbal subject and overt DOM was .22 (SD = .42). With ORCs featuring a pre-verbal subject response accuracy was higher, reaching .67 (SD = .47) with ORCs with no overt DOM, and .48 (SD

⁵²The resulting R model was: `m1 <- glmer (accuracy ~ 1 + type of relative * age + SRT-score + (1 + type of relative|ID), data = spanish_relatives, family = "binomial", control = glmerControl(optimizer = "bobyqa"))`. By running a Likelihood ratio test for model comparison by means of the *anova* function in R, we observed that adding the interaction between age and SRT-score to *m1* did not improve the model fit ($\chi^2(3) = 1.67$, $p = .64$). We also compared a model including the interaction between type of relative clause and SRT-score, and age as main effect (i.e., `m2 <- glmer (accuracy ~ 1 + type of relative * SRT-score + age + (1+type of relative|ID) [...])`) with the above-mentioned model featuring the three-way interaction between type of relative clause, age and SRT. In this case, the latter model improved the fit ($\chi^2(3) = 10.51$, $p = .01$). As a result, *m1* above was selected for the analysis.

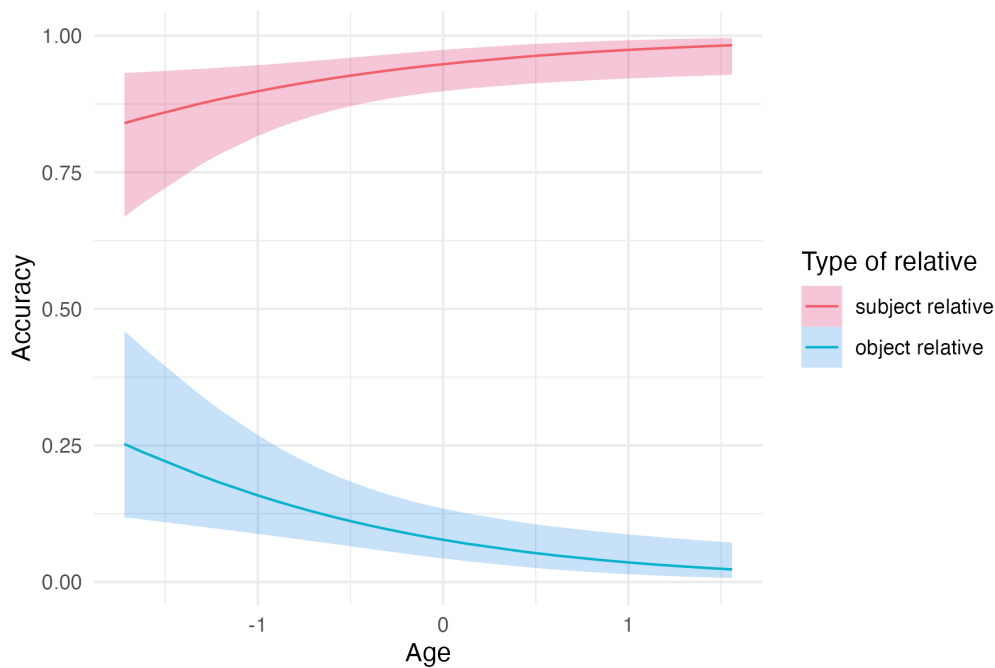


Figure 4.7: Predicted probability related to response accuracy as a function of type of relative clause (SRC vs. ORC) and participants' age (mean-centred values). The shaded lines indicate a 95% confidence interval. The predicted probabilities have been derived by using the `ggpredict()` function in the `ggeffect` package (Lüdtke 2018) and then plotted.

= .50) when DOM was overt. As discussed in Presotto and Torregrossa (2024: 23), we mean-centred the values related to age and DOM-score. We fitted the model with random slopes for subject position by participant.⁵³ We ran a generalised linear mixed-effects model to investigate the extent to which participants' accuracy in the comprehension of ORCs was dependent on the position of the subject (pre-verbal and post-verbal), the presence (vs. absence) of overt DOM, participants' age and performance in the sub-test of the SRT related to the mastery of DOM (i.e. their DOM-score). We did not consider the SRT-score in this case due to the high correlation between the DOM-score and the SRT-score (see above). Moreover, the consideration of the DOM-score is relevant for understanding whether (possible) ameliorating effects of DOM in the comprehension of ORCs are related to the acquisition of DOM. For the type of RC, we considered ORCs with post-verbal subjects and no overt DOM as the reference level.

The results reported in Table 4.7 revealed a significant lower-order effect of the pres-

⁵³The resulting R model was: `m1 <- glmer (accuracy ~ 1 + subject position * DOM * (DOM-score + age) + (1 + subject position|ID), data = spanish_relatives, family = "binomial", control = glmerControl(optimizer = "bobyqa"))`. We did not consider the interaction between DOM-score and age because four-way interactions are notoriously difficult to interpret. Moreover, we had no hypotheses related to this interaction. We did not consider random slopes for presence (vs. absence) of DOM by participant due to convergence issues.

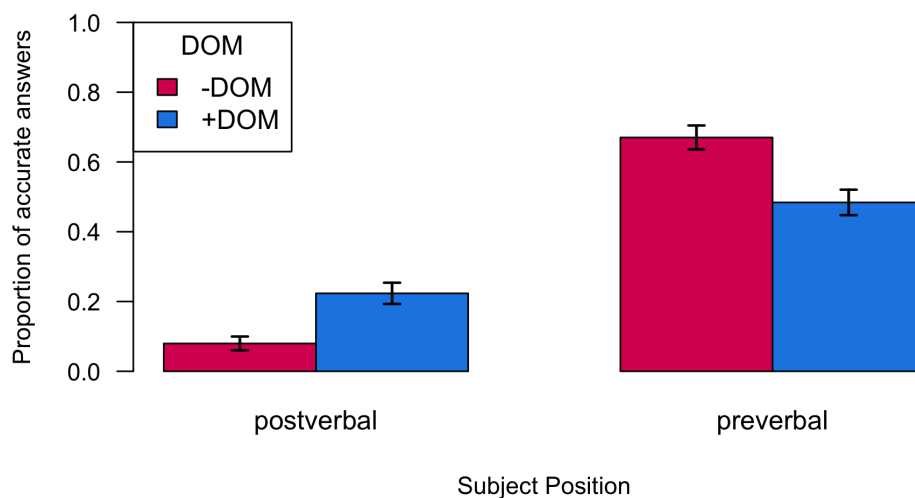


Figure 4.8: Bar graph of the proportion and standard error of accurate answers with ORCs as a function of the position of the subject constituent (pre-verbal vs. post-verbal) and the presence (vs. absence) of overt DOM-marking.

ence of overt DOM: participants were more accurate in the comprehension of ORCs with post-verbal subjects featuring overt DOM than ORCs with post-verbal subjects with no overt DOM, when age and DOM-score were at their mean values. The significant lower-order effect of subject position indicated that among the ORCs with no DOM, participants tended to be more accurate with the ones featuring pre-verbal subjects. However, the significant interaction between overt DOM and subject position indicated that among the ORCs with pre-verbal subjects, response accuracy decreased if the ORC exhibited overt DOM. We also found a significant lower-order effect of age, which replicated the pattern observed in Experiment 3, whereby older children were less accurate with ORCs with post-verbal subjects and no overt DOM. Finally, we observed a significant three-way interaction between DOM, subject position, and DOM-score, which suggested that response accuracy with ORCs with pre-verbal subjects and overt DOM decreased with increasing DOM-scores. This pattern is shown in Figure 4.9. On the one hand, mastery of DOM (as measured by the DOM-score) led to better performance with ORCs with post-verbal subjects. In this case, the corresponding results in the model are not significant but the tendency was as expected (see the positive estimate in association with the interaction between presence of DOM and DOM-score). On the other hand, response accuracy with ORCs with pre-verbal subjects seemed to be affected negatively by participants' mastery of DOM.

Table 4.7: Parameters of the generalised linear mixed-effects analysis with response accuracy in the comprehension of ORCs as the dependent variable and the interaction between subject position, presence of DOM, and age, on the one hand, and DOM-score, on the other hand, as predictors. Significant results are in bold.

Fixed effects	<i>b</i>	SE	<i>z</i>	<i>p</i>
Intercept	−3.00	0.41	−7.33	<.001
DOM (+ DOM)	1.61	0.42	3.90	<.001
Subject position (pre-verbal)	3.76	0.44	8.62	<.001
DOM score	−0.18	0.30	−0.58	.57
Age	−0.91	0.40	−2.277	.02
DOM (+DOM) * Subject position (pre-verbal)	−2.45	0.47	−5.16	<.001
DOM (+DOM) * DOM score	0.51	0.35	1.46	0.15
DOM (+ DOM) * Age	0.70	0.43	1.61	0.11
Subject position (pre-verbal) * DOM score	0.47	0.35	1.35	0.18
Subject position (pre-verbal) * Age	0.80	0.43	1.84	.07
DOM (+ DOM) * Subject position (pre) * DOM score	−0.97	0.43	−2.25	.02
DOM (+ DOM) * Subject position (pre-verbal) * Age	−0.68	0.50	−1.34	.18

4.3.7 Interim discussion

The results of Experiment 3 and their statistical interpretation clearly indicate that SRCs are considerably easier for Spanish children to comprehend than ORCs. This pattern appears robust across all types of ORCs, as even the condition that yielded the highest proportion of accurate responses in Experiment 4 still fell well below the performance observed for SRCs.⁵⁴ As expected, then, a subject–object asymmetry in children’s comprehension of RCs emerges in Spanish, replicating the pattern found in a range of previously studied languages as well as the results with the adults tested in Experiment 1. This finding addresses the first research question, repeated below:

- (60) **[RQ3]** *Is there a subject-object asymmetry in Spanish children’s comprehension of RCs?*

Spanish children comprehend SRCs significantly better than ORCs, expanding the evidence of a cross-linguistic asymmetry between subjects and objects in relativisation.

Additionally, older children and those with higher syntactic abilities performed bet-

⁵⁴Caution is required when comparing results across experiments, even though the same participants were involved. A more reliable assessment of the relative difficulty of SRCs and the four ORC conditions tested in Experiment 4 would require a single experiment combining all these conditions. This design was avoided due to concerns that the task might become too lengthy and taxing for younger participants, particularly 4-year-olds.

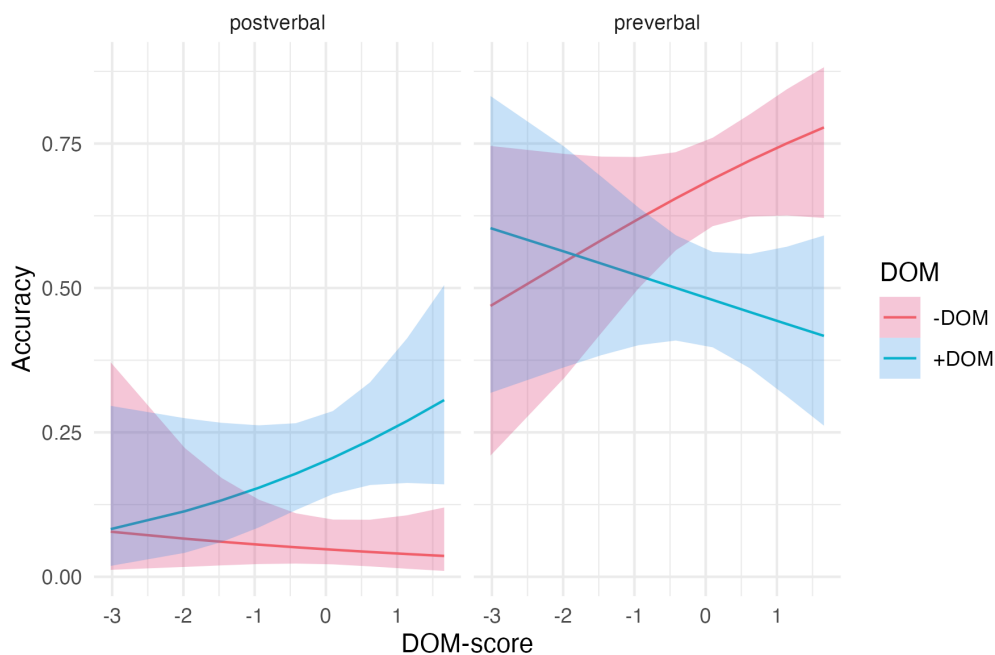


Figure 4.9: Predicted probability related to response accuracy with ORCs as a function of subject position (pre-verbal vs. post-verbal), presence or absence of DOM and participants' mastery of DOM (DOM-score; mean-centred values). The shaded lines indicate a 95% confidence interval. The predicted probabilities were derived with the `ggpredict()` function in the `ggeffect` package (Lüdtke 2018) and then plotted.

ter in the comprehension of SRCs.⁵⁵ More puzzling, however, is the interaction found between age and type of RC, a pattern that also emerged in Experiment 4 (see Figure 4.9 and Table 4.7).

In particular, it is unexpected that older children would perform worse than younger ones in the comprehension of ORCs with post-verbal subjects and *in-absentia* DOM cues. One possible interpretation is that younger children may display a lower degree of commitment to a given interpretation and are more prone to guessing, whereas older children more consistently pursue the (incorrect) SRC reading. In such cases, lower accuracy rates in the older group could reflect adult-like garden-path effects, as discussed in Section 4.2. However, this interpretation conflicts with the observation that the vast majority of children ($n = 45$) produced either 0 or 1 accurate response out of the four items targeting this particular condition (see Table 6 in the Supplementary Materials for a visualisation of response patterns across conditions). An alternative possibility is that the pattern reflects a classic instance of U-shaped developmental trajectory, whereby mastery of a specific category is temporarily disrupted before stabilising again. Evaluating this hypothesis would

⁵⁵Age and syntactic competence naturally tend to correlate. In the present study, this relationship is supported by the observed correlation between SRT-scores and participants' age (see Section 4.3).

require testing an older age group. For now, however, this aspect of the results remains open to further investigation.

Returning to the main finding, the fact that SRCs are more easily understood than ORCs is unsurprising, as it aligns with both empirical and theoretical expectations. Indeed, a theory of intervention locality such as RM explicitly predicts this outcome (see discussion above). As previously noted, RM not only accounts for the presence of intervention effects but also provides a well-articulated set of predictions regarding the conditions under which such effects may be weakened. To reiterate, the degree of disruption is defined as a function of the featural distinctness between the target (here, the relative head) and the intervener (here, the subject), where (a) distinctness is computed based on the set-theoretic relations between the features of the relevant syntactic objects, and (b) only syntactically active (or ‘relevant’) features are considered in this computation. On this view, a mismatch in relevant features between subject and object should facilitate comprehension by triggering an amelioration effect. Yet, it is not entirely clear that the predictions made by RM are borne out by the data. To see why, it is useful to summarise a few key findings from Experiment 4.

First, DOM does contribute to ameliorating the comprehension of ORCs, but only when it is overtly realised: *in-absentia* DOM has no observable effect on performance. Although children’s accuracy remains relatively low even in the presence of overt DOM, a clear improvement is seen relative to ORCs featuring *in-absentia* DOM and a post-verbal subject. Another major determinant of comprehension is word order (WO): when the subject of the RC is preverbal, accuracy improves substantially. Thus, both overt DOM cues and WO cues independently lead to amelioration, with the latter proving far more effective overall. Crucially, however, when both overt DOM and WO cues are present in the same sentence, performance drops significantly.

These findings pose a serious challenge to the predictions of RM. As already argued in Section 4.2.3.4, there are several theoretical issues with treating DOM and WO as RM-relevant features. These concerns apply equally here. Yet Experiment 4 adds a further complication: the observation that two individually ameliorative cues, when combined, result in decreased comprehension. From the perspective of RM, this is deeply puzzling. If both cues contribute to increasing the featural distinctness between target and intervener, their co-occurrence should, at best, lead to additive amelioration or, at worst, yield performance comparable to that of the strongest individual cue (in this case, WO). Instead, the opposite pattern emerges: children struggle more with doubly-cued ORCs than with singly-cued ones: an outcome that, again, cannot be straightforwardly explained by invoking RM.

As a matter of fact, a closer look at the data opens up an alternative and arguably

more straightforward interpretation. Indeed, the results can be naturally accounted for by appealing to general principles of sentence processing and language acquisition. The fact that *in-absentia* DOM does not trigger an amelioration effect, for instance, is largely expected given the nature of this type of cue. Independent evidence suggests that covert elements are particularly challenging for children (Biberauer and Roberts 2012, 2015; Crisma et al. 2020 and references therein). It is therefore likely that *in-absentia* cues lack sufficient robustness, especially in comparison to overt ones. As a result, children may be unable to build on the information conveyed by these cues and thus fail to disambiguate the structure. This would explain why ORCs like (46a) are consistently assigned an SRC-reading, which is, all else being equal, the default or preferred interpretation (see discussion above). As argued in Section 4.2, moreover, these ORCs likely instantiate garden-path structures, where disambiguation only becomes possible upon encountering the final DP. Given that this triggers a noticeable slow-down in adult processing (as shown in Experiments 1 and 2), it is unsurprising that children struggle with this condition. A different outcome would, in fact, have been unexpected.

Another aspect of the results that can be plausibly attributed to domain-general processing factors is the observed decrease in comprehension with ORCs disambiguated by two cues simultaneously. In sentences like (46d), the SRC-reading is ruled out by both subject position and overt DOM, yet children's accuracy drops compared to ORCs disambiguated by WO alone. In Presotto and Torregrossa (2024), we argued that this pattern reflects working memory (WM) limitations in the child parser. Under this view, each grammatical atom contributes a certain computational cost, adding to the overall burden on the parser. Disambiguating cues like DOM and WO are no exception and may lead to a WM-overload when realised together. This effect may be especially pronounced in the context of structurally demanding configurations such as ORCs, which already place a heavy load on processing. If this account is correct, it should not surprise that cues are only beneficial when presented individually. The rationale is as follows: although the integration of ORCs is computationally costly, morphosyntactic cues can help reduce this load by aiding structural disambiguation. However, these cues also introduce additional material for the parser to analyse. Hence, their beneficial effect depends on whether their processing cost is outweighed by the facilitation they offer. If too many cues are present, the parser may be overwhelmed, resulting in decreased comprehension.

Within this perspective, the fact that WO and DOM (when realised individually) yield markedly different accuracy rates (see Figure 4.8) is actually a welcome outcome, as it supports the hypothesis put forth in Section 4.3. As discussed there, grammatical properties that are consolidated earlier are expected to be processed more fluently and with greater automaticity. Conversely, recently acquired aspects of the target grammar tend to

require more cognitive resources, since the parser remains more sensitive to novel or ‘less entrenched’ material. Because WO properties are acquired early (see above), it is entirely expected that WO-based cues yield the most effective amelioration: they aid disambiguation without overburdening the parser. DOM, by contrast, is acquired later and likely imposes a heavier processing cost. In other words, the facilitative power of a cue reflects its timing of acquisition (Tsimpli 2014; Schulz and Grimm 2018; Listanti and Torregrossa 2023). This interpretation is further supported by the adult data: in that group, RTs for ORCs featuring both DOM and WO were not higher, but in fact even slightly lower than those involving either cue alone. This follows naturally from the fact that adults have long mastered DOM-grammar, such that its integration is as automated as that of canonical word order properties, and the co-occurrence of the two does not overload the parser.

At this point, the second research question can be addressed:

(61) **[RQ4]** *Do DOM and WO cues ameliorate the comprehension of ORCs by Spanish-speaking children? Do these cues interact in any way? Is there a difference between in-absentia and in-praesentia DOM cues?*

- Both DOM and WO cues ameliorate Spanish children’s comprehension of ORCs, with the latter triggering appreciably higher accuracy.
- An interaction between the two is observed because DOM enhances children’s comprehension when it occurs in isolation, but hinders it when it co-occurs with WO cues.
- The distinction between *in-absentia* and *in-praesentia* DOM cues does make a difference, as the latter produce significantly better performance.

The third research question concerned the impact of acquiring a cue’s underlying grammar on its ameliorative effectiveness. In other words, the goal was to assess whether the capacity of a morphosyntactic cue to facilitate comprehension is contingent on the acquisition of the grammar that licenses it. The answer to this question emerges from the results of the SRT presented in Section 4.3.6, where a three-way interaction was observed between overt DOM, WO, and DOM-score. As shown in Figure 4.9, the effect of (*in-praesentia*) DOM cues is modulated by the level of DOM-acquisition: as children’s mastery of DOM increases, their accuracy improves in the presence of isolated DOM cues but decreases when such cues co-occur with WO cues. In plain terms, the impact of DOM cues, positive or negative, only emerges among children who have acquired the grammatical knowledge that the particle *a/al*, preceding a [+human, +definite, +specific] DP, marks the latter as a direct object. In the absence of such knowledge, the child parser

appears to ignore the DOM morpheme altogether. This leads to the following answer to the final research question:

- (62) [RQ4.1] *Does the acquisition of the grammar underlying a cue affect its effectiveness?*

The level of mastery of DOM grammar determines whether and to what extent *in-praesentia* DOM cues impact on the comprehension of Spanish ORCs. This suggests that morphosyntactic cues are only effective when acquired.

If this is correct, then the conclusion that grammar plays a decisive role becomes unavoidable. Without the relevant grammatical knowledge, morphosyntactic cues appear to be functionally inert. This is by no means a trivial point. On the contrary, it is essential to the overall interpretation of the results. To see why, I believe it is useful to briefly summarise the preceding discussion. As shown above, an account based solely on RM (which fully modulates intervention locality within the grammar) faces some important empirical challenges. Without substantial revisions, the theory's current set-theoretic computations fail to predict key patterns observed in the data. On the other hand, an account grounded in extra-grammatical constraints (such as processing cost, WM-load, and the parser's receptiveness) seems, in principle, more compatible with the observed results. Yet the fact that cue acquisition is decisive changes the picture: abstract representations play a crucial role, against a purely extra-grammatical reductionist perspective.

The conclusion reached at this point closely mirrors that reached in the discussion of Experiments 1 and 2: a strictly grammatical explanation like RM appears insufficient, but the fact that abstract grammatical representations remain central precludes any purely reductionist interpretation, too. Rather than a contradiction, I consider the coexistence of these opposing pressures to offer genuine support in favour of the Interface Hypothesis. With the results of Experiments 1 and 2 alone, however, I could only ground the claim that access to grammatical representations is key on theory-internal arguments: re-analysis and/or avoidance of garden-path-induced readings presupposes knowledge of the abstract rules governing, e.g., DOM. Moreover, I could not rule out the possibility that participants relied on surface-level statistical heuristics linking DOM morphology with an ORC interpretation. The results of Experiment 4 now allow me to support the claim that access to grammatical knowledge is indeed fundamental on empirical grounds—and, crucially, to fully rule out a frequency-based reductionist account.

Indeed, the issue of whether the acquisition of a cue's grammar affects its effectiveness becomes a diagnostic of the role that grammar plays more generally. Put differently, we could say that to understand the effect of cue acquisition is to understand the scope of grammar in shaping intervention effects. Had DOM improved comprehension irrespective of the child's underlying knowledge, the conclusion would have favoured a view in

which grammar plays little to no role in modulating locality. In the case at hand, however, DOM facilitates comprehension only when its grammar has been acquired, implying that the cue's contribution is not surface-based or heuristic, but rooted in structural representations.

Once again, then, the most reasonable way to interpret the tension observed in intervention effects and their amelioration in Spanish RCs is to locate their source at the Interface, where pressures from both grammatical constraints and domain-general limitations naturally converge. This perspective also accounts for the behaviour of WO and DOM in the child parser more specifically: limitations of the processing system (e.g., WM-load) explain the performance costs associated with cue co-occurrence, while the centrality of abstract structural representations clarifies why these mechanisms operate exclusively on acquired linguistic material.

4.4 Summary

This chapter examined how adult and child native speakers of Spanish process and comprehend SRCs and ORCs. After proposing a syntactic analysis of these structures (Section 4.1), I presented four experimental studies.

Experiments 1 and 2 revealed that adult native speakers find –DOM, POST-VERBAL ORCs particularly challenging. However, their performance improves significantly when more robust and linearly earlier disambiguating cues (i.e., WO and in-praesentia DOM) are provided. In discussing these results I argued that they cannot be accounted for by RM, whose set-theoretic mechanisms fail to predict the observed patterns (unless, perhaps, rather drastic theoretical amendments are pursued). Likewise, reductionist accounts based on the “referential weight” of the subject intervener (Warren and Gibson 2002, 2005) or its similarity to the object (Gordon et al. 2001, 2004) fall short of explaining the data. Instead, I proposed that the difficulty associated with these ORCs stems from their status as classic garden-path sentences.

The presence of clear morphosyntactic cues (e.g., overt DOM or pre-verbal subjects) facilitates comprehension by disambiguating the structure and easing reanalysis. Crucially, such facilitation depends on access to grammatical knowledge—more specifically, to the abstract representations and structural rules governing the language, such as those licensing DOM. I also showed that this account naturally aligns with prominent views on sentence processing and garden-path resolutions, which hold that processing dynamics reflect grammatical constraints and interact strictly with them (Pritchett 1991; Siloni 2014). Clearly, this perspective fits naturally within the Interface Hypothesis as well.

That said, the results of Experiment 2 alone did not rule out an alternative frequency-

based account relying on surface-form heuristics. This possibility, however, is decisively excluded by the results from child participants in Experiments 3 and 4. In particular, Experiment 4 demonstrated that the effect of DOM on comprehension depends on the child's mastery of the underlying grammar: only when DOM's function within the grammar has been acquired does it exert an ameliorating effect. This suggests that the cue's contribution is not superficial or heuristic, but derives from abstract grammatical knowledge.

Overall, then, these findings challenge the adequacy of RM, referentiality-based, similarity-based, *and* frequency-based approaches. The hypothesis defended here better accounts for the data: locality effects and their amelioration emerge at the interface between grammatical constraints and domain-general processing pressures.⁵⁶ On this view, the inherent cost of ORCs is magnified when the structure lacks reliable cues to ease their interpretation. For instance, Spanish –DOM, POST-VERBAL ORCs are not only structurally more complex than SRCs but also affected by a substantial degree of (transient) ambiguity (see discussion above). On top of clearly indicating the essential role of abstract grammatical knowledge, amelioration effects highlight that resolving this ambiguity is central to comprehension. In Chapter 6, I will illustrate that ambiguity is, in fact, something of a common denominator of most (if not all) locality effects.

Let me conclude with one last observation. The results from children in Experiment 4 offer a particularly clear illustration of the interface dynamics: DOM proves facilitative only once its underlying grammar has been acquired, but grammatical knowledge alone is not sufficient. It must operate within the limits of the parser's computational resources. When two morphosyntactic cues co-occur, their combined processing demands may exceed the parser's capacity, ultimately impairing rather than aiding comprehension. This delicate balance epitomises the dynamic equilibrium predicted by the Interface account: grammatical representations support disambiguation, but are simultaneously constrained by processing limitations. The result is an interaction between grammar and domain-general pressures, or in other terms, between first- and third-factor constraints.

⁵⁶In Chapter 6, I will further clarify the details of this idea and how it deviates from RM. In particular, I will reason more explicitly on the reasons underlying the cost associated with intervention-affected configurations.

5

Asymmetries in English Strong and Weak Islands

This chapter investigates extraction out of strong and weak islands. Traditionally, these two domains have been associated with distinct types of locality constraints: strong islands (SIs) with *impenetrability*, and weak islands (WIs) with *intervention* locality (see Section 2.1.1). A long-standing question in the literature is whether these constraints reflect fully independent mechanisms, or whether, despite their surface differences, they are ultimately two manifestations of a unified underlying principle. Here, I focus on two prototypical cases: complex noun phrases, as canonical SIs, and *whether*-clauses, as canonical WIs. Specifically, I present two experimental studies investigating how native speakers of English comprehend and process adjunct extraction from these island types. As I will show, the results reveal an original asymmetry between the two constraints, offering new insights into the nature of islandhood and locality theory more broadly.

The chapter is structured as follows. I begin with a brief overview of the key properties of SIs and WIs. I then present Experiments 5 and 6, which test English native speakers' comprehension and processing of extractions from SIs and WIs using two variants of the same experimental paradigm. An interim discussion follows, and the chapter concludes with a summary of the findings and their theoretical implications.

5.1 Strong and weak islands

Chapter 2 already offers a fairly detailed discussion of island constraints and the theoretical frameworks that account for them. For this reason, I will not elaborate further on the formal underpinnings of islandhood here. To briefly recapitulate and properly introduce the empirical study presented in Section 5.2, however, I will summarise a few key aspects of strong and weak islands, along with a short overview of their typology. In doing so, I draw primarily on the classification adopted in Szabolcsi (2006), Szabolcsi and den Dikken (2014), and Szabolcsi and Lohndal (2017), which provide thorough surveys of island types and competing accounts.

In formal linguistics, the term ‘Island’ is used as a “cover term for nodes which obstruct syntactic movement” (Szabolcsi and den Dikken 2014: 213). However, the degree of obstruction may vary depending on a range of factors, and constraints on extraction from islands are often restricted to a subclass of syntactic elements. For this reason, the theory has, from its very inception, recognised the need to distinguish at least two types of islands: strong and weak. Informally:

(1) a. **Strong Island:**

No extraction is allowed (constructions with an appropriate resumptive pronoun may be allowed).

b. **Weak Island:**

Some phrases can extract, others cannot.

(From Szabolcsi and Lohndal 2017: 2)

In a way, then, SIs can be considered more rigid or inflexible, which is why they are also referred to as *absolute* or, in Postal’s (1999) terminology, “locked.” WIs, by contrast, are taken to be *selective* or “unlocked,” as they restrict extraction only for a specific class of elements.

Distinguishing between the two is far from straightforward, however. While it is true that certain configurations tend to block extraction more consistently, there is by now broad consensus that no island is truly absolute (see discussion in Chapter 2, and a.o. Boeckx 2012). A well-known diagnostic proposed by Cinque (1990) relies on the presence of PP-gaps to differentiate strong from weak islands:

(2) **Cinque’s (1990) diagnostic of Strong and Weak Islands:**

Among the domains that do not allow all standard extractions, those that allow a PP-gap are weak islands, and those that can at best contain a DP-gap are strong islands.

(from Szabolcsi and den Dikken 2014: 214)

This diagnostic follows work by Perlmutter (1972) and Obenauer (1984), and builds, in particular, on the observation that gaps inside SIs (as well as parasitic gaps) are generally restricted to the category DP, whereas PP-gaps are not permitted (3). By contrast, WIs seemingly allow both gap types (4).¹

- (3) a. * About which topic did John ask <who was talking __>?
 b. *? Which topic did John ask <who was talking about __>?
 (Szabolcsi and Lohndal 2017: 4)
- (4) a. % About which topic did John ask <whether to talk __>?
 b. Which topic did John ask <whether to talk about __>?
 (Szabolcsi and Lohndal 2017: 4)

According to Cinque, this distinction arises because gaps in SIs are not traces of movement but rather A-bar bound *pros*—i.e., null resumptive pronouns. The unavailability of PP-gaps would then follow from the general absence of resumptive pronouns of category PP in natural languages. Although this account has been challenged on both theoretical and empirical grounds (see, e.g., Postal 1999 and the discussion in Szabolcsi and Lohndal 2017) it has remained, since its original formulation, a useful tool for classifying strong vs. weak islands. Among the classical SIs are, for example: Complex DPs (5, 6); Subjects (7, 8); Adjuncts (9); Coordinate structures (10); and Left branches (11).

(5) **Complex DP (with relative clause)**

- a. * This kid, you must call < the teacher who punished __ >
 (Szabolcsi and Lohndal 2017: 6)
- b. * On the table, you must call < the teacher who put the book __ >
 (Szabolcsi and Lohndal 2017: 6)
- c. * What size shoes did you call < the man who wears __ >?
 (Szabolcsi and Lohndal 2017: 6)
- d. * How did you call < the man who behaved __ >?
 (Szabolcsi and Lohndal 2017: 6)
- e. * What did you meet the woman [who made __]?
 (Kush et al. 2018: 744)

(6) **Complex DP (with complement clause)**

- a. * This man, I heard < the rumour that my dog bit __ >
 (Szabolcsi and Lohndal 2017: 6)

¹ Szabolcsi and Lohndal (2017) use the percentage sign ‘%’ to indicate inter-speaker variation.

- b. * On the table, I heard the rumour < that you put the book ___ >
(Szabolcsi and Lohndal 2017: 6)
- c. * What size shoes did you hear < the rumour that I wear ___ >?
(Szabolcsi and Lohndal 2017: 6)
- d. * How did you hear < the rumour that I behaved ___ >?
(Szabolcsi and Lohndal 2017: 6)
- e. * What did you make [the claim that John bought ___]?
(Sprouse and Hornstein 2013: 2)

(7) Subjects

- a. * Which man did < his visiting ___ > shock you?
(Szabolcsi and Lohndal 2017: 6–7)
- b. * Which book do you believe < the first chapter of ___ > to be full of lies?
(Szabolcsi and Lohndal 2017: 6–7)
- c. * What did you think that [the recipe for ___] was sitting on the counter?
(Kush et al. 2018: 744)
- d. * What did [the story about ___] impress Mary?
(Sprouse and Villata 2021: 228)

(8) Sentential subjects

- a. * What did [that John wrote ___] offend the editor?
(Sprouse et al. 2016: 310)
- b. * Who did [that Mary kissed ___] bother you?
(Boeckx 2012: 5)
- c. * How would [to behave ___] be inappropriate?
(Cinque 1990: 1)
- d. * Who was [that Ellie would invite ___] absurd?

(9) Adjuncts²

- a. * Which topic did you leave < because Mary talked about ___ >?
(Szabolcsi and Lohndal 2017: 7)
- b. * Who did you leave the party [because Mary kissed ___]?
(Sprouse 2007: 125)
- c. * Which magazine did Sophie play the violin [before she read ___]?
(Brown 2017: 120)
- d. * What do you wonder [how I repaired ___]?
(Manzini 1992: 117)

- e. * Who did Mary cry [after John hit ___]?

(Huang 1982: 503)

(10) **Coordinate structures**

- a. * Which man did you invite < Mary and ___ >?

(Szabolcsi and Lohndal 2017: 7)

- b. * Which man did you invite < Mary and a friend of ___ >?

(Szabolcsi and Lohndal 2017: 7)

- c. * It was Alex you saw Robin and ___ yesterday.

(Chaves 2022: 2)

- d. * What did you eat [a biscuit and ___]?

(11) **Left branches**

- a. * Which man's did you see < ___ picture >?

(Szabolcsi and Lohndal 2017: 8)

- b. * It was Robin's I liked ___ painting the most.

(Chaves 2022: 3)

- c. * Which did John borrow [NP ___ book]?

(Sprouse and Hornstein 2013: 2)

- d. * Whose did you ride [NP ___ bike]?

Providing a list of WIs is more complex, as these typically arise from the interaction of two factors: (a) the type of element undergoing movement, and (b) the type of domain from which it is extracted. Regarding the latter, a well-known observation is that the presence of finite tense in a WI environment tends to worsen the acceptability of extractions.³ For example, in *wh*-islands, finiteness seems to degrade extraction:

- (12) a. What do you wonder [how to repair ___]?

- b. * What do you wonder [how I repaired ___]?

(Manzini 1992: 117)

As for the first factor—the nature of the extractee—the picture is even more intricate, with various properties of the moved element modulating the island effect. Szabolcsi and

²The class of adjunct islands is not as uniform as other classical SIs. Extraction from adjuncts has been shown to be possible under certain conditions (see, e.g., Kush et al. 2018; Müller 2019; Bondevik et al. 2021; McInnerney 2022b; Porta 2024). For a comprehensive review of the factors influencing adjunct extraction, along with original empirical results, see Dal Farra (2020).

³Finiteness effects have been observed across a range of island types, both weak and strong (Ross 1967; Cinque 1990; Manzini 1992; Truswell 2011).

Lohndal (2017) offer an extensive inventory of such cases, illustrating how canonical WI-inducing domains (such as *wh*-islands and negative islands) interact with properties of the extractee, including its referentiality, discourse status, or semantic type. For instance, extraction of a D-linked element from a *wh*-island is considerably more acceptable than extraction of a non-D-linked one (see Chapter 2 for details):

- (13) a. * How many books are you wondering < whether to write __ next year >?
 b. How many books on the list are you wondering < whether to write __ next year >?

(Szabolcsi and Lohndal 2017: 17)

Contrasts of this type have been shown to lack full robustness, however. If D-linking were the true discriminating factor, extraction of *wh-the-hell* elements from a *wh*-island should be categorically blocked, as such elements are “aggressively non-D-linked” (Petsky 1987). Yet Szabolcsi and Lohndal (2017) report examples that run counter to this prediction. For instance, (14) is perceived as fairly natural when uttered by a speaker addressing someone who is rummaging through a dictionary.

- (14) What the hell do you still < not know < how to spell __ > >?

(Szabolcsi and Lohndal 2017: 19)

Building on this, Szabolcsi and Zwarts (1993, 1997) argue that the decisive factor determining the possibility of extraction from a WI is not D-linking, but rather the distinction between individuals and non-individuals. For reasons of space, I will not go into further detail here. Nor will I provide an extensive list of factors that various accounts have identified as relevant to WI-sensitivity, especially since, in Szabolcsi’s own words, the literature on WIs has “mushroomed” since the advent of Relativised Minimality.⁴ Crucially, each theoretical approach has brought with it its own set of data, making it particularly difficult to offer a unified, comprehensive typology of WIs. In this chapter, I will therefore concentrate on what is arguably the clearest case of a WI-violation—namely, adjunct extraction from *wh*-islands. This phenomenon stems from the early observation that, unlike SIs, WIs are sensitive to the argument–adjunct distinction.

5.1.1 The argument–adjunct asymmetry

Ross’s original treatise on islands dealt almost exclusively with SIs, with the sole exception of a few weak *wh*-islands. Before Cinque introduced his diagnostic, the divide

⁴See 2 and the sources cited therein for a bird’s-eye overview of the available accounts. For more detailed discussion, see also Hofmeister and Sag (2010) and the contributions collected in Sprouse and Hornstein (2013).

between strong and weak islands was based on a classic observation by Huang (1982), Rizzi (1982, 1990), Lasnik and Saito (1984, 1992) and Chomsky (1986): whereas SIs block both argument and adjunct extraction (15), WIs block only adjuncts (16, 17, 18, 19).⁵

- (15) a. * Who/Which kid must you call [the teacher who punished ___]?
 b. * Where/In which room must you call [the teacher who put the book ___]?
 (Adapted from Szabolcsi and Lohndal 2017: 5)

- (16) a. Which problem did John ask [how to phrase ___]?
 b. * How did John ask [which problem to phrase ___]?
 (Adapted from Szabolcsi and Lohndal 2017: 15–17)

- (17) a. ? Who/Which girl do you wonder [whether Bill kissed ___]?
 b. * How/In what way do you wonder [whether Bill kissed Sue ___]?
 (Adapted from Boeckx 2012: 17)

- (18) a. ?? Who_i do you wonder [whether we can help t_i]?
 b. * How_i do you wonder [whether we can help Bill t_i]?
 (Rizzi 1990: 81)

- (19) a. What_i did you figure out [how to finance t_i]?
 b. * How_i did you figure out [what to finance t_i]?
 (Suzuki 2004: 4–5)

In surveying the ever-growing body of work on islands, Szabolcsi and Lohndal (2017) note that the sensitivity of WIs to adjunct extraction finds its only formal interpretation in the early account(s) based on the ECP and Subjacency. Consider the following examples, adapted from Szabolcsi and Lohndal (2017: 33).

- (20) Which problem are you wondering how to [t' [phrase t]]?
 (21) How have you [t''' [decided [t'' to [t'[phrase the problem t]]]]]?

The ECP requires that traces (i.e., empty categories left by movement) be properly governed. This condition is automatically satisfied in argument positions, where the trace is lexically governed. Thus, in (20), the trace *t* is properly governed. Since government is strictly local and subject to 0-Subjacency, *t'* is instead not properly governed. However,

⁵Judgements on WI-extracted arguments vary considerably across speakers and contexts, with some sentences reported as fully acceptable and others as markedly degraded. That said, extraction of adjuncts has been consistently reported as more severely degraded.

because proper government is evaluated at S-structure, and intermediate traces, being semantically inert, are deleted before LF, (20) is nonetheless licit. By contrast, non-argumental traces (adjuncts) are subject to proper government at LF. In (21), t' is not properly governed by t'' , as the two are separated by an island boundary. Crucially, t' cannot be deleted at LF, as this would result in t being ungoverned. As a consequence, adjunct extraction from a wh-island necessarily induces an ECP violation.

A partially different explanation is given in Rizzi (1990) and Cinque (1990), where the distinction between arguments and adjuncts is recast in terms of referential vs. non-referential elements. On this view, referential elements possess referential indices, which allow them to establish long-distance dependencies through binding. In contrast, non-referential elements must rely on strictly local government chains, which are disrupted when crossed by an island boundary (Rizzi 1990: 104). Accordingly, Rizzi redefines the relevant property for WI-sensitivity, proposing that only elements bearing an event-participant thematic role are extractable. To this account, Cinque (1990) adds the requirement of being D-linked (see above).

Interestingly, and to the best of my knowledge, no other syntactic, semantic, or pragmatic account directly appeals to the categorial distinction between arguments and adjuncts to explain this asymmetry.⁶ In contrast, Szabolcsi and Zwarts (1997) argue that the relevant factor for WI-sensitivity is not syntactic category but semantic type. On their account, sensitivity arises from a Boolean-algebraic theory of scope: only individual-denoting expressions (type e) can take scope out of an embedded context, while non-individuals (such as degree, amount, or propositional expressions) are confined to their local scope domains.

As already mentioned, many alternative accounts have proliferated, each re(de)fining the factors affecting WI-sensitivity. Still, the distinction between arguments and adjuncts, albeit arguably too coarse, remains a useful heuristic for classifying island effects. It is true that being an argument does not necessarily guarantee extractability (as mentioned, additional conditions such as referentiality, D-linking, or being of type e may also be required). However, judging from the data discussed in the literature, being an adjunct appears, for the most part, sufficient to trigger WI-effects.

There are, of course, exceptions. But these tend to involve specific types of wh-adjuncts that satisfy particular properties. Moreover, these exceptions have only been

⁶A partial exception may be found in a conference paper by Suzuki (2004), which proposes a rather creative account of the asymmetry: only arguments, and only in derivations involving wh-islands, can be base-generated in Spec,CP. However, this approach would pose a problem for most theoretical assumptions about structure-building operations, beginning with θ - and case assignment, which the author only tentatively attempts to address. It should also be noted that the distinction between arguments and adjuncts is itself theoretically problematic. See, e.g., McInnerney (2022a,b) for a critique of treating ‘argument’ and ‘adjunct’ as primitive grammatical categories.

addressed through vague or underspecified interpretive criteria. For instance, if extraction of *how* and *why* is (more) consistently judged as severely degraded, extraction of *when* and *where* is sometimes considered to yield a milder violation. Note, for example, that Szabolcsi and Lohndal (2017: 16) mark the following sentences with one and two “?”, respectively:

- (22) a. ? Where did John ask < whether to put/read this book __ >?
 b. ?? When did John ask < whether to fire him __ >?

Rizzi (1990) hypothesises that, because “events take place in space and time,” event structure may license temporal and locative indices, potentially explaining why such phrases are less sensitive to WIs than manner (*how*) and reason (*why*) adjuncts. The latter may well count as arguments, but since they lack referential θ -roles, they are deemed “inextractable from weak islands” (Szabolcsi and den Dikken 2014: 222). This, however, does not quite amount to a principled explanation, but rather a second-order characterisation of the data. One that is not even particularly accurate, moreover, as *how*-extraction in (23) appears to be permitted, according to Szabolcsi and Lohndal (2017), when supported by a contextually salient checklist of ways to solve the problem.

- (23) a. How are you wondering < whether to solve the problem __ >?
 (OK when choosing from salient checklist)
 b. How did < no student solve the problem __ >?
 (OK when choosing from salient checklist)

(Szabolcsi and Lohndal 2017: 17)

According to the authors, if such a checklist is not contextually available extraction is degraded. This observation, moreover, applies only to *how*, and it remains unclear how Szabolcsi’s approach, based on semantic type distinctions, could account for the comparatively milder degradation observed in (22). Why should *when* and *where* extractions be better than, say, *why* extractions if the only relevant distinction is between individuals and non-individuals? Szabolcsi’s account can only separate *which/what*-phrases from canonical adjunct wh-expressions, whose extraction is typically banned—at least, as she puts it, “when the latter are not individuated by contextual brute force” (Szabolcsi and den Dikken 2014: 225). Szabolcsi does not provide further elaboration on this remark, which reappears, unchanged, in (Szabolcsi and Lohndal 2017: 19). I believe, however, that the intuition behind it is a promising one. In what follows, I will argue that what Szabolcsi presents as an incidental aside may in fact point to a factor that is central to our understanding of islandhood. Specifically, I will propose that context plays a pivotal role in facilitating adjunct extraction from WIs. Not through “brute force,” but by licensing a

structural representation of an island-internal gap, in a predictive way. I return to these aspects in the next section, where I present Experiments 5 and 6. Before that, let me briefly summarise the key points discussed so far.

- According to the literature, SIs block extraction of (almost) all arguments and adjuncts, while WIs allow extraction of (referential, D-linked/type *e*) arguments.
- Traditional accounts based on the ECP and Subjacency ban extraction of all adjuncts from WIs.
- Rizzi (1990) hypothesises that locative (*where*) and temporal (*when*) wh-phrases yield milder violations than, e.g., manner expressions (*how*), because the former can rely on spatio-temporal indices.
- Szabolcsi and Lohndal (2017) argue that *how*-extraction is licit, but only when supported by a contextually specified checklist of alternatives.
- Apparent exceptions to the (rigidity of the) ban on adjunct extraction from WIs have been treated only peripherally, with no fully developed theoretical accounts to explain them.

In the next sections, I will present new experimental evidence testing the extent to which native speakers accept extraction of adjuncts (*how*-, *when*-, and *where*-phrases) from both strong and weak islands. As I will show, the results call for a critical reconsideration of the empirical generalisations reviewed above, and crucially, of the theoretical accounts proposed to explain them.

5.2 Experiments 5–6

As discussed, the theoretical literature on strong and weak islands has expanded significantly, and a growing body of experimental work is now providing important new insights. A major turning point was Sprouse's (2007) dissertation, which introduced a robust and replicable experimental methodology for detecting island effects (see Chapter 2 for a detailed review). This methodology has since been widely adopted, leading to a number of studies that test, and often confirm, classic island effects identified in the theoretical literature (a.o., Sprouse et al. 2011, 2012, 2016; Almeida 2014; Pham et al. 2020). Crucially, this line of research has also enabled cross-linguistic comparison, extending the investigation of island effects beyond English. One recent example is Pañeda et al. (2024), who show that Spanish exhibits wh-island effects similar to English, contrary to earlier claims by Torrego (1984).

While this literature has contributed substantially to our understanding of locality, I see two partial limitations. First, although many strong and weak islands have now been tested, most experimental work on *whether*-islands (a prototypical case of WI) has focused exclusively on the extraction of arguments, not adjuncts. Yet this somewhat overlooks the foundational theoretical observation discussed above: WIs are *selective*, they are expected to permit argument extraction while resisting the extraction of non-arguments. Against this backdrop, it is striking that Sprouse and collaborators have repeatedly found degraded acceptability for sentences like *What do you wonder whether...?*—that is, for argument extraction out of *whether*-islands. In fact, many studies have confirmed a superadditive degradation in the DD-scores for such structures, based on participants' acceptability rates (again, see Chapter 2 for details on this methodology).⁷ This brings me to a second concern. As Pañeda et al. themselves acknowledge, while these experiments can reliably *detect* island effects, they do not by themselves tell us *why* those effects arise. In their words: “While our results and the results of the empirical studies listed above have uncovered island effects where there were presumed to be none, they do not conclusively determine what set of factors bear causal responsibility for the effects themselves” (Pañeda et al. 2024: 36). To put it another way, the observed degradation remains theoretically ambiguous: it may arise from syntactic constraints (e.g. Subjacency, phases, Relativised Minimality), from semantic or pragmatic factors, or from processing-based limitations. Or indeed from an interaction of all three. This connects to the broader, well-known question of what acceptability judgements actually measure. While I cannot pursue this issue in full here, it is important to acknowledge that acceptability reflects the *overall* status of a sentence (cognitively and grammatically) and is thus potentially sensitive to multiple sources of difficulty (see, a.o., Sprouse and Almeida 2011; Sprouse et al. 2013a for discussion). For this reason, we cannot be certain that argument and adjunct extraction from *whether*-islands are degraded for the same reasons. The traditional argument–adjunct asymmetry may still hold, not in terms of whether (or to what extent) extraction is possible, but in terms of *why* the degradation occurs in each case.⁸

A similar, and perhaps more pressing, question concerns whether the effects observed in strong and weak islands arise from distinct underlying causes. This is the focus of Experiments 5 and 6. Before introducing them, however, I want to offer one final reflection

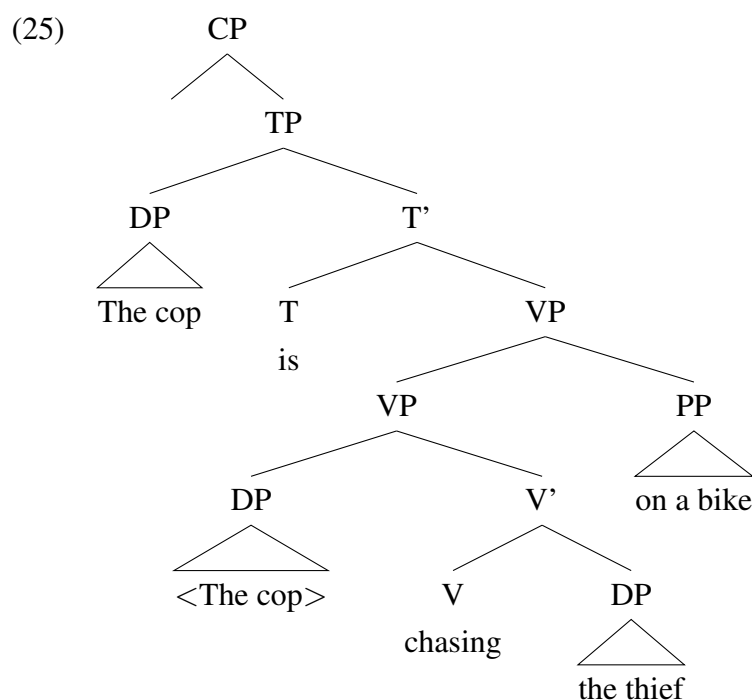
⁷Note, however, that the methodology developed by Sprouse and collaborators relies on relative ratings and standardised difference-in-differences (DD) scores. Consequently, what they identify as an island effect should be interpreted primarily as a measure of how much harder it is for participants to process and accept a particular extraction relative to matched grammatical baselines. Thus, these effects do not directly correspond to traditional binary grammaticality judgements. Instead, they represent gradient, experimentally detectable indications of complexity or interpretive difficulty, which may or may not align neatly with categorical acceptability intuitions for individual speakers or, crucially, linguists.

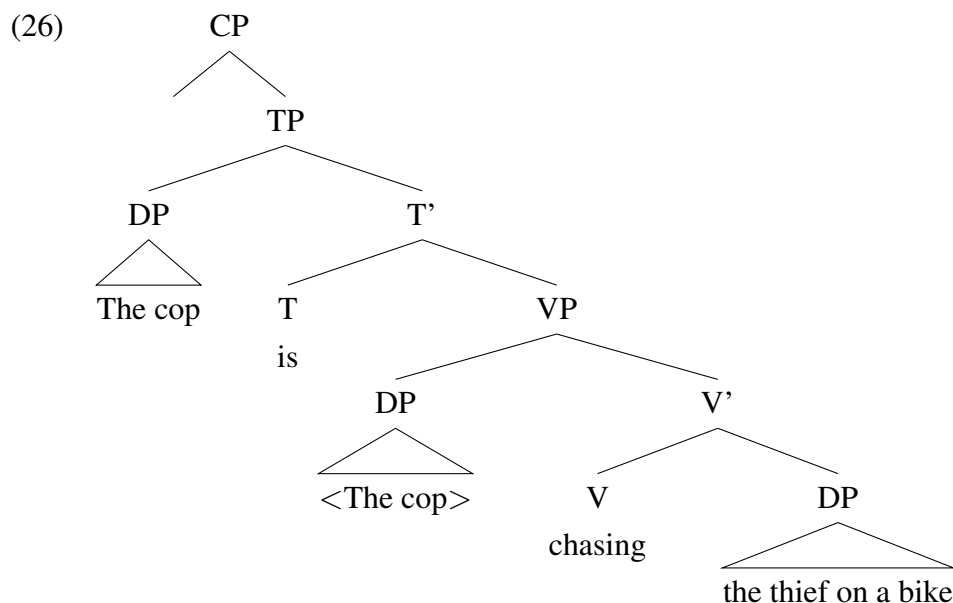
⁸It would also be worth investigating whether the magnitude of degradation differs across extraction types: a natural hypothesis is that adjunct extraction should be penalised more severely.

on the experimental literature just discussed. These studies remain of vital importance: they provide the empirical foundation on which our understanding of locality effects continues to expand. They also shed light on the variability in the gradience of island effects across languages, speakers, and island types—variability that any adequate theory must account for. That said, if we wish to answer more fine-grained questions about the source of locality effects, we may need to adopt different strategies. Experiments 5 and 6 represent an attempt in this direction. I introduce the methodology in the next section. At a glance, however, the intuition behind it is as follows. Let us consider the following sentence:

(24) The cop is chasing the thief on a bike.

Now suppose we present this sentence to two native English speakers and ask them to evaluate the truth of two different paraphrases. One speaker is asked: “*The thief is riding a bike – true or false?*” and the other is asked: “*The cop is riding a bike – true or false?*”. Most likely, both will answer “true”. This is because sentence (24) exhibits classic attachment ambiguity: the adjunct *on a bike* can syntactically attach at different levels of the structure, either modifying the predicate or the object DP. Crucially, although the surface string is identical, the underlying syntactic representations differ, as illustrated in the *very simplified* trees in (25, 26).





Now consider what would happen if we asked these speakers to rate the sentence's acceptability on a 1–7 scale instead. Most likely, we would receive high ratings (e.g., two 7s), but we would learn nothing about which underlying representation each speaker evaluated. In contrast, by eliciting truth-value judgements, we can infer which specific structure the participant is accessing: if someone answers “true” to a paraphrase, then they are effectively licensing a syntactic representation for which the corresponding interpretation holds.

In the case of sentence (24), there is no need to experimentally probe this ambiguity. We already have clear intuitions that both interpretations are equally available, and the reason for this is straightforward: adjuncts are typically free to attach at various points in the structure. But now consider a minimally different case involving adjunct wh-movement:

(27) How is the cop chasing the thief who's running away?

In this sentence, the fronted adjunct *how* could, at least in principle, be interpreted as modifying either the matrix verb *chase* or the RC-internal verb *run away*. Once again, there is a single linear string compatible with two different underlying structures. However, in this case, if a speaker were able to access the second interpretation (where *how* modifies *run away*) this would imply that they successfully extracted an adjunct out of a relative clause. Given that relative clauses, being SIs, are known to block extraction, it would be surprising if this second reading were accessible. A more plausible expectation is that speakers only access the structure in which *how* modifies the matrix verb, since this does not involve violating island constraints.

Building on this logic (presented in detail in the next section), Experiments 5 and 6

test whether gaps inside strong and weak islands are accessible to native speakers. To briefly illustrate how this can be tested, consider the contrast in the following examples:

(28) Where did Britt wonder __? [_{WI} whether Vardy scored __?]

(29) Where did Britt meet __? [_{SI} the player who scored __?]

To ensure that *where* could plausibly modify both the matrix and the embedded verb, such questions must be preceded by a context that makes both readings accessible. In this illustrative case, a context might specify that: (a) Britt is at the pub; (b) she does not know whether Vardy scored in yesterday's England game; (c) the game took place at Wembley; and (d) Vardy, who did score, enters the pub where Britt is. These contextual details support two interpretations for each question: *where* may be understood as modifying either the matrix verb (*wonder/meet*) or the embedded verb (*score*). Thus, if a speaker judges an answer like "at Wembley" as appropriate, it suggests they are accessing a gap inside a strong or weak island.

Note that, if a gap is accessed, this implies that speakers are able to construct and interpret an underlying representation in which a filler is extracted from an island configuration. Crucially, if this is the case, then the degradation observed in standard acceptability judgements must arise from something *other than* a strict ban on island-violating structures in the syntax: arguably, a speaker cannot access a representation that cannot be generated.⁹

In short, this paradigm offers a way of probing whether the source of the degradation effects reported in the literature is genuinely syntactic, or whether it reflects other factors. This is particularly relevant given the recurring claim that SIs and WIs may have distinct underlying causes. While there appears to be an irreducible syntactic aspect to SIs, WIs have been often claimed to be more prone to extra-syntactic explanations. For instance, Szabolcsi and den Dikken (2014: 14) note that theoretical treatments of WIs have become "less and less syntactic," and Boeckx (2012: 55) explicitly entertains "the possibility to account for weak islands in non-syntactic terms."

Based on these considerations, the experiments presented here specifically test adjunct extraction from two types of island environments: (i) SIs (complex DPs and relative clauses) and (ii) WIs, with a focus on *whether*-islands. The focus on adjuncts is motivated by two considerations. First, the literature consistently holds that adjuncts, more than anything, are not extractable from either SIs or WIs (see above for a discussion, and below for a breakdown of the specific island configurations tested). Second, to the best

⁹For a representation to be successfully 'generated,' it must be Spelled-Out. This means that it is both (a) derivable by the syntax and (b) interpretable at the LF/PF interfaces. I will discuss this in more detail in Chapter 6.

of my knowledge, experimental work on *whether*-islands has overwhelmingly focused on argument extraction, not adjuncts. The only exception I am aware of is a single study reported in Hofmeister and Sag (2010), which I briefly review in the next lines.

5.2.1 Hofmeister et al. (2010)

In their Experiment 3, Hofmeister and Sag (2010) tested whether referentially richer adjunct fillers improve the acceptability of extraction from *whether*-islands. The study employed a 2x2 factorial design crossing (i) filler type ('simple' adjuncts like *how long* vs. 'complex' adjuncts like *how many hours*) and (ii) structure type (island vs. non-island). Participants performed a self-paced reading task in which they read a brief context followed by a *wh*-question. After reading the question, they were prompted to judge whether it could be meaningfully answered in the given context. Specifically, participants responded "yes" or "no" to the prompt *Could this question be answered?*

Two illustrative items are reproduced in (30) and (31). In the first case, a "yes" response is appropriate, since the context supplies the relevant information. In the second case, participants should respond "no," as the context fails to support an answer to the embedded question.¹⁰

- (30) **Context:** Julie discerned that the survivor had managed to stay alive for eight days after the crash in the harsh conditions.
- a. SIMPLE: *How long* did Julie observe whether the passenger had survived in the unbelievably harsh conditions?
 - b. COMPLEX: *For what period of time after the crash* did Julie observe whether the passenger had survived in the unbelievably harsh conditions?
 - c. BASELINE: *How long* did Julie observe that the passenger had survived in the unbelievably harsh conditions?
- (31) **Context:** Andrew overheard the daycare staff discussing how they wanted to get away from the children for a few hours.
- a. SIMPLE: *How long* did Andrew hear whether the children had played during the daycare's afternoon recess?
 - b. COMPLEX: *How many hours* did Andrew hear whether the children had played during the daycare's afternoon recess?
 - c. BASELINE: *How long* did Andrew hear that the children had played during the daycare's afternoon recess?

¹⁰A potential concern with this task is that participants may not have needed to parse the target sentence in full. Rather, they could have relied on a shallow strategy, identifying the main and embedded predicates and simply checking whether the context supplied relevant information.

No significant accuracy differences were found across conditions. However, reading times did show an effect of filler complexity: complex adjuncts were read significantly faster than simple ones, replicating the pattern observed in the authors' earlier experiment on argument extraction. Unlike the argument case, however, the reading time differences emerged "just after the clause boundary, rather than at the embedded verb" (Hofmeister and Sag 2010: 400). This suggests that participants may have processed the adjunct as modifying the matrix verb, rather than interpreting it as extracted from within the embedded clause. If so, participants did not engage with the island configuration in the intended way. As the authors themselves speculate (Hofmeister and Sag 2010: 400–401):

[I]f something akin to an active-filler strategy applies to adjuncts as it does for arguments, then the processor would retrieve and attempt to integrate the relevant adjunct information at the first possible site. This would mean that, even if retrieval and integration at some downstream point (say, at an embedded verb) was also possible, retrieval would preferentially take place at any plausible earlier attachment site, for example, a clause boundary

In what follows, I will argue that this idea, though only briefly touched upon by Hofmeister and Sag, is in fact central to the interpretation of the WIs tested in Experiments 5 and 6. Indeed, it effectively anticipates one of the interpretive claims I will advance on the basis of my own data.

5.2.2 Research questions

Before turning to the methodology, let me briefly recapitulate the central points discussed so far and present the research questions guiding this study. While standard acceptability judgements remain an excellent and indispensable tool for studying island effects, teasing apart potential differences in their underlying sources may require a different methodological approach. Experiments 5 and 6 constitute a first step in this direction. They target a central theoretical question in the study of locality:

- (32) **[RQ5]** Do strong and weak island effects originate from a common source? And if so, is this source located within the syntax?

To address this question, I test whether providing an apt context can render a gap accessible in strong and weak islands, respectively. This leads to a more specific empirical question:

- (33) **[RQ5.1]** Can context ameliorate extraction from islands? Is there a difference between SIs and WIs?

The logic is straightforward: if context can ameliorate island violations by making an abstract underlying representation accessible, this excludes the possibility that observed degradations stem from deep, rigid syntactic constraints. Indeed, if a representation is accessed, this presupposes that the grammar was able to generate it in the first place. Moreover, if the same context-based strategy yields different results across island types, this suggests that SIs and WIs must involve constraints that differ in kind, pointing towards distinct underlying sources rather than a unified (syntactic) account. This, once again, follows the intuition that amelioration effects/circumvention strategies can serve as diagnostics for the nature of locality constraints (see discussion in Chapter 2).

5.2.3 Materials and methods

Experiment 5 manipulates two variables: island type (weak vs. strong) and dependency length (short vs. long). The resulting 4 conditions were represented as question–answer pairs, introduced by the same context, as illustrated in the following two examples:

Context: *Luke is queuing up at the till when he remembers that the day before his colleagues told him there could be a demonstration tonight. He is worried and wonders if that means that he'll have to forgo his usual walk after dinner.*

WEAK ISLAND

Q: When does Luke wonder whether he can take a walk?

- a. While queuing (short)
- b. After dinner (long)

STRONG ISLAND

Q: When did Luke hear the rumour that there could be a demonstration?

- a. The day before (short)
- b. Tonight (long)

Context: *Carol is concerned about her little Jimmy. He never wants to eat his meals, and when he does, he insists on standing instead of sitting at the table. Carol reaches out to the new babysitter with a text to ask her if today little Jimmy was standing while eating his lunch. Since the babysitter looks after little Jimmy with great dedication, Carol is confident her child will soon stop acting up during mealtime.*

WEAK ISLAND

Q: How did Carol ask whether little Jimmy ate?

- a. With a text (short)
- b. Standing (long)

STRONG ISLAND

Q: How did Carol reach the babysitter who tends to little Jimmy?

- a. With a text (short)
- b. With great dedication (long)

Each answer, paired with its corresponding question and preceding context, represented one experimental condition. Sixteen such items were created and distributed across four lists using a Latin Square design, such that each participant encountered all four conditions four times, and each item appeared in only one condition per list, with no lexical repetitions (i.e., each participant saw a given context only once). All experimental items involved extraction of an adjunct from either a strong or a weak island. SIs were represented by Complex DPs (containing either a complement clause or a relative clause), WIs by *whether*-clauses. The *wh*-types targeted were *where* (12 occurrences), *how* (10 occurrences), and *when* (10 occurrences). Each list was then supplemented with the same sixteen filler items. These followed the same structure as the experimental items: a context, followed by a question, followed by an answer. Of the sixteen fillers, eight displayed false answers and eight displayed true ones.

The procedure unfolded as follows: participants first read a context at their own pace. Once ready, they pressed the space-bar to proceed. This triggered the disappearance of the context and the appearance of a question, which remained on screen for 4 seconds. The question then disappeared and was replaced by a possible answer, displayed for 3 seconds. During this time, participants judged whether the answer was appropriate by pressing ‘F’ (appropriate) or ‘J’ (inappropriate).¹¹ If no response was recorded within the 3-second window, the message “*Too slow!*” appeared, prompting participants to press the space-bar to move to the next trial. Both their selection and reaction times (RTs, measured as the interval between answer onset and keypress) were recorded.

One possible concern that arose during the design phase was that participants might be biased towards choosing answers corresponding to information in the final sentence of the context, due to a recency effect. To address this issue, a second variant, Experiment 6, was created. This experiment used the same materials and overall structure but modified the timing: both the context and the question remained visible throughout the trial, and participants had 6 seconds to respond after the appearance of the answer. This adjustment was designed to mitigate the recency bias that could potentially affect responses in Experiment 5, by allowing participants to consult the context before answering. To maintain parallelism with Experiment 5, RTs were also recorded and analysed in this variant. However, since responses were not time-constrained in the same way, these measures were treated as additional exploratory data rather than as a reliable index of online processing.

Both experiments were implemented in PcIbex (Zehr and Schwarz 2018) and lasted approximately 20 minutes each.

¹¹The labels “appropriate” and “inappropriate” were chosen instead of “true” and “false” to avoid a potential association between the ‘F’ key and the label “false.” Using the latter pair would have required presenting the options with “false” on the left and “true” on the right—opposite to the conventional order—which could have led to confusion and/or unintended selections.

5.2.4 Participants

For both experiments, native speakers of British English with no reported history of language-related disorders were recruited via Prolific and compensated £2.50 for their participation. An accuracy threshold of 80% on filler items was applied as an inclusion criterion. A total of 104 participants were tested. The final sample included 44 participants for Experiment 5 (mean age = 43.6, SD = 13.76) and 48 participants for Experiment 6 (mean age = 44.4, SD = 13.39).

5.2.5 Results and analysis – Experiment 5

The answers that participants judged as (non-)appropriate functioned as fillers of wh-gap dependencies. Judging an answer appropriate was taken to indicate access to the corresponding gap (see discussion above)—either at the matrix verb (short dependency) or the embedded verb (long dependency). Accordingly, the first dependent variable, i.e. participants' responses, was operationalised as *gap access*.

The plot in Figure 5.1 displays the mean gap access rate across the four conditions. Participants showed the highest level of gap access when the gap modified the matrix verb in a structure involving a SI ($M = 0.85$, $SD = 0.36$). By contrast, gaps within SIs were accessed drastically less reliably ($M = 0.19$, $SD = 0.39$). In WI structures, long dependencies were favoured over short ones: gaps embedded within the WI were more easily accessed ($M = 0.73$, $SD = 0.44$) than those located at the matrix verb level ($M = 0.50$, $SD = 0.50$).

A generalised linear mixed-effects model was fitted using the `lme4` package (Bates et al. 2015) in R. The fixed-effects structure included Island type, Dependency length, and their interaction. Following Barr et al. (2013), the analysis began with a maximal random-effects structure, including by-participant and by-item intercepts and slopes for all predictors and their interaction. The structure was then progressively simplified while preserving theoretically motivated variance components (see below). The final model included by-participant random intercepts and by-item random slopes for dependency length.¹² STRONG ISLAND and LONG DEPENDENCY were set as the reference levels; all fixed-effect estimates are interpreted relative to these baselines. Table 5.1 summarises the results. The main effect of Island type ($\beta = 2.70$, $p < .001$) indicates that long dependencies yielded much higher gap access rates in WIs than in SIs. The main effect of Dependency length ($\beta = 3.60$, $p < .001$) indicates that, in SI-conditions, gap access was substantially higher when the gap modified the matrix verb (short dependency) rather than

¹²Model formula: `m8 <- glmer(Gap_access ~ island_type * dependency_length + (1 | Participant) + (1 + dependency_length | Item), data = data, family = binomial(link = "logit"), glmerControl(optimizer = "bobyqa"))`.

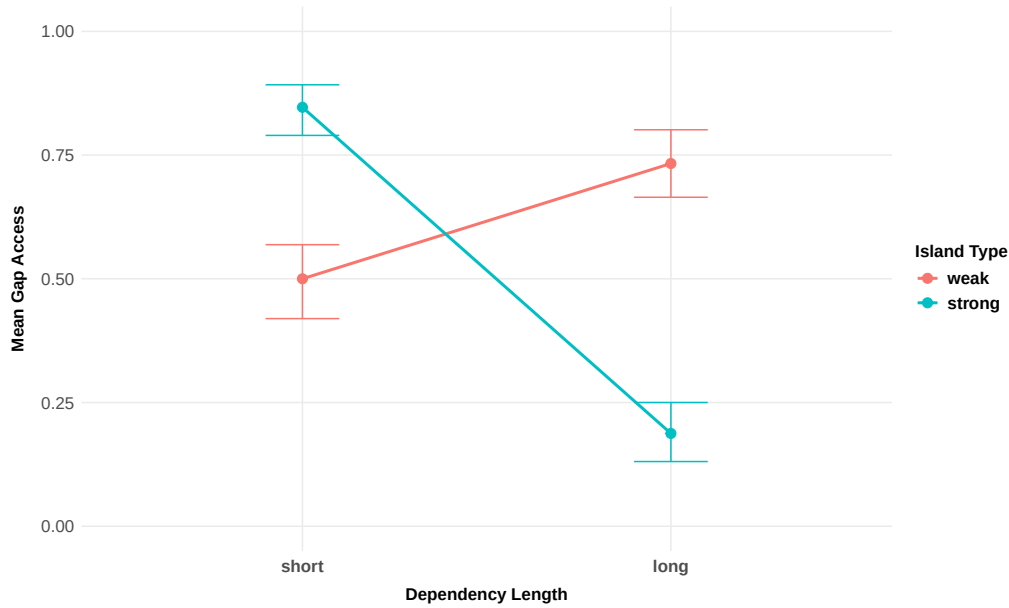


Figure 5.1: Mean gap access across conditions in Experiment 5, as a function of Island type (strong vs. weak) and Dependency length (short vs. long). Error bars represent 95% bootstrapped confidence intervals, computed using the `mean_cl_boot` function in `ggplot2`.

the embedded verb (long dependency). Finally, the significant interaction between Island type and Dependency length ($\beta = -4.72$, $p < .001$) indicates that short dependencies ease gap access in SIs but not in WIs, where they appear to reduce it slightly. The `emmeans` package (Lenth 2025) was used to calculate pairwise comparisons of the estimated marginal means, with Tukey adjustment for multiple comparisons. Results confirmed that the SI-LONG condition yielded significantly lower gap access than all other conditions: SI-LONG vs. WI-LONG, $p < .0001$; SI-LONG vs. SI-SHORT, $p < .0001$; SI-LONG vs. WI-SHORT, $p = .0022$. No significant difference was found between the WI-LONG and SI-SHORT conditions ($p = .22$), which were the conditions with the highest gap access rates. The WI-SHORT condition was significantly different from both SI-SHORT ($p < .0001$) and, to a lesser extent, from WI-LONG ($p = .05$).

Turning to RTs, the boxplot in Figure 5.2 shows that participants were slowest in the WI-SHORT condition ($M = 1608\text{ms}$, $SD = 683$), and fastest in the SI-SHORT condition ($M = 1326\text{ms}$, $SD = 484$). Intermediate values were observed for the SI-LONG ($M = 1525\text{ms}$, $SD = 572$) and WI-LONG conditions ($M = 1510\text{ms}$, $SD = 708$).

RTs were analysed using a linear mixed-effects model with island type, dependency length, and their interaction as fixed effects; by-participant random intercepts and by-item random intercepts and slopes for Dependency length were included as random effects.¹³

¹³Model formula: `m_rts <- lmer(RTs ~ island_type * dependency_length + (1 | Participant) + (1 + dependency_length | Item), data = dataRTs, control =`

Table 5.1: Parameters of the generalised linear mixed-effects analysis with gap access as the dependent variable and type of island in interaction with dependency length as fixed effects (Experiment 5).

Fixed effects	b	SE	z	p
Intercept	-1.59	0.26	-6.13	<.001
Island type (weak island)	2.70	0.29	9.28	<.001
Dependency length (short)	3.60	0.48	7.47	<.001
Island type * Dependency length	-4.72	0.42	-11.18	<.001

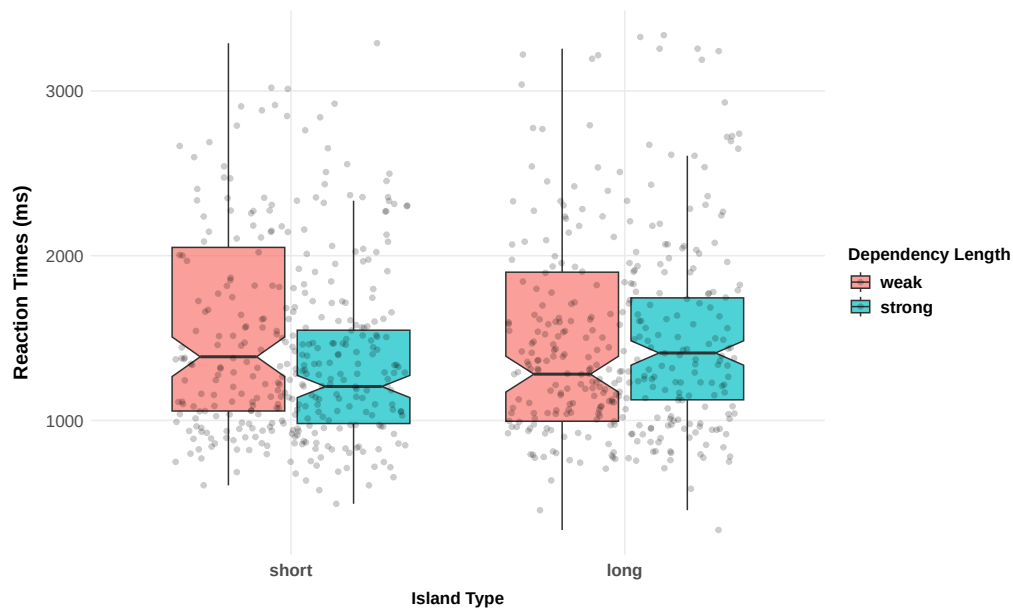


Figure 5.2: Reaction times (ms) by Island type and Dependency length in Experiment 5. Boxplots show median, quartiles, and 95% confidence intervals (notches). Jittered data points represent individual observations.

The model (summarised in Table 5.2) revealed a significant main effect of Dependency length ($b = -181.10$, $p = .020$), with short dependencies eliciting faster responses in SI-conditions. No main effect of Island type was observed ($p = .55$). Crucially, a significant interaction was found between Island type and Dependency length ($b = 262.46$, $p = .001$), reflecting the pattern observed for gap access: short dependencies led to faster RTs in SIs, but actually led to slower RTs in WIs. Post hoc pairwise comparisons further clarify the interaction: while short dependencies were marginally faster than long dependencies in SIs (181ms difference, $p = .087$), this facilitation was absent in WIs, where short dependencies were numerically, but not significantly, slower than long dependencies (-81ms, $p = .690$). The strongest effect emerged when comparing short dependencies directly across island types, with SIs eliciting significantly faster responses than WIs (229ms difference, $p < .001$).

Table 5.2: Parameters of the linear mixed-effects analysis with RTs as dependent variable and type of island in interaction with dependency length as fixed effects.

Fixed effects	<i>b</i>	SE	<i>df</i>	<i>t</i>	<i>p</i>
Intercept	1507.72	64.43	35.47	23.40	<.001
Island type (weak island)	-33.89	56.08	617.76	-0.60	.546
Dependency length (short)	-181.10	73.26	28.63	-2.47	.020
Island * Dependency	262.46	79.37	618.29	3.31	.001

5.2.6 Results and analysis – Experiment 6

The plot in Figure 5.3 displays the mean gap access rates across the four conditions. Participants' responses mirrored the pattern observed in Experiment 5. However, in this case, the difference between the highest and lowest gap access rates was more pronounced: short dependencies in SI-conditions once again yielded the highest rates of gap access ($M = 0.91$, $SD = 0.29$). Gaps embedded in SIs (i.e., long dependencies) remained the least accessible, with even lower access rates than in Experiment 5 ($M = 0.11$, $SD = 0.31$). As in Experiment 5, WI-conditions showed greater gap access with long dependencies ($M = 0.72$, $SD = 0.45$) than with short ones ($M = 0.56$, $SD = 0.50$).

Following the same methodology used for Experiment 5, a generalised linear mixed-effects model was fitted with gap access as the outcome variable and the interaction between Island type and Dependency length as a predictor. The random structure of the final model included by-participant random intercepts and by-item random slopes for de-

```
lmerControl(calc.derivs = FALSE, optimizer = "bobyqa"))).
```

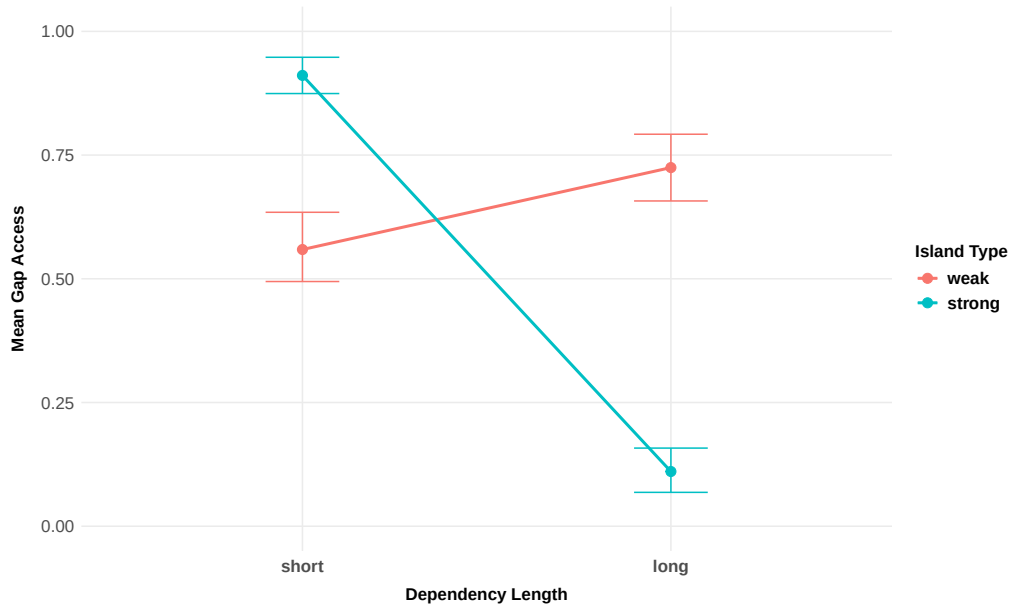


Figure 5.3: Mean gap access across conditions in Experiment 6, as a function of island type (strong vs. weak) and dependency length (short vs. long). Error bars represent 95% bootstrapped confidence intervals, computed using the `mean_cl_boot` function in `ggplot2`.

dependency length.¹⁴ The results are summarised in Table 5.3. As expected from the mean values, the effects mirror the pattern observed in Experiment 5, but with more pronounced estimates. The main effect of Island type ($\beta = 3.43$, $p < .001$) indicates that gap access with long dependencies is significantly greater in WI-conditions than SI-conditions. The main effect of Dependency length ($\beta = 5.27$, $p < .001$) shows that short dependencies in SI-conditions yield much higher gap access rates than long dependencies. Finally, the significant interaction between the two factors ($\beta = -5.99$, $p < .001$) indicates that short dependencies facilitate gap access in SIs but not in WIs, where the opposite pattern holds. Post hoc pairwise comparisons replicated the pattern observed in Experiment 5, with generally larger effect magnitudes. However, unlike Experiment 5, the disadvantage of short dependencies in WIs is no longer significant ($p = .0498$ vs. $p = .578$).

Although responses were not time-limited in the same way, RTs were also recorded in Experiment 6. Participants had more time to make their selection (6s vs. 3s). Moreover, in this variant, the answer to be judged as (non-)appropriate appeared below the question and the context, allowing participants to retrieve all relevant information before responding. For these reasons, the recorded RTs likely do not reflect a reliable online processing measure. I nonetheless report the data, which largely mirror the pattern observed in Ex-

¹⁴Model formula: `m12 <- glmer(Gap_access ~ island_type * dependency_length + (1 | Participant) + (1 + dependency_length | Item), data = data, family = binomial(link = "logit"), glmerControl(optimizer = "bobyqa"))`.

Table 5.3: Parameters of the generalised linear mixed-effects analysis with gap access as the dependent variable and type of island in interaction with dependency length as fixed effects (Experiment 6).

Fixed effects	<i>b</i>	SE	<i>z</i>	<i>p</i>
Intercept	−2.34	0.33	−6.99	<.001
Island type (weak island)	3.43	0.35	9.91	<.001
Dependency length (short)	5.27	0.65	8.05	<.001
Island * Dependency	−5.99	0.50	−12.04	<.001

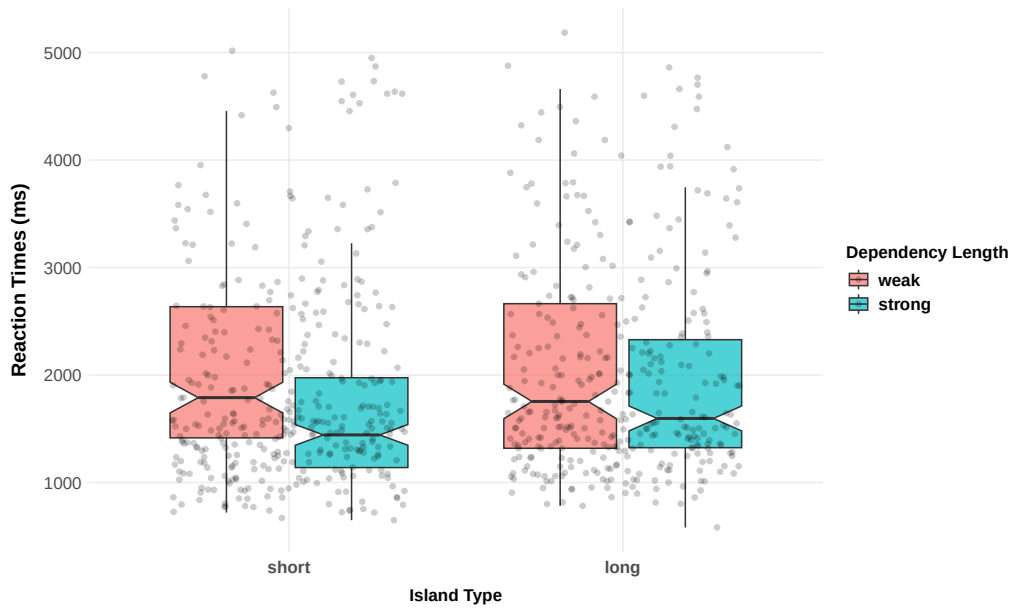


Figure 5.4: Reaction times (ms) by Island type and Dependency length in Experiment 6. Boxplots show median, quartiles, and 95% confidence intervals (notches). Jittered data points represent individual observations.

periment 5—albeit with overall higher values, as expected.

The boxplot in Figure 5.4 confirms that participants were slowest in the WI-SHORT condition ($M = 2152\text{ms}$, $SD = 1078$), and fastest in the SI-SHORT condition ($M = 1768\text{ms}$, $SD = 996$). Again, intermediate values were recorded for the SI-LONG ($M = 2009\text{ms}$, $SD = 1020$) and WI-LONG conditions ($M = 2146\text{ms}$, $SD = 1117$).

The model used to analyse RTs in Experiment 5 was replicated for Experiment 6. A linear mixed-effects model was fitted with island type, dependency length, and their interaction as fixed effects; by-participant random intercepts and by-item random intercepts and slopes for dependency length were included as random effects.¹⁵ The model is

¹⁵Model formula: `m_rts2 <- lmer(RTs ~ island_type * dependency_length + (1 | Participant) + (1 + dependency_length | Item), data = dataRTs2, control = lmerControl(calc.derivs = FALSE, optimizer = "bobyqa"))`.

summarised in Table 5.4.

This time, the main effect of island type did not reach significance ($\beta = 133$, $p = .19$). The main effect of dependency length remained significant, though only marginally ($\beta = -243$, $p = .050$). Unlike in Experiment 5, the interaction between island type and dependency length was not significant, but showed a marginal trend ($\beta = 252$, $p = .078$).

In sum, while gap access showed a *stronger* replication—with larger effect sizes—RTs showed a weaker replication, with the key interaction becoming non-significant. This suggests that the possibility to retrieve information from the context did not affect gap accessibility (if anything, it reinforced the pattern observed in Experiment 5), whereas the extended response window may have attenuated the RT differences observed previously, likely cancelling out any potential online processing effects.

Table 5.4: Parameters of the linear mixed-effects analysis with RTs as the dependent variable and type of island in interaction with dependency length as fixed effects (Experiment 6).

Fixed effects	b	SE	df	t	p
Intercept	2012.42	94.63	62.64	21.27	<.001
Island type (weak island)	133.01	101.60	668.12	1.31	.191
Dependency length (short)	-243.37	119.33	33.07	-2.04	.050
Island * Dependency	252.20	142.77	667.17	1.77	.078

5.3 Interim discussion

In Section 5.2, I introduced two research questions: first, a broad theoretical question concerning the nature of locality constraints; and second, a more specific empirical question about the potential of context to ameliorate extraction from WIs and SIs. In the discussion that follows, I adopt the reverse order: I begin by addressing the empirical findings, and then consider their implications for the broader theoretical debate on locality constraints.

The results from Experiments 5 and 6 demonstrate that context has a crucial impact on adjunct extraction. As discussed, fronted wh-adjuncts in sentences involving an island are potentially ambiguous, as they could in principle reconstruct at the matrix or the embedded verb level. What the present data show is that, when provided with a context containing information that licenses both interpretations, English native speakers can readily access a gap within a WI. This appears to be a novel finding, as it indicates that extracting an adjunct from a WI is perfectly possible. Even more strikingly, such an extraction is not only possible, but actually preferred: participants overwhelmingly chose to

reconstruct the *wh*-moved adjunct *within* a WI rather than outside of it. In both experiments, the mean gap-access rates at the embedded verb level were higher than those at the matrix verb level.

This is of particular theoretical interest for several reasons. First and foremost, these results run counter to traditional generalisations which ban the extraction of adjuncts from WIs. Secondly, although exceptions to such generalisations have been discussed in the literature, they are typically restricted to narrowly defined subcases and accounted for in ways that the current data do not seem to support. For instance, Rizzi's idea that spatial and temporal adjuncts like *when* and *where* could undergo argument-like binding through dedicated indices (see above) does not explain why *how* (which was also included in the conditions of my study) can be equally extracted. A somewhat complementary claim is presented in Szabolcsi and Lohndal (2017), who suggest that among the adjunct class, manner-type adjuncts like *how* could be extracted, but only when paired with a contextually specified checklist of alternatives. This, however, would also not capture the present results: along with *how*, both *where* and *when* were included in the conditions, and, crucially, no contexts presented a checklist of alternatives. Another oft-cited ameliorative factor is tense finiteness, whereby extractions are generally considered to be more acceptable from non-finite clauses. Yet all the islands included in Experiments 5 and 6 featured finite-tense verbs. Finally, a possible extension of the role of D-linking and/or lexical complexity—typically discussed for arguments—to the adjunct class would not hold either, since all tested adjuncts occurred in their bare form.

This does not preclude the possibility that these factors could further enhance adjunct extraction from WIs. It is actually plausible that, for instance, a bare *where* might yield lower gap-accessibility than a lexically enriched variant like *in which place*, echoing the established facilitative role of lexical complexity found in Hofmeister and Sag (2010). Still, the critical finding remains: even in their bare forms, regardless of type (spatial, temporal, or manner), and without a checklist of alternatives, adjuncts were extractable from WIs at rates exceeding 70% across both experiments. Hence, I take this as robust evidence that adjunct extraction from WIs is not ungrammatical. The rationale, as discussed in Section 5.2, is straightforward: if participants can accurately judge the truth-value of (an answer to) a sentence, it means that their mental grammar must have first generated the abstract underlying representation of that sentence.

This leads to the second major empirical finding: gaps embedded in SIs were *not* accessible. In both experiments, SI-internal gaps yielded extremely low access rates, demonstrating a sharp asymmetry between the two island types. That is, whereas providing context containing relevant information successfully circumvented WI-boundaries, it failed to ameliorate adjunct extraction from SIs. This directly answers the empirical

research question:

- (34) **[RQ5.1]** *Can context ameliorate extraction from islands? Is there a difference between SIs and WIs?*

Native English speakers access adjunct-gaps embedded within WIs unproblematically when these are supported by contexts providing information relevant for their interpretation. The same contexts, however, do not make SI-embedded gaps accessible, indicating a clear WI-SI asymmetry.

Before discussing the theoretical implications of this asymmetry, let me first ward off a potential concern regarding the interpretation of these data.

One might think that deeming an answer “appropriate” does not necessarily tap into the mental grammar of a speaker—in minimalist terms, their I-language (Chapter 3). Rather, participants might have been biased towards certain interpretations due to recency effects. For instance, if a gap was associated with information presented in the final sentence(s) of a context, participants might have been facilitated in accessing it simply due to the temporal proximity of that information.

I exclude this possibility on three grounds. First, contexts were constructed to present relevant information as randomly as possible throughout the text. This was not always feasible, however, as it is natural to establish the temporal and spatial setting of an event before providing further specifications. Consequently, information relevant for *where* and *when* gaps was frequently introduced earlier in the context, whereas information relevant for manner adjuncts necessarily followed. Precisely because of this potential confound, two experimental variants were implemented. In Experiment 6, participants could retrieve relevant information from the context before providing their answer, effectively controlling for any potential impact of information arrangement. Experiment 6 not only replicates the pattern observed in Experiment 5 but actually sharpens the contrasts, indicating that recency effects did not drive participants’ gap-access preferences. Last, and most crucially, the robust interaction between island type and dependency length demonstrates that participants systematically rejected answers associated with SI-embedded gaps. If recency were the determining factor, one would need to explain why such effects selectively influenced WI-conditions while failing to impact SI-ones. The systematic nature of this asymmetry strongly suggests that the observed pattern hinges on genuine grammatical factors rather than task-specific biases.

Returning to the asymmetry between WIs and SIs: why could participants access gaps within WIs, but not gaps within SIs? Building again on the logic that gap access presupposes the construction of the underlying structure containing that gap, the data pattern becomes rather self-explanatory. Gaps contained within SIs cannot be accessed

precisely because structures involving extraction from such environments (specifically, complex DPs) cannot be syntactically generated. By contrast, gaps within WIs are readily accessed, indicating that the syntax *can* build a structure where a copy of a wh-moved adjunct is contained in a WI (specifically, a *whether*-clause).

This asymmetry ultimately points to a fundamental distinction: the constraints underlying strong and weak islands likely do not originate from the exact same source. SI effects must stem from a core syntactic constraint, as the syntax cannot generate or license a movement chain originating from within SI-environments. WI-effects, however, cannot be traced to the same structural limitations, given that the syntax demonstrably can build and interpret structures involving extractions from WIs. As far as WI-extractions are concerned, then, the apparent prohibition must arise elsewhere in the grammar.

This addresses the original, broader research question guiding Experiments 5 and 6:

- (35) **[RQ5]** *Do strong and weak island effects originate from a common source? And if so, is this source located within the syntax?*

The degradation associated with strong and weak island effects must originate from distinct grammatical sources. The fact that gaps within SIs cannot be accessed suggests that SI effects emerge because extractions from these environments are categorically ruled out by core syntactic constraints. By contrast, the fact that gaps within WIs can be readily accessed demonstrates that WI-constraints are not strictly syntactic in nature. Rather, the syntax can generate the relevant structures, suggesting that WI-effects arise from conditions that operate subsequent to the syntactic output.

5.3.1 Theoretical consequences and the nature of island effects

These conclusions place themselves in an intriguing theoretical position. As discussed in Section 5.2, linguists have long suspected that the constraints responsible for WI and SI effects might be distinct. Yet simultaneously, many have emphasised the desirability of unifying intervention and impenetrability locality under a single principle, even advancing proposals in that direction (e.g., Starke 2001; Abels 2003; Rizzi 2009; Zeijlstra 2023). I believe that the Interface Hypothesis pursued in this thesis might offer a solution to reconcile these apparently conflicting positions. I will return to this shortly. First, let me emphasise that despite this contrast, most researchers agree that at least for SIs, an irreducible syntactic nature seems to hold (even part of the reductionist literature concurs, see Chapter 2). The data presented here provide strong empirical support for this claim,

exploiting a novel methodology that, incidentally, could potentially be extended to probe “how syntactic” other islands are.

Regarding WIs, however, the situation is notably less straightforward. Different theories have been proposed to account for these effects, ranging from semantic and pragmatic to processing-based explanations. Yet it is important to note that at least one class of accounts attributes the ban on WI-extraction to syntactic constraints. For instance, the prohibition against adjunct extraction from WIs was explained by numerous ECP-based accounts (reviewed above), including the aforementioned proposal by Rizzi (1990) and Cinque (1990), based on the distinction between arguments and adjuncts and their respective (non-)availability of resumptive pros. Again, all these accounts have traditionally explained the degradation of adjunct extraction from WIs as a syntactic effect, crucially marking such examples with an asterisk. Yet the findings reported here cannot support this view, for the reasons already explained: they strongly suggest that the syntax *per se* does permit these types of extractions, suggesting that the degradation traditionally associated with extraction from WIs likely originates elsewhere. The question, then, becomes whether this source is pragmatic, semantic, or processing-based.

5.3.1.1 Possible sources of weak island effects and the argument–adjunct asymmetry

In Chapter 2, I already raised several concerns about semantic-pragmatic accounts, which I will not reiterate here. As for the specific case of adjunct extractions from *whether*-islands, I have pointed out that the solution proposed by Szabolcsi and Lohndal (2017) is not supported by the present data. Nonetheless, as hinted at in Section 5.2, a side remark by Szabolcsi may in fact offer a promising lead for interpreting these findings. As she puts it, adjunct extraction should be impossible unless adjuncts are “individuated by contextual brute force.” The data presented here clearly demonstrate that context plays a crucial role; so her intuition was, in this respect, remarkably accurate. For this reason, I will here attempt to elaborate on this insight.

I propose, in particular, that providing a context like those used in Experiments 5 and 6 opens up a space for multiple interpretations. Crucially, this includes interpretations that build on an underlying structure with a gap inside a (weak) island. In such cases, the context licenses the postulation of an island-embedded gap: it allows the parser to infer a dependency whose interpretation is relevant, salient, or even simply plausible within the discourse. In the absence of such contextual support, however, it is likely that the parser defaults to the simplest attachment strategy, resolving the dependency at the first available site—namely, at the matrix verb. This, in turn, accords with Hofmeister and Sag’s (2010) intuition (reported above) that adjuncts may exhibit processing behaviour

similar to arguments, which are known to be sensitive to the Active Filler Strategy (see Chapter 2). To reiterate, they suggest that adjuncts' "retrieval would preferentially take place at any plausible earlier attachment site." Not only does this account for their own online processing data (where the *wh*-complexity effect emerged at the embedded verb with arguments but at the island boundary with adjuncts, i.e. right after the matrix verb; see above for details) but it also offers an interesting path for explaining my own findings. Let us consider again the example sentence in (28), now repeated in (36).

(36) Where does Britt wonder whether Vardy scored?

Traditional generalisations would likely mark this sentence with an asterisk. Crucially, though, no formal theory would question the grammaticality of this sentence if the gap associated with *where* is interpreted at the matrix level, as in (37).

(37) Where do you wonder __ whether Vardy scored?

That is, if no further context is provided, it seems that the default (and perhaps only) possible interpretation is the one above. In order to extract *where* from within the WI-environment, something must license the parser to infer that it is possible for the adjunct to modify the embedded verb. In a sense, we could say that such an operation makes adjuncts more similar to arguments, in that they become more directly associated with the particular event structure of a particular verb. This, in a nutshell, is what the context in Experiments 5 and 6 does: it makes adjuncts more argument-like.

If this line of reasoning is on the right track, its scope can in fact be broadened to shed light also on the traditional argument–adjunct asymmetry. Specifically, it allows us to advance an interesting prediction: arguments should behave similarly to adjuncts when θ -role assignment is ambiguous. Let me unpack this. The reason why arguments rarely, if ever, encounter the same problems as adjuncts in examples like (36) is that arguments cannot freely attach at any point in the structure. They must satisfy θ -theory requirements. Thus, from a processing perspective, if the parser holds a *wh*-argument filler in working memory, it cannot discharge it arbitrarily; it must attach it to the verb that selects it (and assigns it case). And even if earlier attachment were possible, as in (38) below, a reanalysis would be required as soon as the embedded verb is encountered, which would otherwise be left without one of its obligatory arguments:

(38) Who did you ask (* __) whether Britt made __ happy?

This, then, may well be the actual source of the well-known argument–adjunct asymmetry. Not one based on idiosyncratic formal features of the two categories that endow them with distinct locality constraints, but rather one based on the independent conditions

they satisfy in terms of θ -theory. That is, the fact that arguments have been known to extract more easily might be a by-product of their inability to attach at the matrix verb. Indeed, unless the matrix verb involves a double-object construction, as in (38), it cannot host an additional complement argument, as it already hosts the complement clause.¹⁶ And, as we have seen, even when matrix-level attachment is possible, reanalysis must occur upon encountering the embedded verb.

Adjuncts, by definition, are not subject to θ -criterion requirements, effectively leaving the parser free(r) to discharge them at the first available opportunity, as speculated by Hofmeister and Sag (2010). In these terms, the argument–adjunct asymmetry appears more accidental than structural. Again, we can predict that, under the right conditions, arguments may not differ from adjuncts.

For instance, if we ensure that the embedded verb can have both transitive and intransitive interpretations, we may generate a situation akin to the standard case with adjuncts: a fronted *wh*-element that can reconstruct at two possible sites. Take, for example, the following case:

- (39) Who did you ask whether Ellie ate?

The most immediate interpretation is one in which *who* reconstructs at the matrix verb. In this case, the parser can safely integrate *who* at that level, maintaining the analysis because *ate* can be interpreted intransitively. Yet let us replace *Ellie* with *Grimvour*, a ruthless cannibal:

- (40) Who did you ask whether Grimvour ate?

Suddenly, extraction of *who* from within the *whether*-clause is not only possible but most likely preferred. Here is another example:

- (41) Who did you ask whether Adam was freezing?

An appropriate answer would be: “*Rowan, I asked Rowan whether Adam was freezing (because he was locked outside in his t-shirt).*” Now let’s switch *Adam* with *Frozone*, the ice-spraying superhero from *The Incredibles* (a personal favourite):

- (42) Who did you ask whether Frozone was freezing?

In this case, *who* is likely more felicitously reconstructed at the level of the embedded verb.¹⁷ Thus, just as with adjuncts, modifying the discourse/context has a direct impact

¹⁶The idea of using a double-object construction at the matrix level to illustrate this point arose in discussion with Hedde Zeijlstra, to whom I am grateful.

¹⁷It would be interesting to test these sentences experimentally using the paradigm developed in Experiments 5 and 6. However, preparing the stimuli would likely not be feasible without completely sacrificing lexical variation. Indeed, the stimuli would necessarily involve “who” and “ask” (as a double-object construction verb is needed). Also, the set of verbs that can freely alternate between transitive and intransitive readings (while taking a human complement) is likely quite limited.

on the assumed predicate structure, which in turn has clear consequences for how gaps are reconstructed. Note that this is not merely a restatement of claims acknowledging a role for pragmatics in WI-extractions. Pragmatic accounts typically argue that extraction from such environments compromises a sentence’s felicitousness. What I am proposing, instead, is that context can alter the underlying predicate structure itself, thereby counteracting the parser’s default tendency to resolve wh-dependencies at the earliest possible site.

To summarise the claims so far, I believe it is possible to capture and articulate the intuitions underlying the remarks of both Szabolcsi (Szabolcsi and den Dikken 2014; Szabolcsi and Lohndal 2017) and Hofmeister and Sag (2010). Adjuncts tend to reconstruct at the first available site, adhering to the same Active Filler Strategy observed for arguments. Yet when context is provided, adjuncts can become more argument-like, inducing the parser to infer the (possibility of a) gap within a WI. I would be hesitant to label this strategy as one of “brute force”, however. That term would perhaps be more appropriate if the context heavily biased one interpretation over the other. What Experiments 5 and 6 demonstrate, however, is that this is not necessary for a WI-embedded gap to be accessed. All contexts licensed both the matrix and embedded interpretations equally. Therefore, to access a gap within a WI, it suffices that the context provides information relevant to its interpretation, without necessarily ruling out alternative ones.

5.3.1.2 Amelioration effects and locality’s source

What, then, gives rise to WI effects? Clearly, there is no simple answer, nor a single cause. As we have seen, WI effects vary significantly depending on the type of island and the properties of the extracted element. I will thus narrow the focus to the specific case at hand: adjunct extraction from *whether*-islands. Even here, the answer is not quite self-evident. But I argue that the present data, paired with the insights outlined above, point towards a compelling interpretation: something akin to the Active Filler Strategy appears to be at play—that is, an independent processing pressure favouring early gap resolution. On this view, interpreting a wh-gap within the embedded clause runs counter to this default preference, thereby causing the observed degradation.¹⁸ Yet this processing

¹⁸Note, moreover, that the marginality often ascribed to these structures may be exacerbated by the fact that even early attachment of the adjunct (i.e., at the matrix verb) can yield interpretations that are pragmatically infelicitous. This is particularly true for *whether*-clauses selected by the verb *wonder*, which frequently features in examples in the literature. After all, how plausible is it that someone is genuinely concerned with *how* someone wonders about something? And are there even distinct ways of wondering? One might, perhaps, wonder “obsessively” about a specific idea, or even “while standing on one leg”, but in the absence of an appropriate context, such interpretations seem rather far-fetched. As a result, these sentences can easily end up sounding degraded, regardless of where the adjunct reconstructs. This is clearly reflected in the data, which show a strong shift in preference once a suitable context is introduced: when the embedded attachment becomes plausible, participants overwhelmingly favour it.

pressure can be overridden by providing an appropriate context, as originally suggested by Szabolcsi, effectively ameliorating adjunct extraction from *whether*-islands.

As I did for Experiments 1–4, I would like to entertain the hypothesis that amelioration effects can help reveal the nature of locality constraints themselves (following speculations by Merchant 2001, 2008; Lasnik 2001; Boeckx 2003, 2012). The key insight is this: if a constraint can be ameliorated through a specific (set of) factor(s), it must originate where those factors operate. In this light, the pattern observed here is particularly revealing. The constraint appears to stem from domain-general processing pressures, yet it is crucially sensitive to semantic and contextual factors that influence the construction of underlying syntactic representations. This suggests that the constraint responsible for WI-effects is not operating purely at the level of surface processing, nor is it a rigid syntactic ban. Instead, it must be located in a domain where computational pressures interact systematically with semantic and contextual information. More specifically, context appears to license the activation of abstract grammatical representations—in this case, representations where an adjunct modifies the embedded rather than matrix verb. Once such a representation is licensed and activated, the default processing pressures (Active Filler Strategy) can be overridden, allowing for the construction and interpretation of the WI-embedded structure. This sort of systematic interaction between abstract grammatical representations, contextual licensing, and computational pressures is precisely what we expect if the source of WI-effects lies in a domain where all these factors converge. This domain, I argue, is naturally instantiated by the Interface system described in Chapter 3.

5.3.1.3 A natural fit: the Interface

In a nutshell, I characterised the Interface as a system involving a ‘gradient of pressures,’ where the weight of grammatical and domain-general factors varies along a continuous spectrum (see Section 3.2.1.1). By hypothesis, the left edge of this space corresponds to areas where language-specific factors are more prominent, giving rise to constraints that are more straightforwardly syntactic. Towards the right edge, by contrast, domain-general pressures on sentence processing become increasingly influential, yielding constraints that inevitably reflect interactions between grammatical structure and processing dynamics.

Within this framework, the question of what underlies different locality effects seems more tractable. Specifically, WI-effects can be naturally characterised as emerging at the right edge of the Interface, where processing pressures such as the Active Filler Strategy are expected to shape the interpretation of syntactic structures. In the next chapter, I will argue that this is precisely the domain where the constraints responsible for subject–object relative clause asymmetries also arise, thus allowing us to unify distinct instances of in-

intervention locality under a single conceptual source. To anticipate the argument, I believe there is a meaningful parallel to be drawn between the findings on Spanish ORCs (Experiments 1–4) and those on English WIs. Theoretically, both involve structures affected by intervention locality. Empirically, both sets of experiments reveal that the observed degradation arises in the presence of some form of ambiguity. In the case of ORCs, speakers were initially misled by a garden path effect favouring a SRC interpretation, but were able to resolve the ambiguity via *acquired* morphosyntactic cues. In the *whether*-island cases discussed here, a similar process appears to be at work: the parser imposes a default attachment strategy, but this can be overridden when contextual cues activate an alternative underlying representation. In both cases, then, a default parse can be redirected by appropriate cues. So what appears to unify these (and arguably other) instances of intervention locality is a form of garden-path ambiguity.

Turning to SIs, these are naturally construed as effects rooted at the *left* edge of the Interface. There, the influence of language-specific constraints is predominant, giving rise to what we might call more ‘genuinely syntactic’ locality constraints. One might wonder why such effects are not simply located outside the Interface altogether; that is, to the left of its leftmost boundary. As discussed in Chapter 2, this would imply that impenetrability effects are caused by a derivational constraint. However, building on the proposal in Boeckx (2012), I will argue that such a view is too rigid, and that these effects are better captured by an interface-based representational account.¹⁹ Moreover, I will show that even in the case of syntactic impenetrability, ambiguity appears to play a central role, thereby suggesting a link between impenetrability and intervention effects.

This brings me to a final consideration, tied to the question raised in Section 5.2: should weak and strong island effects be treated as fundamentally distinct, or unified under a general locality principle? The Interface Hypothesis offers a way to resolve this apparent tension rather naturally. Although strong and weak island effects emerge from distinct underlying pressures, they can both be traced to a shared interactional space. That is, SIs and WIs, though arising at different points along the Interface gradient, are captured within a single explanatory system.

5.4 Summary

This chapter examined how native speakers of English process and interpret adjunct extraction from strong and weak islands. After reviewing the key properties and traditional

¹⁹Thus, when I refer to the impossibility of ‘generating’ structures involving SI violations, I mean that such structures cannot be successfully spelled out. WIs, by contrast, can be spelled out but are later obstructed by right-edge pressures. I return to this point in detail in Chapter 6.

typology of island constraints (Section 5.1), I presented two experimental studies investigating extraction from prototypical SIs (complex DPs) and WIs (*whether*-clauses).

Experiments 5 and 6 revealed a striking asymmetry between the two island types. Participants readily accessed gaps within WIs when provided with contextually relevant information, but gaps embedded within SIs remained inaccessible across conditions, yielding extremely low access rates even when the same contextual information was provided. This clear SI-WI asymmetry persisted across both experimental variants, ruling out alternative explanations based on recency effects or task-specific biases.

In discussing these results, I argued that they cannot be accommodated by traditional syntactic accounts that treat both island types as reflecting syntactic constraints (albeit different ones). Similarly, the specific amelioration factors discussed in the literature (D-linking, lexical complexity, finiteness effects, or contextual checklists) do not account for the present findings, as none of these factors were included in the experiments.

Instead, I proposed that the observed asymmetry reflects distinct underlying sources for SI and WI effects. The finding that gaps within SIs cannot be accessed suggests that SI effects emerge because extractions from these environments are categorically ruled out by core syntactic constraints: speakers cannot access representations that the grammar cannot generate. By contrast, the ready accessibility of gaps within WIs demonstrates that WI-constraints are not strictly syntactic in nature. Rather, the syntax can build the relevant structures, indicating that WI-effects arise from conditions operating subsequent to syntactic licensing. I further argued that WI-effects stem from processing pressures akin to the Active Filler Strategy, which favours early gap resolution. Context can override this default preference by licensing the postulation of an island-embedded gap, making adjuncts more ‘argument-like’ through direct association with particular event structures. Moreover, this account extends naturally to explain the traditional argument–adjunct asymmetry: arguments resist early attachment due to θ -theory requirements, whereas adjuncts can be discharged at the first available site. Under appropriate conditions (i.e., ambiguous θ -role assignment) arguments behave similarly to adjuncts, thereby indicating the ‘accidental’ nature of the asymmetry.

Finally, I framed these findings within the Interface Hypothesis guiding this thesis. On this view, WI and SI effects emerge from distinct sources within the same Interface system: SI effects arise at the left edge, reflecting a strict syntactic constraint; WI-effects arise at the right edge, where processing pressures interact more directly with grammatical representations.

This framework, I argued, naturally resolves the apparent tension in the ongoing debate over the source of impenetrability and intervention locality effects. Although strong and weak islands reflect different underlying pressures, they are both captured within a

single interface-based explanatory system where grammatical representations and processing strategies interact along a continuous spectrum.

6

Discussion and Implications of the Interface Hypothesis

Chapters 4 and 5 examined specific instances of intervention and impenetrability locality, respectively. By analysing patterns of (non-)amelioration in these domains, I presented arguments in support of an interface-based account. In this chapter, I discuss in more detail how these results support the Interface Hypothesis introduced in Chapter 3 and help shape its internal workings. I begin with intervention locality (Section 6.1), before turning to impenetrability (Section 6.2). Section 6.3 provides an interim summary connecting the model outlined in Chapter 3 with the empirical evidence from Experiments 1–6. I then consider some broader theoretical implications of the Interface Hypothesis and possible avenues for extension. Finally, in Section 6.5, I return to the main research questions guiding this thesis: what is the source of locality, and which approach offers the best explanation for its multifaceted nature?

6.1 Intervention locality

There is an important property that wh-islands and ORCs have in common. More generally, in fact, this property characterises WIs and (most) object A-bar dependencies: they all instantiate an intervention configuration. Descriptively, such configurations can be defined as follows: in $X \dots Z \dots Y$, an element Z , similar to X , intervenes between X and Y , with X and Y forming a syntactic dependency (see Section 2.1.1). In the case of object A-bar dependencies, the intervener Z is the subject, which occurs between the merge-site of the object (Y) and its target position (X). In WIs, the intervener Z can take different forms (see Section 5.1); in the case of wh-islands, for instance, Z is the wh-element introducing the embedded clause, while Y and X are, respectively, the origin and the target of the extracted element.

The theoretical motivation for formalising this configuration stems from the fact that structures exhibiting it typically incur a penalty. For instance, such structures may delay acquisition (see Section 4.3), hinder processing (see Sections 2.2 and 4.2), or compromise acceptability to varying degrees (see Sections 2.1.1 and 5.1). As discussed in Chapter 2, Relativised Minimality constitutes one of the earliest and most influential formal accounts of intervention. And to date, it remains the most established framework for describing these effects. RM is built on two core conditions: intervention and similarity. The intervention condition requires that ‘being in between’ be understood structurally, not linearly: Z must c-command Y but not X . The similarity condition requires that Z be *syntactically* similar to X . As reviewed in Section 2.1.1, the notion of syntactic similarity has been formalised in different ways across versions of RM. I will not repeat such differences here. In brief, however, in its recent formulations the similarity between target X and intervener Z is computed on the basis of “active” or “relevant” syntactic features (Rizzi 2018). Additionally, a set-theoretic computation calculates the overlap between feature sets, and an evaluation procedure assigns penalties accordingly. Thus, in its later developments, RM expanded beyond Rizzi’s (1990) original formulation to include principles and procedures aimed not only at capturing the difficulties caused by intervention effects, but also at predicting their amelioration at a more granular level.

The results of Experiments 1–4, however, show that these additional set-theoretic mechanisms are not optimal: they fail to predict the findings and can plausibly be replaced by independently motivated factors regulating sentence processing (see Chapter 4 and discussion below). Nevertheless, intervention configurations are consistently associated with some form of degradation, and this fact clearly requires explanation. In Chapter 4, I argued against RM’s set-theoretic approach to amelioration in favour of a disambiguation-based account. This account captures when amelioration occurs, but it

does not yet explain explicitly why intervention-affected structures are difficult in the first place. In what follows, I discuss two possible ways to address this gap: one more conservative, the other more radical.

6.1.1 Re(lati)vised Minimality

A first possibility is to integrate RM into the Interface system outlined in Chapter 3. This would be a reasonable avenue, especially considering the substantial body of evidence supporting RM's core intuition: when a structurally intervening element matches, in some relevant way, the target of a filler–gap dependency, the resulting structure is ‘more difficult.’ The innovation of the present hypothesis rather lies in its focus on the *source* of such effects. That is, the central question is where intervention locality (i.e., the principles defining RM) is encoded. Under the Interface Hypothesis, it cannot be reduced to a grammar-internal phenomenon dependent on derivational constraints in narrow syntax. Instead, it must be located at the Interface, where both language-specific and domain-general pressures interact.

Before illustrating how RM can be integrated into this framework, let me first address a crucial point (which I have only touched upon in earlier chapters). Technically, RM is a representational constraint rather than a derivational one (see Rizzi 2013b for discussion). As such, it already qualifies as compatible with the Interface system described in Chapter 3 (see especially Figure 6.2). Specifically, it could be treated as a principle operating at the leftmost edge of the spectrum, i.e. at LF.¹ This solution would not be ideal, however. The very point of conceptualising the Interface as a spectrum is to capture a ‘gradient of pressures,’ to keep the term used in Section 3.2.1.1. If RM were confined to the LF edge, we would lose the prediction that intervention effects are directly modulated by domain-general processing constraints: by hypothesis, that end of the Interface hosts rigid, language-specific constraints largely unaffected by processing considerations. Yet the results of Experiments 1–4 show that amelioration in Spanish ORCs can only be explained if we factor in general processing principles (such as those underlying garden-path effects). Thus, if ameliorations reflect the nature of the effects themselves (see discussion in Section 3.1), then RM should be located in a region of the spectrum where processing pressures exert direct influence—namely, towards the right edge.

A further consideration is that locating RM at LF would place both intervention and impenetrability locality at the same locus, while still treating them as distinct constraints.² Now, not only would this create redundancy in the system (with two distinct yet partially overlapping locality principles encoded at the same level), it would also blur the (by now

¹In principle, it could also be placed at PF, but I see no compelling reasons to prefer this option.

²I discuss the arguments for locating impenetrability at LF/PF in Section 6.2.

established) empirical distinction between the two: SIs are rigid and can hardly be circumvented, whereas WIs are comparatively flexible, with a number of factors facilitating extraction out of them. To preserve this asymmetry, RM is better situated further along the spectrum, consistent with all the empirical and theoretical arguments advanced in Chapters 4 and 5. Figure 6.1 illustrates where RM should be encoded in the Interface space defined in Chapter 3.

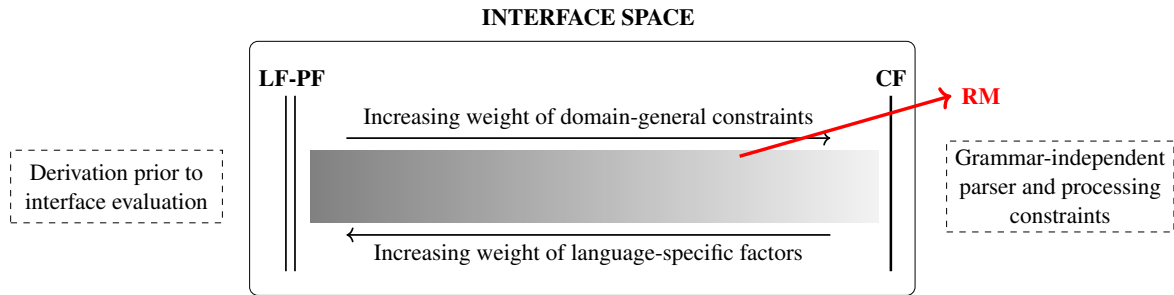


Figure 6.1: A conceptual illustration of the Interface space, as defined in Chapter 3. The red arrow indicates the location of RM under an Interface-compatible reinterpretation.

On this view, then, RM’s original intuition remains valid, but with two caveats: (i) intervention locality is primarily determined by general constraints on processing, rather than grammaticalised as a fixed constraint at LF; and (ii) the elaborate set-theoretic apparatus proposed to regulate amelioration effects is dispensed with. Such machinery not only generates incorrect predictions (see Sections 4.2 and 4.3) but also adds unnecessary theoretical complexity with dispensable grammatical primitives. A more minimal (and minimalist) approach ought to pursue principled explanations, with as little postulation as possible. The account that I argued for in Chapters 4 and 5 is an attempt in this exact direction: amelioration of intervention effects do not result from grammatically encoded computational procedures, but from the parser exploiting abstract grammatical representations to resolve ambiguity.

In sum, if we pursue this first possibility, RM would essentially be restated in an Interface-compatible version, reduced to its basic original tenets: intervention is structural, and interveners must be similar to targets. No further specifications involving featural computations or set-theoretic relations would be needed. Yet, of course, the question of *how* similarity should be determined remains unresolved. Syntactic features could in principle still be invoked, but their selection would need to rest on a definition more precise than the current one, which relies on rather ad-hoc assumptions about the features “licensing applications of external and internal merge” (Rizzi 2018: 347) and often leads to incorrect predictions (see Section 4.2.1). And even if precise, testable criteria for identifying ‘relevant’ features were found, it would still be necessary to determine what kind of feature matching qualifies as ‘similarity’—crucially, without running into

the same empirical challenges of the current system (see discussion in Chapter 4). I certainly do not exclude the possibility of an adequate solution to these issues, but at present I see no principled way to resolve them. This, then, brings me to the consideration of a more radical alternative, in which RM need not be considered a primitive of UG, but can rather be derived from independent properties of the parser.

6.1.2 RM as a non-primitive

I explained the amelioration effects observed in Spanish ORCs and in English *whether*-islands in a similar way (see Chapters 4 and 5). In both cases, I appealed to an independently motivated principle of sentence processing: the Active Filler Strategy. This, as I showed, not only fits naturally with the Interface Hypothesis, but also corroborates the original insight of Hofmeister and Sag (2010). The idea is straightforward: maintaining a filler in working memory is costly, so the parser attempts to discharge it as soon as possible. In ORCs, this leads the parser to initially attach the relative head in subject position, yielding a misanalysed SRC-reading. In *whether*-islands, the extracted adjunct is reconstructed as a modifier of the matrix verb, which is the first potential attachment site. Moreover, I showed that the parser can be assisted by robust early morphosyntactic cues (Chapter 4) or contextual gap-licensing (Chapter 5), which reduce the likelihood of such garden-path effects. This allowed me to support specific hypotheses about the source of intervention effects and the conditions under which they are ameliorated. What remained implicit, however, was how these findings bear on the formalisation of intervention itself. In other words, up to that point I had argued about where intervention effects originate, without committing explicitly to what principle causes them. In the previous section I reasoned that such a principle might still be RM (albeit with the caveats considered above). Though on closer inspection, there is probably no need to assume an independent formal principle at all: the garden-path account advanced in Chapters 4 and 5 suffices to explain both the penalty of intervention-affected structures and their amelioration.

To see this, consider once again the two core conditions of RM. As far as the intervention condition is concerned, this is naturally derived by the sheer workings of the parser. Specifically, the parser must discharge a filler X at the first available position. This proceeds unproblematically when only one such position exists, i.e. the origin Y. If, instead, another potential attachment site is encountered along the way, i.e. the position hosting Z, the parser will attempt to attach X there, only later revising its analysis. This revision-process explains the observed penalty.³ Additionally, the fact that intervention must be structural and not linear (see Section 2.1.1) is readily accounted for in the Inter-

³In this light, it is not surprising that such penalties typically manifest as processing difficulty or mild disruption rather than full ungrammaticality, at least in adult populations.

face system: the parser operates on *syntactic structures*, not linear strings. This is a point I extensively emphasised (and supported empirically) in Chapters 4 and 5, and which, as discussed there, aligns with seminal garden-path models (Pritchett 1991; Siloni 2014).

Concerning the similarity condition, the explanation is likewise natural. In particular, Z qualifies as an intervener only if it occurs in a licit attachment site for X. Similarity, then, is not a primitive (i.e., a grammatically defined status) but simply follows from whether X and Z share the properties required for selection in that position; for instance, both being DPs. On this view, two DPs are always ‘similar’ enough to generate potential garden-path effects, as they have the same distributional properties. Furthermore, RM attempts to explain the amelioration (or absence) of intervention effects by appealing to fine-grained feature distinctions between X and Z, based on language-specific criteria that determine which features are eligible for RM-computation. The system proposed here renders this additional machinery unnecessary: intervention effects are avoided whenever (morpho-syntactic or contextual) cues provide sufficient disambiguation, allowing the parser to reanalyse or avoid the misparse entirely. Importantly, this does not mean that feature-based differences between X and Y are disregarded. The proposed approach naturally predicts such distinctions to be relevant, but only insofar as features function as cues that reduce ambiguity, rather than as inputs to set-theoretic operations internal to the grammar.

On this account, then, intervention effects do not result from a grammatical constraint encoded in UG, whether represented within narrow syntax or as a representational ‘filter’ at the Interface. Rather, they are *emergent* effects modulated by the strategies employed by the parser to resolve dependencies. These, as seen, are constrained by principles of processing economy (Active Filler Strategy) and ambiguity resolution. In other words, this perspective treats the difficulties associated with intervention not as the outcome of feature-based evaluation procedures within the grammar, but as the by-product of how abstract grammatical representations interact with domain-general pressures at the Interface. In this sense, we could say that RM preserves a generally valid descriptive power, but need not be postulated as a primitive.

As I reflected in Section 3.1.1, reductions of this sort should be favoured if we pursue the definition of a minimal grammar. And reinterpreting intervention effects along these lines is, I believe, fully consistent with the spirit of Minimalism and the SMT, which aim to derive as many linguistic phenomena as possible from the interaction of grammar (1st factor) with domain-general properties (3rd factor) rather than by postulating stipulative rules of UG. In this sense, we could say that the proposal outlined here amounts to a ‘minimalist re-interpretation’ of intervention effects.

Before concluding this section, let me clarify two important points. First, the perspective pursued here does not entail discarding the general intuition behind RM-related

constraints. Derivational principles such as the Minimal Link Condition may still play a role in regulating movement. That is, the MLC requires dependencies to target the closest available element, thereby capturing intervention-like effects in the derivation itself (see Section 2.1.1). What I have argued is rather that the difficulty associated with intervention configurations in spelled-out structures should not be attributed to RM as an independent grammatical primitive. This does not undermine the theoretical soundness of its derivational counterpart: intervention effects emerge at the Interface, while the MLC can independently restrict movement in narrow syntax.

Second, and finally, this account does not aim to resolve all phenomena traditionally grouped under the label of intervention effects. It leaves room for other explanations of certain WIs. For instance, as discussed in Chapter 2, negative islands seem to have an irreducible ‘semantic nature,’ and are therefore plausibly best explained by semantic-pragmatic accounts (e.g., Szabolcsi and Zwarts 1993, 1997; Abrusán 2014). These are fully compatible with the Interface system defended here, which does not introduce innovations concerning structures’ interpretability or the conditions on their pragmatic felicitousness.

6.2 Impenetrability locality

In Chapter 5, I addressed the long-standing question of what distinguishes SIs and WIs. The empirical differences between the two are often evident (see Section 5.1), but whether, and to what extent, their underlying causes should be treated in a unified way has been widely debated. On the one hand, there is consensus on treating WIs and SIs as substantially separate phenomena. On the other, there have been attempts to unify intervention and impenetrability under a single theory, in the quest for a common ontology of locality-based phenomena (see Chapters 2 and 5; Boeckx 2012 for discussion).

Experiments 5 and 6 tested the distinction between WIs and SIs using contextual gap-licensing, which I argued can serve as a diagnostic for assessing ‘how syntactic’ a constraint is. In short, if context manipulation can render an island-embedded gap accessible for interpretation, the underlying structure must be syntactically derivable and interpretable at LF/PF (see Section 5.2 for details). The results show a clear asymmetry: gaps in WIs were consistently accessible, whereas SI gaps remained overwhelmingly inaccessible. I argued that this pattern follows naturally if intervention is treated as a flexible, emergent constraint that applies *after* Spell-Out, whereas impenetrability reflects a rigid grammar-internal ban on extraction. In the last section I showed how and where intervention should be interpreted in the Interface system. I will now turn to impenetrability.

In essence, two questions arise: where should impenetrability be encoded, and what causes it? Starting with the first, two possibilities are compatible with the empirical findings of Chapter 5. For the reasons just given, let us assume that the inaccessibility of SI-embedded gaps stems from the impossibility of generating a structure involving such extractions. This implies either that the syntax cannot derive them, or that the LF/PF interfaces fail to interpret them and reject the attempt at Spell-Out. Referring again to the representation in Chapter 3, the constraint(s) responsible for SI effects must therefore be located at the derivational level, prior to interface evaluation, or, at the latest, as an LF/PF filter on the output of Merge.

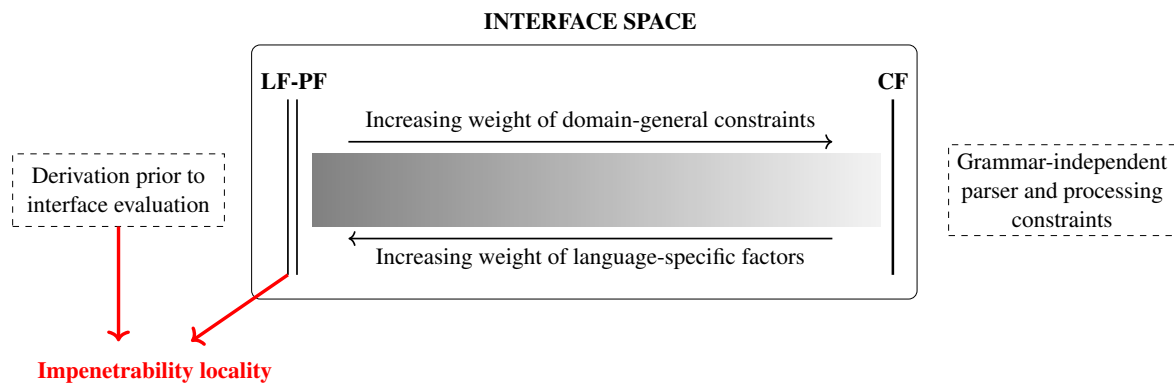


Figure 6.2: A conceptual illustration of the Interface space, as defined in Chapter 3. The red arrows indicate the two possible loci for Impenetrability locality.

This should already suggest that the questions of what causes SI effects and where they originate are inextricably linked. One cannot identify the locus of impenetrability locality without also explaining what underlies its effects. In fact, this takes us to a long-standing question in the theory of locality, where two opposed views have been offered to explain the opacity of SIs. On the one hand, derivational approaches construe strong islandhood as the outcome of restrictions on structure-building operations in narrow syntax. On the other, representational approaches identify extractions from SIs as structures not legible at the interface.⁴

Phillips (2013b: 158) characterises this contrast as follows (quoted in Boeckx 2012: 55–56):

[F]ormal grammatical constraints could exert their effects in different ways. On the one hand, a constraint could apply as a constraint on structure generation, meaning that the constraint makes it impossible for speakers to construct mental representations that violate the constraint. On the other hand, a constraint could apply as a

⁴Hornstein et al. (2007) refer to these as, respectively, “limits on rule application” and “conditions on the output(s).”

filter on generated structures, meaning that speakers are able to mentally represent structures that violate the constraint, but that they somehow also encode the fact that those representations are grammatically illicit.

However, the distinction is not always clear-cut. Indeed, whether formal constraints should be understood in derivational or representational terms often reduces to how one chooses to conceptualise the grammar itself (see Chomsky 1995, Rizzi 2001, Brody 2002 for discussion). For example, Brody (2002) argues that a purely derivational approach to syntax is actually redundant, since representational conditions at the interface are independently required, and ultimately unachievable, since even derivational systems must treat intermediate outputs as transparent representations. In this regard, he writes (Brody 2002: 23):

It is clear that derivational theories must be at least weakly representational. Take an object z , the result of merging x and y . At some later step move can only apply to y if z is a transparent rather than an opaque object, since otherwise y would not be accessible or even visible for this operation. Notice that even if move is reduced to merge and an interpretive linking operation (as in the theory of distributed chains, [Brody 1998]), the same conclusion would still hold: the interpretive link between x and y could not be established if z was opaque. The derivational theory therefore is at the same time a (weakly) representational theory with multiple (weakly) representational stages instead of just one at the interface.

Extricating all the details of this technical debate falls outside the scope of this section. The crucial point, however, is that a rigid syntactic constraint formally encoded in the grammar does not allow speakers to fully interpret or represent a structure that violates it. This holds independently of whether such a constraint is formalised as a limit on the derivation or as a ‘filter’ at the interface: in both cases, the structure cannot be Spelled Out. That said, in the case of SIs, there may still be empirical and theoretical reasons to favour one approach over the other. I turn to this in the next section. First, however, let us recall what the scope of impenetrability locality actually is.

In Section 3.2.3, I pointed out that the CED essentially reduces to a ban on extraction out of non-complements, or, following Boeckx (2012: 55), out of “adjoined domains.” These include adjuncts, complex DPs (relative clauses and sentential complements to nouns), and subjects (sentential or otherwise). At the heart of impenetrability locality, then, lies the question of why extraction is possible from complements (1a), but not from non-complements (1b, 1c).

- (1) a. Who did you see [pictures of ___]?
- b. *Who did [pictures of ___] cause Bill to leave the room?

- c. *Who did Bill arrive [after seeing (pictures of) ___]?

(from Boeckx 2012: 55)

The crucial question is what distinguishes these specific domains (i.e. non-complements) such that they give rise to impenetrability effects. In light of the debate outlined above, moreover, we should ask whether the relevant constraint applies during the derivation or acts instead as a filter at the LF/PF interfaces. Put simply: where does the CED originate, and what triggers it?

6.2.1 Deriving the CED from the PIC

In Chapter 2, I discussed how early attempts to derive the CED within the minimalist framework centred on the role of phases. Since phasehood closely resembled earlier notions of bounding nodes and barriers (see Section 2.1; Boeckx and Grohmann 2007 for discussion), it initially seemed like a natural way to capture impenetrability locality. Indeed, the idea that phases block (or severely restrict) extraction is built into the PIC.⁵

I already provided the formal definition of the PIC in Section 2.1, but the basic intuition is as follows. Syntactic derivations proceed in cycles for reasons of computational efficiency. Once a phase is completed, its content is Spelled Out and effectively ‘put aside:’ the complement of the phase head becomes inaccessible to later derivational stages, while the phase edge, i.e. the head and its specifier(s), remains available. Before Spell-Out, however, material in the phase complement may move to the edge, where it remains accessible to later operations in the derivation. This movement is triggered by general movement-inducing features at the phase edge, often called EPP-features (though various labels have been used over the years; see Section 2.1), and it is crucial. Without it, even simple cases of object wh-movement could not be derived. Under this view, then, long-distance movement is essentially an illusion: “what looks like a long-distance movement rule is in fact the conjunction of small movement steps” (Boeckx 2012: 58).

This neat picture, however, runs into a serious problem when viewed against the backdrop of the CED (Sections 2.1.3 and 3.2.3). The CED posits that complements are transparent for movement, whereas non-complements are opaque. Phase theory, by contrast, predicts the opposite: it treats the complement of a phase head as inaccessible, but keeps material at the phase edge (including specifiers) available for extraction. In other words, the PIC would predict specifiers to be more transparent than complements, thereby reversing the CED’s generalisation. As already mentioned in Chapter 2, Boeckx (2012: 62) bluntly remarks that “[t]he PIC simply gets the CED *backward*” (emphasis mine).

⁵It is not entirely clear, however, whether Chomsky explicitly intended the PIC to capture (strong) islandhood. Boeckx (2012) argues that he did (see in particular footnote 5 of Chapter 3 in his book).

Still, the minimalist drive to link the PIC and the CED remained too appealing to abandon without attempting a reconciliation. Ideally, the opacity of subjects and adjuncts should follow from independently motivated properties of the grammar, rather than stipulative rules or pre-minimalist notions like bounding nodes, barriers, L-marking, and government (see Section 2.1). And what better candidate than phases for this role? With their inherent notion of local domain, phasehood and the PIC offered a seemingly natural way to derive the CED from independent principles.

What is perhaps the most influential and explicit attempt in this direction is the proposal advanced by Müller (2010).⁶ As discussed in Chapter 2, however, this account has faced important criticism (Boeckx and Gallego 2011; Boeckx 2012). For reasons of space, I cannot review all the technical details here, but I will summarise Müller's main ideas as well as the key weaknesses highlighted in this debate.

Müller starts from four assumptions. First, all syntactic operations are feature-driven. In particular, he extends the standard assumption that Agree and Internal Merge are feature-driven to instances of External Merge.⁷ This, he argues, must also be triggered by “subcategorisation” features (see, a.o., Svenonius 1994; Collins 2002). He further assumes that features come in two types: *structure-building features* and *probe features*. The former, represented as [$\bullet F \bullet$], trigger (internal and external) Merge operations,⁸ while the latter, represented as [$*F*$], are the features involved in Agree operations. These features must abide by the *Last Resort* condition, which requires that “every syntactic operation must discharge either [$\bullet F \bullet$] or [$*F*$]” (Müller 2010: 38).

Second, features on lexical items are ordered. This means that syntactic operations triggered by such features must proceed strictly top-down: the feature at the top of the stack is discharged first and removed, then the next one becomes active, and so on. This ordering determines when a phase head becomes *inert*: once all its structure-building features are used up, it can no longer be modified.⁹

Third, all phrases are phases. As such, “the PIC forces intermediate movement steps to all phrase edges” (Müller 2010: 40).

Fourth, edge features can be added only while the phase head is still *active*. That is, when a phase head has discharged all its features, it becomes inert (or “complete” in the

⁶For a review of alternative approaches, see Müller (2011: Ch.2).

⁷He characterises External Merge as “Merge applying to items from the numeration, or from the workspace of the derivation” (Müller 2010: 38).

⁸These subsume the movement-inducing features often labelled as edge features or EPP-features (see above). In particular, [$\bullet F \bullet$] can select a specific category, say [$\bullet wh \bullet$], but can also be “blind to the properties of the attracted item” (Müller 2010: fn.7), effectively turning [$\bullet X \bullet$] into a bona fide EPP-feature.

⁹Accordingly, Müller revises the Last Resort condition as follows:

Last Resort (revised) (Müller 2010: 40)

- a. Every syntactic operation must discharge either [$\bullet F \bullet$] or [$*F*$].
- b. Only features on the top of a feature list are accessible.

sense of Chomsky 2000). This, paired with the third assumption, defines the *Edge Feature Condition*:

(2) **Edge Feature Condition** (Müller 2010: 42)

An edge feature [$\bullet X \bullet$] can be assigned to the head γ of a phase only if (a) and (b) hold:

- a. γ has not yet discharged all its structure-building or probe features.
- b. [$\bullet X \bullet$] ends up on top of γ 's list of structure-building features.

Given these premises, if an XP is last-merged in a phase, the latter's head has become inert, so no edge feature can be added to license successive-cyclic movement through the edge of XP. As a result, extraction out of that XP is impossible, effectively turning it into a 'barrier.' This is the case for external arguments and adjuncts, which are thus predicted to be opaque for movement.¹⁰ Complements, by contrast, are predicted to be transparent because they are merged earlier, leaving the phase head still active to license edge features. In sum, only non-complements ban *sub*-extraction.

Importantly, under this account the CED is not a primitive constraint but a property naturally arising from how the PIC interacts with edge feature insertion.¹¹ This is an elegant explanation and fits well with the minimalist pursuit of a 'simpler' UG (Section 3.1.1). Yet, despite its appeal, the system runs into serious empirical and theoretical challenges. I summarise below what I take to be the most prominent ones, drawing on the discussion in Boeckx (2012: 62–73).

6.2.1.1 Some challenges for Müller's (2010) system

A first question concerns the exact dynamics by which the Subject Condition should be derived. Again, once a phase head has exhausted its features, it becomes inert and cannot assign edge features to last-merged specifiers. This explains why β is blocked within α in (3) (see fn. 10).

$$(3) \quad [_{VP} [\alpha \dots \beta \dots] \vee [_{VP} V [\gamma]]]$$

Let us now proceed with the next step of the derivation assuming T is merged. Upon merger of T (a phase head in Müller's system) the complement of the previous head is transferred to Spell-Out, yielding (4).

¹⁰More technically: let α be a category last-merged in Γ , Γ the maximal projection of phase head γ . At the point in which α is merged, γ has discharged all its [$\bullet X \bullet$]-features and has thus become inert. As such, no edge feature can be assigned to α , and a category β that is part of α can therefore not move to an outer specifier of Γ . Hence, when the derivation proceeds by merging Γ with a new element from the numeration, the PIC makes β in α inaccessible (Müller 2010: 42–43).

¹¹I refer directly to Müller (2010) for further technicalities of the system and how it can be expanded to other phenomena like "freezing effects" and "melting effects." See also Müller (2011) for an extended treatment.

- (4) [TP T [vP [α ... β ...] v {~~vP~~-V{ γ } }]]

At this point α is visible to T, as it sits at the edge of the vP phase. Given this configuration, Boeckx (2012: 66) wonders what would prevent T, an active phase head, from being assigned an edge feature capable of attracting β to Spec,TP, as in (5).

- (5) [TP [β] T [vP [α ... β ...] v {~~vP~~-V{ γ } }]]

If this were possible, then β would be visible to C, which could then trigger movement of β to its specifier (6), effectively contravening the Subject Condition.

- (6) [CP [β] C [TP [< β >] T {~~vP~~-[α ...< β >...]-v {~~vP~~-V{ γ } }]]]

Now, under Müller's account, the step that Boeckx envisages in (5) should in fact not be possible. Albeit active, the phase head T cannot target β directly, as β is *not* at the edge of vP: it is at the edge of α , which itself occupies the edge of vP.¹² For β to be visible to T, it would have had to move to Spec,vP at the previous derivational stage, but such a movement is made impossible by the inert status of v by the time α is merged.

Regardless of these details, however, I believe Boeckx's criticism is on point. Indeed, under Müller's account, external arguments are predicted to be *always* opaque *qua* last-merged specifiers. This, however, runs counter to crucial facts from Romance. Boeckx discusses an example from Uriagereka (1988: 118) showing that Spanish subjects block sub-extraction in pre-verbal position (7a) but allow it when they remain in situ (7b).

- (7) a. * De qué conferenciantes_i C te parece que [TP [las propuestas
of what speakers CL-to-you seem-3.SG that the proposals
t_i]_j T_S me_z van a impresionar_v [vP t_j v t_z t_v]] ?
CL-to-me go-3.PL to impress-INF

- b. ? De qué conferenciantes_i C te parece que [TP T_S me_z
of what speakers CL-to-you seem-3.SG that CL-to-me
van a impresionar_v [vP [las propuestas t_i] v t_z t_v]] ?
go-3.PL to impress-INF the proposals
'Which speakers does it seem to you that the proposals by will impress me?'

(Adapted from Boeckx 2012: 67)

I should add that a similar asymmetry can also be observed in Italian, where sub-extraction from in-situ subjects is permitted (8b), whereas pre-verbal subjects act as irremediable 'barriers' to extraction (8a).

- (8) a. * Di chi [il figlio ___] ha telefonato?
of whom the son has phoned

¹²Müller (2010: fn.13) clarifies that his system "presupposes a nonrecursive notion of edge, such that the edge β of a category α is not part of the edge of a phase Γ even if α is part of the edge of Γ ."

- b. Di chi ha telefonato [il figlio ___]?
 of whom has phoned the son
 ‘Whose son called?’

As is standardly assumed, external arguments, including the subjects in (7) and (8), are (last-)merged as specifiers in Spec,vP (see, a.o., Larson 1988, 1990, 2014; Koopman and Sportiche 1991; Chomsky 1995; Kratzer 1996; Belletti 2001, 2004; Adger 2003; Hoot and Ebert 2021 for discussion). Under Müller’s account, then, the extractions in (7a) and (8a) should invariably be banned. In other words, Müller (2010) treats subject specifiers as uniformly opaque, but the data do not support this: the examples above show that being a (last-merged) specifier is not, by itself, sufficient to trigger impenetrability locality. I return to this point below, as it plays a crucial role in formally accounting for the CED.

A second issue arises when considering the case of DOM in Spanish (see Section 4.1 and references therein for an overview). Following Torrego (1998) and Rodríguez-Mondoñedo (2007), DOM-marked internal arguments are raised from Comp,VP to a specifier position in vP, above the external argument.¹³ In the system advanced by Müller, this movement should be triggered by an [$\bullet$ X $\bullet$] on v, which becomes inert after discharging it. In other words, DOM-marked internal arguments, by moving to Spec,vP, instantiate a configuration matching that of external arguments and are therefore predicted to ban sub-extraction. However, Demonte (1987: 153) shows that such domains are, in fact, transparent (example taken from Boeckx 2012: 67):

- (9) [_{CP} De qué amigo_i C entregaste [a la hija t_i] a la policía]?
 of what friend gave-2SG to the daughter to the police
 ‘What friend did you give the daughter of to the police?’

(From Boeckx 2012: 67)

I have additionally checked with several other (Peninsular) Spanish speakers, who agree that sentences like (10, 11) are acceptable.

- (10) [De qué equipo] conociste [al futbolista ___]?
 of what team met-2sg DOM.the footballer
 ‘Which team’s footballer did you meet?’
- (11) [De quién] conociste [al hijo ___]?
 of whom met-2sg DOM.the son
 ‘Whose son did you meet?’

¹³I return to this in more detail in Section 6.2.2.2.1, where I illustrate an alternative analysis proposed in López (2012). As I will discuss, sub-extraction from DOM-marked objects remain problematic even when adopting the insights presented in Boeckx’s proposal.

A further empirical challenge for Müller's system concerns extraction from adjuncts. Müller (2010: 46) argues that "[t]he barrier status of adjuncts follows immediately if we assume that adjuncts are to be reanalyzed as last-merged specifiers of functional projections (see Alexiadou 1997; Cinque 1999)." That is, since adjuncts are analysed as derivationally and configurationally akin to subjects, they are predicted to equally block sub-extraction. This is illustrated in the examples in (12), taken from Müller (2010: 46).

- (12) a. * Who did you get jealous [_{CP} because I talked to ___]?
 b. * To whom did they leave [_{CP} before speaking ___]?
 c. * Who did they leave [_{CP} before speaking to ___]?

(from Müller 2010: 46)

However, it has been observed that adjuncts do permit sub-extraction under certain circumstances (see Section 5.1). For instance, Truswell (2007) discusses the following cases:

- (13) a. What are you working so hard [in order to achieve ___]?
 b. Who did John go home [after talking to ___]?
 c. What did John drive Mary crazy [whistling ___]?

As Boeckx notes, these apparent exceptions might still be captured within Müller's account if one adopts Truswell's (2007) analysis, which treats such adjuncts as merged low in the vP structure (see Boeckx 2012 fn. 14 and 15 of Ch. 3). If the adjuncts in (13) are not late-merged, they are expected to be transparent, just like other non-late-merged complements. Nevertheless, Boeckx maintains that this analysis cannot easily be extended to other cases, such as tensed adjuncts, which seem to remain transparent despite being late-merged:¹⁴

- (14) [Which book_i did you say [that [if Quinn ever read t_i] he would abandon linguistics]]?

(Boeckx 2012: 68)

In addition to these empirical challenges, Boeckx (2012) argues that Müller's account rests on assumptions that are difficult to reconcile with the minimalist drive for *principled* explanations. I cannot review all details here, but Boeckx's main criticisms target the assumptions concerning structure-building features and their hierarchical arrangement. In this regard, he writes (Boeckx 2012: 69):

¹⁴See Boeckx (2012: 68) and references therein, including Etxepare (1999) and Hornstein (2001), for further discussion.

Unfortunately, Müller (2010) offers no motivation for the specific features he uses, nor for their hierarchy. This failure is, I think, fatal, for Müller's deduction of CED effects in fact relies, not on the PIC, but on those features and their orderings. This is *very* disappointing in a minimalist context, where the goal is not merely to describe linguistic patterns (after all, these have been described fairly well in previous frameworks), but rather to understand what they follow from: why they are this way.

Boeckx further highlights a problem concerning edge features, one that Müller (2010: fn. 3) himself acknowledges: the addition of edge features to otherwise active heads violates the *Inclusiveness Condition* (Chomsky 1995), according to which the output of a syntactic derivation must not contain any elements that were not already present in the lexical items selected for the derivation. He also questions two additional assumptions: (i) that all phrases are phases, and (ii) the precise syntactic nature of edge features, which he claims depart significantly from Chomsky's original, supposedly more "minimal," formulations (see Boeckx 2012: 68–72). Based on these observations, then, Boeckx concludes that Müller's analysis relies on too many stipulations, and rejects the possibility to truly *derive* the CED from the PIC (Boeckx 2012: 72):

[A]t least for the time being, it seems clear to me that [Müller's] analysis rests on stipulations that are no better than those used in a Barriers-based framework. [...] The clearest conclusion that can be drawn from Müller's (2010) approach is that the PIC is not directly related to CED effects.

In sum, Müller's proposal offers a coherent and elegant attempt to integrate the CED into broader phase-based dynamics, yet the PIC does not seem to be the most suitable candidate for interpreting SI effects. As we have seen, indeed, this account raises both empirical and theoretical issues. And while some of Boeckx's critiques may be overly uncompromising, the broader point stands: we may need to look beyond the PIC and derivational dynamics for a principled explanation of impenetrability locality. As I discuss in the next section, there are reasons to believe that an interface-based approach may offer a more promising way forward.

6.2.2 Deriving the CED from the interface-systems

One of the earliest accounts to explicitly factor in the role of interface systems is presented in Uriagereka (1999b). In a nutshell, Uriagereka builds his proposal on the observation that Kayne's (1994) LCA requires syntax to generate strictly monotonic structures. That is, in order to map hierarchical representations onto linearisation statements in a straightforward way, PF must be fed trees in which each element is introduced as a single unit and one step at a time. Problems arise, however, when a complex constituent such as an

adjunct is merged with the main spine: in such cases, multiple nodes are integrated simultaneously, yielding a non-monotonic configuration that PF cannot interpret. To escape this impasse, the interface system forces early Spell-Out of the offending structure, allowing it to be “compactified” into an atomic unit (see Boeckx 2012: 77) and subsequently merged as a single object with the rest of the derivation. In other words, to manage linearisation efficiently, the derivation transfers certain domains early, thereby indirectly imposing locality restrictions on them. Early Spell-Out renders such environments ‘frozen,’ preventing further syntactic operations from applying to their content. This, in turn, accounts for the strong islandhood of subjects, complex left branches, and adjuncts: if not atomised via early transfer, each of these would yield an intractable configuration for PF.

This approach appears to be overly restrictive, however. If early Spell Out is triggered whenever a complex syntactic object is adjoined to the ongoing derivation, this entails that all non-complements are deemed indiscriminately opaque. But as I have already emphasised above, there is evidence showing that in some cases SIs *can* be circumvented.¹⁵ More generally, Boeckx (2012) takes the possibility of “repairing” island effects as a clear indication that their origin is to be sought outside the derivation proper, even when the derivation is “influenced” by interface requirements, as in Uriagereka’s account.¹⁶

The rationale is straightforward: if movement out of an island can be repaired, then such an operation must have been constructed in the first place. As Boeckx (2012: 79) puts it:

Island repair poses a serious problem for derivational approaches to islandhood, [...] for [these] types of accounts are designed to prevent the island-violating extraction from taking place. It is thus a very serious puzzle for such approaches if it indeed turns out that the relevant extraction can be shown to have taken place, only to be legitimized by a subsequent, post-syntactic operation. How can something be repaired if it was never constructed in the first place? How can something be legitimized in certain contexts if it is predicted never to happen in any context? It is phenomena like island repair that seem to me to tip the balance in favor of representational, filter-based accounts of island effects, according to which, instead of banning the application of certain rules outright, certain outputs of perfectly possible syntactic operations are eventually deemed illicit.

¹⁵As I mentioned in Section 3.1 (fn. 1), the terms *circumvention*, *repair*, *amelioration*, and *obviation* can be treated as synonyms, although they point to (partly) different theoretical traditions.

¹⁶Boeckx suggests that Uriagereka’s proposal lies somewhere between a derivational and a representational account, and this seems consistent with Uriagereka’s own interpretation (Uriagereka 1999b: 276). That said, it is not entirely clear to me how Uriagereka’s approach differs substantially from purely representational accounts like Boeckx’s (detailed below): in both cases, islandhood arises because certain syntactic configurations cannot be interpreted by the interface systems, thereby triggering early Spell-Out. The main difference, as I see it, is that Uriagereka construes the interfaces as ‘too sensitive’ or overly restrictive, incorrectly banning an entire class of domains (adjoined non-complements) from being merged unless they are first atomised via early transfer.

Once again, I can emphasise that the amelioration of island effects (a recurring theme throughout this thesis) plays a crucial role in shaping our understanding of locality. Beyond the cases of island obviation discussed above (see especially Chapter 5), Boeckx reviews four specific “repair strategies” which, starting from Ross (1967), have been shown to salvage SI effects across a wide range of structures and languages: pied-piping, sluicing, resumption, and *wh*-in-situ.¹⁷

Pied-piping is perhaps the least surprising. The key observation is that although movement of a specific element out of an island is banned (15a), if the entire island domain pied-pipes along with it, the dependency becomes licit (15b). Essentially, this shows that impenetrability locality constrains *sub-extraction from*, rather than *extraction of*, an island domain.

- (15) a. *Whose did you buy [___ book]?
b. [Whose book] did you buy?

(Boeckx 2012: 81)

Sluicing (IP-ellipsis) phenomena are also known to improve acceptability in island contexts (Ross 1969; Lasnik 2001; Merchant 2001, 2008). When the island domain is contained within the ellipsis site (16b), the degradation observed for ‘overt’ extraction (16a) largely disappears.

- (16) a. *Who did John refute [the claim that Peter saw ___]?
b. John refuted the claim that Peter saw someone, but I can’t remember who
[elided: <John refuted the claim that Peter saw ___>]

(Boeckx 2012: 82)

Resumption is another well-attested repair strategy: the presence of an overt pronoun instead of a gap has long been observed to improve acceptability cross-linguistically.¹⁸ In English, for instance, although typically perceived as substandard, the sentence in (17b) is judged more acceptable than (17a):

- (17) a. *Which woman did John started laughing [after ___ kissed Bill]?
b. Which woman did John started laughing [after *she* kissed Bill]?

(Adapted from Boeckx 2012: 81)

¹⁷Importantly, these strategies are not universal, and Boeckx concedes that his arguments remain “very tentative,” since “we need to understand better the nature of all the situations leading to island repair before we can be sure that our interpretation of the remarkable phenomenon of island alleviation can be said to be on the right track” (Boeckx 2012: 96).

¹⁸As discussed in Section (2.3), however, Hawkins (1999) shows that the preference for overt pronouns depends on the inherent complexity of the dependency, as determined by inferential complexity hierarchies.

Finally, Boeckx (2012: 87–88) reasons that *wh*-in-situ should also be included among island-repair strategies. Cross-linguistic research has shown that these structures are often ‘immune’ to island effects, as observed, for instance, in Chinese (see a.o. Huang 1982; Lasnik and Saito 1984; Pesetsky 1987 for discussion). Arguably, this parallels what happens with echo-questions in languages like English (18) or Italian (19): although pragmatically marked, in-situ questions are perfectly acceptable compared to their ex-situ counterparts.

- (18) a. * Who didn’t you see [the boy who kissed ___]?
 b. You didn’t see the boy who kissed *who*?
- (19) a. * Chi non hai visto [il ragazzo che ha baciato ___]?
 who not have.2SG seen the boy who hae.3SG kissed
 b. Non hai visto il ragazzo che ha baciato *chi*?
 not have.2SG seen the boy who have.3sg kissed who
 ‘You didn’t see the boy who kissed who?’

Let me clarify why these repair strategies are theoretically crucial. If sluicing, resumption, and *wh*-in-situ can all be traced back to derivations involving movement, then we can reasonably assume that the ban underlying SIs should not be a derivational one. Otherwise, it would remain mysterious why the same (type of) movement is blocked in some cases but licensed in others.¹⁹ A more parsimonious explanation is that movement is unconstrained, but certain representations are filtered out by the interfaces. This is precisely the line of argument pursued by Boeckx (2012). Following Lasnik (2001), Merchant (2001, 2008), Boeckx (2003) and Fox and Lasnik (2003), he argues that both sluicing and resumption involve movement. In fact, he suggests that sluicing can be analysed as a special case of resumption with a silent *pro* (Wang 2007; Boeckx 2008a). Moreover, drawing on Watanabe (1992) and Cable (2010), he proposes that *wh*-in-situ is best analysed as a case of “reverse resumption” (see Boeckx 2012: 81–96 and references therein for details). Hence, under this unified analysis, sluicing, resumption, and *wh*-in-situ instantiate a single underlying mechanism involving movement.

This unification is critical for two reasons. First, if movement is indeed involved, we have strong grounds to reject derivational bans on island-crossing (see above). Second, if repair strategies work uniformly, SIs themselves must be unified phenomena. As Boeckx puts it:

I would like to point out for now that the possibility of a unified configuration of island repair suggests that there ought to be an underlying unity to (strong) islands.

¹⁹Repair effects via pied-piping are compatible even with derivational accounts, since the entire spelled-out constituent moves.

Wouldn't it indeed be really puzzling if genuinely different conditions on transformations could be uniformly repaired by a single strategy? It seems much more plausible to assume that the reason that generalized resumption works to circumvent all strong islands is because these all have something in common.

Incidentally, this reasoning chimes with the one I adopted in Chapter 5, albeit 'in reverse.' There, I argued that if the same amelioration strategy yields different effects across strong and weak islands, then we should conclude that the constraints underlying the two are also different.

Back to the problem of the CED, maintaining that it must be accounted for in representational terms is only the first step. The next, obvious step is to identify what exactly causes the interface system to ban extraction from SIs. Again, repair phenomena provide an instructive insight: all reviewed strategies somehow imply that there is no genuine gap inside the 'repaired' island. With pied-piping, no island-internal gap arises because the whole island is moved. With resumption and *wh*-in-situ, the would-be gap is filled by a pronoun or a *wh*-element. And in sluicing, the entire island is unpronounced. Because these strategies either replace the gap with pronounced material or place it inside a non-pronounced domain, many scholars have concluded that SI effects are best explained in terms of a PF filter that is sensitive to the presence of gaps in certain domains.

6.2.2.1 The CED and PF

The core idea of PF-oriented proposals is that gaps created by movement inside islands are problematic for PF's linearisation task (Lasnik 2001; Merchant 2001, 2008; Fox and Pesetsky 2005; Hornstein et al. 2007). For example, Fox and Pesetsky (2005) argue that island effects arise because PF cannot resolve contradictory linearisation statements.²⁰

Their account starts from the assumption that the relative order of elements in the derivation is fixed at the completion of each phase: once a phase is Spelled Out, its linear order cannot be altered. Consider a simple object *wh*-movement structure such as (20):

(20) [_{CP} What_i did John_j [_{vP} <John_j> eat <what_i>]]

Focusing on the relative order between *what* and *eat*, we see that it differs across phases: in vP, the order is *V* > *Obj*, whereas in CP it is *Obj* > *V*. Since both orders are fixed at Spell-Out, PF must attempt to linearise two contradictory statements, which it cannot do. To avoid this problem, Fox and Pesetsky (2005) argue that the *wh*-object must first move to the edge of vP, as shown in (21):

(21) [_{vP} <what> John eat <what>]]

²⁰Note, however, that the implications for SIs emerge within a broader programmatic approach; impenetrability effects are not the direct focus of the authors' argument.

This way, the relative order between the object and the verb is consistently $\text{Obj} > \text{V}$ across phases. Essentially, then, successive cyclicity follows from PF's requirement to preserve ordering: intermediate "escape hatches" are not primitives of the theory but emerge as a by-product of linearisation constraints.²¹ This view also explains why islands "repaired" via sluicing and resumption are unproblematic. In sluicing, PF does not need to linearise the elided domain; in resumption, the pronounced pronoun matches the PF ordering directly (and the same holds for *wh*-in-situ).

However, the central question remains: why are 'regular' islands problematic in the first place? For Fox and Pesetsky (2005), the problem with SIs is that movement to the edge (like in 21) cannot take place, which renders the structure uninterpretable at PF. But this simply pushes the puzzle back, so to speak: why are phase edges inaccessible in (all and only) SIs? On this crucial point, Fox and Pesetsky (2005) leave the issue largely unaddressed. In a way, then, the real cause of impenetrability "remains a mysterious syntactic fact" (Boeckx 2012: 87). According to Boeckx, this explanatory gap is characteristic of most linearisation-based approaches, which is why he ultimately turns to alternative interface dynamics to explain the CED.

6.2.2.2 The CED and LF

If the problem with island-contained gaps does not stem from PF requirements, a natural alternative is to locate it at LF. Boeckx (2003), building on insights from Richards (2001), argues that gaps within SIs cannot be reconstructed (i.e., interpreted at LF). In his view, the uninterpretability of such gaps arises from the mechanics of movement itself. Following Chomsky (2000), he assumes that movement involves several ingredients: matching, Agree, copying and remerging. He then suggests that SIs specifically block configurations where agreement is required. Since movement dependencies can sometimes occur without the need to initiate an Agree operation, we occasionally observe movement across islands. However, when interpretation demands agreement in addition to matching, copying and remerging, SIs intervene, rendering reconstruction unavailable.²²

This system, though well-argued, raises several problems, which Boeckx himself acknowledges in his (2012) book. In short, he emphasises that his (2003) account relies too heavily on assumptions about feature relations and the internal decomposition of Merge, and that such assumptions are neither empirically supported nor, crucially, 'minimal enough' (see Boeckx 2012: 106–107 for details). For this reason, Boeckx proposes a different account of the CED, one that still treats movement as central but avoids building

²¹Crucially, though, even in Chomsky's (1995) original formulation the PIC and escape hatches were never conceived as true primitives but rather as derived constraints motivated by principles of economy.

²²I refer directly to Boeckx (2003) for the specifics of this account.

the explanation on a fine-grained decomposition of its ingredients.

He builds on work by Takahashi (1994), Rizzi (2006), Rizzi and Shlonsky (2008) and Boeckx (2008a) and shifts the focus from the internal mechanics of (re)Merge to how its implementation affects the interface legibility of the resulting chain. A central starting point is that independent theoretical work has indicated A-chains to be particularly taxing for the external systems that take syntactic representations as their input (Chomsky 2007, 2008; Richards 2007; Epstein et al. 2012). In particular, the A-relation established when the subject moves to Spec,TP is characterised as especially complex. To alleviate this burden and ensure the derivation proceeds efficiently, such movement is argued to trigger early Spell-Out of the relevant syntactic domain. Boeckx elucidates this as follows (Boeckx 2012: 113):

The claim I would like to make is that this obligatory step of spell-out *indirectly* leads to [...] (strong) island effects [...]. Specifically, I claim that if the portion of the tree being spelled-out contains a unit that can be regarded as independent and ‘complete,’ such as a complete A-chain, or an adjunct phrase, such a unit will be processed and considered ‘closed’ by the external systems interacting with syntax, leading to its opacification. If at a later stage in the processing of the syntactic derivation the external systems receive a portion of structure that demands reprocessing of the structure that was considered closed, an island effect will result. To put it concisely: early closure, induced by early spell-out, leads to islandhood.

Interestingly, this excerpt also points out that Boeckx treats A-chains and adjuncts as closely related: both induce early Spell-Out, and for similar reasons. So, before I address why A-chains are considered so syntactically complex, let me first spend a few words on adjunction.

In this respect, Boeckx argues that adjuncts trigger early Spell-Out to resolve otherwise intractable labelling problems. Following Chomsky (2004a), he adopts the distinction between Set-Merge and Pair-Merge. Whereas the former produces asymmetric structures with unambiguous labels, the latter gives rise to complex labels of the form $\{\langle A, B \rangle, \{A, B\}\}$, effectively creating a “doubly rooted” configuration. Because adjunction produces such labels, immediate Spell-Out is required to “simplify” the structure (see also Narita 2011). The adjunct, once spelled out, can then re-enter the derivation as an atomic syntactic object. Importantly, assuming adjuncts are phases, this transfer involves the entire adjunct, including both its complement and its edge (Raposo 2002). As a result, adjuncts become opaque: once transferred, their content is fully inaccessible to further operations. In a way, “it is as if adjuncts were treated like independent clauses; closed units, hence islands from the perspective of the independent systems” (Boeckx 2012: 113–114).

In later works, Chomsky abandons the distinction between Set- and Pair-Merge, arguing that there is only one type of binary, set-forming Merge (Chomsky 2013, 2015; Chomsky et al. 2023; see also Oseki 2015 for discussion). Yet the original insight that adjuncts are built on a “separate plane” (Chomsky 2004a: 118) is still largely endorsed. More generally, the idea that derivations proceed by building up structures in multiple ‘tiers’ or workspaces is by now a widespread assumption in formal syntax (Müller 2004a; De Belder and van Craenenbroeck 2015; Collins and Stabler 2016; Chomsky et al. 2019; Krivochen 2023; Pinzin and Mattiuzzi 2025). Overall, then, whether the explanation appeals to labelling or to more general computational properties of the merge algorithm, linguists converge on the idea that adjuncts are “completed” substructures attached to the main derivation (see Milway 2022 for discussion). Incidentally, similar intuitions can also be extended to complex DPs, which can be seen as syntactic chunks treated as wholes in the main derivation (Bošković 2015).

To reiterate: adjuncts are merged to the ongoing derivation as spelled-out, monolithic syntactic objects impermeable to later sub-extraction. This explains the ‘adjunct part’ of the CED. Let us now return to the ‘subject part.’ As we have seen, Boeckx (2012) argues that A-chains, much like adjuncts, trigger Spell Out in order to reduce derivational complexity. According to him, the source of this effect lies in the system of *Feature Inheritance* developed by (Chomsky 2007, 2008). Specifically, Chomsky maintains that the A-type features (ϕ , EPP, etc...) responsible for attracting the subject to Spec,TP are *inherited* by T from C (see also Richards 2007; Ouali 2008; Gallego 2014). Crucially, this very mechanism of inheritance marks the completion of the CP phase and thus triggers Spell-Out of C’s complement.²³

Epstein et al. (2012), however, reason that if movement of the subject to Spec,TP can only take place after C is merged (and ‘passes on’ its A-features to T), then this movement does not target the root node. This, they claim, would result in a counter-cyclic derivation and violate the *Extension Condition*.²⁴

To avoid counter-cyclicity, Epstein et al. (2012), propose that C-to-T feature inheritance yields a doubly-rooted structure, where C stands at the same hierarchical level of T, as shown in (22), adapted from Epstein et al. (2012: 256).²⁵

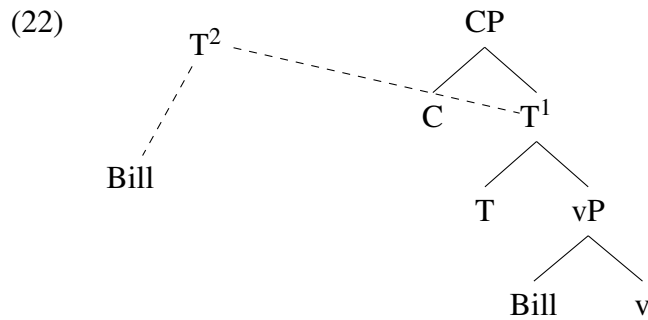
²³Boeckx does not elaborate this point, simply observing that “Chomsky’s reasoning is quite complex” (Boeckx 2003: 114); a remark with which it is hard to disagree.

²⁴The Extension Condition (Chomsky 1995) requires that Merge always apply at the root. A similar idea was already expressed in Chomsky’s (1973) *Strict Cycle Condition*, which, however, applied to *all* syntactic operations, not just Merge:

Strict Cycle Condition (Chomsky 1973):

No operation can apply to a domain dominated by a cyclic node α in such a way as to affect solely a proper subdomain of α dominated by a node β which is also a cyclic node.

²⁵In particular, the authors argue that the counter-cyclic internal merge operation undergone by the raised subject generates a “two-peaked object”. Note that they refer to (22) as an “informal graph-theoretic”



The complexity of this structure, according to the authors, leads to immediate Spell Out of TP, which results in the opacification of the subject. Boeckx's describes this process as follows (Boeckx 2012: 114):

Spelling out the complement of C will have the effect of spelling out the entire A-chain formed by raising of the subject phrase. This, I claim, is what turns the latter into an island. Like adjoined structures, A-chains are complete enough for the external systems to process as independent units [...], closing them as soon as they are processed, thereby turning them into islands.

He makes the point even more explicit when discussing the effect of Spell-Out on the subject chain (Boeckx 2012: 115) (emphasis mine):

[The] immediate spell-out of TP [...] will have the effect of *trapping the subject chain into the portion of the tree being spelled-out*, leading to subject island.

So, on this view, because the entire TP is transferred to the interface systems, further syntactic operations cannot look into the subject phrase, as this is embedded within the TP. This, however, differs in a subtle yet crucial way from the generalisation Boeckx proposes just a few lines below (emphasis mine):

(23) **Boeckx's (2012) CED Generalisation**

Structures that induce immediate spell-out (e.g., doubly rooted, adjoined structures) are islands *for those elements* (and the elements that these properly contain) *that led to the construction of such structures* and to which the external systems can assign independent, autonomous, complete status (A-chains, adjoined phrases).

representation. Formally, this operation “will produce two distinct but intersecting set-theoretic syntactic objects, SOs, which happen to share a term, namely, $T1 = \{T, vP\}$. That is, as a result of this ‘counter cyclic’ IM [internal merge], NS [narrow syntax] yields two intersecting sets: $\{Bill, T1\}$ and $\{C, T1\}$.” And thus, they “argue that these intersecting set-theoretic SOs do not have a single root; there is no single category that (reflexively) contains all terms” (Epstein et al. 2012: 256).

In some sense, it seems as if the opacification (and the ensuing island effect) were *relativised* to the elements that induce Spell-Out. Take the case of subjects: by moving to Spec,TP, they yield a structure that is too complex for the external system, and thus trigger Spell-Out. Because it is the subject that triggers Spell-Out, it is the subject that is turned into an island. However, it is not clear how exactly this relativisation is achieved formally: if the entire complement of C is spelled out, then the entire TP should be inaccessible, including its complement. This is crucial, because the whole point of the CED is, of course, to explain the opacity of non-complements while keeping complements transparent. I will return to this issue in the next section, where I take into account this and other potential challenges to the system envisaged by Boeckx. First, let me highlight some interesting theoretical reflections put forward by Boeckx in illustrating his account.

Just after the presentation of the generalisation in (23), Boeckx clarifies the core underlying insight of the system he proposes. Essentially, he seems to suggest that the interface systems tasked with interpreting syntactic structures operate under the same computational constraints described for language processing models. In other words, when reading off the output of narrow syntax, these external systems behave much like a parser: they are sensitive to domain-general processing limitations. As such, this ‘processing procedure’ is naturally expected to display properties familiar from general parsing dynamics. And this, I believe, is exactly what emerges from Boeckx’s description of the interaction between A- and A-bar chains, where he makes direct reference to ‘garden-path-like’ dynamics (Boeckx 2012: 115) (emphasis mine):

The claim being made here is that trying to integrate an A-bar chain into one of the structures that the external systems have already processed and considered closed would give rise to *a species of garden path*: a unit that the external system had previously closed would have to be re-opened to accommodate an extension (the A-bar chain), causing the external system to ‘gag.’

This link to parsing becomes even clearer when Boeckx mentions how his hypothesis is compatible with reductionist perspectives on locality (Boeckx 2012: 118):

I believe that the type of solution sketched here could dovetail with reductionist accounts of island effects based on processing factors [...]. It strikes me as quite plausible indeed to entertain the idea that doubly rooted structures that pose problems for chain-formation create on-line processing problems, problems that may be alleviated in certain circumstances if the parser can find an alternative parse or can resort to a heuristic strategy to recover the intended meaning [...].

As I will discuss in Section 6.2.4, I find it more plausible that on-line processing is not *directly* sensitive to highly abstract grammatical categories such as A- and A-bar features. But I share with Boeckx the broader idea that the systems responsible for assessing

the well-formedness of syntactic representations (LF and PF) are constrained by the same computational pressures that affect ‘later’ stages of processing. In other words, while LF/PF and the parser may be sensitive to different grammatical categories and/or factors, they are both driven by a common thrust towards efficiency. This, ultimately, imposes constraints designed to minimise the risk of forming potentially ambiguous interpretations/parses. I will return to the notion of ambiguity in Section 6.2.4, drawing again on Boeckx’s familiarity with the historical development of generative grammar to show that this concept has, in fact, been central to the theory since its earliest formulations.

6.2.2.2.1 Some challenges for Boeckx’s (2012) system

As I have just emphasised, Boeckx’s proposal is highly compatible with the Interface system presented in Chapter 3. Indeed, his broader idea of giving “priority to the interfaces” has been a direct source of inspiration for the hypothesis defended in this thesis. That said, the technical details of his account of the CED raise several non-negligible issues.

First, as mentioned earlier, it remains unclear how the Spell-Out mechanism Boeckx proposes can restrict opacification specifically to the subject. To recap the core idea: C percolates its A-features to T, which attracts the subject to satisfy the EPP; this movement forms an A-chain, which in turn triggers Spell-Out of TP. Compared to Müller’s (2010) proposal, this has an immediate advantage: in-situ subjects are predicted to remain transparent, since they do not form A-chains and therefore do not trigger early Spell-Out. However, if the entire complement of C is transferred, then a direct object in Comp,VP should be as ‘locked in’ as the subject in Spec,TP. Boeckx claims that complements are “immune” to the Spell-Out problem because they do not establish A-chains: no chain, no (early) Spell-Out. But, again, this only works if the opacification effect can somehow be relativised to the syntactic object that causes it. Exactly *how* and *why* this relativisation obtains, however, is not explored, leaving unclear why objects remain transparent when subject movement triggers Spell-Out of the whole TP.

A related problem emerges when we examine this ‘relativisation’ idea more closely. In (23), Boeckx claims that “[s]tructures that induce immediate Spell-Out [...] are islands for those elements (and the elements they properly contain) that led to the construction of such structures.” So, in the case at stake in (22), the doubly-rooted TP/CP should be an island for the subject, i.e. the ‘culprit’ for this complex structure. And following Boeckx’s definition, not only *sub-extraction from* but also *extraction of* the subject should be banned, which is, of course, problematic. To put it another way, the doubly-rooted CP/TP triggers immediate Spell-Out, and therefore becomes an island *for the element that led to its construction*. But if the TP/CP is an island for the subject, we would wrongly predict that extracting the entire subject should also be banned. This is clearly not the case, as shown by the well-known contrast:

- (24) a. [Whose pictures] caused a riot?
 b. *Who did [pictures of ____] cause a riot?

Boeckx himself discusses this type of contrast explicitly when addressing pied-piping repair strategies (see above). But if his system can predict the acceptability of (24a), this is left implicit, as is the question of how it would do so.²⁶

A third issue concerns objects more directly. Boeckx notes that objects naturally form A-relations (for instance, when they are assigned case). But crucially, they typically do not form A-chains, since they normally do not undergo A-movement. As he puts it (Boeckx 2012: 116):

Because no A-chain need be formed in the case of direct objects, no doubly-rooted structure [...] is formed, hence, according to [Epstein et al. 2012], no immediate spell-out needs to take place. Since spell-out does not take place at that point, no early closure leading to islandhood will ensue.

However, when objects do form A-chains, Boeckx predicts that they should behave “just like subjects” and become islands. He illustrates this using Exceptional Case Marking (ECM) contexts such as:

- (25) * Who did John believe [pictures of ____] to have caused the riot?
 (from Boeckx 2012: 116)

I am not sure whether this is an optimal example to illustrate the point, however. The crucial property of ECM is that the subject DP of a complement clause is *exceptionally* assigned accusative case by the matrix clause. In (25), *pictures of who* is first merged as the subject of the embedded clause, and its islandhood may therefore simply follow from its (early) subject status rather than from ECM per se.²⁷

A clearer testing ground comes from Spanish DOM, which, incidentally, Boeckx himself discusses when evaluating Müller’s account (see Section 6.2.1.1). Specifically, he adopts Torrego’s assumption that DOM-marked objects raise to Spec,vP, forming a configuration structurally analogous to that of raised subjects. On this view, DOM objects should behave like subjects for sub-extraction purposes, and the prediction is straightforward: sub-extraction should be banned. As I discussed earlier, however, Boeckx considers an example from Demonte (1987) showing that this prediction is incorrect: sub-extraction

²⁶I set aside the possibility that transparency of the entire subject constituent follows from the accessibility of phase edges under the PIC, since Boeckx explicitly rejects any substantive connection between the PIC and islandhood (see discussion above).

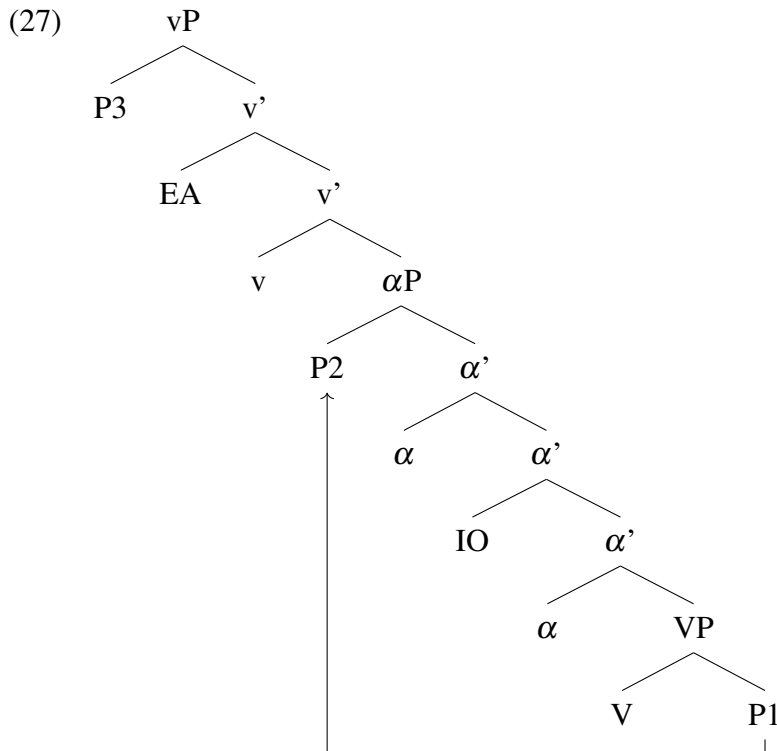
²⁷Under some analyses, this DP later raises into the matrix vP to check case, but under others, it remains in situ and is valued via Agree, effectively never occupying an object position (see, a.o., Postal 1974; Chomsky 2000, 2001; Adger 2003).

from DOM-marked objects is acceptable (see 9 above). Crucially, though, this is problematic not only for Müller's proposal but also for Boeckx's own system. Consider the example in (11), repeated here as (26).

- (26) [De quién] conociste [al hijo ____]?
 of whom met-2sg DOM.the son
 'Whose son did you meet?'

Under analyses like Torrego (1998) and Rodríguez-Mondoñedo (2007), the direct object *el hijo de quién* is first merged in Comp,VP and then raised via A-movement to Spec,vP, where it is DOM-licensed (hence the realised form *al hijo*; see Section 4.1 for further details on Spanish DOM). Crucially, this movement forms an A-chain, exactly parallel to the subject case discussed by Boeckx. If A-chain formation triggers Spell-Out and consequent opacification, sub-extraction from DOM-marked objects should be impossible. But, as the examples show, it is not.

As a matter of fact, a possible way out for Müller comes from the more recent analysis proposed by López (2012). López argues that unmarked objects remain in a low position (P1), whereas DOM-marked objects raise to an intermediate projection (P2), and P3 is reserved for even higher A-movement configurations, such as Germanic-type object shift (see López 2012: 44ff for discussion). The relevant structure is sketched in (27):



Under this view, DOM-marked objects only raise as far as P2, meaning that the *v* head is not yet inert and thus can still license sub-extraction (see above). This resolves

the problem for Müller's analysis. For Boeckx's account, however, the difficulty remains: regardless of how high they raise, DOM-marked objects still form an A-chain under either Torrego's or López's analysis, and thus should be rendered opaque by the very mechanism that predicts subject islands.

Finally, there are some broader theoretical concerns that, while not undermining Boeckx's core insights, may render his implementation less straightforward than it might initially appear.

First, Boeckx claims that sub-extracting from a 'closed' A-chain is problematic at both the phonological and semantic interfaces, i.e. that the source of island effects lies simultaneously at PF and LF (Boeckx 2012: 117). This is, I believe, an intriguing and promising idea, and I will adopt a related view later. However, while it is reasonable (and even expected) that LF would be sensitive to the distinction between A- and A-bar features/chains, it remains unclear why exactly PF should distinguish between the two. For purposes of linearisation, after all, A- and A-bar movements are equivalent. So without further motivation, this assumption remains rather puzzling.

Second, Boeckx repeatedly stresses the need for principled explanations, viewing the minimisation of stipulations and the grounding of locality effects in independently motivated mechanisms as essential requisites for an adequate theory. Yet several core components of his account face challenges similar to those he identifies in competing proposals. For instance, the feature-inheritance mechanism plays a central role in his proposal but actually lacks a clear independent justification, effectively making it no less 'abstract' and 'stipulative' than the assumptions in, e.g., Müller (2010). Similarly, the adoption of doubly rooted structures from Epstein et al. (2012) introduces significant architectural complexity without a clear connection with independent interface principles. Sure, immediate Spell-Out is conceived as a kind of 'rescue strategy' to facilitate efficient computation by avoiding the (simultaneous) interpretation of an overly complex doubly rooted structure. Yet the reason *why* such taxing configurations arise in the first place is grounded almost entirely in theory-internal considerations. Finally, the crucial notion of 'relativisation,' intended to explain why Spell-Out-induced opacity applies only to the elements that trigger it, remains underspecified and would benefit from explicit formalisation to distinguish it from more ad-hoc stipulations.

In conclusion, I broadly endorse Boeckx's insights and share his interface-oriented perspective, but I believe the technical implementation of his account faces several empirical and theoretical challenges. Given these limitations, I will explore two potential routes to address the issues raised above. One possibility is to abandon Boeckx's sharp distinction between A- and A-bar chains altogether and derive the CED from independent properties of Merge. The other is to retain Boeckx's distinction but refine its implement-

ation. The following sections are structured around these two options.

I first present the theory proposed by De Belder and van Craenenbroeck (2015), where the relevant locality effects follow from independently motivated properties of Merge. As I will show, this account neatly sidesteps the problem of sub-extraction from DOM-marked objects but faces different challenges of its own.

I will then propose my own approach to the problems posed by the CED in Section 6.2.4. Rather than a full-fledged theory, this is intended as a possible way forward that builds constructively on Boeckx's insights while incorporating refinements that (a) integrate more seamlessly with my overall Interface Hypothesis and (b) attempt to resolve the empirical issues highlighted here.

Since the treatment of adjuncts is relatively uniform across accounts, analysing them as 'monolithic,' impermeable objects, I will henceforth focus primarily on the 'subject part' of the CED.²⁸

6.2.3 Deriving the CED from the properties of Merge

Boeckx's (2012) approach, we have seen, faces some issues. Let us put aside, for the time being, concerns regarding the (partial) ad-hocness of his system; I will return to those in the next section. In this section, I will focus instead on what is arguably the major empirical drawback highlighted above: Boeckx's account incorrectly predicts the impossibility of sub-extracting from DOM-marked direct objects. In fact, the prediction is that any XP (whether subject or object) undergoing A-movement should block sub-extraction.²⁹ This raises a natural question: perhaps a system based on a strict distinction between A- and A-bar chains is simply too restrictive. As mentioned, then, a possible route forward is to shift focus away from chain types and towards the specific (structural) properties of external arguments. After all, this component of the CED was originally formulated as the *Subject Condition*. I believe that a promising starting point in this direction is the system proposed by De Belder and van Craenenbroeck (2015).

A first clear advantage of De Belder and van Craenenbroeck's proposal is that it was not originally formulated to address locality restrictions. Their account is primarily concerned with the properties of lexical roots, which they explain through a technical revision of the Merge algorithm. Crucially, this revision has welcome side-effects, one of which is the derivation of the opacity of non-complements. Thus, because the assumptions under-

²⁸For transparent adjuncts, Truswell (2007) claims that these are those that do not introduce their own event substructure and instead denote a single event with the main predicate. Such adjuncts may behave more argument-like and merge closer to vP as non-spelled-out, accessible constituents. See Dal Farra (2020) and Porta (2024) for discussion.

²⁹It would be interesting to see whether and to what extent this prediction holds when exploring other instances of A-movement. I must leave this inquiry to (hopefully near) future research.

pinning the system are independently motivated by phenomena involving lexical roots, De Belder and van Craenenbroeck’s approach escapes, in principle, concerns regarding ad-hoc postulation that may affect the accounts reviewed so far: the CED falls out as a by-product of principles required elsewhere. Moreover, it achieves this result by appealing solely to the mechanics of Merge, which represents a conceptual advantage in minimalist terms, at least insofar as we assume that the CED retains some properties attributable to UG. As discussed in Section 3.2, any such property should, as much as possible, be derived from, or reducible to, the workings of Merge itself. In what follows, I present the relevant assumptions on Merge they adopt and how these can explain the opacity of non-complements.

6.2.3.1 De Belder and van Craenenbroeck (2015)

The authors start by considering the definition of Merge. Since Chomsky (1995), this has been standardly conceived as a binary function that combines two syntactic objects into a set containing them: $\text{Merge}(\alpha, \beta) = \{\alpha, \beta\}$ (see, e.g., Chomsky 1995, 2008, 2013, 2015; Collins 2017; Epstein et al. 2012). This operation is perfectly symmetric, as sets do not impose ordering on their members: $\text{Merge}(\alpha, \beta) = \text{Merge}(\beta, \alpha) = \{\alpha, \beta\}$.³⁰ However, Chomsky (1995: 243) argues that the interface systems interpret different types of constituents differently, and that this distinction should be encoded in syntax (De Belder and van Craenenbroeck 2015: 634). Specifically, Merge is ‘sensitive to’ the distinction between heads and non-heads: when a phrase is merged with a head, the latter projects its label onto the newly formed constituent. This, therefore, introduces a crucial asymmetry into the function: when a head α merges with a non-head β , the result is $\text{Merge}(\alpha, \beta) = \{\alpha, \{\alpha, \beta\}\}$, where α is the label of the complex syntactic object.

Following Langendoen (2003), De Belder and van Craenenbroeck remark that the output $\{\alpha, \{\alpha, \beta\}\}$ is set-theoretically equivalent to the ordered pair $\langle \alpha, \beta \rangle$.³¹ Given that one of the two elements combined by Merge must project, producing an asymmetric output, the authors argue that the proper characterisation of the function should be Pair Merge rather than Set Merge (see above). Moreover, they build on insights by Jaspers (1998) and argue that Merge should be reconceptualised as unary, not binary. This, they claim, explains why one of the elements taken as input at any instantiation of Merge is always the ongoing derivation, which in Chomsky’s system must be imposed as an

³⁰For a relevant distinction between *members* and *terms*, see Chomsky et al. (2023: 19), where it is specified that “X is a *term of* Y iff X is either (i) a member of Y or (ii) a member of a term of Y.”

³¹In some theories, this equivalence is used to derive ordered pairs as non-primitives. Essentially, a given order can be associated with a set of sets (or collection) $\mathcal{C}$ “in such a way that the given order can be uniquely recaptured from $\mathcal{C}$ ” (Halmos 1974: 23). This is achieved by defining the ordered pair $\langle a, b \rangle$ as the set $\{\{a\}, \{a, b\}\}$, where the asymmetric structure of nested sets encodes the ordering information: the element appearing in both the singleton and the pair is identified as the first element (Kuratowski 1921).

axiom. They also claim, following Zwart (2009, 2011), that this resolves concerns about the assumption of Merge’s binarity. Although Chomsky (2007: 5) defines binarity as “the simplest case,” they argue that “one is the absolute minimum of elements an operation can manipulate” and that “an implementation of Merge that can reach this minimum is to be preferred over one that uses two syntactic objects as its input” (De Belder and van Craenenbroeck 2015: 635).³²

Finally, they specify that the objects selected by Merge are not bare items but sets of features (Chomsky 1995; Drury 1998; Fortuny 2008). Bringing these insights together, the authors propose the following definition:³³

(28) **Unary Merge** (De Belder and van Craenenbroeck 2015: 636)

Merge selects a single subset from a resource (e.g., $\{\alpha\}$), includes it in the object under construction (δ), and yields an ordered pair (e.g., $\langle\{\alpha\}, \delta\rangle$, assuming that α projects).

This definition has important consequences for a long-standing puzzle in the theory of Merge: its very first application. The initial step of the derivation, often called *Primary Merge*, poses a challenge for formalising the algorithm. Under standard binary Merge, the operation requires two arguments: the ongoing derivation and a feature set from the resource, selected via *Select*. But at the start of the derivation there is no ongoing structure to merge with, creating an apparent paradox.

De Belder and van Craenenbroeck reason that this imposes a substantial degree of arbitrariness to the first step of the algorithm. Either *Select* exceptionally takes two elements from the resource, or it takes one element, but does not feed it immediately into Merge, waiting until Merge is applied a second time, when two elements of the resource become available. In both cases, special provisions must be built into the algorithm to avoid overgeneration at later derivational stages, making *Primary Merge* an idiosyncratic exception (De Belder and van Craenenbroeck 2015: 636).

They argue that this problem can be avoided altogether by taking (28) literally: Merge always targets a feature set α from the resource and adds it to the ongoing derivation δ .

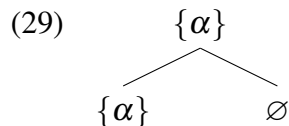
³²I have two observations in this respect. First, it seems to me that claiming that Merge always adds an element to the ongoing derivation is equivalent to saying that Merge is binary and (always) takes the workspace as one of its arguments. In both cases, the operation has two arguments. Second, Chomsky et al. (2023) argue that two is in fact the minimal number of arguments Merge can take, and that its binarity can be derived from third-factor principles of economy of computation (Chomsky et al. 2023: 17):

To build any relevant syntactic structure at all, Merge must target a minimum of *two* objects. By parsimony, then, the simplest form of Merge is binary; Merge targets two and only two elements of the WS [workspace] on any given application. Two is necessary; by parsimony, we seek a system where two is sufficient.

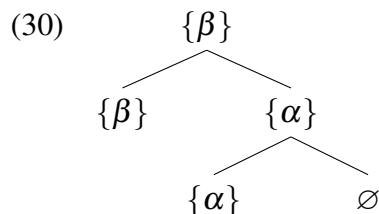
³³Here, *resource* is intended to indicate what has been alternatively defined as *lexicon*, *Numeration*, or lexical (*sub*)*array* (De Belder and van Craenenbroeck 2015: fn19).

At the initial step, however, there simply is no derivation yet, so δ is equivalent to the empty set. As a result, Primary Merge works just like any other application of Merge, except that $\delta = \emptyset$ by necessity. In sum, in De Belder and van Craenenbroeck’s proposal, Primary Merge runs as follows: $\text{Merge}(\alpha) = \langle \{\alpha\}, \emptyset \rangle$.³⁴

Building again on the logic that $\{\alpha, \{\alpha, \beta\}\} = \langle \alpha, \beta \rangle$, the authors propose that the left-hand member of an ordered pair projects (De Belder and van Craenenbroeck 2015: 637). As a result, Primary Merge of α yields:³⁵



De Belder and van Craenenbroeck provide a clear argument for why $\emptyset$ cannot project. If projecting a label involves sharing/copying features from a daughter node to its mother node, it follows that under no circumstances could $\emptyset$ project, as it is, by definition, empty and, therefore, featureless (De Belder and van Craenenbroeck 2015: 637). A similar reasoning was already adopted by Zwart (2009, 2011) and especially Fortuny (2008). However, De Belder and van Craenenbroeck’s proposal differs from the latter in assuming that $\emptyset$ does not ‘disappear’ from the derivation, but rather constitutes a bona fide terminal node that will be assigned a phonological exponent via Late Insertion. These terminals, they argue, are where roots are inserted. Therefore, merging another element β from the resource to (29) will yield $\langle \{\beta\}, \langle \{\alpha\}, \emptyset \rangle \rangle$, represented in (30).³⁶



The hypothesis that lexical roots are inserted post-syntactically into terminals hosting $\emptyset$ derives several empirical properties of roots (see De Belder and van Craenenbroeck

³⁴ α is represented as $\{\alpha\}$ because it is a set of features, see above.

³⁵Despite appearances, this is not an actual theory of labelling. The authors claim that the left-hand element of an ordered pair labels the complex constituent, but this only begs the question of what determines the position of the members in the first place. Intuitively, if Unary Merge ‘piles up’ elements from the resource by adding them one at a time to the derivational root, then the last-merged element should occupy the left-hand position of the complex ordered pair. If so, this means that the LA should proceed as follows: the newly merged element projects. It is not clear, however, whether De Belder and van Craenenbroeck adopt this assumption. In fact, it seems they reject it: “being the left-hand member of an ordered pair correlates with projection, not with being the most recently merged element” (De Belder and van Craenenbroeck 2015: 637). I set this question aside, as it does not directly affect the arguments relevant for the derivation of the CED.

³⁶Again, the authors must assume that β is a projecting element (De Belder and van Craenenbroeck 2015: 638), see fn. 35.

2015: 639–642). For present purposes, however, what matters are the theoretical consequences this hypothesis bears on the derivation. Let us examine these.

Clearly, the ultimate output of syntactic operations combined with (late) vocabulary insertion will yield a structure containing several lexical roots (e.g., the verb and its nominal arguments). By hypothesis, for each root, there must be an occurrence of $\emptyset$, and $\emptyset$ can only arise via Primary Merge. Since by definition there is only one instance of Primary Merge per derivation, each occurrence of $\emptyset$ must originate in an independent (sub-)derivation or, following De Belder and van Craenenbroeck’s terminology, “derivational workspace.” Ultimately, this logic entails a one-to-one correspondence between the number of lexical roots in a structure and the number of workspaces required to derive it. The authors formalise this as follows:

- (31) **One Derivational Workspace, One Root** (De Belder and van Craenenbroeck 2015: 642)

In every derivational workspace, there is exactly one root, and for every root there is exactly one derivational workspace.

6.2.3.1.1 Layered derivations and the opacity of non-complements

If every root requires its own workspace (WS), it means that a phrase like *eat the cookie* results from two WSs, and a whole sentence like *The child eats the cookies* from three WSs (De Belder and van Craenenbroeck 2015: 642–643). To allow for the construction of multiple-roots sentences, then, the derivation must proceed in a multi-layered fashion, allowing for the opening and completion of several independent layers/WSs. To achieve this, De Belder and van Craenenbroeck build once again on ideas from Zwart (2009, 2011) to define the properties of multi-layered derivations, where the output of one derivation can serve as an atomic input to the next. Let me clarify how these procedures unfold.

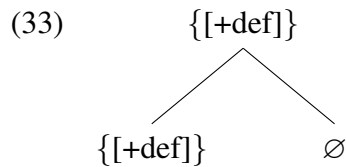
It all starts with a resource *R* containing (sets of) features. Merge targets these features and builds up complex structures within a WS. The output of this WS can then be readmitted to the resource, so that it will appear as a possible atomic input to the next instances of Merge.³⁷ Let us now examine how this works, starting with *eat the cookie*.³⁸ The derivation starts from the resource in (32), which only contains functional features (roots will be inserted post-syntactically to replace occurrences of $\emptyset$, see above).

³⁷I think this procedure requires no special justification. Given the (uncontroversial) assumption that human language is a recursive system, it is in fact the null hypothesis. If we accept this foundational property of language, then multi-layered derivations simply reflect the natural consequence of recursivity: the output of one derivational process can serve as input to another instance of the same process. In the present case, the syntactic operation that builds structure within one WS can be applied again to combine the outputs of multiple WSs, creating the layered derivational architecture that De Belder and van Craenenbroeck propose.

³⁸For reasons of space, I will only sketch the relevant parts of the derivation. For further details see De Belder and van Craenenbroeck (2015: 643–648).

$$(32) \quad R = \{v, [+def]\}$$

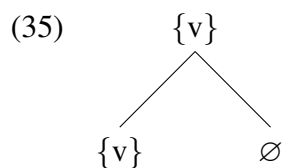
First, Merge applies to $\{[+def]\}$, selected from R , and combines it with the empty derivational WS, i.e., $\emptyset$, yielding $\langle\{[+def]\}, \emptyset\rangle$. This is represented as follows:



At this point, (33) is Spelled-Out and readmitted as an atom to R , which now looks as in (34).

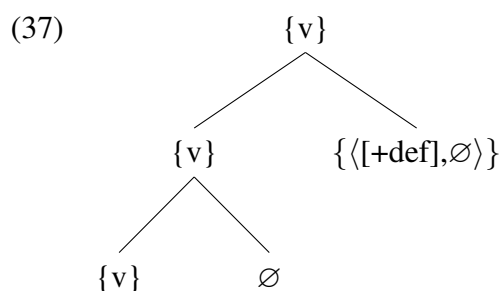
$$(34) \quad R = \{v, \langle\{[+def]\}, \emptyset\rangle\}$$

Because this readmittance operation has cleared the WS, the next Merge operation will be an instance of Primary Merge. The derivation now targets $\{v\}$ from (34) and combines it with the empty WS, yielding $\langle\{v\}, \emptyset\rangle$, represented as (35).



The complex element created in (33) is now the only element available in R (36), so that the derivation can now select it and merge it as a unit to the current WS, yielding:

$$(36) \quad R = \{\langle\{[+def]\}, \emptyset\rangle\}$$



In the postsyntactic component, the occurrences of $\emptyset$ will be assigned the respective exponents, resulting in the surface form *eat the cookie*.

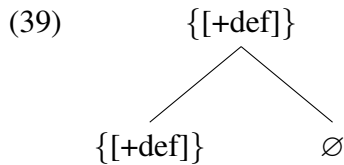
Having illustrated how layered derivations work in De Belder and van Craenenbroeck's system, I can now focus on how they explain opacity effects. To see this, consider again the tree in (37). There, the complement of v is, crucially, $\{\langle[+def], \emptyset\rangle\}$, not $\langle[+def], \emptyset\rangle$. Because it has been selected as a complete, atomic element from the resource, it does not appear in the derivation as a regular output of Merge, i.e., an ordered/asymmetric pair of

type α, α, β (see above). As such, this syntactic object is interpreted not as a complex hierarchical structure but rather as an atom. In other words, the process of readmittance of a WS to the resource, followed by its later merger into a new one, creates atomised, ‘compactified’ (see above) units that are, therefore, *opaque* to further syntactic operations (De Belder and van Craenenbroeck 2015: 645).

At this stage, however, the result is problematic: in (37), this atomised, opaque syntactic object is a complement—namely, *the cookie* in the phrase *eat the cookie*. But complements are expected to be transparent. To derive the correct empirical pattern (i.e. transparency of complements and opacity of non-complements), De Belder and van Craenenbroeck propose to “change [...] the order in which the various subderivations are readmitted to the resource” (De Belder and van Craenenbroeck 2015: 645). To illustrate this, they consider a sentence with two arguments: *The child eats the cookie*. Since this sentence features three roots (*child*, *eat*, *cookie*), it involves a three-layered derivation, i.e. one with three WSs.

The derivation proceeds as follows. Starting from the resource in (38), the subject is created by merging $\{[+def]\}$ with $\emptyset$, just as in (33). This yields the structure in (39).

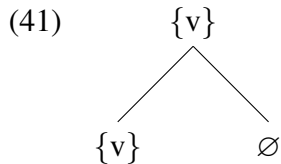
$$(38) \quad R = \{v, [+def], [+def]\}$$



The structure just created is then readmitted to R , which now looks as in (40).

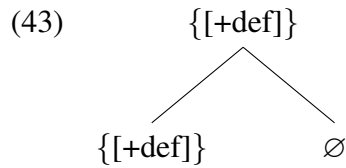
$$(40) \quad R = \{v, [+def], \langle \{[+def]\}, \emptyset \rangle\}$$

At this point, the ongoing derivation is empty, and the verbal structure can be built by merging $\{v\}$ from the resource with $\emptyset$. This yields the following structure:



By readmitting this output to the resource, we end up once again with an empty WS. At this point, Merge targets $\{[+def]\}$ (i.e., the second nominal) from the resource in (42), and yields (43).

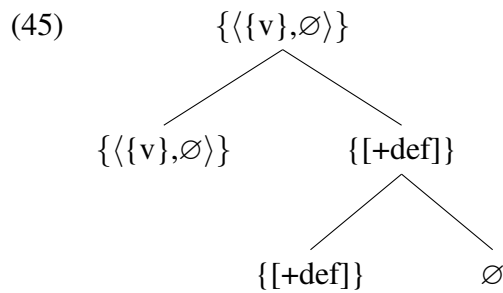
$$(42) \quad R = \{[+def], \langle \{[+def]\}, \emptyset \rangle, \langle \{v\}, \emptyset \rangle\}$$



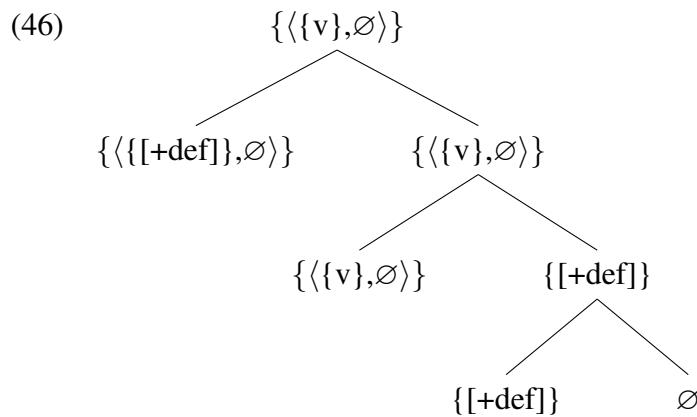
Crucially, the object just created is *not* readmitted to the resource, which at this point contains only the complex outputs of the previous sub-derivations (44).

(44) $R = \{\langle\{[+def]\}, \emptyset\rangle, \langle\{v\}, \emptyset\rangle\}$

From here, the derivation proceeds by merging $\{\langle\{v\}, \emptyset\rangle\}$ with the current WS, yielding (45).



Finally, the subject (i.e., the remaining $\{\langle\{[+def]\}, \emptyset\rangle\}$ in 44) is merged, yielding (46).



With the vP phase completed, the entire structure can be spelled out. Then, “[a]t the level of Vocabulary Insertion, the syntactic terminals of (46) are matched up with their respective phonological exponents and the structure is spelled out as *The child eats the cookie*” (De Belder and van Craenenbroeck 2015: 647), where, in particular, the occurrences of $\emptyset$ will host the lexical roots. What is crucial for the present discussion, however, is that in the derivation outlined above the object *the cookie* was not readmitted to the resource. And since the tree in (46) represents the WS within which the object was built, the latter appears as a complex constituent rather than an atomised unit (i.e., $\langle\{[+def]\}, \emptyset\rangle$)

not $\{\langle\{[+def]\}, \emptyset\}\}$). As a result, such a constituent is predicted to be transparent for extraction. By contrast, both the subject and the verbal complex are predicted to be opaque, as both were readmitted to the resource and thereby rendered inaccessible, atomised syntactic objects.³⁹

In sum, the multi-layered derivational system adopted by De Belder and van Craenenbroeck (2015) offers a straightforward way to account for the CED: phrases built in separate WSs and subsequently readmitted to the resource are opaque to the current derivation. Since objects are last-merged, they do not need to be readmitted and therefore remain transparent: the ‘main’ derivation is precisely the WS in which they are assembled. Importantly, because objects are always predicted to be transparent, independently of whether they undergo A-movement, the problem of sub-extraction from DOM-marked elements that affects Boeckx’s account is fully avoided here. Other issues arise, however.

6.2.3.2 Some challenges for De Belder and van Craenenbroeck’s (2015) system

The system is elegant, but not without issues. It is true that it derives the CED from independent properties of lexical roots rather than by positing explicit ad-hoc constraints, yet it still relies on assumptions that are, in practice, not so different from those highlighted in other accounts reviewed above. Let me explain why.

To reiterate, the mechanics of the system are, in brief, as follows: phrases are constructed in independent WSs and can be readmitted to the resource to serve as inputs in later WSs. A phrase that has been built in a separate WS and merged into the current derivation behaves like an atomic unit and is therefore opaque. To ensure that objects remain transparent, they must be merged last, so that the ‘main derivation,’ so to speak, coincides with the WS in which they are assembled. This, however, raises a crucial question: what imposes the order of operations in the derivation just sketched? In particular, what forces complements to be merged last? If they were merged earlier, they would wrongly be predicted to be opaque, just like any other constituent built in a separate WS. De Belder and van Craenenbroeck (2015) are aware of this issue, but, by their own admission, the solution they propose is only partial and “leaves several questions unanswered” (De Belder and van Craenenbroeck 2015: 648).

On the empirical side, even though the problem of DOM-marked objects is resolved,

³⁹Interestingly, this appears to reflect a broader architectural principle, as De Belder and van Craenenbroeck (2015: 647) explain:

[T]he approach just developed suggests that everything but the most deeply embedded internal argument is spelled out first, in a separate derivation. It is only when the most deeply embedded internal argument is reached that all subderivations are brought together into one syntactic representation.

another serious issue arises. Much like in Müller's (2010) account, De Belder and van Craenenbroeck's theory predicts that subjects are uniformly opaque. In fact, anything that is not last-merged is necessarily opaque to the 'main' derivation. This is a strong and problematic prediction, since it cannot explain why in-situ subjects are often transparent (see above). Moreover, it remains unclear how the system could accommodate the class of transparent adjuncts (see above, and especially fn. 28). One would need to explain why these adjuncts should count as last-merged, possibly even in configurations where other complements are present (see, e.g., 13c). Similarly, certain cases of island repair may be difficult to reconcile with this account. For example, sub-extraction from subjects is judged more acceptable when resumption is involved (see above), a pattern not expected if subjects are fully opaque under the multi-layered derivation approach.

Because of these challenges, I propose an alternative account that builds on Boeckx's core intuitions and revives the idea, in a new light, that opacity effects may arise from how the external systems interpret syntactic chains.

6.2.4 A possible way forward: the CED within the Interface Hypothesis

In the previous sections, I have examined several crucial aspects of opacity effects, which now provide a clearer basis for establishing the criteria an adequate account of the CED should fulfil. In particular, based on the discussion so far, such an account should satisfy two main sets of requirements.

From a theoretical perspective, CED effects should emerge from principles encoded in the grammar that can block the generation of illicit structures, in contrast to the more flexible constraints responsible for intervention effects (see Section 6.2). Second, these principles should be conceptualised as interface filters rather than derivational bans (see Section 6.2.1.1). Third, the account should minimise ad-hoc stipulations and instead derive opacity effects from independently motivated principles (see Section 6.2.2.2). In a minimalist perspective, this independent justification should rely as much as possible on domain-general properties of the interface systems, i.e. third-factor explanations (see Section 3.1.1).

From an empirical perspective, the account must explain why sub-extraction from non-complements is (largely) ungrammatical. As I already pointed out, there is substantial consensus on the opacity of adjuncts. The more challenging issue concerns subjects. In this respect, the accounts reviewed so far are too restrictive, wrongly predicting opacity for all A-chains (including DOM-marked objects), banning sub-extraction from in-situ subjects, or even excluding the possibility to extract the entire subject constituent from Spec,TP. An adequate account of the 'subject part' of the CED should instead capture the

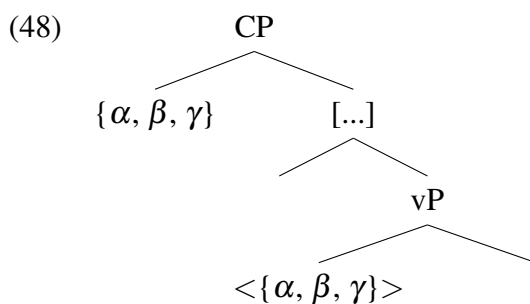
following generalisation:

- (47) sub-extraction is blocked only from moved subjects.

The aim of this section is to explore an account that meets these requirements, both on the empirical and theoretical sides. I develop this in the following sections. First, though, it is worth taking another look at (47). Viewed from a different angle, the generalisation essentially means that opacity arises only when two conditions are simultaneously met: (i) the element is an A-moved subject, and (ii) sub-extraction is attempted from it. That is, both are necessary, but neither alone is sufficient. I will now consider each of these in turn, beginning with sub-extraction.

6.2.4.1 The ingredients of opacity: sub-extraction and PF

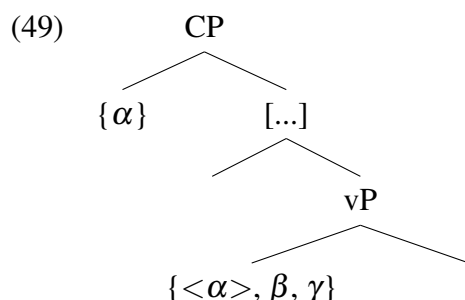
Ever since Chomsky (1993, 1995), syntactic theory has adopted the *Copy Theory of Movement*, according to which movement leaves behind a full copy (rather than a trace) in the original position. As a result, after movement both the higher (moved) and lower (base) positions contain the same element, forming a chain of identical copies. At the PF interface, however, two identical copies pose a problem for linearisation: the same item would both precede and follow another element, which is not linearisable (Kayne 1994). To resolve this, PF applies so-called *Chain Reduction* (Nunes 2004), deleting all but one copy in each chain. In the standard case, PF realises the highest copy and deletes the others.⁴⁰ This is illustrated in (48), which provides a simplified abstract representation of subject movement. Since the present discussion focuses on subject configurations, I will henceforth consider only operations affecting external arguments.⁴¹



⁴⁰For details on the technical aspects of deletion and an overview of alternative configurations, see, a.o., Chomsky (1993, 1995), Bobaljik (1995, 2002), Brody (1995), Nunes (1995, 1999, 2004), Richards (1997) and Abels (2003).

⁴¹Following Grimshaw's (2005) extended projection hypothesis and phase-based models of Transfer (Chomsky 2000, 2001; Abels 2003; Svenonius 2004), in the trees below I assume that PF treats an extended nominal projection as a unitary feature bundle, encompassing all its internal (sub-)constituents. This assumption is conceptually natural, given that the very notion of *sub*-extraction presupposes the existence of a larger extracted-from unit. Regarding notation: curly brackets '{ }' represent feature sets, while '<>' mark unpronounced copies. When referring directly to the deletion operation discussed in Nunes (2004) and Bošković and Nunes (2007) below, deleted material is shown with strike-through, e.g. {~~α~~, ~~β~~, ~~γ~~}.

sub-extraction arises when only a sub-part of a given constituent is moved, as in (49).



From the perspective of PF, the configurations in (48) and (49) are handled differently. In (48), PF encounters a chain whose links are two identical feature bundles. In this case, chain reduction proceeds straightforwardly: the lower copy is deleted, and the higher copy is pronounced. In (49), by contrast, the chain is ‘non-uniform:’ its links do not contain the same set of features, because in the derivation only α has been extracted from the larger feature bundle $\{\alpha, \beta, \gamma\}$. At an intuitive level, non-uniform chains are more complex: to linearise them, PF must keep (at least) two links ‘active’ (i.e., not fully deleted), realising a subset of the original feature set upstairs and the remainder downstairs. Arguably, then, configurations involving non-uniformity make a given derivation more costly at PF.

Independent support for this claim comes from Bošković and Nunes (2007), who discuss operations of “scattered deletion” that closely resemble the non-uniform configurations just described. In particular, they examine an interesting contrast between Bulgarian and Macedonian clitic sequences (first noted by Bošković 2001). In Bulgarian, independent phonological constraints on clitic ordering make it impossible to pronounce the entire verb–clitic cluster, which forms a complex head, in a single position when moved to the sentence-initial field (Bulgarian clitics must encliticise). To avoid a PF crash, the grammar resorts to scattered deletion, pronouncing the verb in the higher copy and the clitics in the lower one, thereby splitting the chain into two sub-parts. In Macedonian, by contrast, no such constraint arises (Macedonian has proclitics), and PF opts for the *simpler* full-deletion strategy, pronouncing the entire clitic sequence in the higher position (see Bošković and Nunes 2007: 20–22 for details). Crucially, Bošković and Nunes conclude that scattered deletion is “a costly option” and a “last-resort operation,” tolerated only when uniform deletion would lead to an unpronounceable structure. Further evidence comes from their analysis of similar configurations in English, where scattered deletion is shown to be systematically disfavoured due to its inherent derivational cost: the default, economical strategy is to delete the entire lower copy. In other words, allowing different pieces of a chain to be pronounced in two places (i.e., pronouncing a split-up chain) violates economy conditions unless there is no other way for the derivation to converge. As they put it, “[scattered deletion] will be employed just in case competing derivations that

employ fewer applications of deletion violate additional requirements of the phonological component so that they do not converge” (Bošković and Nunes 2007: 27).

Now, in the cases discussed by Bošković and Nunes, the two links of the chain are assumed to be identical, and what differs is the operation of deletion, which selectively targets different sub-parts of the copies (i.e., the underlying syntax remains the same). Extending this logic to (49) would mean that the same feature set $\{\alpha, \beta, \gamma\}$ occurs both upstairs and downstairs, but deletion targets different subparts, yielding $\{\alpha, \beta, \gamma\}$ upstairs and $\{\alpha, \beta, \gamma\}$ downstairs. Although this is not compatible with standard syntactic assumptions on sub-extraction, the crucial point is that, for PF, the computational burden is similar: multiple links must remain ‘active’ to permit the phonological realisation of their respective subparts.

In sum, PF is subject to economy constraints that ensure the mechanism underlying deletion applies as few times as possible (Nunes 1995, 1999, 2004). This naturally points to a broader tendency of PF to prefer consistent, uniform chains (48) over non-uniform ones (49). Importantly, however, scattered deletion is not ungrammatical. As Bošković and Nunes (2007) show, it is a marked strategy, typically employed only as a last resort, but nonetheless a legitimate option provided by UG. Similarly, non-uniform chains such as those in (49) are also grammatical, and in fact, cases of sub-extraction are widely attested cross-linguistically.⁴² For instance, an actual instance of (49) can be seen in sub-extraction from post-verbal subjects in Italian, as in (8b), repeated below in (50). These are precisely the kinds of configurations whose grammaticality several accounts discussed above fail to capture:

- (50) Di chi ha telefonato [il figlio ____]?
 of who has phoned the son
 ‘Whose son called?’

In conclusion, sub-extraction yields non-uniform chains that impose a greater computational burden on PF. Despite this cost, however, PF tolerates sub-extraction, even when it involves external arguments.

6.2.4.2 The ingredients of opacity: movement chains and LF

In discussing their hypothesis on scattered deletion, Bošković and Nunes pursue an interesting analogy. They claim that the mechanisms underlying scattered deletion at PF

⁴²If the reasoning pursued here is correct, a natural prediction is that, while attested, sub-extraction should be typologically more marked than pied-piping. A comprehensive investigation is still needed, but existing cross-linguistic patterns seem to support this: some languages overwhelmingly leave *wh*-elements in situ (e.g., Chinese), others require pied-piping (e.g., Romance), and some allow both stranding and pied-piping (e.g., Germanic). Interestingly, however, there appears to be no language that allows stranding but lacks pied-piping, whereas the reverse situation is common (Perpiñán 2008).

effectively mimic certain workings of LF, in particular those proposed in Chomsky (1993, 1995), where LF is fed configurations in which “parts of a chain are interpreted in one position and other parts in a different position” (Bošković and Nunes 2007: 22). As they remark, the idea that the two interfaces share similar mechanisms should not be surprising but rather “a welcome result from a conceptual point of view.”

I think this is on the right track. Under the most minimal hypothesis, PF and LF are subject to the same constraints, which are independently determined by domain-general limitations. This is essentially the claim of the SMT: third-factor considerations shape the grammar (see Section 3.1.1). Rather, the difference between the two interfaces is determined by the categories and properties they are sensitive to. PF’s task is linearisation, and it will therefore be sensitive to everything that concerns the way chains will be pronounced. LF, by contrast, is concerned with the ‘meaning side’ of the output of narrow syntax. As such, it is expected to be sensitive to the properties relevant for the interpretation of syntactic structures, and all the categories these involve.

As far as movement chains are concerned, syntax essentially distinguishes between two types: A-chains and A-bar chains. An A-bar chain is interpreted at LF as an operator–variable dependency, crucial for the semantics of questions, relativisation, focus, and related phenomena. An A-chain, by contrast, is typically interpreted as a single argument chain, with the moved element and its lower copy essentially treated as one ‘spread-out’ argument and without introducing a new operator scope. Simplifying somewhat, A-bar movement generally has an overt semantic effect (e.g., turning a declarative into a question or focusing a constituent), whereas A-movement is typically semantically inert, serving mainly a well-formedness role (e.g., Case licensing or EPP satisfaction).

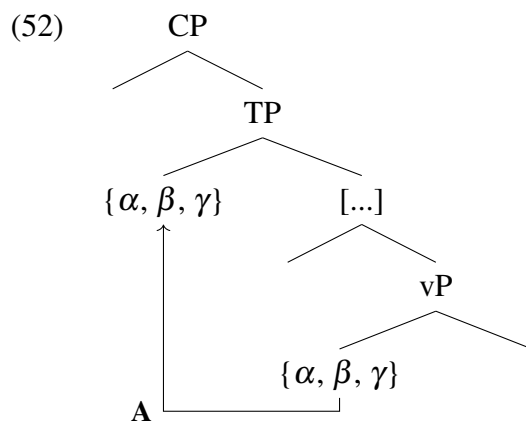
It follows that LF must treat A- and A-bar chains differently and recognise them as ontologically distinct objects: for interpretative purposes, LF must keep track of chain type, and assign them different values. As we have seen, this is precisely the reasoning pursued by Boeckx (2012), who argues that the external systems distinguish between A- and A-bar chains and are, in fact, hindered when these intersect. Crossing an A-bar chain with an A-chain creates an ambiguous configuration that, in his account, causes the interface systems to crash (see Section 6.2.2.2). As I have shown, this is how he derives the opacity of subjects.

Interestingly, this view finds support in recent minimalist suggestions that “Subject Island effects, accounted for by CED, should now be explained under segregation of A- and A-bar systems” (Chomsky et al. 2023: 59). However, although these remarks are intuitively appealing for the reasons just sketched, they remain underdeveloped. In particular, I have noted that Boeckx contends that the ambiguity between A- and A-bar chains is problematic for both LF and PF, yet he provides no independent motivation for why PF

should care (see above). I will build on these insights and propose instead that this ambiguity is problematic *only* at LF, but essentially for the same reasons that sub-extraction is problematic at PF.

Let us start by considering the simplest case: the syntax builds an A-chain through EPP-movement of the subject. This movement occurs, for instance, in any simple SVO sentence in English, such as (51), schematically represented in (52).⁴³

(51) Britt's pictures caused a riot.



In this case, LF recognises a movement chain with two (identical) links. In particular, it identifies that the lower copy has been re-merged through A-movement and thus straightforwardly analyses the chain as an A-chain.⁴⁴

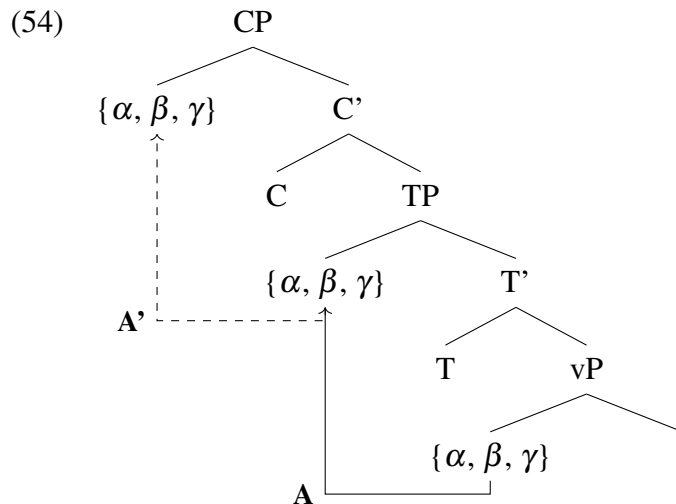
A more complex configuration arises when the subject further raises to Spec,CP via A-bar movement, as in (53), represented schematically in (54).⁴⁵

(53) Pictures of whom caused a riot?

⁴³In this and following trees, I add the labels 'A' and 'A'' to indicate the chain type, with A-bar chains additionally distinguished by dashed arrows.

⁴⁴Here, I use terms such as 'copy', 'movement', 're-merge', and 'internal merge' somewhat loosely. These are not equivalent, as they presuppose different assumptions about the nature of syntactic operations, particularly in the context of the representational vs. derivational debate (see above). I abstract away from these technical distinctions here for the sake of a simpler exposition.

⁴⁵Naturally, the same considerations apply to derivations involving *whose-picture*-type pied-piping.



In this case, LF receives a more complex input. The feature bundle $\{\alpha, \beta, \gamma\}$ occurs three times. Assuming a bottom-up computation, LF first encounters its occurrence in Spec,vP and recognises that it has been A-moved to Spec,TP, thus completing an A-chain. The computation then proceeds and finds that the same bundle in Spec,TP has been further moved to Spec,CP, this time via A-bar movement. As a result, the node in Spec,TP is simultaneously the head of an A-chain *and* the tail of an A-bar chain.⁴⁶

This situation closely resembles the case described in (49) and the instances of scattered deletion analysed in Bošković and Nunes (2007). In all these cases, the interface cannot simply ‘close’ a chain and, in Boeckx’s terms, assign it an “independent, autonomous, complete status.” Instead, multiple links must be kept ‘active’ for proper computation. At PF, this means different links spell out different sub-parts of a chain during linearisation. In the LF configuration at hand, something like a mirror image arises: a single link must serve a double purpose, simultaneously belonging to both an A-chain and an A-bar chain. In both cases, the result is an “ambiguous” configuration that places additional demands on the interface systems (see above and Boeckx 2012: 155ff.).

Now, according to Boeckx (2012), the crossing of an A-chain with an A-bar chain is enough to make “the external systems ‘gag’” (Boeckx 2012: 115), as though they were led into a kind of garden path when forced to ‘re-open’ an A-chain to compute an overlapping A-bar chain. However, for this reason (along with other issues related to spell-out-inducing, doubly rooted structures reviewed above), Boeckx’s account appears to predict that full extraction of subjects from Spec,TP to Spec,CP should be impossible. This problem, already noted in Section 6.2.2.2.1, is evident when considering (53), which is perfectly grammatical.

However, from the perspective developed here, which draws a parallel between PF and LF limitations, this is an expected and welcome result. Just as configurations like

⁴⁶I assume a bottom-up computation here only for expository purposes, but nothing crucial hinges on this. Even under a top-down approach, the same ambiguity arises upon analysis of the node in Spec,TP.

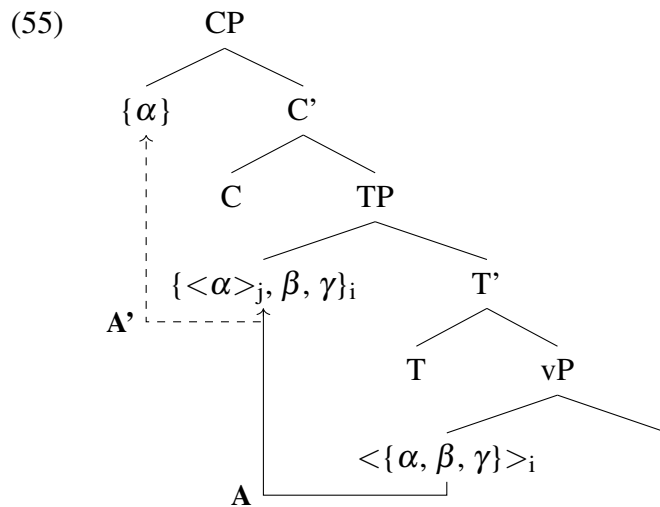
(49) are costly for PF but do not result in ungrammaticality, configurations like (54) are likewise expected to be costly for LF, yet do not trigger a derivational crash.

As I argue below, such configurations are tolerated when considered in isolation: the interface systems can accommodate a certain degree of ambiguity. It is only their *combination* that gives rise to opacity effects.

6.2.4.3 The emergence of opacity: a double ambiguity

I have shown that sub-extraction alone is costly at PF but tolerable, and that the crossing of an A- and an A-bar chain is costly at LF but also tolerable. Note that these configurations each represent one of the two conditions that, together, underlie opacity effects in subjects (see 47). As stated above, both are necessary but not sufficient conditions: for opacity to arise, they must co-occur. We are now in a position to explain why this is the case.

Let us consider a derivation that involves both (i) sub-extraction, and (ii) a configuration where an A-chain and an A-bar chain intersect. This syntactic scenario is represented schematically in (55).



From the perspective of PF, this configuration matches that in (49): a non-uniform chain must be linearised, with a subset of features ($\{\beta, \gamma\}$) realised downstairs (in Spec,TP) and another subset ($\{\alpha\}$) realised upstairs (in Spec,CP). From the perspective of LF, this structure involves an ambiguous chain like that in (54): the node in Spec,TP is simultaneously the head of an A-chain and the tail of an A-bar chain.

Again, each of these configurations is individually costly but tolerable for the respective interface. Yet feeding the external systems a derivation involving *both* of these burdens at once is, I argue, simply too demanding: the computational overload causes the derivation to crash. In other words, the interface systems cannot simultaneously handle PF non-uniformity and LF chain-type ambiguity, and thus reject the derivation at Spell-Out.

At this point, we should ask what kind of sentence corresponds to a configuration like (55). Any such structure must simultaneously involve sub-extraction and a chain-type intersection. Of course, we have already encountered exactly these cases: sentences where sub-extraction targets a raised (pre-verbal) subject—namely, instances of subject opacity *par excellence*. A representative example was presented in (1b), which I repeat here in (56).

(56) *Who did [pictures of ____] cause Bill to leave the room?

On the present view, the ungrammaticality of (56) follows straightforwardly, given that it involves both sub-extraction (yielding non-uniformity) and an A-to-A' chain-shift (yielding chain-type intersection). More generally, this account captures phenomena of subject opacity as a natural consequence of independent computational limitations: LF and PF simply cannot be burdened simultaneously.

In fact, I believe this conclusion has an even broader scope. To see this, let me repeat the crucial insight: non-uniformity and chain-type intersection impose independent computational costs at their respective interfaces; when these configurations co-occur, they overload the external systems, which therefore reject the attempt at Spell-Out. What is particularly significant is that these costs combine in a *superadditive* fashion. Their independent costs do not merely accumulate; they interact in a way that overwhelms the external systems, forcing them to reject the structure entirely. It seems only natural, then, to interpret this very combination as the source of the superadditivity observed in experimental work by Sprouse and colleagues (see Sprouse et al. 2012, 2013b; Sprouse and Villata 2021, and Section 2.2.3 for discussion). This interaction, I argue, constitutes the foundation of impenetrability locality effects in the traditional formulation of the Subject Condition.

6.2.4.3.1 Advantages and challenges of the account

To recap, I propose that PF non-uniformity (from sub-extraction) and LF chain-type intersection (from A-to-A' chain-shift) are individually tolerable but jointly superadditive: when both burdens co-occur, as in sub-extraction from moved subjects, derivations exceed interface capacity, yielding the effects described in the Subject Condition. This proposal offers several advantages.

First, unlike the accounts of Müller (2010) and De Belder and van Craenenbroeck (2015), it does not predict the ungrammaticality of sub-extraction from in-situ subjects. The attested examples in languages such as Spanish and Italian (see above) are therefore not problematic but rather expected under this approach.

Second, cases of full extraction of the entire subject constituent from Spec,TP to Spec,CP are correctly ruled in, in contrast to the predictions emerging from Boeckx

(2012).

Third, the system defended here builds on minimal stipulations. Arguably, it is almost ‘stipulation-free.’ Unlike Boeckx (2012), there is no need to assume elaborate mechanisms of feature inheritance, invoke doubly rooted structures, or postulate some mechanism to relativise spell-out-induced opacity effects to subjects alone (see Section 6.2.2.2.1). Nor does the account require the ad-hoc assumptions on edge features and their hierarchical distribution proposed by Müller (2010). And although fully compatible with the system of De Belder and van Craenenbroeck (2015), this account does not need to derive the transparency of complements via stipulative constraints on derivational order (i.e., forcing objects to be last-merged). Instead, all necessary assumptions follow independently from current SMT views on grammatical architecture: the interface systems are constrained by domain-general computational limitations. On this account, there is a single mechanism, derived from principles of economical computation, which applies uniformly to the relevant categories at PF and LF.

Fourth, the identified source of opacity aligns exactly with experimental findings that identify island-effects as a superadditive interaction: LF and PF burdens are tolerated individually but become disruptive when combined.

Fifth, at a conceptual level, this account integrates multiple strands in the literature on the CED. It accommodates both PF-based approaches (e.g., Uriagereka 1999a; Richards 2001; Fox and Pesetsky 2005) and LF-based accounts (e.g., Rizzi 2006; Rizzi and Shlonsky 2008; Stepanov 2007; Boeckx 2012), and naturally adopts the increasingly accepted view that islands are *emergent* phenomena, arising from the interaction of independent grammatical pressures rather than from fixed properties of specific nodes (Boeckx 2012: 121–13). Crucially, it also ties opacity to interface *ambiguity*, fitting theoretical arguments (Boeckx 2012: 121) as well as the account on intervention locality defended above (see also Section 6.3).

Sixth, because opacity effects are explained as the result of a derived ‘filter’ emerging from superadditively costly configurations, this account retains the main benefits of representational analyses. For example, the challenge raised by Boeckx (2012) for derivational approaches, particularly regarding the existence of ‘repair’ strategies (see above), receives a straightforward explanation. On the present view, such strategies follow naturally: sluicing repairs island effects because PF is unaffected by non-uniform chains when these are contained within material that is not linearised (i.e., PF is not concerned with unpronounced domains, as argued by, a.o., Lasnik 2001 and Merchant 2008). Similarly, pied-piping avoids sub-extraction and therefore generates uniform chains, which PF can compute without difficulty.⁴⁷

⁴⁷ As for *wh*-in-situ and resumption, these are typically discussed as strategies alleviating extraction from

There remains, however, one problematic point. The issue concerns sub-extraction from DOM-marked objects, a problem that also affects Boeckx (2012) and, depending on one's analysis of DOM, possibly Müller (2010) as well (see discussion in Section 6.2.2.2.1). If LF's sensitivity to chain types is based on a simple distinction between A- and A-bar chains, then sub-extraction from a DOM-marked object, as in (11), should be predicted to cause the same difficulty as sub-extraction from subjects: in both cases, sub-extraction co-occurs with A-to-A' chain-shift.

A possible way forward may come from a suggestion in Chomsky et al. (2023). As noted earlier, they endorse the idea that island effects should be explained via the segregation of A- and A-bar systems. However, a few pages later, they observe that the A/A-bar distinction might ultimately be eliminable, "with its core properties being fully attributable to the featural properties of IM landing sites" (Chomsky et al. 2023: 63).⁴⁸ If the interfaces are not inherently sensitive to A- vs. A-bar chains, but rather to the features of the landing sites targeted by movement, it may become possible to restrict the relevant segregation. Specifically, we can assume that EPP-driven movement of subjects to Spec,TP (defined in terms of EPP and A-type features responsible for nominative assignment) is the only chain type that truly 'stands out' from the perspective of the external systems. As discussed, LF must distinguish chain types for interpretive purposes. It makes sense, therefore, that LF would segregate movements that have semantic reflexes from those that are semantically neutral. A-bar chains, of course, systematically have semantic effects. But, on closer inspection, many A-chains do as well: scrambling often behaves as an intermediate case between A and A-bar movement (Bennis and Hoekstra 1984; Webelhuth 1988, 1989, 1990; Déprez 1989; Mahajan 1990; Sternefeld 1990; Bayer and Kornfilt 2011), object shift frequently correlates with definiteness or specificity effects (Holmberg 1986, 1999; Déprez 1989; Diesing 1992; Thráinsson 2001), and, crucially, DOM-marked objects are also commonly linked to interpretive properties concerning specificity, definiteness, and animacy (Bossong 1991; Brugè and Brugger 1996; Leonetti 2004, 2008; López 2012; Fábregas 2013). By contrast, one of the very few cases of truly semantically neutral movement appears to be the canonical raising of subjects to Spec,TP (at least in English). In these cases, LF must essentially 'undo' the chain, treating the lower and higher copies as the same argument without introducing any new interpretive content.

If this refinement is on the right track, subject chains can be 'isolated' and treated as distinct from all other chain types.⁴⁹ As such, they would constitute the only move-

adjuncts or complex DPs. Since I assume the standard view that adjuncts and complex DPs are opaque because they are merged as 'monolithic' or 'compactified' domains (see above), the account defended here makes no specific predictions regarding their alleviation. That said, *wh*-in-situ can be seen as the mirror image of pied-piping: it preserves chain uniformity by realising the entire chain at its lower link.

⁴⁸For further discussion, see Obata and Epstein (2011), Urk (2015) and Fong (2022).

⁴⁹Rizzi (2006) and Rizzi and Shlonsky (2008) appear to point in a similar direction. Their Subject

ment configurations that create ambiguity when intersecting other (semantically relevant) chains. This would resolve the apparent problem with DOM-marked objects and, ideally, restrict opacity effects to moved subjects. That said, this remains a tentative proposal, and I must leave the question open for now.

6.3 Interim summary: the (non-)similarities between intervention and impenetrability

I am now in a better position to draw some conclusions on the properties of intervention and impenetrability locality. Based on the empirical evidence from Chapters 4 and 5, I argued that (much of) the literature is correct in treating these as distinct constraints, thereby setting an ontological divide between WIs and SIs. In this chapter, I further explored how these two instances of locality should be understood theoretically.

Regarding intervention, I argued that its effects do not reflect a grammatical constraint encoded in UG, whether within narrow syntax or as a representational filter at the Interface. Instead, they are *emergent* effects shaped by the strategies employed by the parser to resolve syntactic dependencies. As shown, these strategies are constrained by principles of processing economy (e.g., the Active Filler Strategy) and ambiguity resolution. In other words, difficulties associated with intervention reflect the interaction between abstract grammatical representations and domain-general pressures at the Interface, not feature-driven computations within the grammar itself (see Section 6.1).

With respect to impenetrability, I argued that it should instead be analysed as a genuine syntax-based constraint arising at the level of PF and LF. Crucially, however, I did not posit a fixed grammatical principle that explicitly bans extraction from specific domains. Rather, I contended that opacity of non-complements *emerges* due to early spell-out mechanisms or, in the case of subjects, when the derivation simultaneously involves two computationally costly configurations, overloading PF and LF and creating a super-additive burden.

Taken together, then, these considerations suggest that intervention and impenetrability are largely distinct phenomena. Within the Interface Hypothesis developed here, impenetrability arises at the leftmost edge of the Interface spectrum, where PF and LF interpret the derivational output of narrow syntax. By contrast, intervention effects emerge further to the right, where domain-general constraints on parsing dominate. This is summarised in figure 6.3.

In other words, locality effects span the entire Interface space, and their position along

Criterion highlights the special status of subject positions, suggesting that the subject chain is structurally and interpretively distinct from other movement chains.

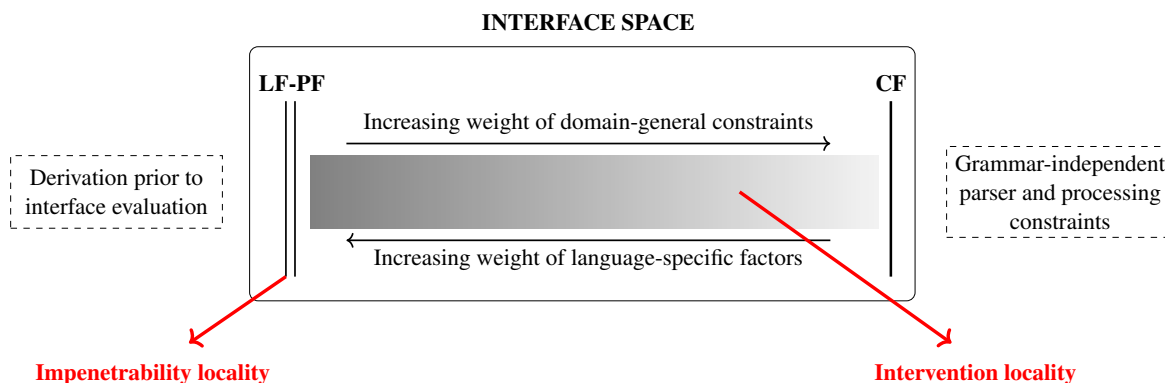


Figure 6.3: A conceptual illustration of the Interface space, as defined in Chapter 3. The red arrows indicate the location where intervention and impenetrability effects emerge, respectively.

this continuum determines, so to speak, their ‘nature.’ This, as I have emphasised throughout the thesis, is also reflected directly in amelioration effects. In particular, pinpointing the factors that determine the amelioration reveals whether a given constraint arises from structural properties of syntactic representations or from right-edge parsing pressures exerted on such representations.

That said, although intervention and impenetrability differ systematically, they also exhibit certain similarities. First, both are emergent, not fixed UG principles. They arise from the interaction of independent factors: a “grammatical conspiracy,” in the sense of Boeckx (2012: 121), according to which “islands can no longer be [...] dissociated from the process that constructs them. That is to say, islands are dynamic, emergent entities.” Second, both types of effects ultimately arise from configurations that create ambiguity at the Interface (see also Section 6.5). Third, as a direct consequence, both can be alleviated or circumvented when strategies are involved that reduce or eliminate the relevant source(s) of ambiguity.

In sum, intervention and impenetrability are conceived here as distinct but related emergent constraints that share key properties. Crucially, both their differences and their commonalities follow naturally from the Interface Hypothesis: they arise at different points within a single, shared interactional space. Their position on this spectrum determines which specific factors modulate them, while their common location at the Interface predicts their uniform sensitivity to ‘ambiguity.’

Stretching this line of reasoning somewhat, we can actually view locality as a fundamentally uniform phenomenon, whose effects ultimately emerge from a single, deeper principle: reduce ambiguity. As we have seen, both intervention and impenetrability can be traced back to configurations that create parsing or representational uncertainty, i.e. ‘ambiguous configurations’ that lead the external systems to experience some kind of

garden-path effects (see Chapters 4, 5 and Sections 6.1, 6.2.4). And quite revealingly, both effects can be ameliorated through strategies that resolve such ambiguity, whether via overt morphosyntactic cues or by avoiding non-uniformity and chain-type shifts at PF or LF.

Now, the reason why ambiguity avoidance plays such a prominent role should not be too surprising. After all, reducing ambiguity is arguably a necessary condition for any efficient computational system. And, in turn, principles of economy and efficient computation represent a large part of those *third-factor pressures* shaping the I-language (Chomsky 2017, 2020, 2021). In this light, then, it is understandable that principles centred on some sort of ambiguity avoidance have played a recurrent role in shaping theories of natural language throughout the generative tradition.

Already in the early years, Hankamer's (1973) influential article *Unacceptable Ambiguity* made this intuition explicit as a proposed universal principle. Somewhat simplifying, he proposes the generalisation that whenever a syntactic rule could in principle yield an ambiguous string (i.e., one compatible with multiple underlying sources) natural languages uniformly disallow such derivations, either by not having the relevant rule at all or, crucially, by imposing additional constraints, as formulated in his *No Ambiguity Condition*. And this connection between syntax (or grammar, more generally) and ambiguity avoidance resurfaces repeatedly, also with respect to (intervention) locality specifically. For instance, Rizzi (2004: 224) explicitly notes that "RM can be intuitively construed as an economy principle in that it severely limits the portion of structure within which a given local relation is computed" and that this principle "reduces ambiguity in a number of cases." But the roots of this idea go back even further, as eloquently shown in Boeckx (2012: 121):

[...] I find it pleasing that certain elements of the earliest approaches to islands remain as part of the explanations being offered today [...]. Recall that the A-over-A condition was based on the idea that "transformations must be unambiguous," and Chomsky (1973) suggested that conditions like his Specified Subject Condition have the effect of "reducing ambiguity." Ambiguity is a concept that Kayne (1984) also incorporated into his path-based theory of islands (cf. his notion of "ambiguous path"). Likewise, Rizzi (1990) pointed out that "the functional correlate of this formal principle [RM] is the reduction of ambiguity in government relations." Finally, it is worth pointing out that Chomsky's earliest generative treatment of syntax (Chomsky 1955/1975:309) contained the following remark: "[t]he first requirement of grammatical transformations is that the result of a transformation must be unambiguous."

Moreover, as Boeckx goes on to argue, his own account (based on the ban on crossing

A- and A-bar chains, see above) also follows the same core principle of ambiguity avoidance, and traces it back to insights from Chomsky (1965: 14) about the unacceptability of self-embedding (quote from Boeckx 2012: 122):

[W]e might assume, for example, that the perceptual device has a stock of analytic procedures available to it, one corresponding to each kind of phrase, and that it is organized in such a way that it is unable (or finds it difficult) to utilize a procedure ϕ while it is in the course of executing ϕ .

And again, if ambiguity avoidance itself emerges from a deeper, domain-general principle that drives the grammar towards efficiency, then its pervasive role is ultimately predicted (see, e.g., Gibson et al. 2019; Pijpops et al. 2018; Pijpops and Zehentner 2022 for discussion). Indeed, independently of ‘how syntactic’ a given phenomenon is, if we assume the validity of the SMT, we are to expect that (Chomsky et al. 2023: 65):

[...] extra-syntactic factors [...] can have a significant effect in ostensibly syntactic phenomena. For islands, a range of different extra-syntactic principles could conceivably be relevant. Given islands’ characteristically complex empirical profile, progress depends on identifying the precise contribution of independent syntactic and extra-syntactic mechanisms, especially where judgments are marginal.

To conclude, these observations suggest that locality effects, however distinct they may appear on the surface, are deeply intertwined with the pressure to minimise ambiguity. Within the Interface Hypothesis, this pressure provides a unifying explanation for why intervention and impenetrability arise, why and how their surface effects differ, and what factors are predicted to support their amelioration. In Section 6.5, I return to this point and address the main research questions of this thesis more directly. Before doing so, I briefly consider some broader implications of the Interface Hypothesis and outline possible directions for extending, refining, or challenging the proposal in future research.

6.4 Consequences and prospects of the Interface Hypothesis

An obvious way to extend the hypothesis developed in this thesis is to test its coverage across a broader range of empirical domains and evaluate it against both existing and novel experimental data. Ideally, further evidence should support the core predictions emerging from Chapters 4 and 5. For instance, we would expect to find that WIs cannot be fully reduced to processing difficulties independent of abstract grammatical representations;

that SIs and WIs remain sensitive to distinct amelioration strategies; or that both arise whenever the underlying structures involve ambiguous configurations of some sort.

Nevertheless, this does not necessarily imply that *all* locality phenomena must be explained as emergent constraints within the Interface space. It remains possible (and, indeed, likely) that some phenomena are better analysed as purely derivational or as fully grammar/syntax-independent (i.e., falling beyond the left or right edge of the Interface spectrum). This does not undermine the Interface Hypothesis itself, however. Within Minimalism, the guiding principle should be to maximise explanatory coverage with ‘minimal means:’ attempt to derive as many locality phenomena as possible from independently motivated, interface-driven principles, and reserve ad-hoc explanations only for a restricted subset of genuinely exceptional cases. Under this approach, even if certain locality effects resist integration into an Interface-based model, its explanatory gain remains, I believe, significant. The Interface Hypothesis avoids unnecessary stipulations by grounding locality effects in principles of domain-general computational economy as applied to language-specific categories. As such, it should be preferred as the ‘default approach,’ so to speak, while phenomena that cannot be unified under it can be addressed separately. Once again, this perspective follows naturally from the logic of explanatory adequacy within the SMT. As Chomsky (2021: 10) puts it:

Conformity to SMT — hence genuine explanation — is achieved when an account relies on “third factor” principles such as computational efficiency, understood in this context as natural law. These can be taken off the shelf when needed, with no cost.

In short, the Interface Hypothesis offers a principled unification of diverse locality effects by treating them as the product of interactions between independently motivated grammatical properties and domain-general constraints. It thus provides, I believe, a robust explanatory baseline, capable of accommodating exceptions without compromising the broader architecture. Following Chomsky, we might say that an interface-based explanation of locality can be “taken off the shelf” at no theoretical cost, while more ‘costly’ explanations should be invoked only when clear empirical evidence demands them (see also Dillon and Hornstein 2013; Presotto 2024b).

That said, I believe the Interface Hypothesis offers advantages not only from an epistemological perspective, but also in opening a new interpretative window on some long-standing issues raised by locality effects. I briefly explore two such examples in the following paragraphs.

6.4.1 Language variation and language change

When I reviewed the main families of locality accounts in Chapter 2, I showed that each has been challenged by several issues. Among them, what appears particularly thorny is the question of cross-linguistic variation in locality effects. From a reductionist perspective, it is not clear why languages should differ in what movements are licit and which are banned. If movement were constrained exclusively by domain-general cognitive limitations, and assuming speakers across languages share the same cognitive system, it is difficult to explain why cross-linguistic variation in (the intensity of) island effects should arise at all (see Section 2.2.3). Similar concerns arise if we consider grammar-based approaches: although variation can always be accounted for by positing language-specific parameters (Section 2.1.3), this is clearly not ideal under the minimalist agenda (see above and Section 3.1.1).

However, in discussing how cross-linguistic differences impact resource-based approaches, Hofmeister et al. (2013: 60) claim that:

[t]here is no reason to assume that the cost of, say, forming a dependency between a *wh*-word and an element inside a *wh*-island is the same from language to language. One language may utilize features and cues to aid retrieval that another does not, e.g. resumptives.

They then go on to consider the case of Superiority violations (e.g., **I wonder what who saw*, now explained in terms of MLC/RM, see Chapter 2), and how these are analysed by Arnon et al. (2006). According to Arnon et al., the severity of Superiority effects varies as a function of the availability of case cues. For instance, Superiority effects are larger in English than in German, while apparently entirely absent in Russian (but see Häussler et al. 2015 for partially different results). Arnon et al. link this pattern to the increasingly richer morphological paradigms of German and Russian compared to English, and conclude that there is a universal difficulty associated with processing of these structures, but “the extent of the difficulty will depend on speakers’ ability to draw on additional information” (Arnon et al. 2006, quoted in Hofmeister et al. 2013: 60). It is unclear, however, how to properly formalise this intuition within a purely reductionist approach. As Hofmeister et al. stress, “it remains a challenge for processing accounts to identify what information varies across languages along with island sensitivity” (Hofmeister et al. 2013: 60).

I do not see this as an insurmountable problem. It seems relatively uncontroversial that languages that have (more) cues to efficiently parse a computationally costly structure will display fewer and/or less severe locality effects. This is the intuition at the base of, e.g., Pritchett (1991) or Hawkins (1999) (see Section 2.3). It is also the rationale underlying

the generalisation on amelioration effects in RCs presented in Section 4.2: whenever identification of θ -roles is supported by morphosyntactic cues, locality effects are alleviated. The deeper issue, I believe, lies at the conceptual level. Within a purely processing-based account, why should the abstract grammatical function of a cue be so critical? As discussed extensively in Chapter 4, these cues cannot be superficial or (just) heuristic: their underlying function must be *acquired* in order to support the parser and mitigate locality effects. This is naturally predicted under the Interface Hypothesis, where the parser directly exploits abstract grammatical representations, but it is far less straightforward under a purely resource-based model (see Section 4.3.7).

In sum, what appears problematic for resource-based approaches follows naturally from the Interface Hypothesis. Cross-linguistic variation in, e.g., intervention effects is straightforwardly predicted, since these arise from the interaction of (i) domain-general constraints on the parser *and* (ii) language-specific morphosyntactic properties. Another advantage of the Interface Hypothesis is that, unlike purely grammar-based accounts, it does not have to resort to ad-hoc parametrisation: variation in the strength or absence of intervention effects is expected whenever the parser can exploit the language-specific ‘toolkit’ available to it. Ultimately, this reduces to the broader principle of ambiguity avoidance discussed above, and to how this principle is shaped within the Interface space, where grammatical representations and third-factor constraints are assumed to interact dynamically.

Interestingly, this logic is not limited to intervention effects and WIs. If the account advanced in Section 6.2.4 is on the right track, then some degree of cross-linguistic variation is also expected in subject opacity effects, though to a lesser extent. Let me clarify. As discussed, both impenetrability and intervention can be connected, loosely speaking, to ‘garden-path effects,’ which arise when the external systems are confronted with ambiguous configurations. Since languages vary considerably in the grammatical properties that can facilitate θ -role identification, a substantial amount of cross-linguistic variation in the strength of these effects is expected (Berwick and Weinberg 1984; Pritchett 1991, 1992; Berwick et al. 1992; Hawkins 1999, 2004, 2014, 2022; Polinsky et al. 2012; Lau and Tanaka 2021; Levshina 2021; Presotto and Torregrossa 2024). With respect to subject opacity, however, I argued that its emergence crucially depends on two ingredients: EPP-driven movement to Spec,TP and the possibility of sub-extraction. Both are highly abstract grammatical properties that serve deep derivational purposes and, as such, appear to characterise a large number of languages. Nevertheless, if a language lacks obligatory EPP movement, or if its grammar disallows splitting movement chains via sub-extraction, the prediction is that subject opacity effects will not arise.⁵⁰

⁵⁰It has been claimed that Goidelic languages lack an obligatory EPP feature on T, or at least behave

Crucially, this kind of variation does not require positing language-specific parameters on locality constraints. Instead, it follows directly from differences in the functional lexicon of individual languages. If a language lacks an EPP feature or the relevant features responsible for attracting specific sub-parts of a constituent, the structural conditions giving rise to subject opacity are simply not met. In this way, cross-linguistic variation in subject opacity naturally reduces to the featural make-up of a language's lexicon, fully in line with the Borer–Chomsky Conjecture (Borer 1984; Chomsky 1995; Baker 2008).

Speaking of conjectures, I would now like to pursue one myself, albeit briefly. In reviewing the main approaches to locality in Chapter 2, I dedicated a section to so-called grounded approaches. In that context, I emphasised that these accounts offer an interesting perspective by bringing together insights from both processing-based and grammar-based theories (see Section 2.3). In this sense, the Interface Hypothesis is not, on balance, so distant from grounded proposals: both assign a central role to the interplay between grammatical representations and cognitive constraints. But whereas grounded theories propose that grammar adapts to the needs of sentence processing (Berwick and Weinberg 1984; Hawkins 1999, 2004), the mechanisms just described define a sort of inverse relation: the strength of locality effects is modulated by the different shapes that human grammars can take. These perspectives, however, need not be in conflict. More plausibly, they represent two sides of the same coin, or perhaps even two stages in the ‘evolution’ of island constraints.

If islands are indeed “dynamic, emergent entities” (see above), it seems reasonable to analyse them as the evolving outcome of a continuous interaction between cognitive pressures and grammatical representations. Possibly, even the different strength of locality constraints might in fact reduce to the distinction between different ‘snapshots’ of such dynamic evolution. As I noted in Section 2.3.1, Hawkins (1999) suggests that “[g]rammatical conventions are ‘frozen’ processing preferences.” Pursuing this insight further, we could entertain the idea that certain SIs may, descriptively at least, reflect ‘frozen’ computational strategies. Take adjuncts, for example. As we have seen, many accounts converge on the view that these are opaque because they are built in a separate WS and undergo early Spell-Out, making them inaccessible to further derivational steps. It is conceivable that such early Spell-Out mechanisms originated as some kind of processing optimisation: given that merging two open WSs imposes excessive demands on the external systems, grammars may have ‘frozen’ early Spell-Out as a default imposition on the Merge algorithm. If so, the depth at which such frozen strategies become embedded within the grammar architecture could correlate with the strength of an island

as though the EPP is inactive in the same way as in English or other languages (see, a.o., Carnie 1995; McCloskey 1996, 2001 for discussion). These languages may offer an interesting testing ground for the hypothesis developed here.

constraint: the more abstract and root-level the constraint, the fewer options remain to circumvent it.

I leave these remarks as sheer conjectures. My aim here was simply to illustrate that the insights of grounded theories are not only compatible with the Interface Hypothesis but can even be expected if grammars are directly shaped by the ongoing interplay between interface pressures and language-specific factors.

6.5 Taking stock: the source of locality

I began this thesis by raising several long-standing questions that have shaped the theoretical debate on locality over the years. The first obvious question is: why do locality phenomena exist in the first place? There is broad consensus that locality reflects the grammar's response to a need for efficient computation, avoiding overly costly (i.e., non-local) relations among elements involved in dependencies. Less clear, however, is why this general principle manifests itself in so many different ways, both within and across languages. Numerous accounts have attempted to explain these issues, ultimately locating the source of locality effects in different places. It is precisely this theoretical contrast that informed the core research questions of this dissertation: what is the source of locality? Is it uniform? And why have so many, and so different, accounts been able to explain locality effects only partially? Why does locality remain so 'elusive' and variable?

The findings and arguments developed in this thesis converge on a simple but, I believe, far-reaching conclusion: the source of locality lies at the Interface. Locality effects emerge from the interaction between grammar-internal representations and domain-general pressures, but the balance of these factors varies along a continuum. Some phenomena, such as intervention-driven WIs and the comparative difficulty of ORCs, are shaped primarily by processing constraints that dominate at the right edge of the Interface. These constraints privilege shorter, more 'local' parses of the relevant abstract representations, but their effects can be mitigated when morphosyntactic cues reduce ambiguity and force the parser to avoid misparses. By contrast, impenetrability locality as observed in SIs reflects restrictions arising at the left edge of the Interface: PF and LF cannot simultaneously compute ambiguous or misleading syntactic structures.

This perspective also explains why so many seemingly incompatible accounts have been partially successful. The factors emphasised by interface-based grammatical theories, semantico-pragmatic approaches, and *principled* processing accounts are all naturally integrated within the Interface model. Crucially, especially towards the right end of the Interface, any factor that can hinder or facilitate the parser is expected to play a role, be they morphological markers, word-order properties (Chapter 4), or context-driven struc-

tural cues (Chapter 5).

At a more conceptual level, we could say that the Interface Hypothesis subsumes the insights of the three main approaches reviewed in Chapter 2. Like grammar-based accounts, it assumes deep, abstract grammatical representations; like reductionist approaches, it incorporates the pressure imposed by domain-general processing constraints. Finally, it resonates with grounded approaches by suggesting that a single, unified principle of ambiguity avoidance (or, more generally, computational efficiency) pervades the grammatical architecture. On this view, some locality constraints may arise as reflexes of processing preferences, progressively extending to increasingly abstract features at deeper levels of representation, thereby resembling ‘frozen’ rules and yielding stronger effects.

At the same time, locating the source of locality at the Interface has strong predictions: it excludes purely derivational accounts that derive locality exclusively within narrow syntax, as well as purely processing-based approaches that rely on superficial heuristics disconnected from abstract grammatical representations.

In conclusion, the Interface Hypothesis provides a principled resolution to the puzzle raised in the Introduction: a unified account has remained elusive not because locality lacks a common source, but because its source is inherently composite, distributed across the Interface between grammatical representations and the cognitive systems that interpret them.

Conclusions

In this thesis, I have examined the nature and sources of locality effects through a combination of theoretical analysis and experimental evidence. I have argued that locality originates at the Interface: a composite source where different locality effects distribute across the interactional space between grammar and the cognitive systems that interpret it. The end of Chapter 6 already provides a detailed summary of the findings and analyses supporting this hypothesis, together with its implications and possible future developments. Rather than repeating them here, I use this concluding section to offer two final considerations.

First, the proposals advanced here should not be seen as definitive. I rather intend this work as a first step towards reconciling the vast literature on locality with the programmatic ambition of linguistic minimalism to achieve a minimally specified grammatical system. Concretely, I have sought to ground the Interface Hypothesis in clear empirical evidence, using experimental results to motivate a framework that (a) situates locality effects within a plausible minimalist architecture and (b) accounts for specific phenomena, such as weak islands, intervention effects in A-bar dependencies, and the opacity patterns traditionally captured by the CED. Since many other central aspects of locality effects could not be addressed, I see this work as a starting point rather than an endpoint.

Second, the Interface Hypothesis has the welcome result of integrating insights from diverse accounts, tracing a ‘red thread’ through grammar-based, reductionist, and grounded approaches to locality. That said, it is not an all-embracing framework into which all theories can be assimilated. As emphasised throughout Chapters 3–6, it has clear theoretical consequences, substantially limiting the explanatory scope of purely resource-based and derivational accounts. More generally, I have tried as much as possible to move beyond sheer descriptive adequacy, formulating explanations that generate clear, falsifiable predictions. Whether these ideas ultimately point in the right direction for understanding

locality remains an open question, one that future research will need to settle by testing these predictions. Further examination of the Interface Hypothesis across diverse empirical domains will be essential to refine both its scope and its explanatory power.

Appendix

The experimental materials and data are available in the Open Science Framework (OSF) repositories linked below:

- **Materials for Experiments 1–4**

(click or scan the QR code)



- **Materials for Experiments 5–6**

(click or scan the QR code)



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